

QIAGEN

QIAstat-Dx

Global Product Support QIAstat-Dx Service Manual SM-5000-0035-002, QFS, QTS, QSTAT



Service Manual

SM-5000-0035-002, QFS, QTS, QSTAT

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1 Introduction

1.1 General Information

QIAGEN Global Product support (GPS) is responsible for the creation and maintenance of Instrument Service documentation. It is the policy of QIAGEN to improve products as new techniques and components become available. QIAGEN reserves the right to change specifications at any time. In an effort to produce useful and appropriate documentation, we welcome your comments and suggestions on this publication.

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INFORMATION 	If an error is found in this Service Manual please contact Global Product Support immediately (see contact details above).
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1.2 QIAstat-Dx Service Manual Overview

This Service Manual represents an integral part of the QIAGEN QIAstat-Dx and DiagCORE. All information and directives also apply to DigCORE analyzers. This manual is intended for the service and repair of general IVD 98/79/EC products. This service manual provides detailed information regarding the operation, service and testing of QIAstat-Dx and DiagCORE analyzers. The instructions contained in the manual regarding operation, service and the test set-up are to be strictly observed. Pay particular attention to any safety information. The information is presented in the following chapters:

1. Introduction (this chapter)
2. Hazards and Precautions
3. General Information
4. Installation
5. QIAstat-Dx Modules
6. Replacement Procedures
7. Calibration Procedures (CP)
8. Verification Procedures (VP)
9. Annual Inspection
10. Software and Software Tools
11. Troubleshooting
12. Spare Parts
13. Electronic Schematics
14. ISI Notes

1.3 Intended Use

1.3.1 QIAstat-Dx Expected Use

The QIAstat-Dx is intended to be used only in combination with QIAGEN kits indicated for use with QIAstat-Dx for the applications described in the kit handbooks.

If the product is used with kits other than QIAGEN kits, it is the user's responsibility to validate the performance of such product combination for any particular application.

The QIAstat-Dx Analyzer is intended for in vitro diagnostic use and provides diagnostic results. All analytical steps are fully automated using molecular diagnostic QIAstat-Dx cartridges and realtime PCR detection. The QIAstat-Dx Analyzer system is intended for professional use only and is not intended for self-testing.

1.3.2 Requirements for working with the QIAstat-Dx

This section covers the general level of competence and training necessary for installation, use, daily maintenance, cleaning, preventive maintenance and servicing of the product. Appropriate training for all levels can be arranged with QIAGEN.

Task	Personnel	Knowledge
Delivery	No special requirements	No special requirements
Installation	Customer and QIAGEN authorized personnel.	The analyzer can be installed by the customer but the IQOQ must be performed by QIAGEN certified representatives.
Routine Use (running protocols)	Laboratory technicians or equivalent	Appropriately trained and experienced personnel, familiar with use of lab equipment and computers
Routine Maintenance	Laboratory technicians or equivalent	Performed in accordance with HB-2636-001, Chapter 8.
Servicing	QIAGEN authorized personnel.	QIAGEN representatives certified to service the QIAstat-Dx.
Annual Inspection	QIAGEN authorized personnel.	QIAGEN representatives certified to perform the annual inspection (AI).

Table 1.3-1 Requirements for personnel

2 Hazards and Precautions

2.1 Introduction

Before servicing the QIAstat-Dx, it is essential that you read this Service Manual as well as the current User Manual carefully and pay particular attention to the safety information. The handbooks contain information and warnings that must be followed by the user and service engineer to ensure safe operation of the instrument and to maintain the instrument in a safe condition.

Possible hazards that could harm the user or result in damage to the instrument are clearly stated at the appropriate places throughout this manual and HB-2636-001, Chapter 2.

The following safety conventions will be used throughout this manual:

WARNING 	The term WARNING is used to inform you about situation that could result in personal injury to you or other persons. Details about these circumstances are given in a box like this one. w0
CAUTION 	The term CAUTION is used to inform you about situations that could result in damage to the instrument or other equipment. Details about these circumstances are given in a box like this one. c0
INFORMATION 	The term INFORMATION is used to inform you about hints and tricks. Details about these circumstances are given in a box like this one.

The advice given in this handbook is intended to supplement, not supersede, the normal safety requirements prevailing in the user's country.

2.2 Safety

WARNING 	You must observe all safety requirements when installing and operating the product. Failure to install or use the system in a manner specified by QIAGEN may result in personal injury and / or equipment damage. w1
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The product shall be installed in accordance with the *Installation Guideline detailed* in chapter 4 of this service manual and HB-2636-001, Chapter 2.

The mains plugs shall be easily accessible in case of the need of quickly disconnecting the equipment from the mains power.

The product shall be used in accordance with the instructions provided through dialogs appearing on the Operator's PC screen, the manuals, instructions for use, and by QIAGEN Technical Service, and within limits set by the technical specifications.

Follow all generally accepted laboratory safety procedures and applicable laws and regulations. Use appropriate safety precautions when handling biological material.

2.3 Proper Use

WARNING 	RISK OF PERSONAL INJURY AND MATERIAL DAMAGE w2 Improper use of the QIAstat-Dx may cause personal injuries or damage to the system. The QIAstat-Dx must only be operated by qualified personnel who have been appropriately trained. Servicing of the QIAstat-Dx must only be performed by QIAGEN Field Service Specialists.
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- Use only QIAGEN components, otherwise your right to make a claim under the guarantee may be invalidated.
- Perform the maintenance as described in Section 8 of HB-2636-001.
- QIAGEN will charge for repairs that are required due to incorrect maintenance, negligence, or misuse.

WARNING 	RISK OF PERSONAL INJURY AND MATERIAL DAMAGE w3 The QIAstat-Dx is too heavy to be lifted by one person. To avoid personal injury or damage to the instrument, do not lift the instrument alone.
---	--

WARNING 	RISK OF PERSONAL INJURY AND MATERIAL DAMAGE w4 Do Not attempt to move the QIAstat-Dx during operation.
--	--

2.4 Electrical Safety

Disconnect the line power cords from the power outlet before servicing.

WARNING 	ELECTRICAL HAZARD w5 Disconnect the mains power cords from the power outlet before servicing. Lethal voltages inside the instrument When the instrument is connected to mains power, terminals may be live, and opening covers or removing parts is likely to expose live parts.
---	---

WARNING 	ELECTRICAL HAZARD w6 Any interruption of the protective conductor (earth/ground lead) inside or outside the instrument or disconnection of the protective conductor terminal is likely to make the instrument dangerous.
---	--

To ensure satisfactory and safe operation of the QIAstat-Dx, follow the advice below:

- Make sure the line power cord is connected to a line power outlet that has a protective conductor (earth/ground).

INFORMATION 	The ground connector on the mains plugs must not be removed and the plugs should only be connected to grounded outlets as per local and national regulations. Use of power supplies that disable the grounding connections to the instrument is dangerous and prohibited.
---	--

- Keep mains plugs easily accessible in case the equipment needs to be disconnected quickly from the mains power.
- Use only power supplies and cords supplied with the system.

- Power supplies and cords should be inspected periodically and replaced if damaged or have unauthorized alterations.
- If water or reagent has spilled inside the instrument, switch off the instrument and disconnect it from the line power supply. Contact Global Product Support.

INFORMATION 	Note that the power switch does not disconnect the equipment completely from mains power. The mains plug must be removed from the power source to ensure power is completely disconnected.
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- Servicing must be performed by trained QIAGEN service engineers.
- If the instrument becomes electrically unsafe for use, make the instrument inoperative by removing the mains power cord and secure it against unauthorized or unintentional operation. Contact Global Product Support.

The instrument is likely to be electrically unsafe when it:

- Shows visible damage.
- The line power cord shows signs of damage.
- Has been stored inappropriately for a prolonged period of time.
- Has been subjected to severe transportation stresses.

2.5 Biological Safety

When handling biological material, use safe laboratory procedures as outlined in publications such as Biosafety in Microbiology and Microbiology and Biomedical Laboratories (HHS Publication Number CDC 88-8395) (www.cdc.gov/od/ohs/biosfty/biosfty.htm).

WARNING 	Biological materials w7 Handle biological material with the greatest of care and in accordance with the required safety regulations. Always wear the appropriate personal protective equipment which may include safety glasses, gloves, and a lab coat; always follow the lab's established safety protocols. The responsible body (e.g., laboratory manager) must take the necessary precautions to ensure that the surrounding workplace is safe, and that the operators are suitably trained and not exposed to hazardous levels of infectious agents as defined in the applicable Material Safety Data Sheets (MSDSs) or OSHA*, ACGIH†, or COSHH‡ documents. For more information, visit: www.qiagen.com/support/msds.aspx. Venting for fumes and disposal of wastes must be in accordance with all national, state, and local health and safety regulations and laws.
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* OSHA: Occupational Safety and Health Administration (United States of America).

† ACGIH: American Conference of Government Industrial Hygienists (United States of America).

‡ COSHH: Control of Substances Hazardous to Health (United Kingdom).

2.6 Chemical Safety

WARNING 	Hazardous chemicals w8 The specimen cartridge contains reagents proper personal protection; lab coat, nitrile gloves, and eye protection should be worn in case of leakage from the cartridge. The responsible body (e.g., laboratory manager) must take the necessary precautions to ensure that the surrounding workplace is safe, and that the operators are suitably trained and not exposed to hazardous levels of infectious agents as defined in the applicable Material Safety Data Sheets (MSDSs) or OSHA*, ACGIH†, or COSHH‡ documents. For more information, visit: www.qiagen.com/support/msds.aspx. Venting for fumes and disposal of wastes must be in accordance with all national, state, and local health and safety regulations and laws.
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* OSHA: Occupational Safety and Health Administration (United States of America).

† ACGIH: American Conference of Government Industrial Hygienists (United States of America).

‡ COSHH: Control of Substances Hazardous to Health (United Kingdom).

2.7 Mechanical Hazards

The door of the QIAstat-Dx Instrument must remain closed during operation of the instrument.

WARNING 	Moving parts w9 To avoid contact with moving parts during operation of the QIAstat-Dx, the instrument must be operated with the lid closed. If there is a problem with the QIAstat-Dx, great care should be taken if the instrument is operated with covers off during service.
WARNING 	Pinch and impact hazards w10 The Analytical Module lid is motorized. Ensure fingers or objects are not in the path of the lid when it closes.

2.8 Heat Hazard

CAUTION 	Risk of overheating c1 To ensure proper ventilation, maintain a minimum clearance of 15 cm (5.9 in.) at the sides and rear of the QIAstat-Dx. Slits and openings that ensure the ventilation of the QIAstat-Dx must not be covered.
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2.9 Consumables

CAUTION 	Unsupported Consumables c2 Do not connect or use any consumable, accessories, or external equipment other than those specified in the QIAstat-Dx manuals.
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3 General Information

3.1 Technology - Principle

The QIAstat-Dx is designed to provide automated processing of human biological samples in QIAstat-Dx cartridges using real-time PCR to detect pathogen nucleic acids.

3.2 Product General Description

The QIAstat-Dx Analyzer consists of an Operational Module and up to 4 Analytical Modules. The Operational Module includes elements that provide connectivity to the Analytical Module and enable user interaction with the QIAstat-Dx Analyzer. The Analytical Module contains the hardware and firmware for sample testing and analysis.

3.3 Definitions

3.4 Instrument Names

The QIAstat-Dx and DiagCORE are the only models currently available.

3.5 Instrument Operation

Samples are inserted into a QIAstat-Dx assay cartridge that contains all reagents necessary to isolate and amplify nucleic acids from the sample. Detected real-time amplification signals are interpreted by the integrated software and are reported via an intuitive user interface.

3.6 Software Operation

The QIAstat-Dx operating software is resident in the Operational Module and is accessed via the touch screen with user interface software. The Analytical Module contains the firmware for processing sample cartridges. Service, calibration and diagnostics are accessed via the printer plug and CAN bus connector on the back of the Analytical Module independent of the Operational Module.

3.7 Instrument Overview

3.7.1 External features of the product

The Operational Module houses the touch screen, USB port and bar code reader; the Analytical Module external features include the magnetic screw covers and cartridge door on top.

3.7.2 Computer and Software

The QIAstat-Dx is shipped with one Operational Module and up to four Analytical Modules. The operational software is preloaded in the Operational module and cartridge handling firmware in the Analytical Module. Software updates to the Operational Module and firmware updates Analytical Module are installed via the USB port on the Operational Module using menu driven input on the touch screen. Service and diagnostics software are run on the Analytical Module using an external computer connected via a USB port or CAN BUS port on the rear of the module.

3.8 Instrument Modules

The QIAstat-Dx has two modules, Operational module (OM) and up to 4 Analytical Modules (AM).

The Operational Module provides the operator access for setup and input for processing sample cartridges. The module includes a touch screen for input and control, a barcode reader to identify cartridges for processing, LAN connector for a LIMS connection, and a USB port for data transfer. The power switch is for starting and shutting down the analyzer software. Internally the OM consists of two boards: the Mamba Lizard Board for the touch screen and display and the Gaudi Gecko Board for data processing, the bar code reader, USB port and power.

The AM, controlled by the OM, is a self contained PCR processing platform. Each analyzer can have up to four Analytical Modules connected by a bridge assembly in a “Daisy Chain” configuration on the using bridge connections on the back of the analyzer. Internally there are several major modules and assemblies to process the sample cartridge: TC and Clamping module to hold the cartridge and position the cartridge piston, Bead Beater to mix the reagent beads in the cartridge, Thermal Unit for thermo-cycling the cartridge, Voice Coil for mixing, Pressure Pump, qPCR Module, Syringe Module for vacuum, and a power supply,

3.8.1 Design Parts (Covers)

The function of the covers is to protect the user from moving parts and energized circuits. There are no user serviceable parts inside the covers. Covers will only be removed by authorized service technicians.

3.8.2 Modules

Operational Module

The Operational Module provides user control and informational interface with the Analytical Module(s). It consists of a touch screen, bar code reader, USB ports and power switch

Analytical Module

Controlled by the Operational Module, the Analytical Module contains all the firmware and components for handling and processing the sample cartridges. Sample cartridges are inserted and extracted through the top door. Only approved cartridges should be inserted in the analyzer; use of other cartridges could cause damage to the analyzer or a hazard to the operator/user.

3.8.3 External Personal Computer

External computer is only used for diagnostics and testing only.

Component	Specification
Processor	Intel Core Duo, 2 GHz (or higher)
Operating System	Windows 10 (or higher)
RAM	1 GB
Free hard disk space (at least)	1 GB
Graphics card	Supporting the resolution of the monitor
Monitor	1280 x 800
Ethernet Port	1 x Instrument, 1 x LAN
Input Device	Keyboard
Pointer device	Mouse or similar
Interfaces	USB port

Table 3.8-1 Computer Specifications

3.8.4 Product Specification

Introduction	QIAstat-Dx
Short description	The product is a self contained fully automated with real-time PCR detection.

Table 3.8-2

Technical data	QIAstat-Dx
Regulatory	Electromagnetic Compatibility Directive IEC 61326 Class A
Ambient conditions	- Ambient air temperature range: 20°C-68°F / 25°C-77°F - Air humidity range: 20-80%rH, non-condensing - Maximum Altitude: 2200 m/ ft. - Position away from air conditioning and heating vents or exchangers - Keep at least 10 cm clearance on each side for adequate ventilation - Flat horizontal surface with no angles or tilts - Keep instrument out of direct sunlight
Instrument dimensions	Operational Module - Width: 234 mm / 9.3 Inches - Height: 326 mm / 12.0 Inches - Depth 517mm / 23.5 Inches - Weight: 5 kg / 11.0 lbs Analytical Module - Width: 153 mm / 6.1 inches - Height: 307 mm / 12.2 inches - Depth: 428 mm / 16.5 inches - Weight: 16 kg / 35.5 lbs
Capacity	Each Analytical Module can process one cartridge
System requirements	QIAstat-Dx
Power requirements	90-264 VAC, 50/60 Hz - Power consumption: Maximum 8 Amps
Software	The QIAstat-Dx is shipped with the software loaded internally, analytical protocols are loaded via the USB port on the Operational Module
Clearance Space	QIAstat-Dx installed - Sides 155 mm / 6.5 inches - Above: 336 mm / 13.5 inches - Behind: 203 mm / 8.0 inches

Table 3.8-2

3.9 Tools

This section will describe the standard and special tools necessary for working with the QIAstat-Dx.

3.9.1 Standard Tools

Standard tools are defined as tools that QIAGEN expects all trained engineers to have on hand during each service visit.

Standard Tools	Comment	Picture
Angled TORX Keys	T6, T10, T15, T20	
TORX Driver	T6, T10, T15, T20	
Angled Hex Key	2.0 mm	
Adjustable wrench	4.5/ 13mm	
Needle nose pliers	medium size	
Side cutter	standard	
Cable tie(s)	small/ medium/ large	
Flat Head Driver	Small / Medium	

Table 3.9-1 Tools

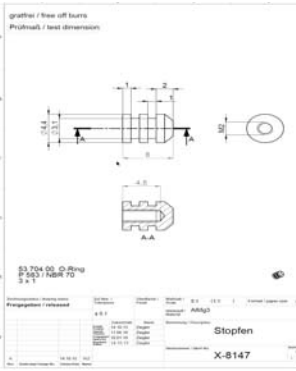
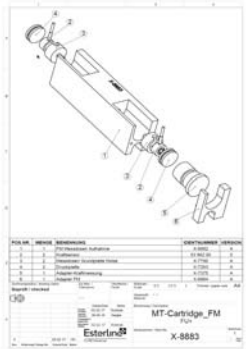
Standard Tools	Comment	Picture
Digital Multimeter	Calibrated	
10.0 mm Socket/Wrench	Standard	
Torque Wrench 4.0 mn	Calibrated	
Dark Cloth measuring approximately 1 meter X1 meter	Procured locally	

Table 3.9-1 Tools

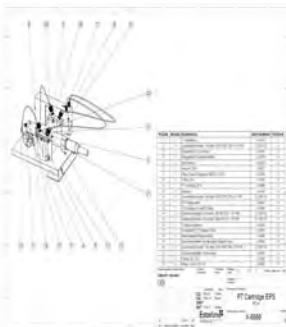
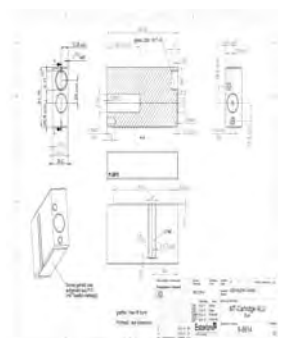
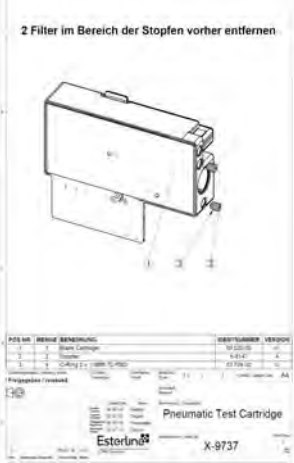
3.9.2 Special Tools

SAP #	Name	Comment	Picture
9026426	Test Mean qPCR Align2 (Cal)	Do not drop cartridge	
9026427	Test Mean qPCR_Align1 (Cal)	Do not drop cartridge	

Tab. 3.9-2 Special tools.

SAP #	Name	Comment	Picture
9026428	Plug , AM ,QSTAT		 graphical / free off hours Profilmaß / best dimension S3 70x400, G-Ring P 353 / KBR 70 1:1 Stopfen X-8147
9026429	Measuring box adapter, AM, QSTAT	Fixture for clamp pressure testing	 MT-Cartridge, FM Esterlin X-8883
9026430	Test Mean Liquid Empty		
9026431	Test Mean Liquid Filled		

Tab. 3.9-2 Special tools.

SAP #	Name	Comment	Picture
9026432	PT Cartridge EPS FU		
9026433	Cartridge End test/ Leak test		
9026434	MT-Cartridge		
9026435	Cartridge with contacts	Pneumatic test cartridge	

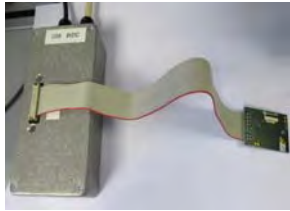
Tab. 3.9-2 Special tools.

SAP #	Name	Comment	Picture
9026436	Distance support angle		
9026437	Position Lifter		
9026438	Pressure plate	Part of the clamp test fixture	

Tab. 3.9-2 Special tools.

SAP #	Name	Comment	Picture
9026439	Adapter force measurement	Part of the clamp test fixture	
9026440	Baseplate force measurement	Part of the clamp pressure test fixture	
9026441	TC-Positioning tool	Test tool with micrometer for testing plunger distance into the cartridge	

Tab. 3.9-2 Special tools.

SAP #	Name	Comment	Picture
9026442	Fixture TU module test	Power distributor for TU Test	
9026443	USB-TC 01	USB Adapter for Temp measurement	
9026444	Fixture QPCR Optic Alignment	Power distributor	
9026445	Control unit TC1-TC2-CC-TU Lever	Power distributor	
9026446	Control device for valves		

Tab. 3.9-2 Special tools.

SAP #	Name	Comment	Picture
9026447	Load Cell, Force Measurement	Force measurement tool, USB connected	
902	Belt Tension Meter	For TC1 Belt Tension	

Tab. 3.9-2 Special tools.

4 Installation

4.1 System delivery and installation

The unpacking and installation of the product should be carried out by certified QIAGEN personnel. A person who is familiar with the laboratory and computer equipment should be present during the installation.

4.1.1 Product - Box 1 Packing List

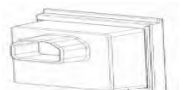
Parts			Quantity
Part #	Part	Picture	
9002814	QIAstat-Dx Analytical Module		1
110002	DiagCORE Analytical Module		1
9026745	Termination Bridge		1
9026562	Accessory Bag, QSTAT		1
90xxxxx	Power cord, US, Europe, Australia, UK. and Switzerland		1
9026744	Analytical/Analytical Module Bridge		1

Table 4.1-1 QIAstat-Dx Shipping Box 1 Contents

4.1.2 Product - Box 2 Packing List

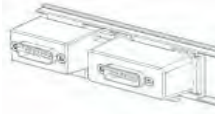
Parts			Quantity
Part #	Part	Picture	
9002813	QIAstat-Dx Operational Module		1
110001	DiagCORE Operational Module		1
9026743	Analytical/Operational Module Bridge		1

Table 4.1-2 QIAstat-Dx Shipping Box 2 Contents

4.2 Installation of the QIAstat-Dx

Refer to the QIAstat-Dx installation check sheet; ICS-5000-0013 or IQOQ-5000-0006.

4.2.1 Requirements

4.2.1.1 Site

The product is recommended be installed in a normal laboratory environment (20°C to 25°C) for optimal performance. Refer to [“Product Specification” on page 16](#) for more instrument specific information.

The product must be located out of direct sunlight, away from heat sources, and away from sources of vibration and electrical interference. The site of installation should also be level, immovable, dry, clean and free of excessive drafts, excessive moisture, excessive dust, and not subject to large temperature fluctuations. Ensure the surface is completely stable and check for level using a bubble level.

Approximately 336 mm (13.5 in.) clearance above the analyzer is required to accommodate the QIAstat-Dx with the lids open. Allow at least 203 mm (8.0 in.) of free space behind and to the sides of the instrument for cooling and cabling. Space, 155 mm (6.5 in.) is required to either side of the instrument.

The QIAstat-Dx must be placed within approximately 1.5 m (59 in.) of a properly grounded (earthed) AC power outlet. The power line to the product should be voltage regulated and surge protected.

INFORMATION 	It is recommended to plug the instrument directly into its own power outlet and not to share the power outlets with other lab equipment.
CAUTION 	Risk of overheating To ensure proper ventilation, maintain a minimum clearance of 15 cm (5.9 in.) at the sides and rear of the product. Slits and openings that ensure the ventilation of the product must not be covered. c2

4.2.1.2 Environment

INFORMATION

The QIAstat-Dx must be installed on a level, fixed immovable surface or the AM will generate tilt errors. Check for level using a bubble level. Refer to IQOQ-5000-0006

- Temperature: 15°C to 30°C
- Humidity: 20% to 80% relative humidity
- Draft-free location not close to window
- Analyzer must be located on a smooth, vibration free, level surface
- Keep the instrument out of direct sunlight

* Recommend that the instrument is run in normal room temperature (20°C to 25°C) for optimal performance

4.2.1.3 Power

The product operates at: 90–264 V AC, 50–60 Hz, current draw is dependant on the number of analytical modules installed:

1 Analytical Module; 264 VA, 2.4 Amps

2 Analytical Modules: 528 VA, 4.8 Amps

3 Analytical Modules: 729 VA, 6.7 Amps

4 Analytical Modules: 1056 VA, 9.6 Amps

Ensure that the voltage rating of the product is compatible with the AC voltage available at the installation site. Mains supply voltage fluctuations are not to exceed 10% of nominal supply voltages.

4.2.1.4 Grounding

To protect operating personnel, the product must be correctly grounded (earthed). To preserve this protection feature, do not operate the product from AC power outlet that has no ground (earth) connection.

4.2.2 Installation of the QIAstat-Dx Modules

4.2.2.1 Carefully unpack the QIAstat-Dx Analyzer.

1. Open the Analytical Module box and remove the accessory pack (A) applicable power cord (B), termination bridge (C), and AM to AM bridge (D).



Picture 4.2-1 Analytical box accessories.

2. Remove the Analytical Module from its box, remove the plastic bag, and peel off the protective plastic sheets.



Picture 4.2-2 Analytical module in the box.

3. Place the analytical module on the installation surface.
4. Remove the Operational Module from its box and remove the plastic bag and protective plastic sheets. The operational module will not stand alone and must be laid down or connected to the AM.



Picture 4.2-3 Operational module in the box.

4.2.2.2 Connect the Modules

1. Use the protective cover removal tool to remove the four magnetic protective covers from the sides of the analytical module. Insert the tool in the slot and twist.



Picture 4.2-4 Magnetic protective cover.

2. Set the Operational Module to left side of the Analytical module with the two modules touching and aligned together. Support the operational module so it does not fall over.



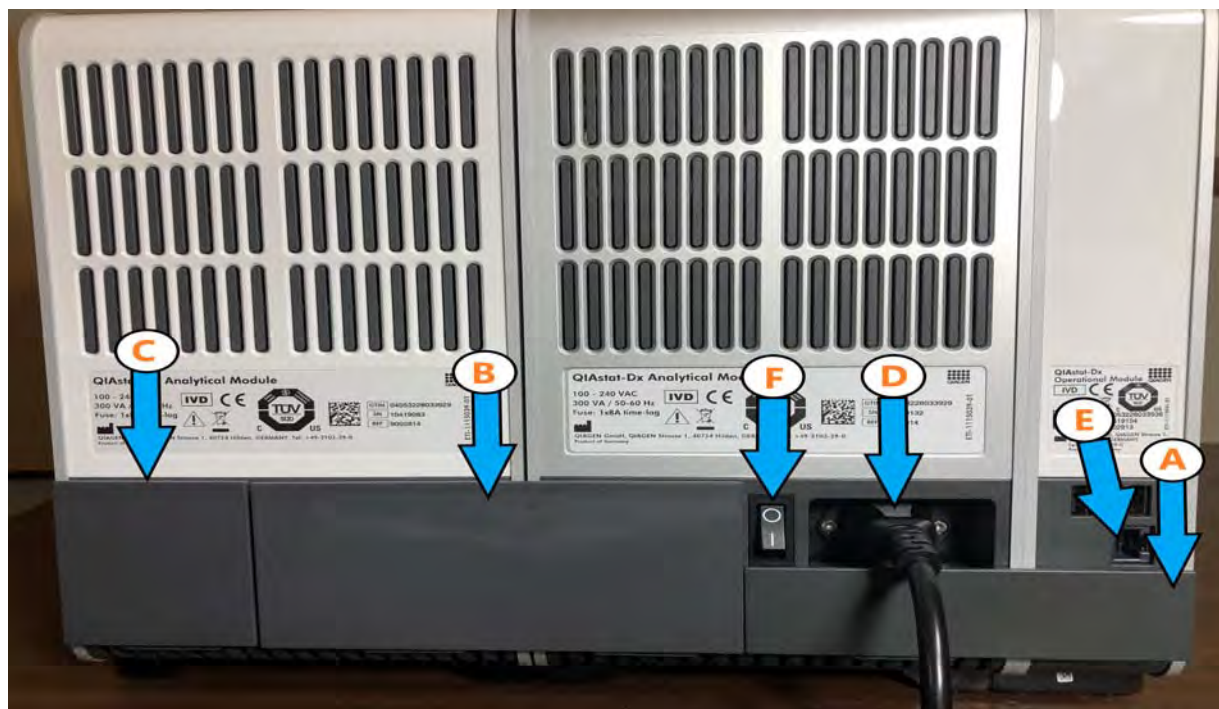
Picture 4.2-5 Operational module and analytical module placement.

3. Locate the two holes housing the recessed screws. Using the Analytical-Operational Module Assembly Tool carefully align and tighten the two screws to hold the modules together. Attach additional analytical modules if applicable. The screws only need to be snug, DO NOT apply excessive torque to the screws.



Picture 4.2-6 Module attachment tool.

4. Install the two magnetic protective covers on the Analytical Module right side.
5. Install the Operational/Analytical Module Bridge between the Operational Module and Analytical Module (A). Additional modules will be bridged together sequentially with the Analytical/Analytical Module Bridge (B).
6. Install the Termination Bridge on the last Analytical Module (C).

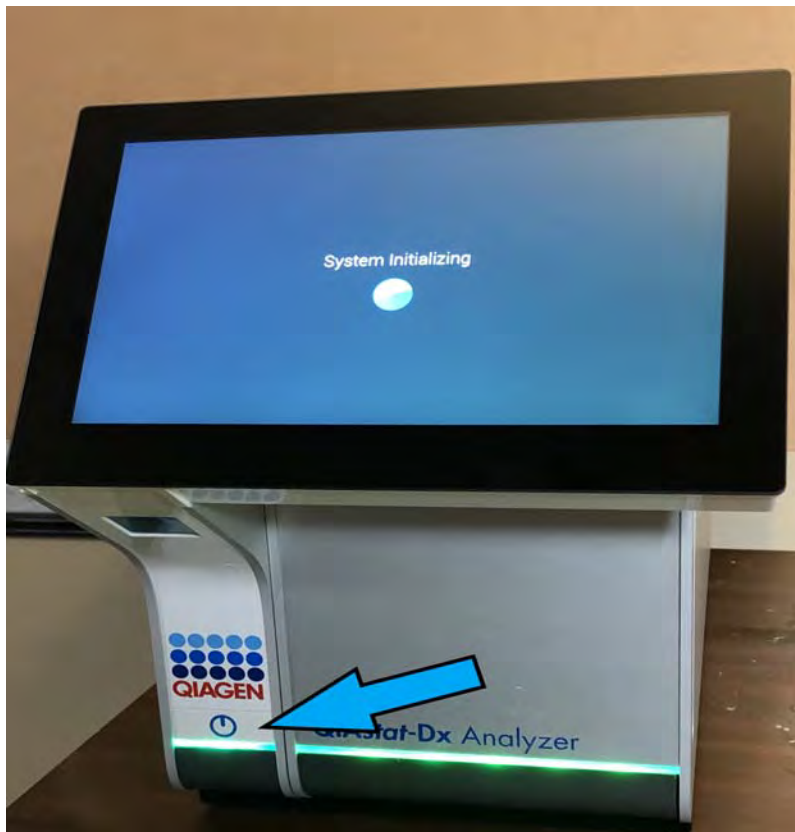


Picture 4.2-7 Module bridges.

7. Connect the power cord to the analytical module next to the operational module (D). This is the power connection for the entire analyzer regardless of the number of analytical modules; up to 4.
8. If the customer uses a LIMS system, connect the LAN cable to the Operational Module LAN port (E).

4.2.3 Analyzer Software Setup

9. Set the power switch on analytical module 1 to "ON" (F).
10. The green LED strip on the analytical module(s) will blink during initialization and self test; it change will to steady green when the analyzer is initialized and ready, approximately 2 minutes. The LED on the operational module will be blue. A red LED signals a fault condition and the module is not ready.
11. Once the analytical module is initialized press the start/power button on the OM, the LED will turn green.



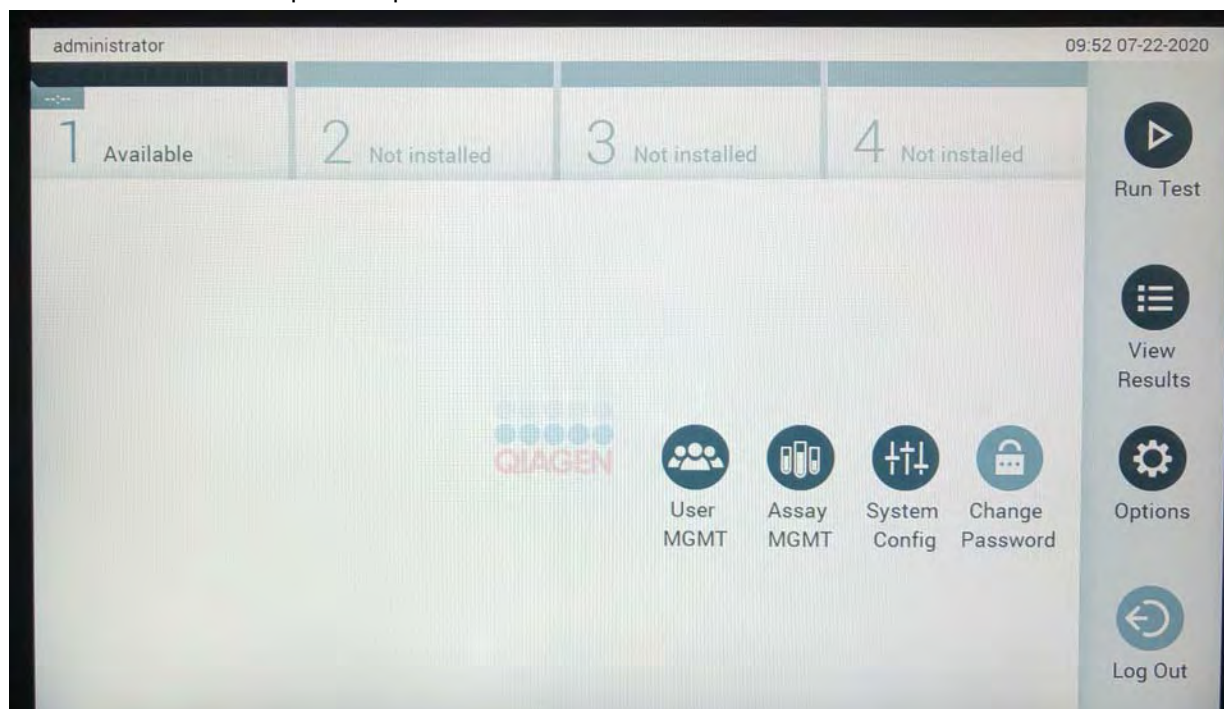
Picture 4.2-8 OM power button and status LEDs.

12. The analyzer will take about 5 minutes for system initialization, then the main screen will be displayed. Note the analytical module status on the top of the screen.



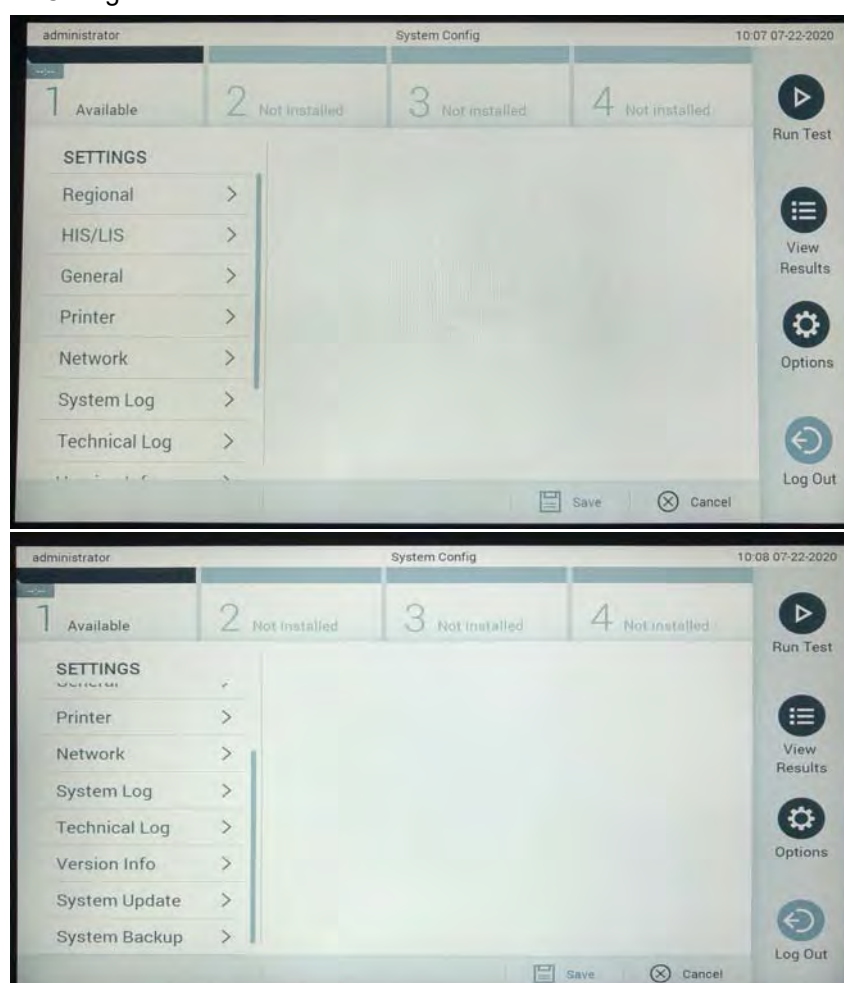
Picture 4.2-9 Main screen AM status

13. On the touch screen press “Options”.



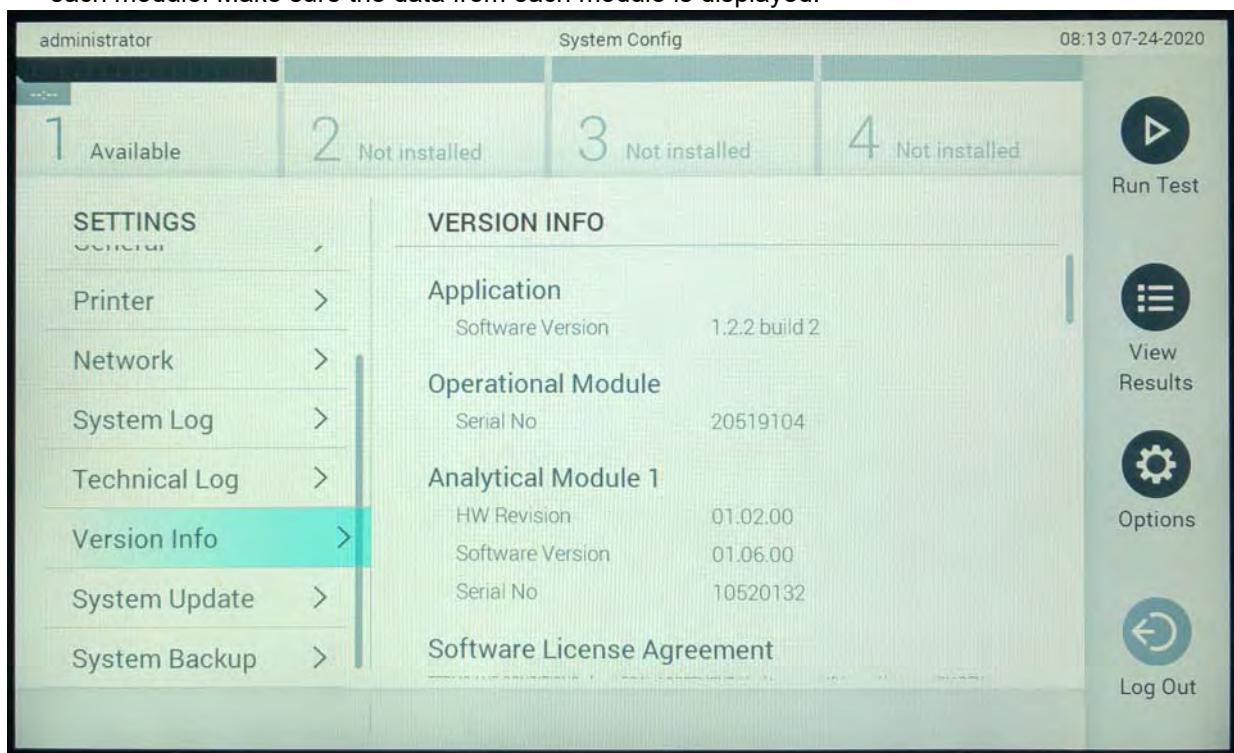
Picture 4.2-10 Options screen.

14. Select “System Config”.



Picture 4.2-11 System Config screen.

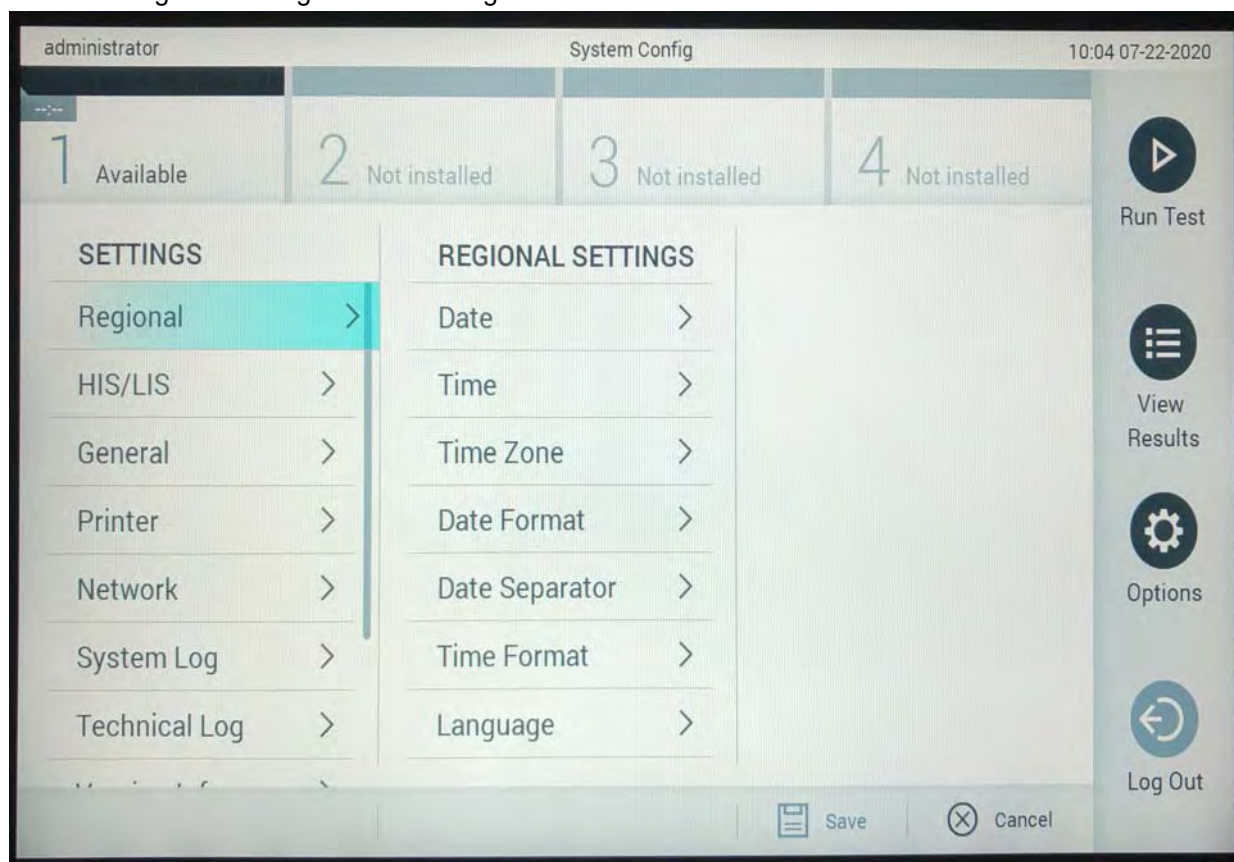
15. Select "Version Info". This displays the module data: serial number, software and firmware revision of each module. Make sure the data from each module is displayed.



Picture 4.2-12 Version info screen.

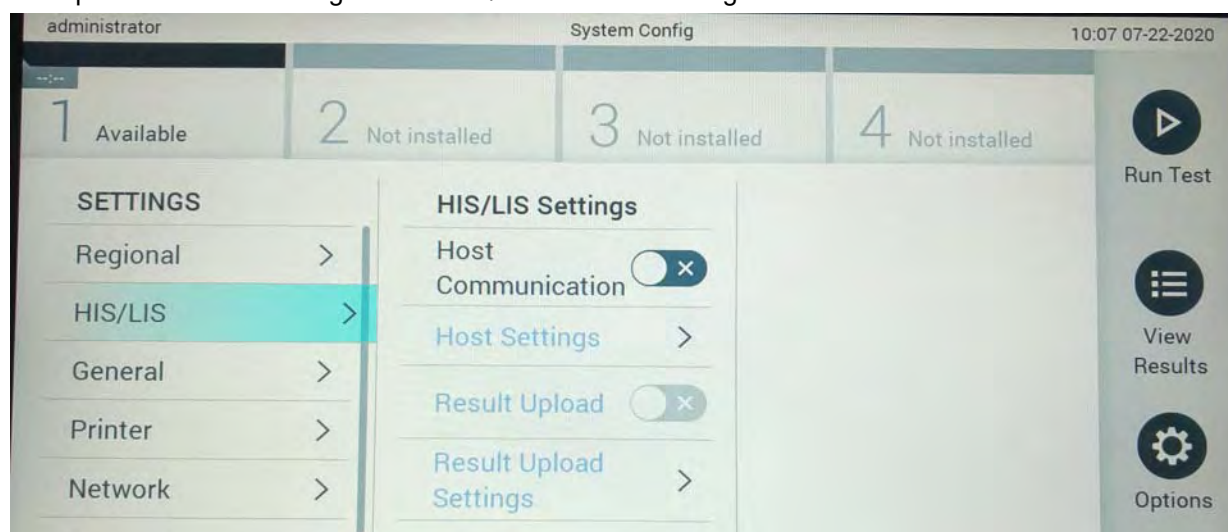
16. If a software update is required see ["Install System Update" on page 214](#).

17. Select regional settings and set configuration for the local area.



Picture 4.2-13 Regional settings.

18. HIS/LIS operation is optional and can be customized by the customer; see the table below for explanation of the settings. Select HIS/LIS form the “settings” in the left column.

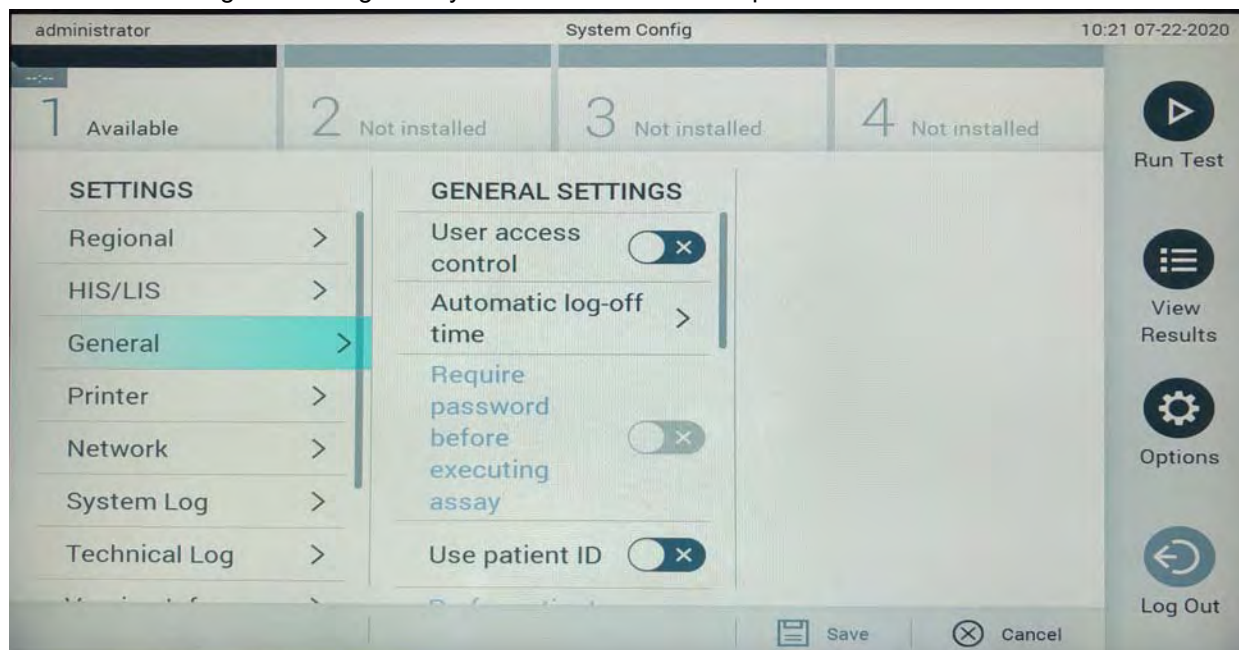


Picture 4.2-14 HIS/LIS screen.

Setting	Description
Host Communication	Enables the HIS/LIS connectivity. This option is disabled by default.
Host Settings	Only active if Host Communication is enabled. This setting defines the host address and port of the host. The host address allows both an IP and a name value of the host. The IP value must be 4 numbers (N.N.N.N) and N must be between 0 and 255. The transfer protocol is currently compatible with HL7 (future versions will also be with POCT1A). Hospital name is an exclusive name to define a DMS or LIS. The default Timeout is configured as 5 seconds and can be extended up to 60 seconds. This is the maximum time the QIAstat-Dx Analyzer 1.0 will wait for a message from the host. Messages queued is an indicator of the number of messages waiting in the queue. The Check connectivity button validates the connection between the QIAstat-Dx Analyzer 1.0 and the host with the IP and port filled.
Result Upload	Enables the functionality of sending results from the QIAstat-Dx Analyzer 1.0 to the host. This option is disabled by default.
Results Upload Settings	Only active if Result Upload is enabled. Results uploading can be performed in two modes: automatic and manual. When automatic mode is enabled, as soon as a test is complete the results are sent to the host. If automatic mode is disabled, the results can be sent manually by pressing the Upload button in the Result Summary and View Results screens. Automatic is disabled by default. Expire Time is the number in days that a test can be sent to the host. When set to zero, this option is disabled so the results will never expire. Reset Uploading clears the queue of messages waiting to be sent. This option can be helpful when many results have been sent but for various reasons the transmission needs to be canceled. Retry resends results that are in upload status “Error”. Authorization can be set to a role to allow uploading of results. As default, only the Administrator role has this authorization enabled.
Test Orders	Enables the functionality of running a test based on a book order created in the HIS/LIS. This option is disabled by default.
Order Settings	Only active if Test Orders is enabled. Force Order enables running a test even if communication with the host is unavailable or if there is no book order associated with the entered sample ID. Force Order is disabled by default.

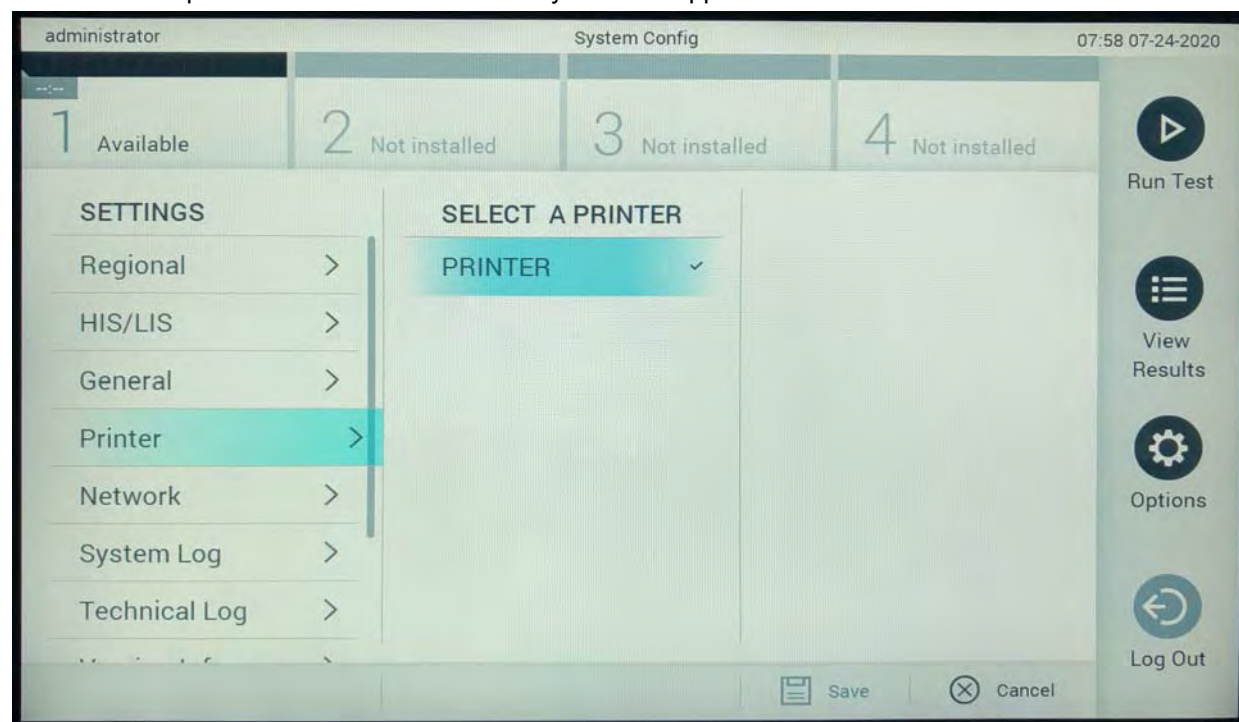
Picture 4.2-15 HIS/LIS settings

19. General settings are configured by the customer to their requirements.



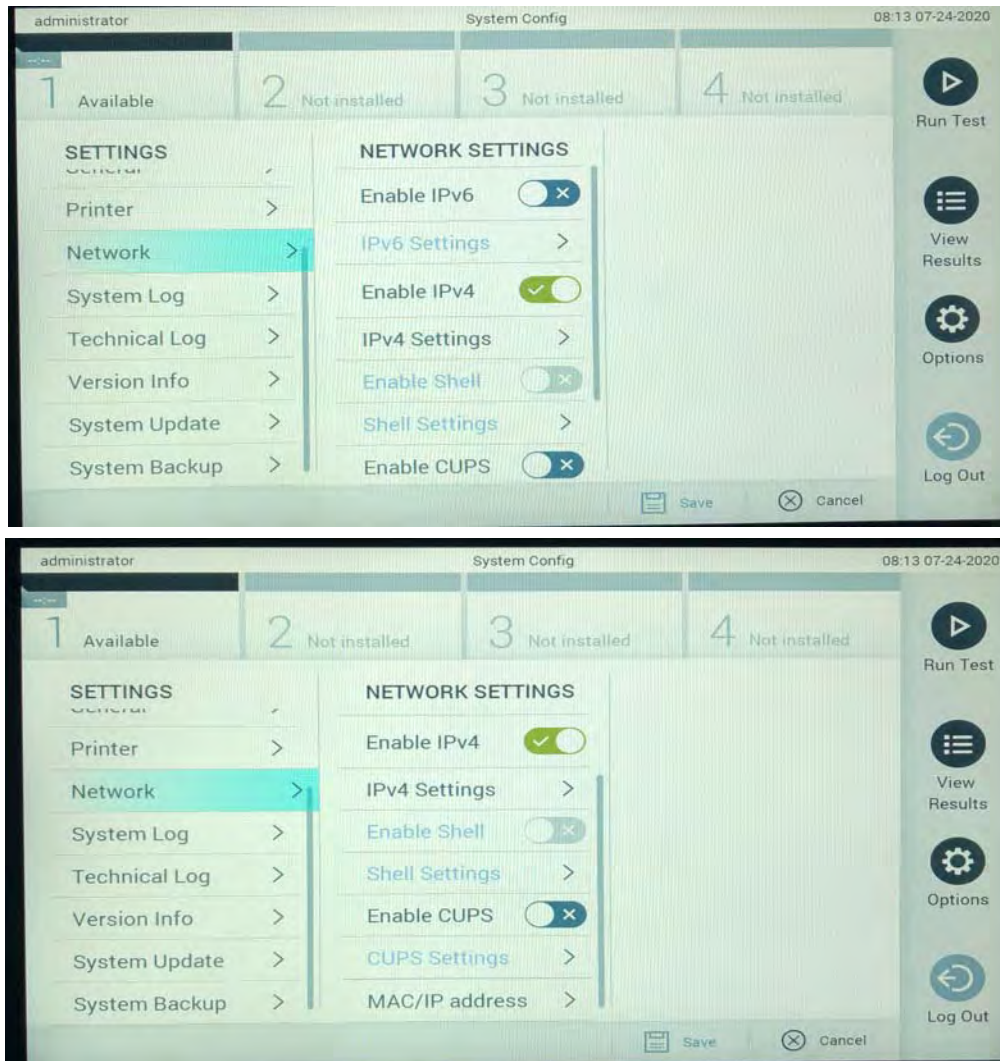
Picture 4.2-16 General settings

20. Printer-If a printer is connected to the analyzer it will appear here. Select it from the list.



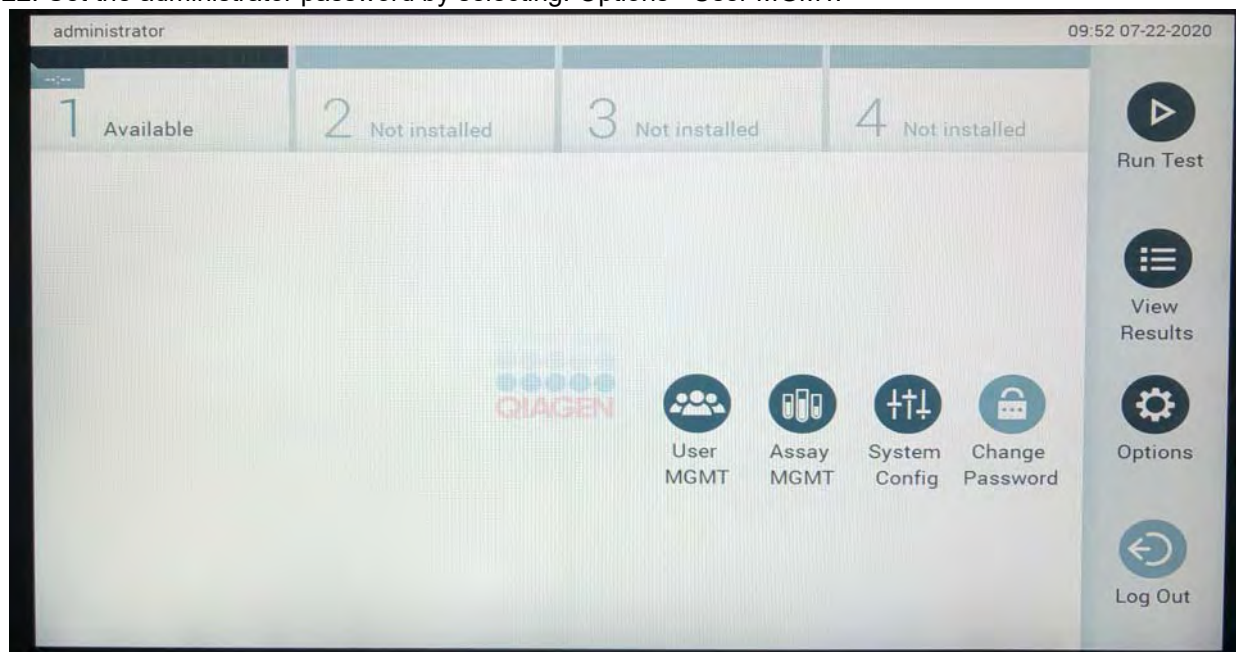
Picture 4.2-17 Printer setup.

21. Network-Set using the customers network requirements.



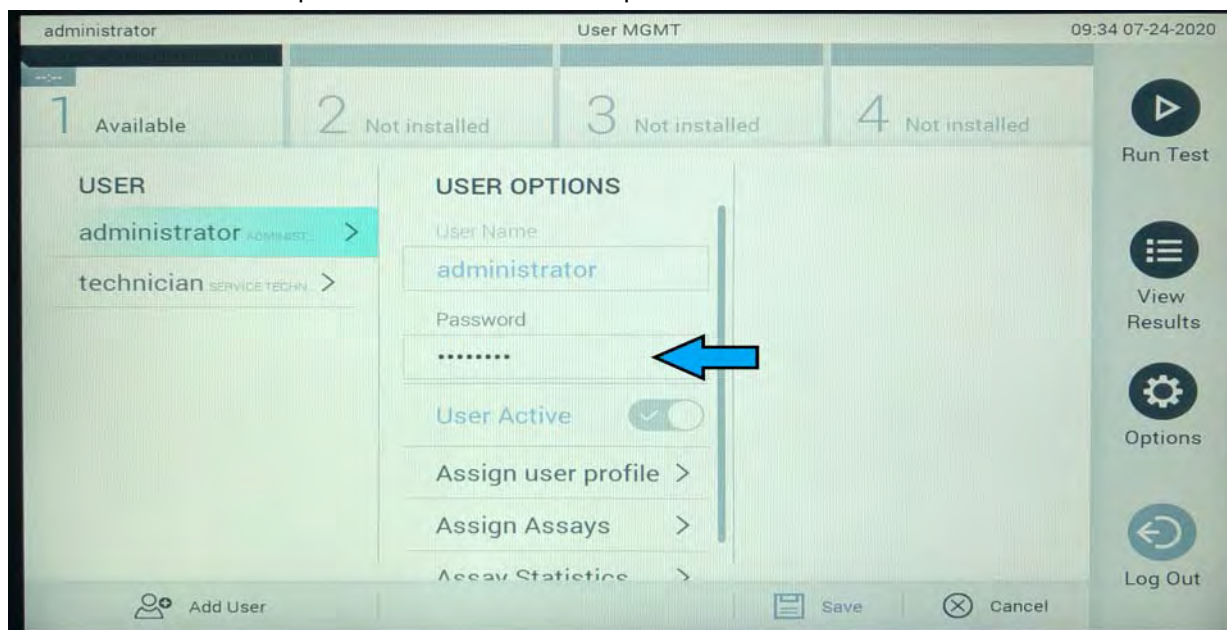
Picture 4.2-18 Network setup.

22. Set the administrator password by selecting: Options> User MGMT.



Picture 4.2-19 Select User MGMT

23. Set the administrator password to the customers preference.



Picture 4.2-20 Set administrator password

24. Complete ICS-5000-0013 and upload to SFDC.

4.3 Installation Test

Before using the QIAstat-Dx the analyzer must pass the Analyzer Check Cartridge work flow test, [“VP7 - Analyzer Check Cartridge Work Flow Test”](#) on page 169, or successfully process a sample cartridge.

5 QIAstat-Dx Modules



Picture 5.0-1 QIAstat-Dx analyzer.

This chapter contains detailed information about the module in the QIAstat DX. Each section will contain the following information:

- Functional Description/Principle of Operation
- Components

5.1 Operational Module

5.1.1 Principle of Operation

The Operational Module includes elements that provide connectivity to the Analytical Modules and provides user interface with the QIAstat-Dx Analyzer. The analyzer is started by pressing the power switch on the bottom of the module. Using the touch screen the user can configure the analyzer setup and enter data for assay testing. The bar code scanner for sample, patient and QIAstat-Dx assay cartridge identification and USB ports for assay and system upgrades, document export and printer connectivity (one in front, three in back). An Ethernet connector for network connectivity is located on the back.

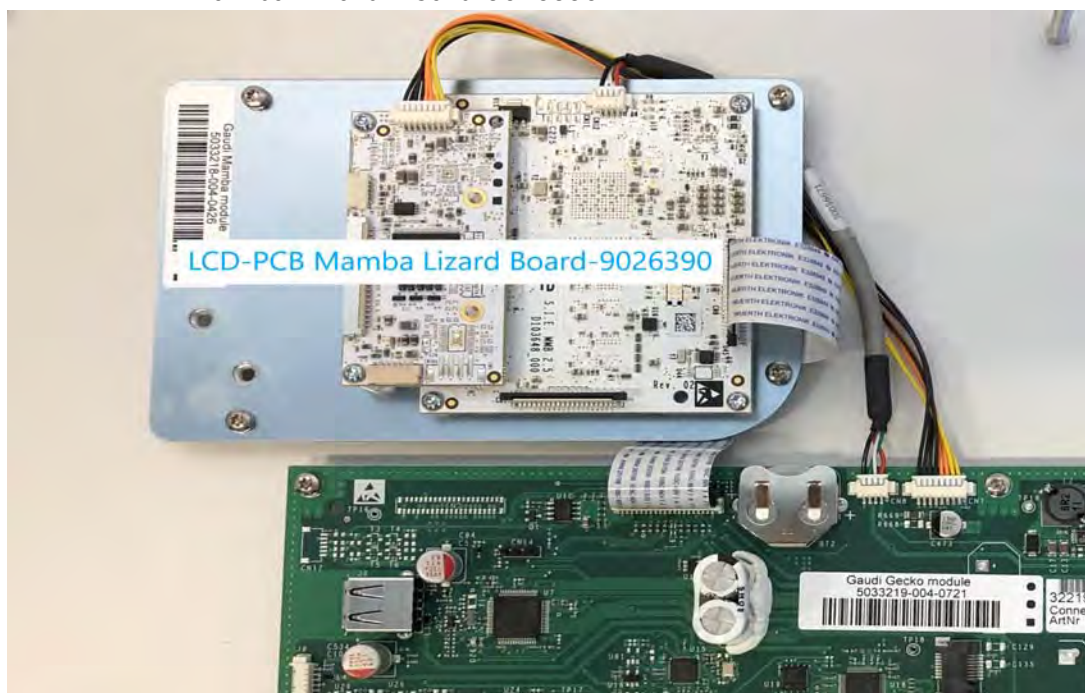
5.1.2 Components - Operational Module

5.1.2.1 Touch Screen-9026389

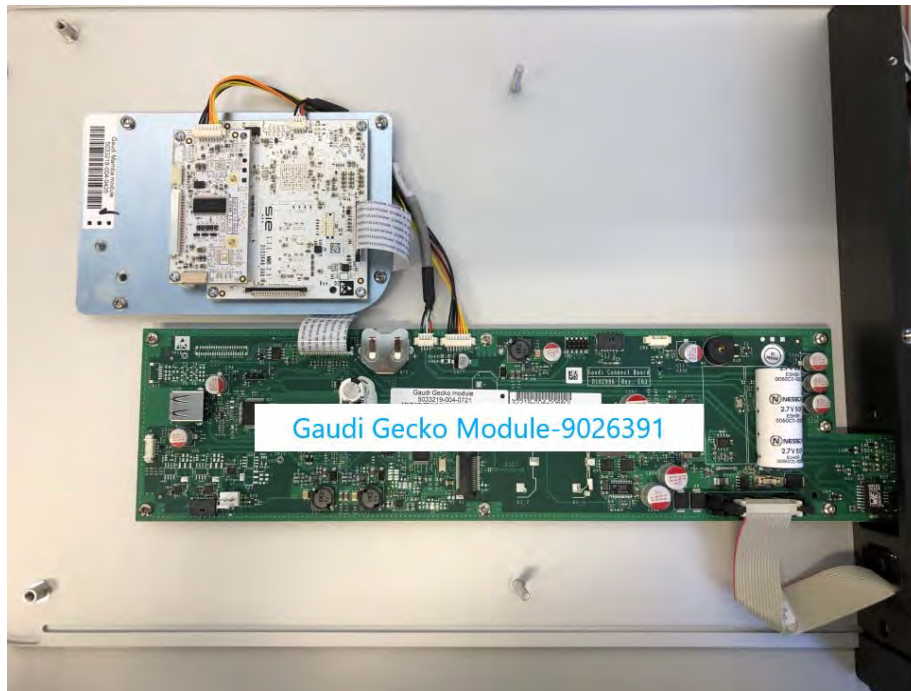


Picture 5.1-1 Touch screen.

5.1.2.2 LCD-PCB Mamba Lizard Board-9026390



Picture 5.1-2 Mamba Lizard board.

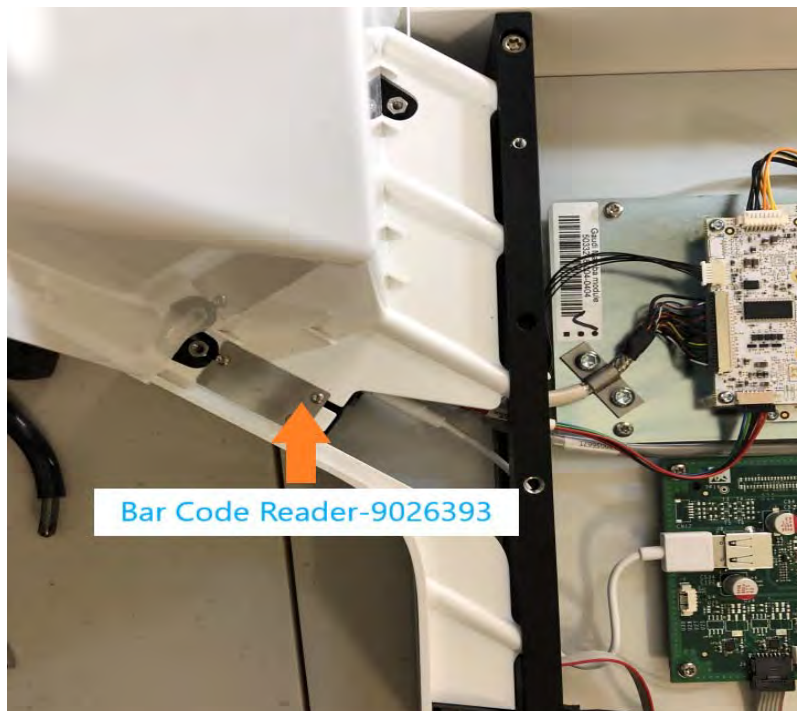
5.1.2.3 Gaudi Gecko Module-9026391

Picture 5.1-3 Gaudi Gecko module.

5.1.2.4 Gaudi Front USB Module-9026392

Picture 5.1-4 Gaudi Front USB module.

5.1.2.5 Bar Code Reader-9026393



Picture 5.1-5 Bar code reader.

5.1.2.6 Front USB Module-9026392



Picture 5.1-6 Front USB module

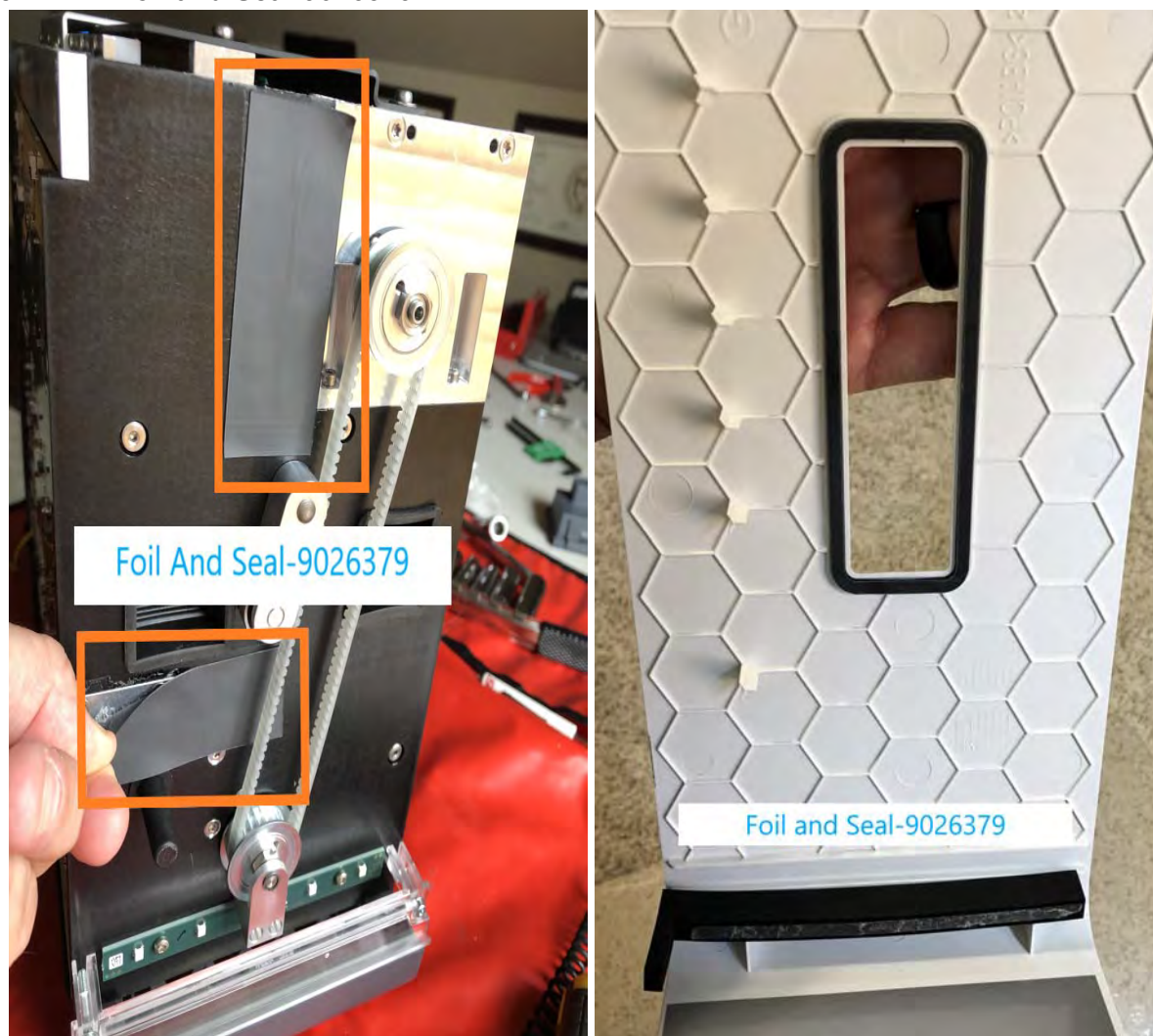
5.2 Analytical Module

5.2.1 Principle of Operation

The Analytical Module has the analyzer main power switch and contains hardware and firmware for sample testing and analysis. The assay cartridge is inserted in the door and does not require any additional user interaction to complete the assay process. The reagents in assay cartridge is pneumatically and mechanically manipulated and do not make contact with any actuators. The Analytical Module houses air filters for both incoming and outgoing air, further safeguarding the environment. After testing, the QIAstat-Dx assay cartridge stays hermetically closed at all times, greatly enhancing its safe disposal.

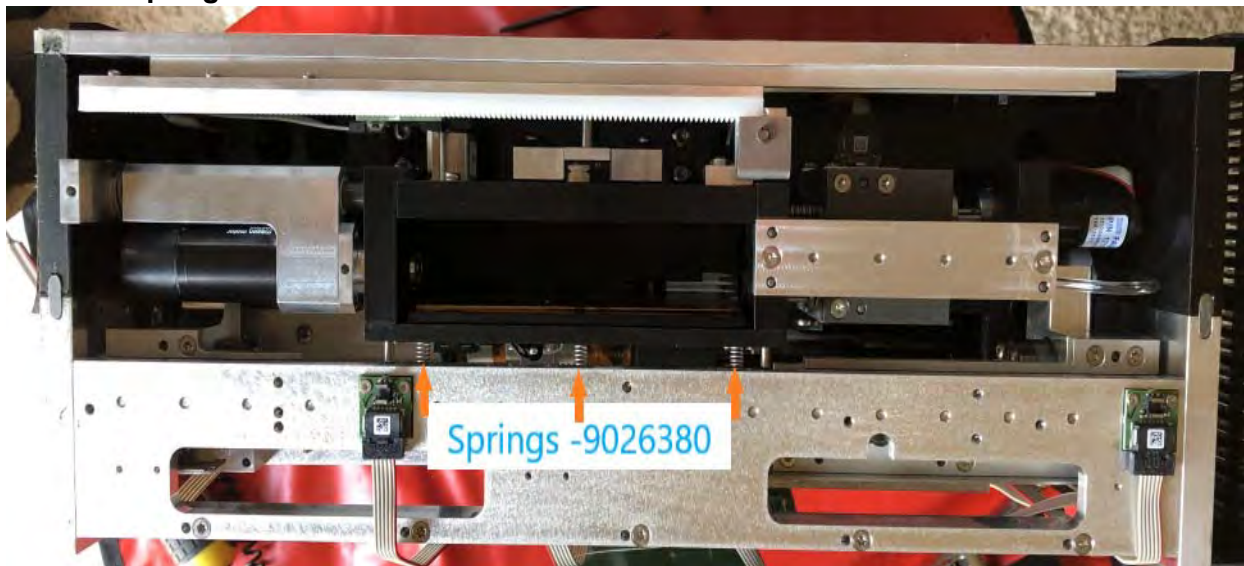
5.2.2 Components - Analytical Module

5.2.2.1 Foil and Seal-9026379



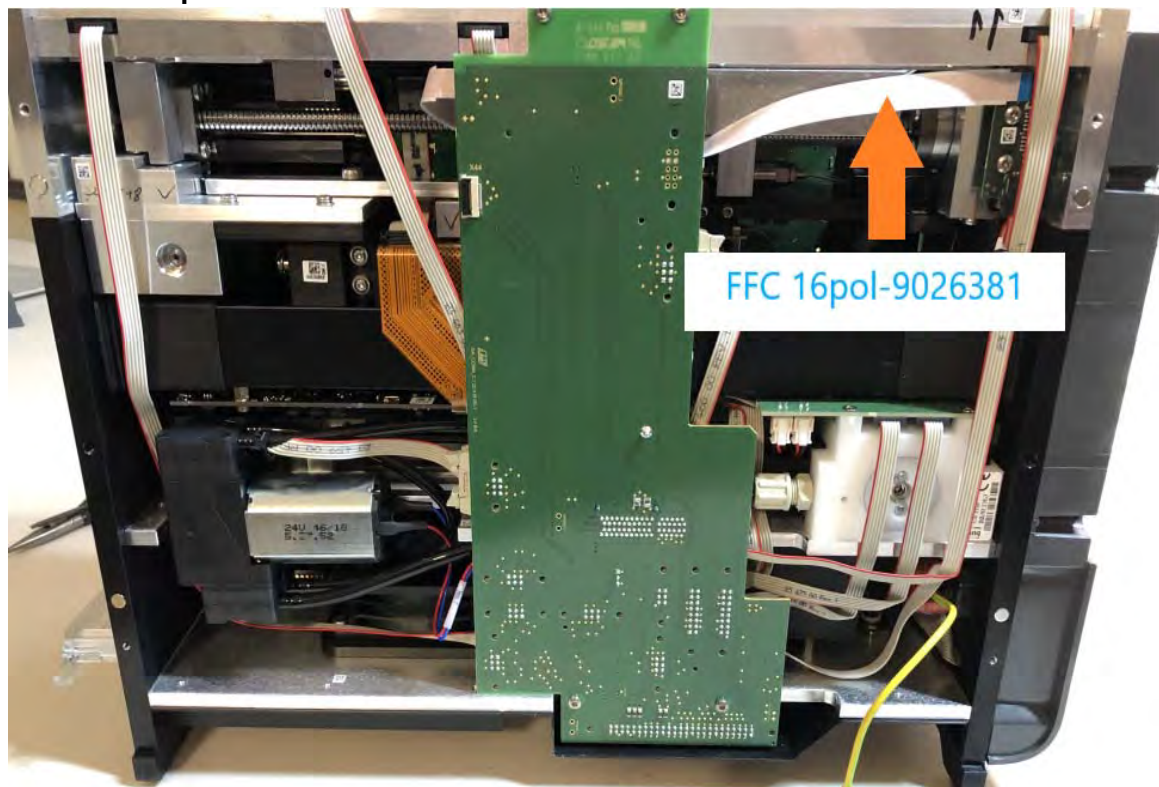
Picture 5.2-1 Foil and seal.

5.2.2.2 Springs-9026380



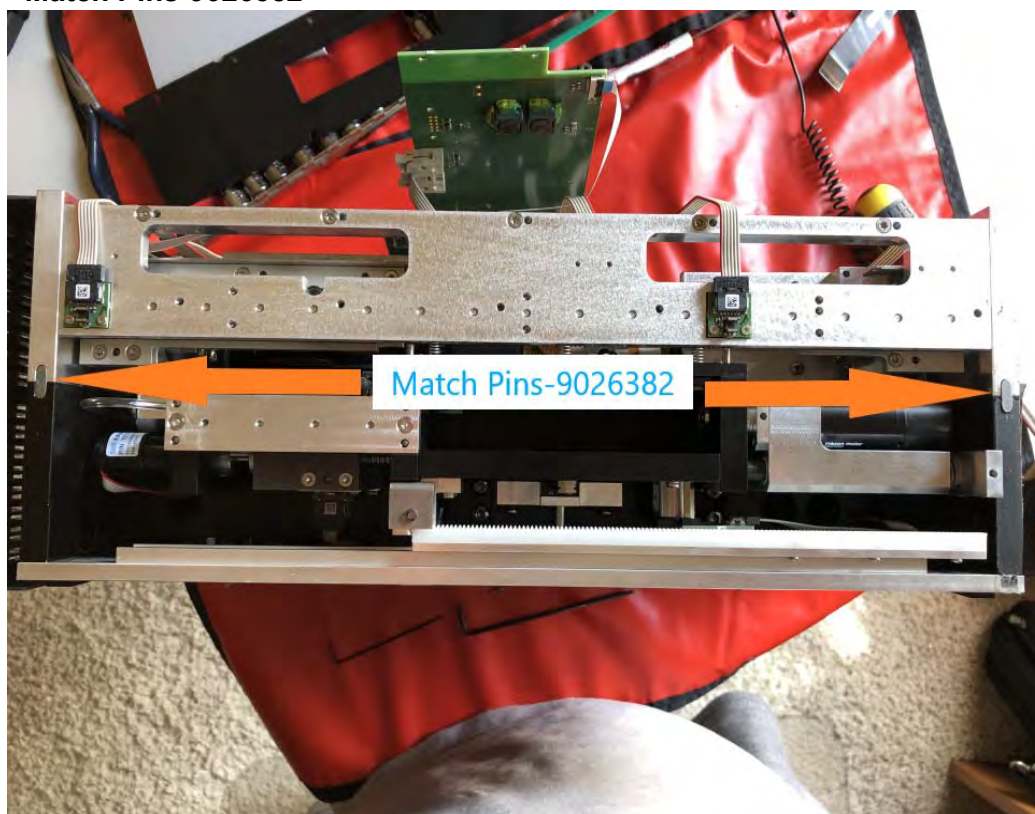
Picture 5.2-2 Springs.

5.2.2.3 FFC 16pol-9026381



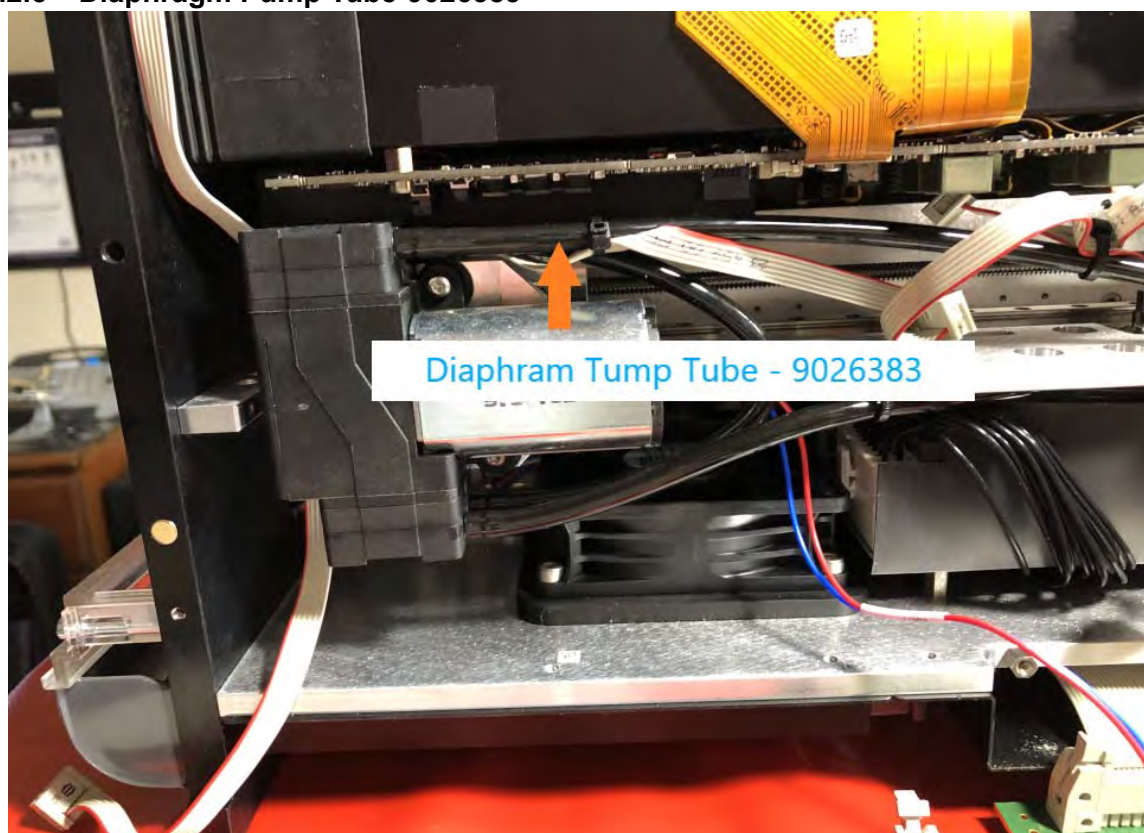
Picture 5.2-3 FFC 16pol flex cable.

5.2.2.4 Match Pins-9026382



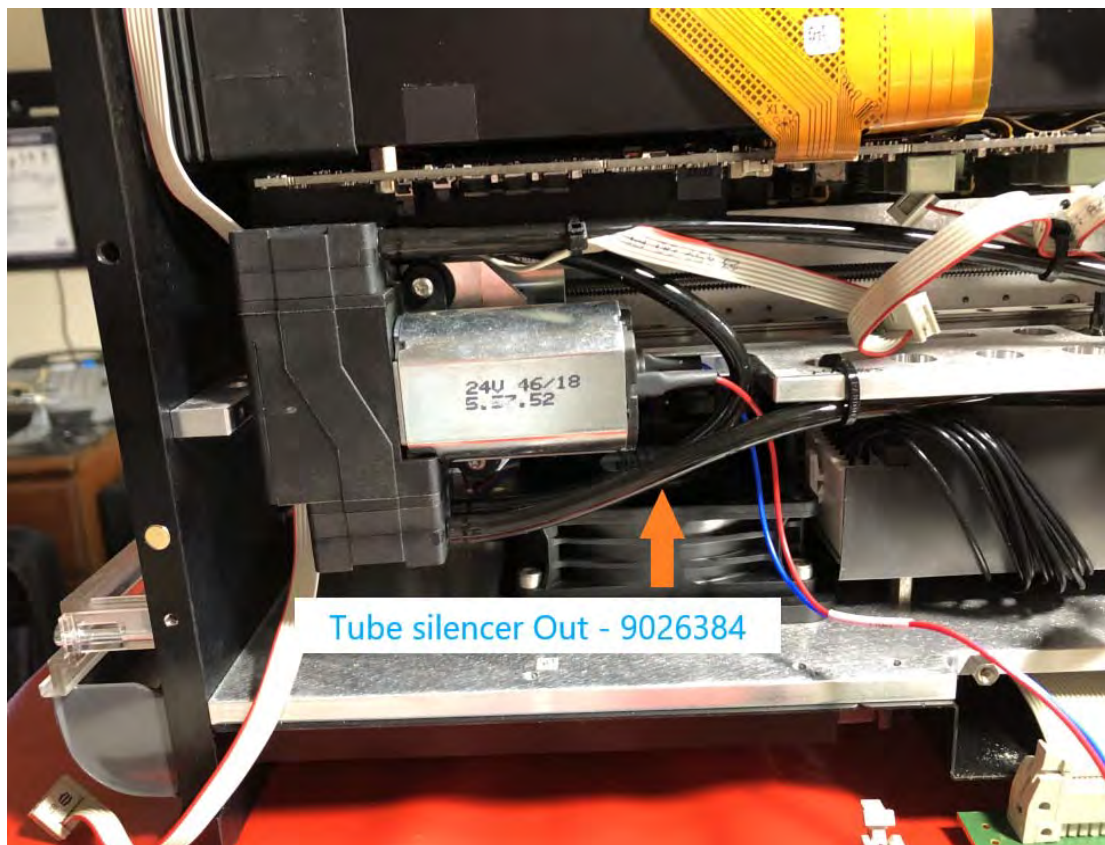
Picture 5.2-4 Match pins

5.2.2.5 Diaphragm Pump Tube-9026383



Picture 5.2-5 Diaphragm pump tube.

5.2.2.6 Tube Silencer Out-9026384



Picture 5.2-6 Tube silencer out.

5.2.2.7 Tube Silencer In-9026385



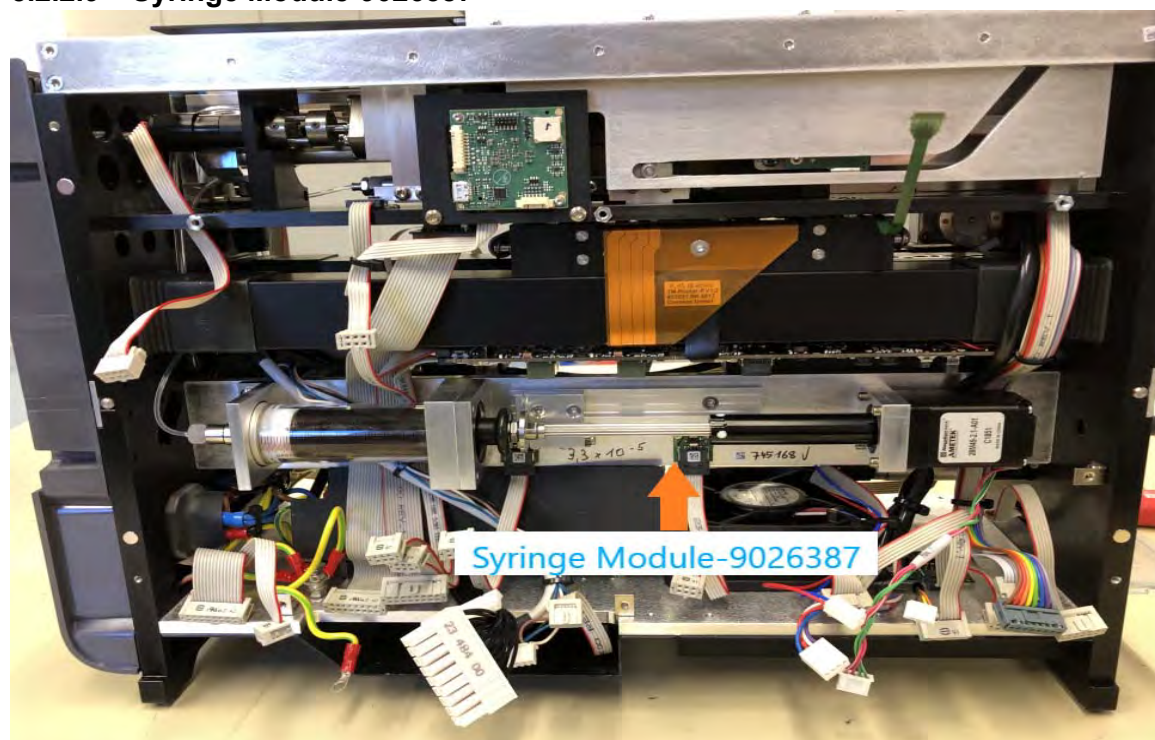
Picture 5.2-7 Tube silencer in.

5.2.2.8 QPCR Complete-9026386 (Not available as a spare part)



Picture 5.2-8 qPCR complete.

5.2.2.9 Syringe Module-9026387



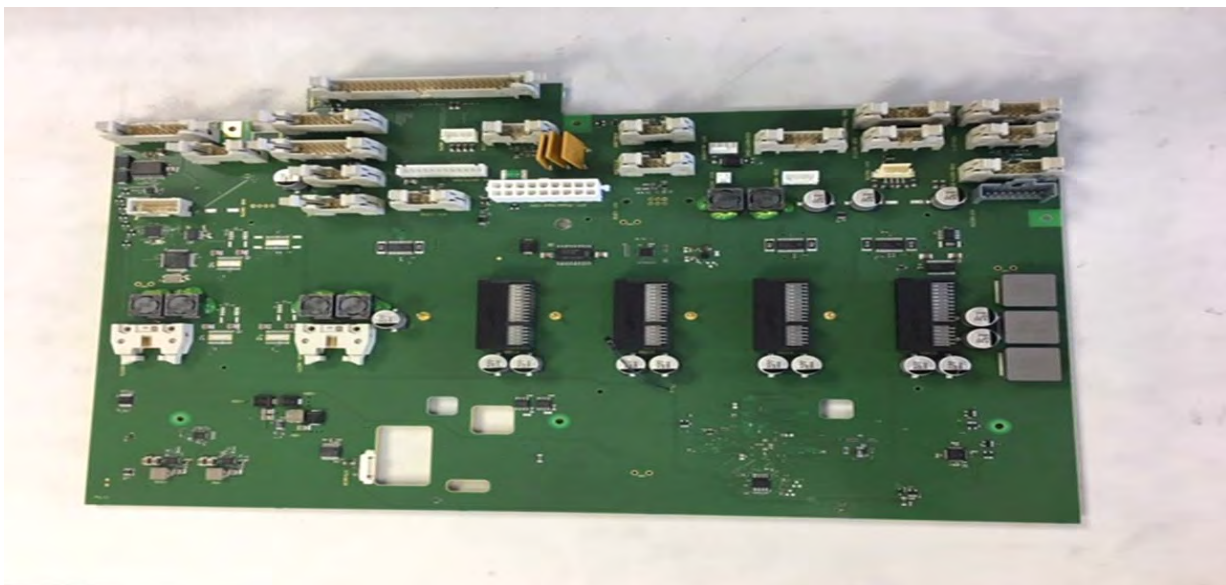
Picture 5.2-9 Syringe module.

5.2.2.10 Lid Mechanism



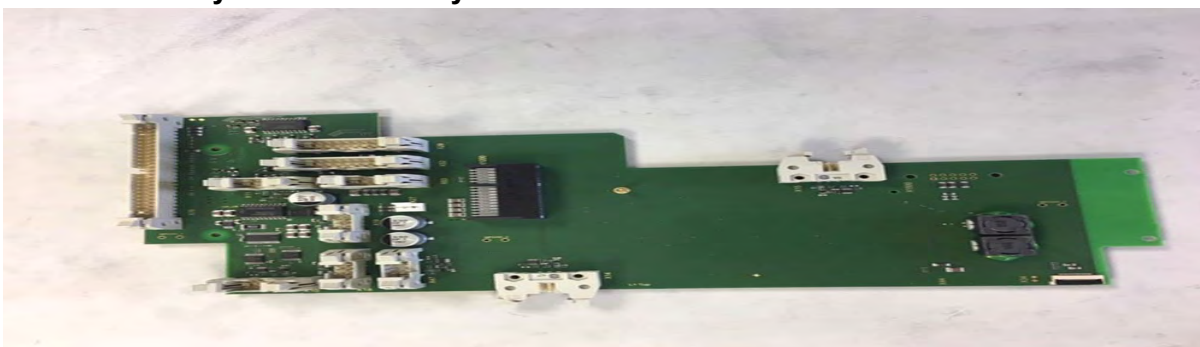
Picture 5.2-10 Lid mechanism.

5.2.2.11 Main Board Assembly-9026541



Picture 5.2-11 Main board assembly.

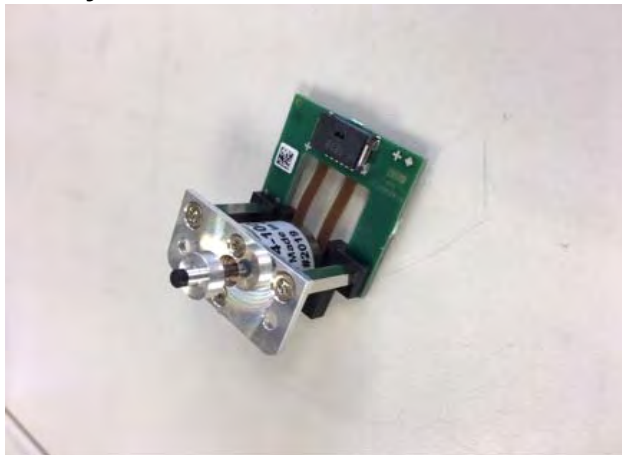
5.2.2.12 Axillary Board Assembly-9026542



Picture 5.2-12 Axillary board assembly.

5.2.2.13 Internal Bar Code Reader-9026543

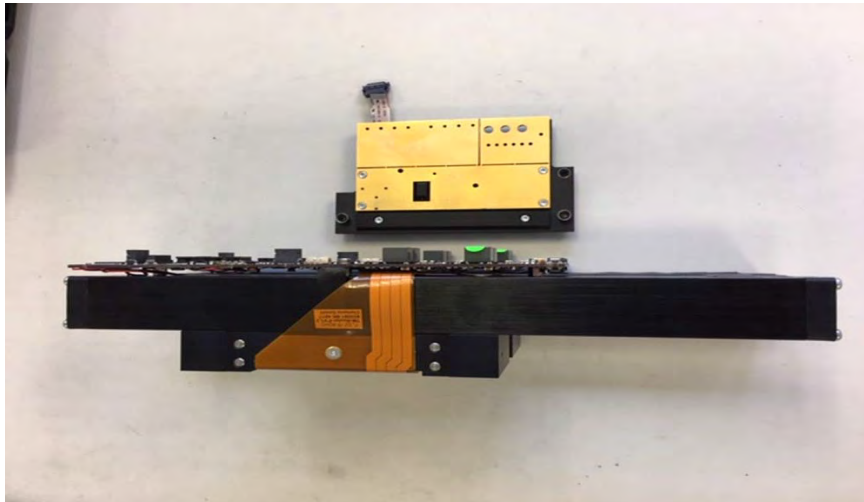
Picture 5.2-13 Internal bar code reader.

5.2.2.14 Voice Coil Assembly-9026544

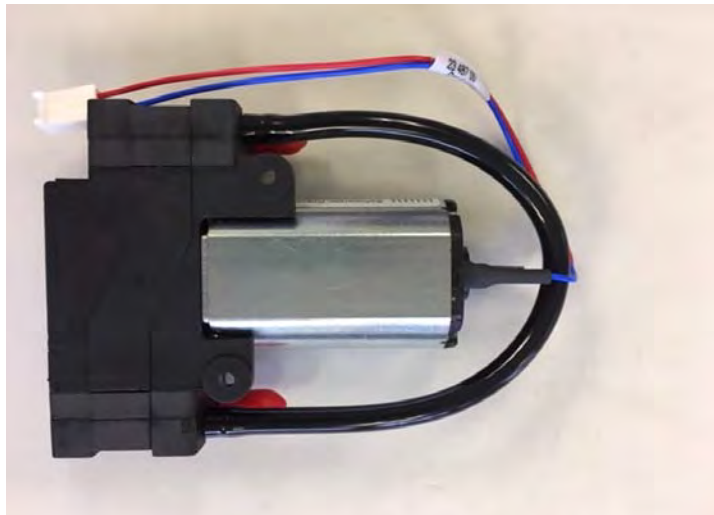
Picture 5.2-14 Voice coil Assembly.

5.2.2.15 TC and Clamping Module-9026545

Picture 5.2-15 TC and clamping module.

5.2.2.16 Thermal Unit Complete-9026545

Picture 5.2-16 Thermal unit.

5.2.2.17 Pressure Pump-9026547

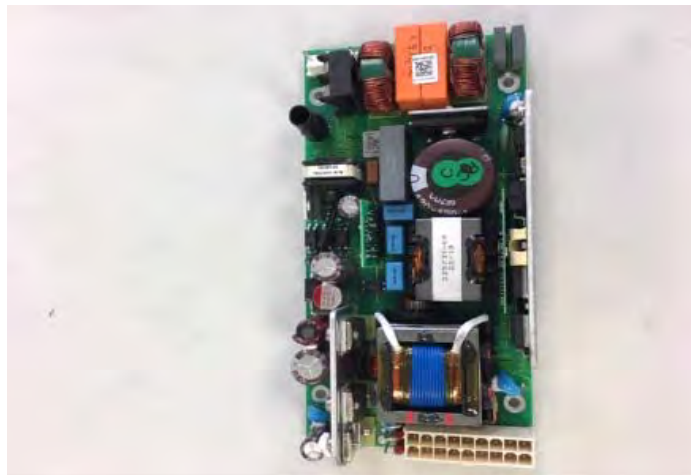
Picture 5.2-17 Pressure pump.

5.2.2.18 Pressure Manifold-9026548

Picture 5.2-18 Pressure manifold.

5.2.2.19 Tilt Sensor-9026549

Picture 5.2-19 Tilt sensor.

5.2.2.20 Power Supply-9026550

Picture 5.2-20 Power supply.

5.2.2.21 Chassis Fan-9026551

Picture 5.2-21 Chassis fan.

6 Replacement Procedures

This chapter provides the removal and replacement procedures for the QIAstat-Dx.

6.1 Protective Measures

In order to prevent possible exposure to electrical, mechanical or biological hazards, make sure to disconnect the instrument power cord and to use appropriate hand, eye, and body protection.

Material	Comment
Eye protection	Safety glasses
Hand protection	Gloves (rubber, nitrile, etc.)
Body protection	Lab coat

Table 6.1-1 Protective Material

WARNING 	Electrical Hazard Disconnect the line power cords from the power outlet before servicing. Lethal voltages inside the instrument When the instrument is connected to line power, terminals may be live, and opening covers or removing parts is likely to expose live parts.
WARNING 	Moving parts Avoid contact with moving parts during operation. If there is a problem with the QIAstat-Dx, great care should be taken if the instrument is operated with covers off.
WARNING 	Pinch and impact hazards The drive belt and drive shafts may move at any time.
WARNING 	Biological materials Biological residue from assay cartridges may be present. Always wear safety glasses, gloves, and a lab coat. For more information, visit: www.qiagen.com/support/msds.aspx .
INFORMATION 	Great care should be taken to prevent the modules from exterior scratches and damage. Use a soft surface when working with the Operational Module and store the Analytical Module covers in a safe place when they are removed

6.2 RP1 - Separate Analytical Module from Operational Module

Part Required	N/A
Tool(s) Required	Protective cover removal tool, AM-OM Hex removal tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing
Replacement Time	0.5 Hours
Calibration/Testing Time	N/A
Total Repair Time	0.5 Hours

Table 6.2-1 RP1 Analytical module/operational module.

6.2.1 RP1 - Separate Analytical Module from Operational Module - Removal Procedure

1. Disconnect the power cord and remove the module bridge (Rear Panel).



Picture 6.2-2 AM back view.

2. Using the protective cover removal tool remove the magnetic protective covers.



Picture 6.2-3 Analytical module protective cover.

3. Using the Analytical-Operational module removal tool loosen the front and rear connecting screw.



Picture 6.2-4 Analytical module right side.

6.2.2 RP1 - Separate Analytical Module from Operational Module - Replacement Procedure

1. Using the protective cover removal tool remove the magnetic protective covers from the left and right side of the Analytical Module.
2. Place the Analytical Module next to the Operational Module and using the Analytical-Operational removal tool lightly tighten the connecting screws (tighten just enough to hold the modules together).
3. Replace the magnetic protective covers on the right side of the Analytical Module.
4. Connect the power cord and the module bridge (Rear Panel).
5. Initialize the analyzer, [see “VP1 - System Initialization” on page 167](#).

6.3 RP2 - Separate Analytical Module from Analytical Module

Part Required	N/A
Tool(s) Required	Protective cover removal tool, Analytical-Operational removal tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	0.5 Hours
Calibration/Testing Time	N/A
Total Repair Time	5 Minutes

Table 6.3-1 RP2 Analytical module.

6.3.1 RP2 - Separate Analytical Module from Analytical Module - Removal Procedure

1. Disconnect the power cord and remove the module bridge (Rear Panel).
2. Using the protective cover removal tool remove the magnetic protective covers.
3. Using the Analytical-Operational removal tool loosen the connecting screws.

6.3.2 RP2 - Separate Analytical Module from Analytical Module - Replacement Procedure

1. Using the protective cover removal tool remove the magnetic protective covers from the left and right side of the Analytical Module.
2. Place the Analytical Module to the right of the existing Analytical Module and using the Analytical-Operational removal tool tighten the connecting screws.
3. Replace the magnetic protective covers on the right side of the Analytical Module.
4. If installed, remove the termination bridge and install the module bridge between modules.
5. Initialize analyzer, [see “VP1 - System Initialization” on page 167](#).
6. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168](#).
7. Run ACC workflow and liquid sensors test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169](#).

6.4 RP3 Remove Operational Module (OM) Cover

Part Required	9026537 Side Cover Left
Tool(s) Required	T10 TORX
Calibration Required	N/A
Testing Required	VP1 - System Initialization
Replacement Time	5 Minutes
Calibration/Testing Time	N/A
Total Repair Time	5 Minutes

Table 6.4-1 RP3 Operational module right cover.

INFORMATION 	Great care should be taken to prevent the modules from exterior scratches and damage. Use a soft surface when working with the Operational Module and store the Analytical Module covers in a safe place when they are removed
---	--

6.4.1 RP3 - Operational Module Right Cover - Removal Procedure

1. Using a T10 TORX wrench, remove the four T10 TORX screws from the right cover.



Picture 6.4-2 Operational module right cover screws.

6.4.2 RP3 - Operational Module Right Cover - Replacement Procedure

1. Place the cover on the module and secure it with the four T10 TORX screws.
2. Initialize the analyzer, [VP1 - System Initialization](#).

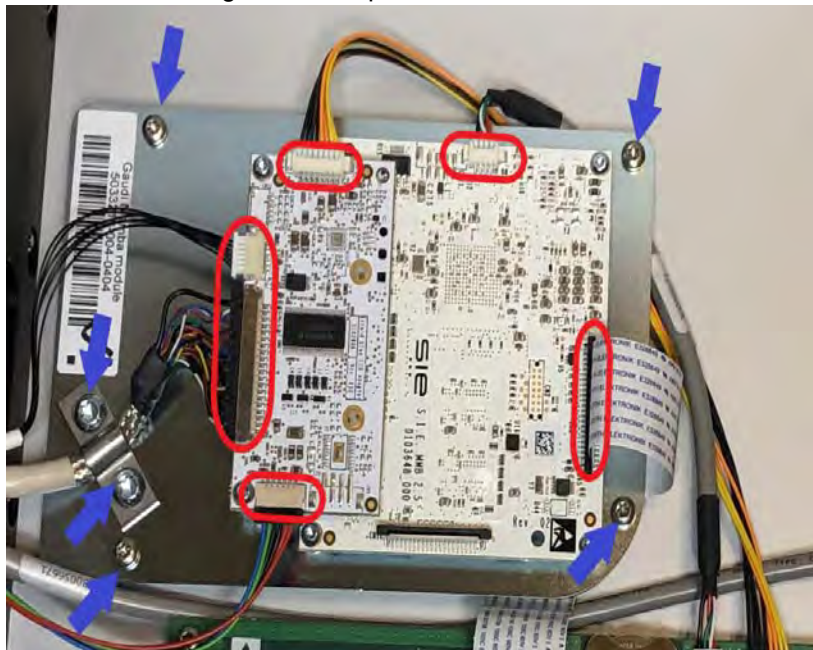
6.5 RP4 Mamba Lizard Board

Part Required	9026390 - Mamba Lizard Board
Tool(s) Required	T10 TORX
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP12 - Operational Module Functional Check , VP8 - Set Operational Module Serial Number
Replacement Time	30.0 Minutes
Calibration/Testing Time	10.0 Minutes
Total Repair Time	40 Minutes

Table 6.5-1 RP4 Mamba lizard board.

6.5.1 RP4 - Mamba Lizard Board- Removal Procedure

1. Using a T10 TORX wrench, remove the right cover, see [“RP3 - Operational Module Right Cover - Removal Procedure” on page 56](#).
2. Disconnect the six connectors and ground clamp for the video connector.



Picture 6.5-2 Mamba lizard board.

3. Use a T10 TORX to remove the four module mounting screws.

6.5.2 RP4 - Mamba Lizard Board - Installation Procedure

1. Install the board and secure it with the four T10 TORX mounting screws.
2. Connect the video connector and install the grounding clamp.
3. connect the remaining four connectors.
4. Install the module cover, see [“RP3 - Operational Module Right Cover - Replacement Procedure” on page 56](#).
5. Initialize the analyzer, [VP1 - System Initialization](#).
6. Set OM serial number, [VP8 - Set Operational Module Serial Number](#).
7. Check OM functions, [VP12 - Operational Module Functional Check](#).

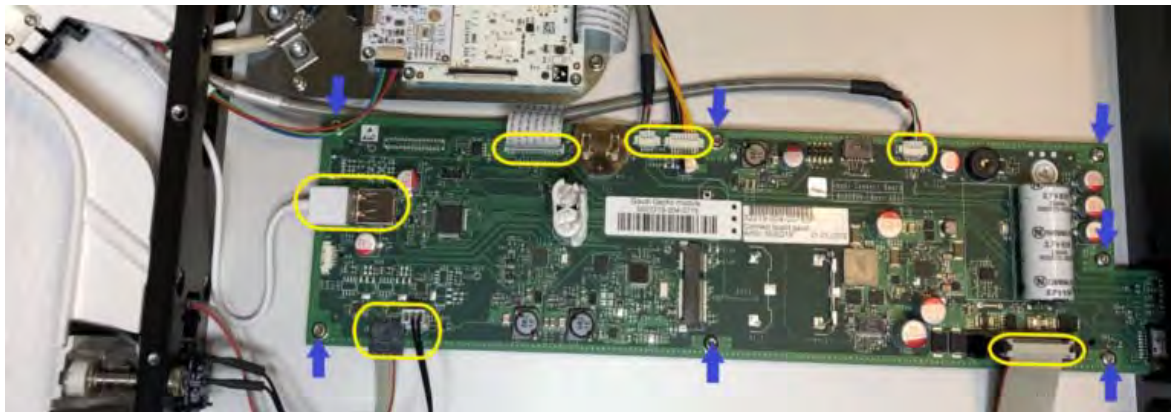
6.6 RP5 Gaudi Gecko Module

Part Required	9026391 - Gaudi Gecko Module
Tool(s) Required	T10 TORX
Calibration Required	Touch Screen
Testing Required	VP1 - System Initialization , VP12 - Operational Module Functional Check
Replacement Time	30 Minutes
Calibration/Testing Time	10 Minutes
Total Repair Time	40 Minutes

Table 6.6-1 RP5 Gaudi gecko module.

6.6.1 RP5 - Gaudi Gecko Module - Removal Procedure

1. Using a T10 TORX wrench, remove the right cover, [see "RP3 - Operational Module Right Cover - Removal Procedure" on page 56.](#)
2. Disconnect the eight connectors.



Picture 6.6-2 Gaudi gecko module.

3. Using a T10 TORX remove the seven mounting screws.

6.6.2 RP5 - Gaudi Gecko Module - Replacement Procedure

1. Install the new module and secure it with the six T10 TORX mounting screws.
2. Connect the eight connectors.
3. Install the module cover, [see "RP3 - Operational Module Right Cover - Replacement Procedure" on page 56.](#)
4. Initialize the analyzer, [VP1 - System Initialization.](#)
5. Check OM functions, [VP12 - Operational Module Functional Check.](#)

6.7 RP6 Bar Code Reader OM

Part Required	9026393 - Bar Code Reader OM
Tool(s) Required	T10 TORX, T6 TORX
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP12 - Operational Module Functional Check
Replacement Time	30 Minutes
Calibration/Testing Time	30 Minutes
Total Repair Time	1 Hour

Table 6.7-1 RP6 Bar code reader OM.

6.7.1 RP6 - Bar Code Reader OM - Removal Procedure

1. Using a T10 TORX wrench, remove the right cover, [see "RP3 - Operational Module Right Cover - Removal Procedure" on page 56](#).
2. Disconnect the USB connector from the Gaudi Gecko Module.



Picture 6.7-2 Bar code reader connection on the gaudi gecko module.

3. Using a T6 TORX remove the three mounting screws and remove the bar code reader.



Picture 6.7-3 Bar code reader screws.

6.7.2 RP6 - Bar Code Reader OM - Replacement Procedure

1. Install the new bar code reader and secure it with the three T6 TORX screws.
2. Connect the USB connector to the Gecko Module.
3. Install the module cover, [see “RP3 - Operational Module Right Cover - Replacement Procedure” on page 56.](#)
4. Initialize the analyzer, [VP1 - System Initialization.](#)
5. Check OM functions, [VP12 - Operational Module Functional Check.](#)

6.8 RP7 - Touch Screen

Part Required	9026389 - Touch Screen
Tool(s) Required	T10, T20 and T6 TORX tool, 6mm Deep Socket
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP12 - Operational Module Functional Check
Replacement Time	1 Hour
Calibration/Testing Time	30 Minutes
Total Repair Time	1 Hour and 30 Minutes

Table 6.8-1 RP7 Touch Screen.

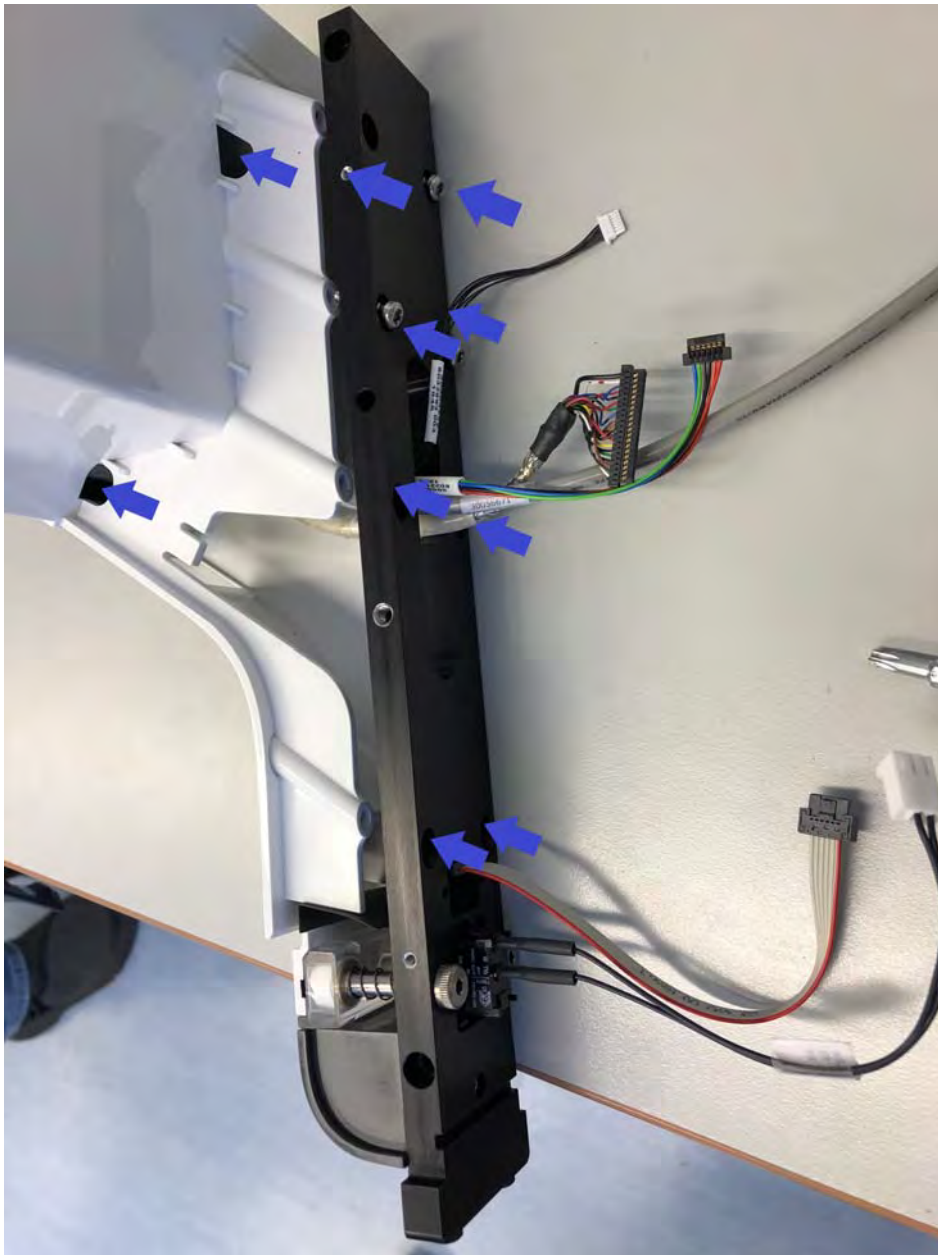
6.8.1 RP7 - Touch Screen- Removal Procedure

1. Use a T10 TORX to remove the right side cover, [see “RP3 - Operational Module Right Cover - Removal Procedure” on page 56.](#)
2. Disconnect the 3 connectors and ground clamp from the Mamba Module.
3. Disconnect the 2 connectors from the Gecko Module.
4. Remove the Bar Code Reader.
5. Remove the two T20 TORX Screws then pull up to detach the front plate from the lower panel.



Picture 6.8-2 Touch screen connections and screws.

6. Remove the eight T20 TORX screws and two 5/16" tower nuts holding the screen assembly on the front plate.



Picture 6.8-3 Touch screen mounting screws.

6.8.2 RP7 - Touch Screen - Installation Procedure

1. Guide the cables through the front plate and install the Touch Screen module to the front plate with the eight T20 TORX screws.
2. Place the front plate on the guide pins and secure it with two T15 TORX screws.
3. Install the bar code reader.
4. Make the electrical connections and install the video cable ground clamp.
5. Install the module cover, [see "RP3 - Operational Module Right Cover - Replacement Procedure" on page 56](#).
6. Initialize the analyzer, [VP1 - System Initialization](#).
7. Check OM functions, [VP12 - Operational Module Functional Check](#).

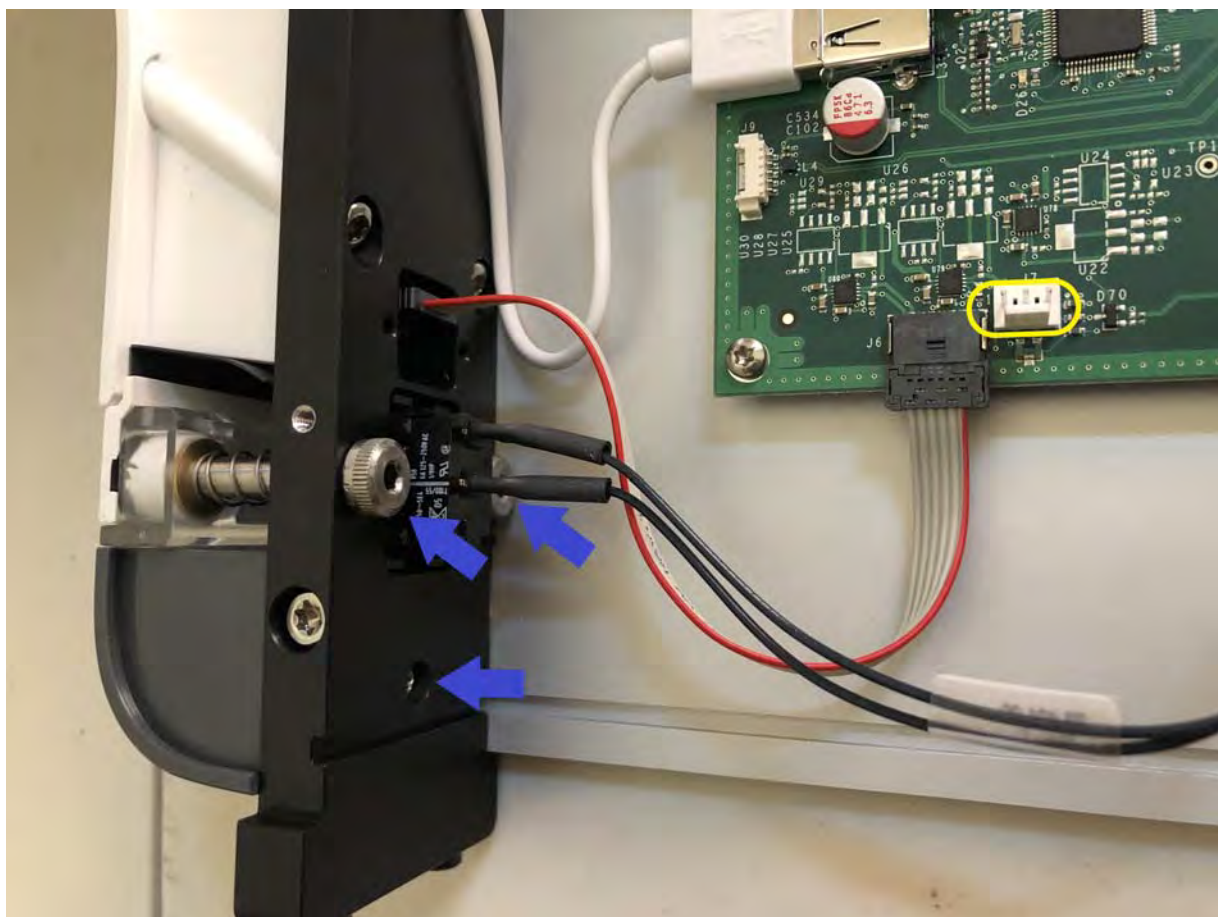
6.9 RP8 - OM Power Switch

Part Required	Non-Stock listed-OM Power Switch
Tool(s) Required	T10 and T15 TORX tool, 17mm Socket
Calibration Required	N/A
Testing Required	VP1 - System Initialization
Replacement Time	1.0 Hours
Calibration/Testing Time	30 Minutes
Total Repair Time	1 Hour and 30 Minutes

Table 6.9-1 RP8 OM power switch.

6.9.1 RP8 - OM Power Switch- Removal Procedure

1. Use a T10 TORX to remove the OM right side cover., [see "RP3 - Operational Module Right Cover - Removal Procedure" on page 56.](#)



Picture 6.9-2 Power switch spring screws.

2. Disconnect the cable from the Gecko Board (J7).
3. Use a T10 TORX to remove the lower front curved cover plate screw and lift it out.
4. Use a T15 TORX to remove the two spring-loaded screws from the back side of the switch and lift the switch cover out.



Picture 6.9-3 Switch retaining nut.

5. Using a 17mm socket or pliers loosen the switch retaining nut and remove the switch.

6.9.2 RP8 - OM Power Switch - Installation Procedure

1. Install switch in the front plate.
2. Push the switch T20 TORX plate screws through the front plate and slide the springs over the screws.
3. Slide the front plate in place and tighten the the T20 TOTX screws.
4. Install the lower cover plate and secure it with the T10 TORX screw.
5. Install the module cover, [see "RP3 - Operational Module Right Cover - Replacement Procedure" on page 56](#).
6. Initialize the analyzer, [VP1 - System Initialization](#).
7. Check OM functions, [VP12 - Operational Module Functional Check](#).

6.10 RP9 - Analytical Module (AM) Covers

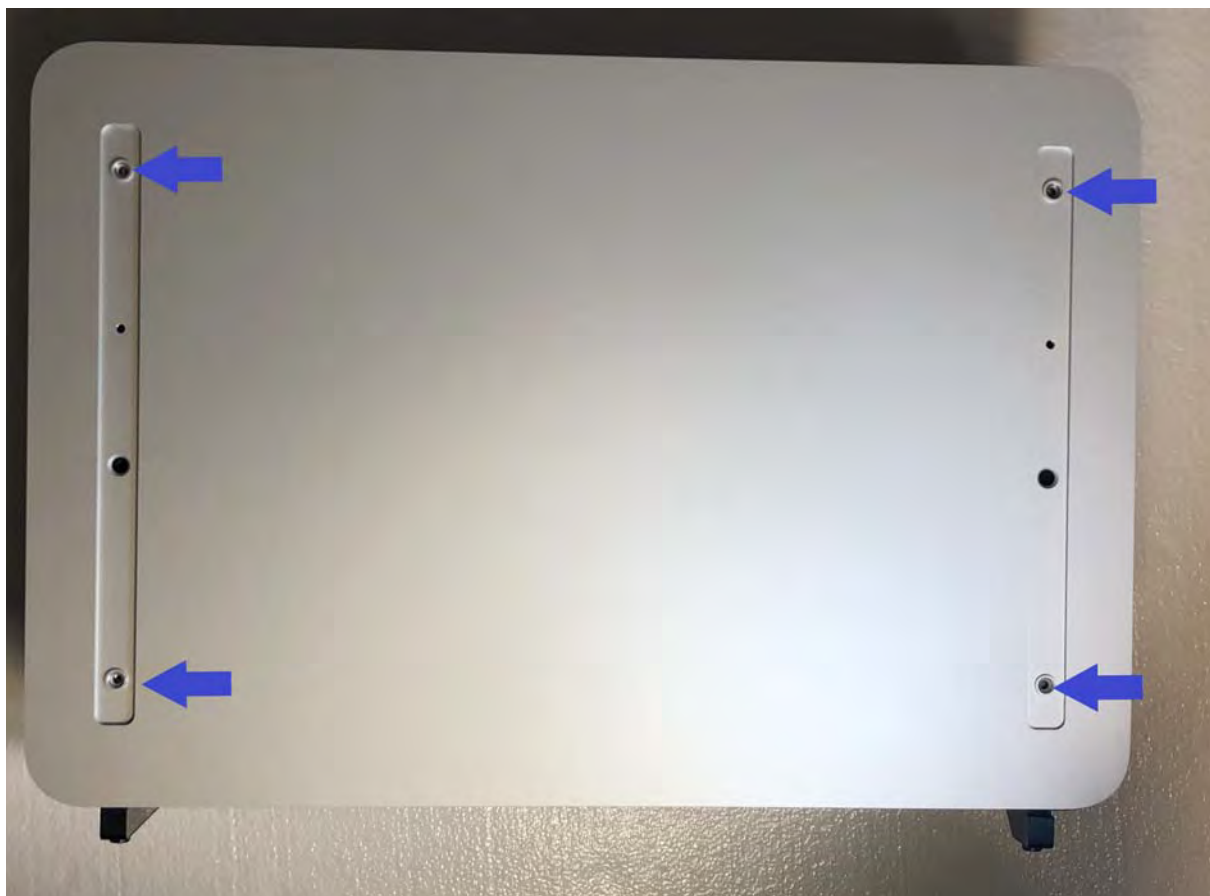
Part Required	9026539 (Right); 9026540 (Left); 9026538 (Top) - AM Module Covers
Tool(s) Required	T10 TORX tool, 7mm Socket
Calibration Required	N/A
Testing Required	N/A
Replacement Time	30 Minutes
Calibration/Testing Time	N/A
Total Repair Time	30 Minutes

Table 6.10-1 RP9 Analytical module covers.

INFORMATION 	Great care should be taken to prevent the modules from exterior scratches and damage. Use a soft surface when working with the Operational Module and store the Analytical Module covers in a safe place when they are removed
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6.10.1 RP9 - AM Covers - Removal Procedure

1. Use a T10 TORX remove the right/left side covers by removing the four T10 TORX screws on each panel then with a 7mm socket remove the ground inside.



Picture 6.10-2 Analytical module cover screws.

2. Remove the Filter Tray Assembly.
3. Lift the top cover by holding the lower front side and gently lifting upward.



Picture 6.10-3 AM top cover.

4. Remove the four T10 TORX screws holding the rear panel/filter assembly.



Picture 6.10-4 AM back cover.

5. Ensure you retain the two O-rings.



Picture 6.10-5 O-rings.

6.10.2 RP9 - AM Covers- Installation Procedure

1. Ensure the two O-Rings are in place and install the rear panel/filter assembly and secure it with the four T10 TORX screws.



Picture 6.10-6 Back cover screws.

2. Inspect the foam seal inside the top cover, replace if needed, and install the top cover, [RP28 - Foil and Seal- Removal Procedure](#).



Picture 6.10-7 Top cover, foam seal.

3. Install the right/left side covers by attaching the ground and four exterior T10 Torx screws on each side.
4. Install the filter assembly.

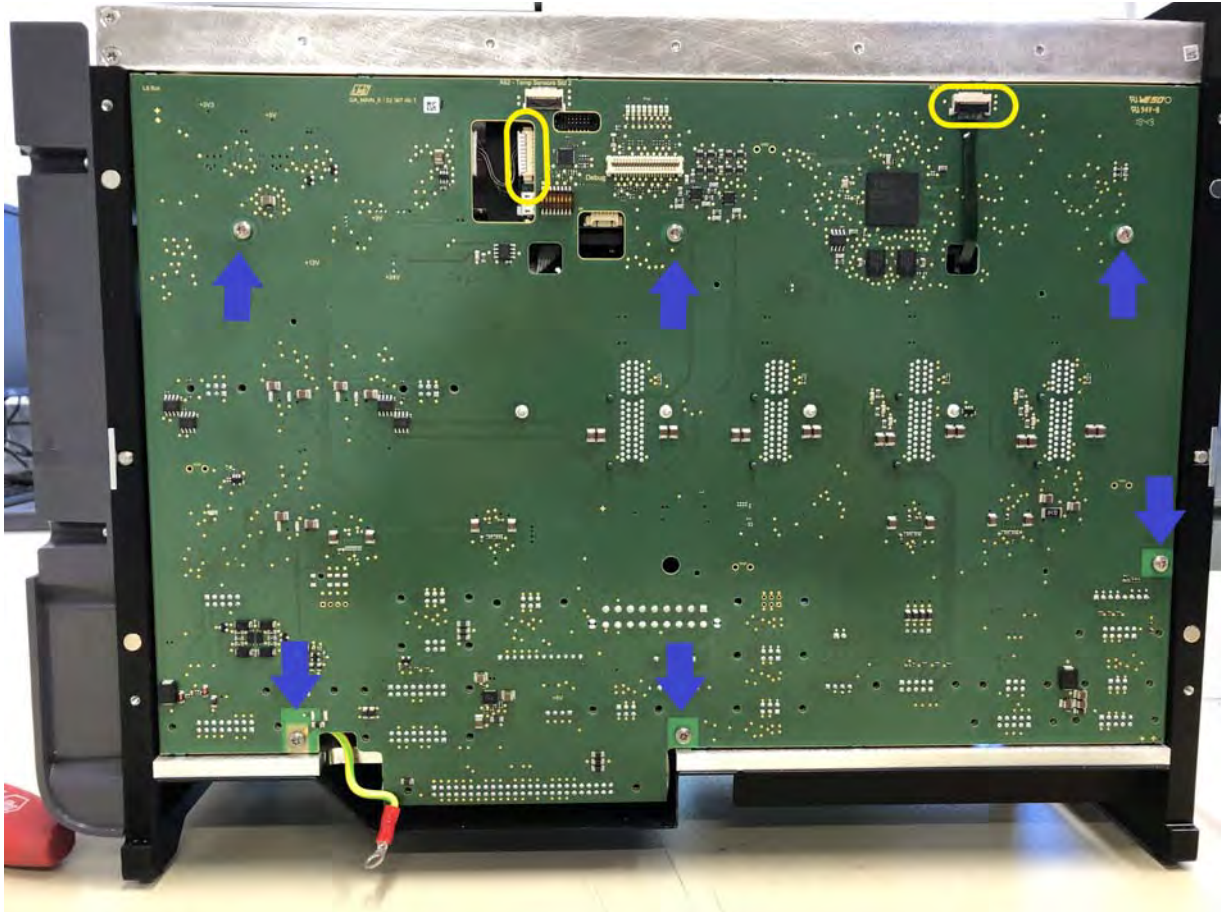
6.11 RP10 - Main Board

Part Required	9026541 - Main Board
Tool(s) Required	T10 TORX tool
Calibration Required	CP1 Automated Mechanical Calibrations
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hour and 30 Minutes
Calibration/Testing Time	2 Hours
Total Repair Time	3 Hours and 30 Minutes

Table 6.11-1 RP10 Main board.

6.11.1 RP10 - Main Board - Removal Procedure

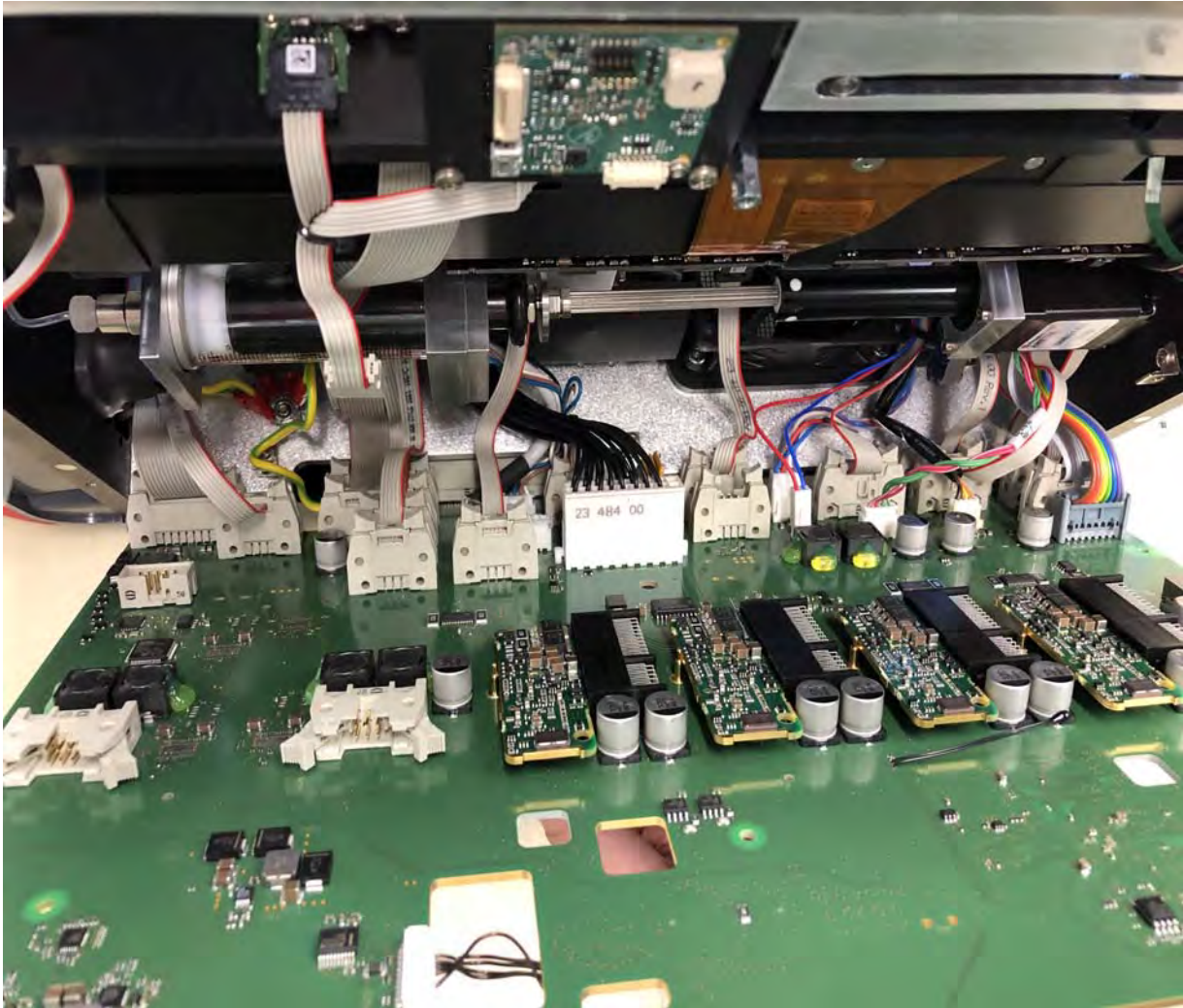
1. Remove the module covers, [see "RP9 - AM Covers - Removal Procedure" on page 64.](#)
2. Remove the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly- Removal Procedure" on page 106.](#)
3. Disconnect temperature sensor ribbon cable and bar code reader connector.



Picture 6.11-2 Main board screws and outer connections.

4. Use a T10 TORX to remove the six mounting screws.

5. Disconnect the electrical connectors.



Picture 6.11-3 Main board inner connections.

6. Remove the four 2 Amp Technosoft Driver Modules to transfer to the new Main Board.

6.11.2 RP10 - Main Board - Installation Procedure

1. Install the four Technosoft Driver Modules in the same position.
2. Connect the electrical connectors.
3. Connect the bar code reader connector and flex cable.
4. Carefully push the board in place guiding the cables to avoid damage.
5. Install 6 T10 TORX mounting screws.
6. Install the module covers, [see "RP9 - AM Covers- Installation Procedure" on page 67.](#)
7. Install the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly - Installation Procedure" on page 107.](#)
8. Initialize the analyzer, [see "VP1 - System Initialization" on page 167.](#)
9. Prepare for testing, [see "VP2 - Preparation For Testing" on page 168.](#)
10. Run ACC test, [see "VP7 - Analyzer Check Cartridge Work Flow Test" on page 169.](#)

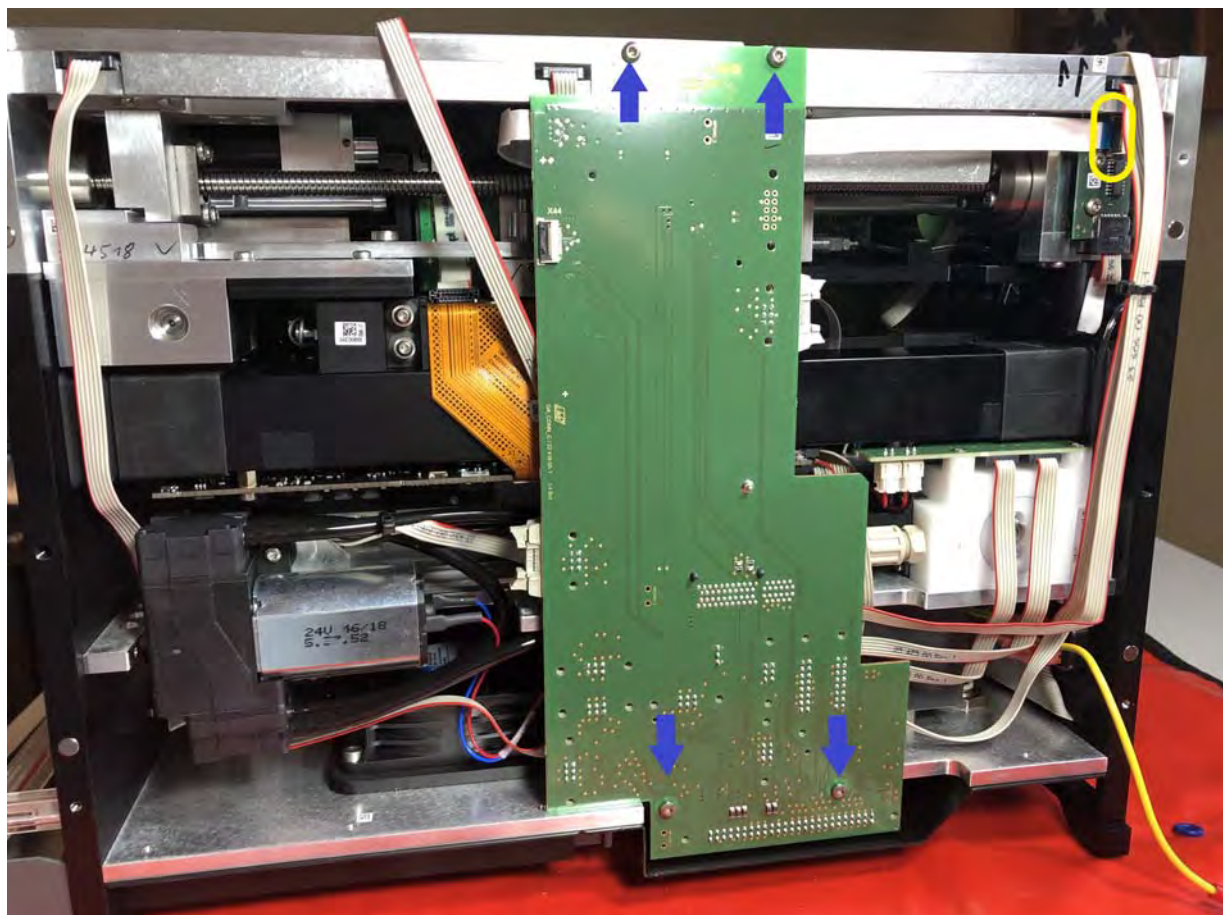
6.12 RP11 - Auxiliary Board

Part Required	9026542 - Auxiliary Board
Tool(s) Required	T10 TORX tool
Calibration Required	CP1 Automated Mechanical Calibrations
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hour
Calibration/Testing Time	3 Hours
Total Repair Time	4 Hours

Table 6.12-1 RP11 Auxiliary board.

6.12.1 RP11 - Auxiliary Board - Removal Procedure

1. Remove the module covers, [see "RP9 - AM Covers - Removal Procedure" on page 64.](#)
2. Remove the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly- Removal Procedure" on page 106.](#)
3. Disconnect the flex cable.



Picture 6.12-2 Auxiliary board screws and flex cable.

4. Use a T10 TORX to remove the 5 mounting screws.
5. Disconnect the connectors from the board.
6. Remove the 4 Amp Technosoft Driver Module.

6.12.2 RP11 - Auxiliary Board - Installation Procedure

1. Install the 4 Amp Technosoft Driver Module.
2. Connect the electrical connectors.
3. Connect the flex cable.
4. Secure the board with the four T10 TORX screws.
5. Install the module covers, [see “RP3 - Operational Module Right Cover - Replacement Procedure” on page 56.](#)
6. Install the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly - Installation Procedure” on page 107.](#)
7. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
8. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
9. Run ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169.](#)

6.13 RP12 - Cartridge Lifter Assembly

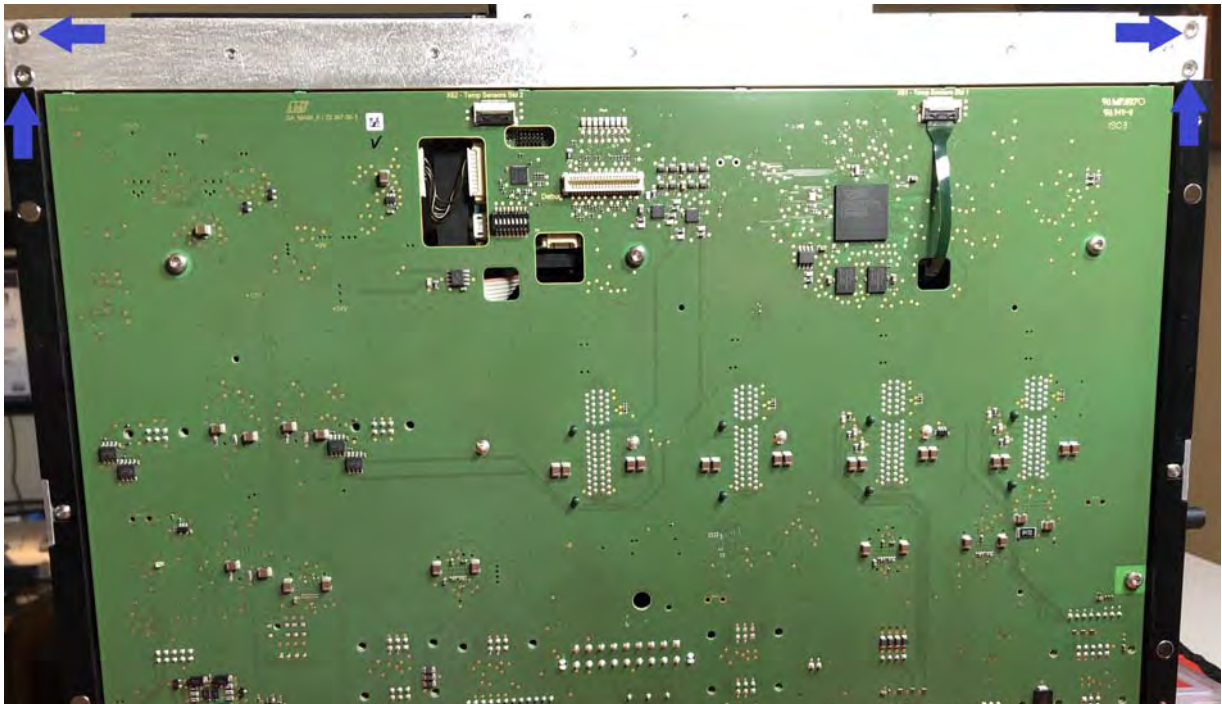
Part Required	Non-Stock listed-Cartridge Lifter Assembly
Tool(s) Required	T10 TORX tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP3 - Actuators Status Test
Replacement Time	1 Hour
Calibration/Testing Time	30 Minutes
Total Repair Time	1 Hour and 30 Minutes

Table 6.13-1 RP12 Cartridge lifter assembly.

6.13.1 RP12 - Cartridge Lifter Assembly - Removal Procedure

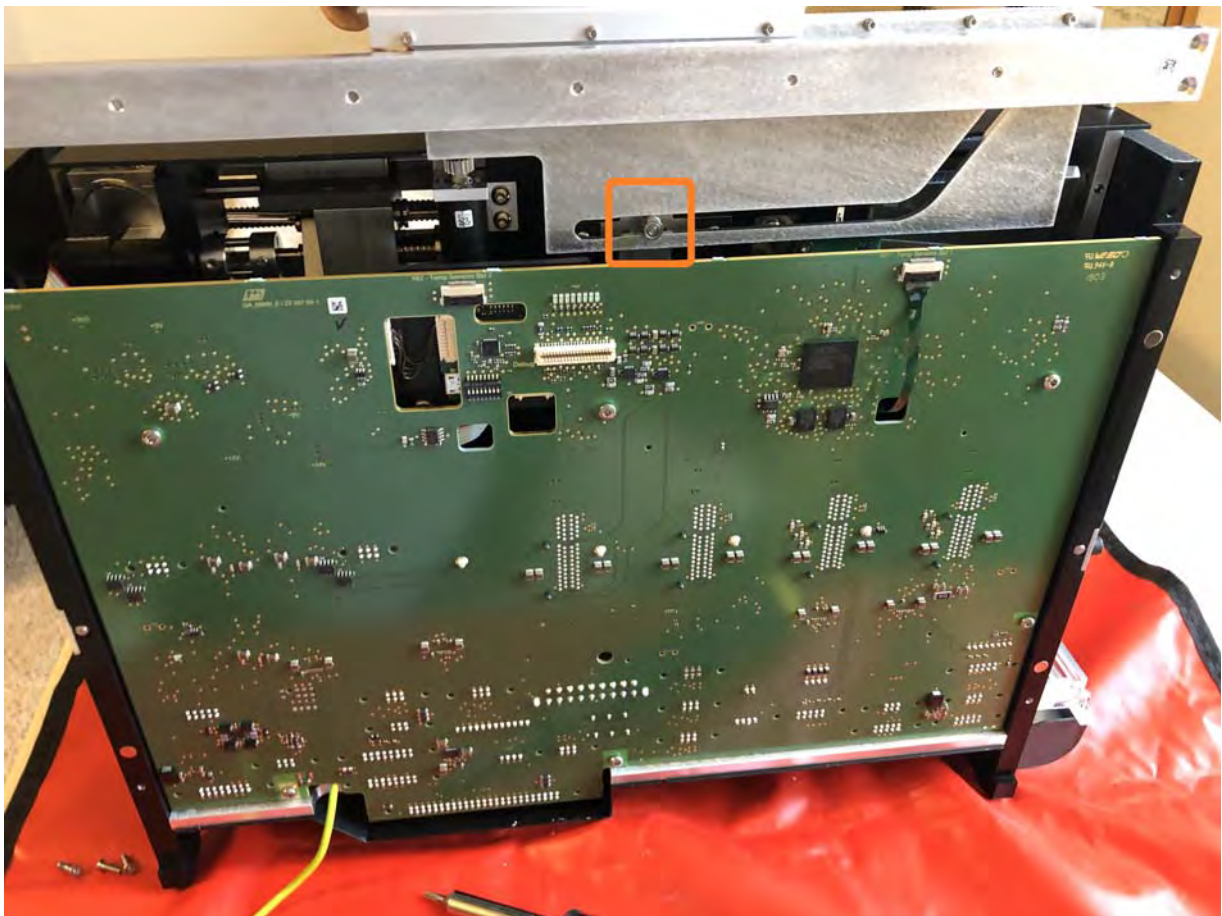
1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)

3. Remove the four T10 TORX mounting screws.



Picture 6.13-2 Cartridge lifter screws.

4. Move the lifter upward and forward to disengage the cartridge lifter pin and remove the cartridge lifter assembly.



Picture 6.13-3 Cartridge lifter pin.

6.13.2 RP12 - Cartridge Lifter Assembly - Installation Procedure

1. Lift the cartridge lifter pin and guide it into the lifter slot.
2. Slide the ends of the module into place and secure it with the four T10 TORX screws.
3. Install the covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
4. Install the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly - Installation Procedure” on page 107.](#)
5. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
6. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
7. Run actuators status test, [see “VP3 - Actuators Status Test” on page 168.](#)

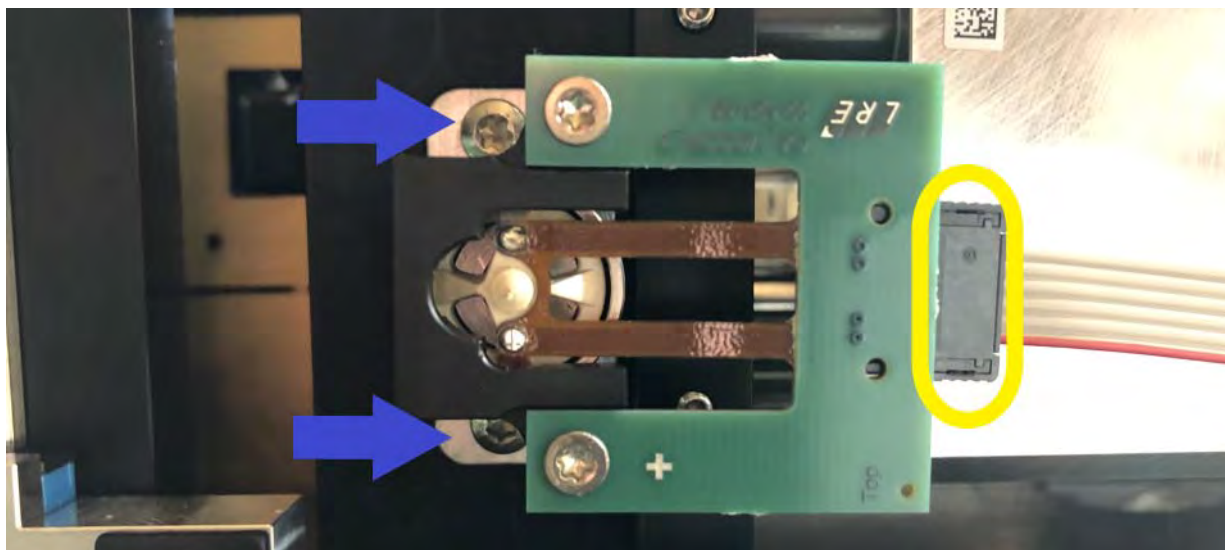
6.14 RP13 - Voice Coil

Part Required	9026544 - Voice Coil
Tool(s) Required	T10 TORX tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hour
Calibration/Testing Time	30 Minutes
Total Repair Time	1 Hour and 30 Minutes

Table 6.14-1 RP13 Voice coil.

6.14.1 RP13 - Voice Coil - Removal Procedure

1. Remove the covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)
3. Remove the Main Board, [see “RP10 - Main Board - Removal Procedure” on page 69.](#)
4. Use a T10 TORX to Remove the main board mounting screws.
5. Disconnect the Voice Coil connector.



Picture 6.14-2 Voice coil.

6. Remove the two T10 TORX mounting screws.

6.14.2 RP13 - Voice Coil - Installation Procedure

1. Mount the Voice Coil with the two T10 TORX screws and connect the electrical connector.
2. Install the Main Board, [see “RP10 - Main Board - Installation Procedure” on page 70.](#)
3. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
4. Install Filter Tray Assembly, [see “RP35 - Filter Tray Assembly - Installation Procedure” on page 107.](#)
5. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
6. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
7. Run ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169.](#)

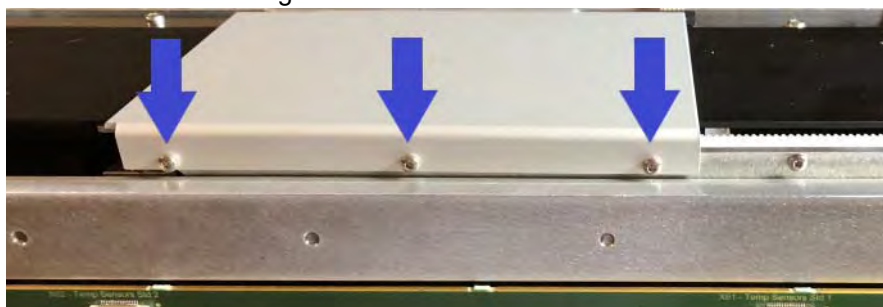
6.15 RP14 - Lid Mechanism

Part Required	9026388 Lid Mechanism
Tool(s) Required	T6, T10 TORX tool, X-9900 Alignment Tool
Calibration Required	N/A
Testing Required	VP9 - Lid Mechanism Alignment , VP1 - System Initialization
Replacement Time	1 Hour
Calibration/Testing Time	30 Minutes
Total Repair Time	1 Hour and 30 Minutes

Table 6.15-1 RP14 Lid mechanism.

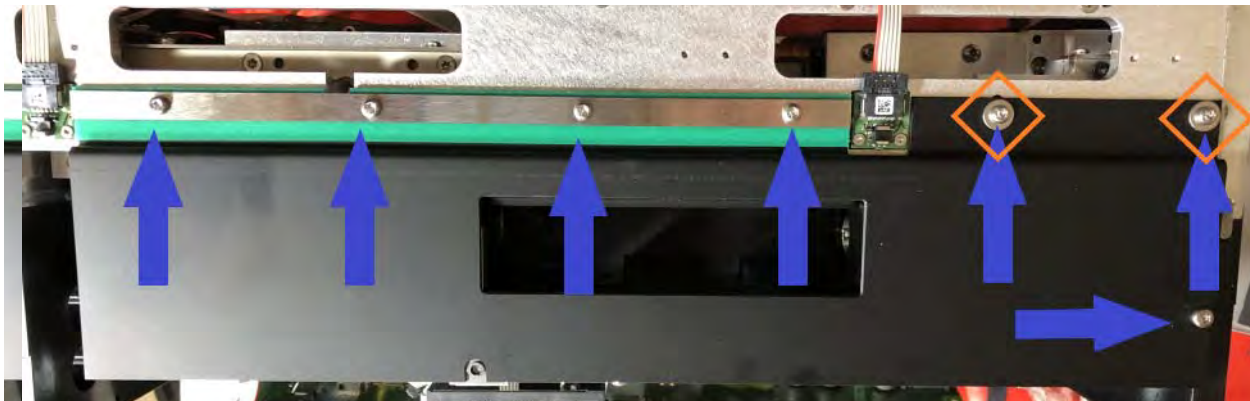
6.15.1 RP14 - Lid Mechanism - Removal Procedure

1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)
3. Using a T6 TORX remove the three screws holding the door.



Picture 6.15-2 Lid screws.

4. Remove the seven T10 TORX mounting screws and lift the assembly, Note screws with washers.



Picture 6.15-3 Lid mechanism screws.

6.15.2 RP14 - Lid Mechanism - Installation Procedure

1. Install the Lid Mechanism but leave the 7 screws loose.
2. Align the opening to ensure cartridges move through without binding.
3. Install the door using the three T6 screws.
4. Install the module covers, [see "RP9 - AM Covers- Installation Procedure" on page 67.](#)
5. Install the filter tray assembly, [see "RP35 - Filter Tray Assembly - Installation Procedure" on page 107.](#)
6. Adjust the lid, [VP9 - Lid Mechanism Alignment.](#)
7. Initialize the system, [VP1 - System Initialization.](#)

6.16 RP15 - TC and Clamping Module

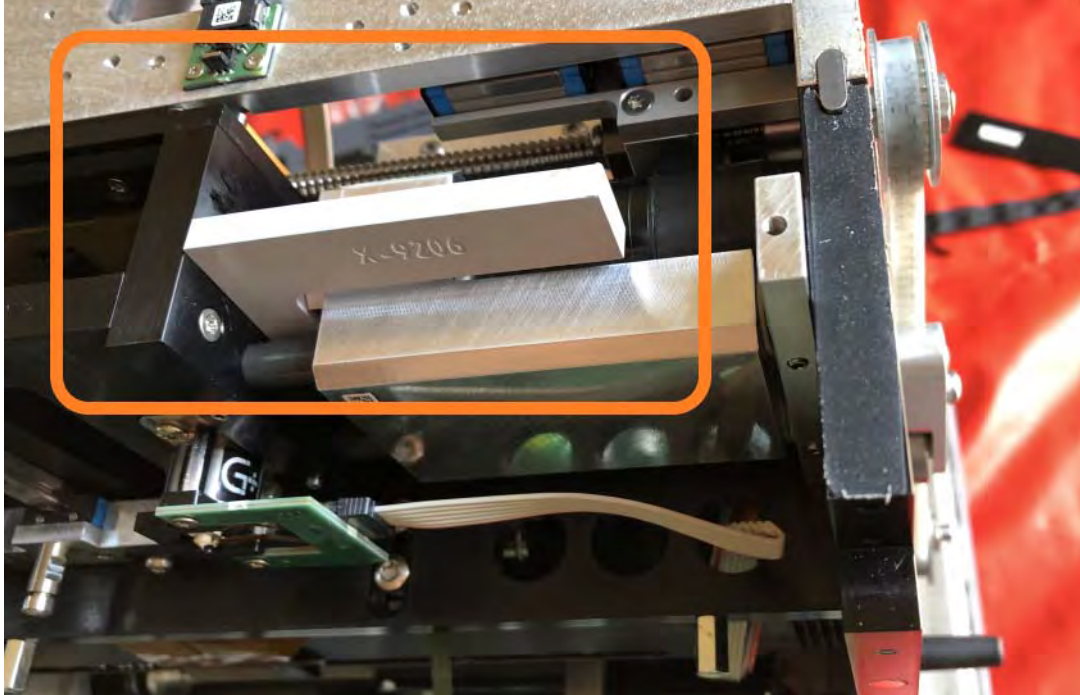
Part Required	9026545 - TC and Clamping Module
Tool(s) Required	T8,T10,T20 TORX tools, 10mm socket and wrench, 8mm Open End Wrench, and 9026436, X-9206 Distance Support Angle
Calibration Required	CP1 Automated Mechanical Calibrations
Testing Required	VP10 - Belt Tension Adjustment , VP1 - System Initialization , VP2 - Preparation For Testing , VP4 - Pneumatics Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	2 Hours and 30 Minutes
Calibration/Testing Time	2 Hours and 30 Minutes
Total Repair Time	5 Hours

Table 6.16-1 RP15 TC and clamping module.

6.16.1 RP15 - TC and Clamping Module - Removal Procedure

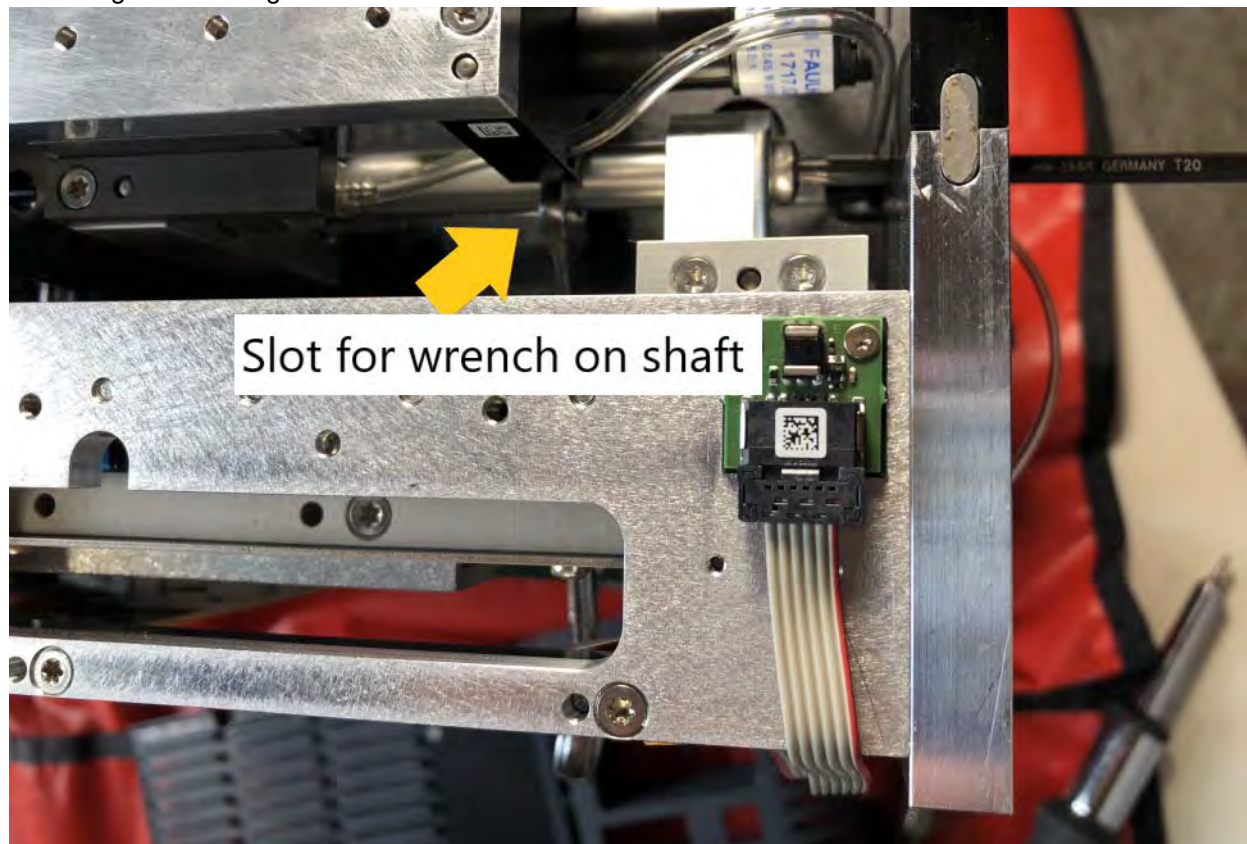
1. Remove the module covers, [see "RP9 - AM Covers - Removal Procedure" on page 64.](#)
2. Remove the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly- Removal Procedure" on page 106](#)
3. Remove the Cartridge Lifter Assembly, [see "RP12 - Cartridge Lifter Assembly - Removal Procedure" on page 72.](#)
4. Remove the Lid Mechanism, [see "RP14 - Lid Mechanism - Removal Procedure" on page 75.](#)

5. Install the X-9206 angled spacer in the Bead Beater Assembly.



Picture 6.16-2 Distance support angle placement.

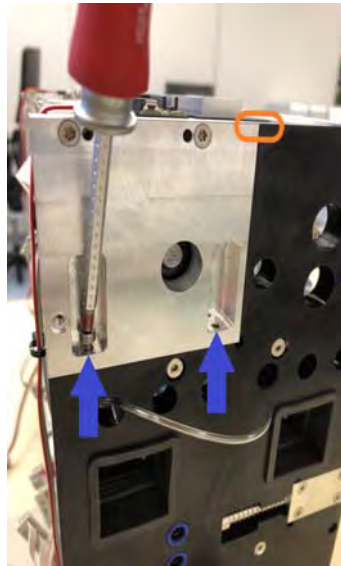
6. Remove the Main Board, see [“RP10 - Main Board - Removal Procedure”](#) on page 69.
7. Remove the Auxiliary board, see [“RP11 - Auxiliary Board - Removal Procedure”](#) on page 71.
8. Using a T20 TORX and 8mm Open End Wrench remove the Plunger Assembly. Extract the pusher through the cartridge slot.



Picture 6.16-3 Loosen plunger screw.

9. Remove the Top Pulley and Belt, see [“RP24 - Belt and Pulleys - Removal Procedure”](#) on page 90.
10. Remove the six electrical connectors.
11. Remove the four T10 TORX mounting screws on the ends and lift out the module.

12. Remove the Match Pins, [see “RP30 - Match Pins - Removal Procedure” on page 100.](#)



Picture 6.16-4 Module screws and match pins.

6.16.2 RP15 - TC and Clamping Module - Installation Procedure

1. Mount the module and insert the two match pins, [see “RP30 - Match Pins - Installation Procedure” on page 100.](#)
2. Install the plunger assembly and secure it using the 8mm wrench and T20 TORX end screw.
3. Connect the six electrical connectors.
4. Remove the X-9206 spacer.
5. Install the Lid Mechanism, [see “RP14 - Lid Mechanism - Installation Procedure” on page 76.](#)
6. Install the Cartridge Lifter Assembly [see “RP12 - Cartridge Lifter Assembly - Installation Procedure” on page 74.](#)
7. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
8. Install the Filter Tray Assembly, [see “RP12 - Cartridge Lifter Assembly - Installation Procedure” on page 74.](#)
9. Adjust the belt, [see “VP10 - Belt Tension Adjustment” on page 173.](#)
10. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
11. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
12. Run pneumatics test, [see “VP4 - Pneumatics Test” on page 168.](#)
13. Run ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169.](#)

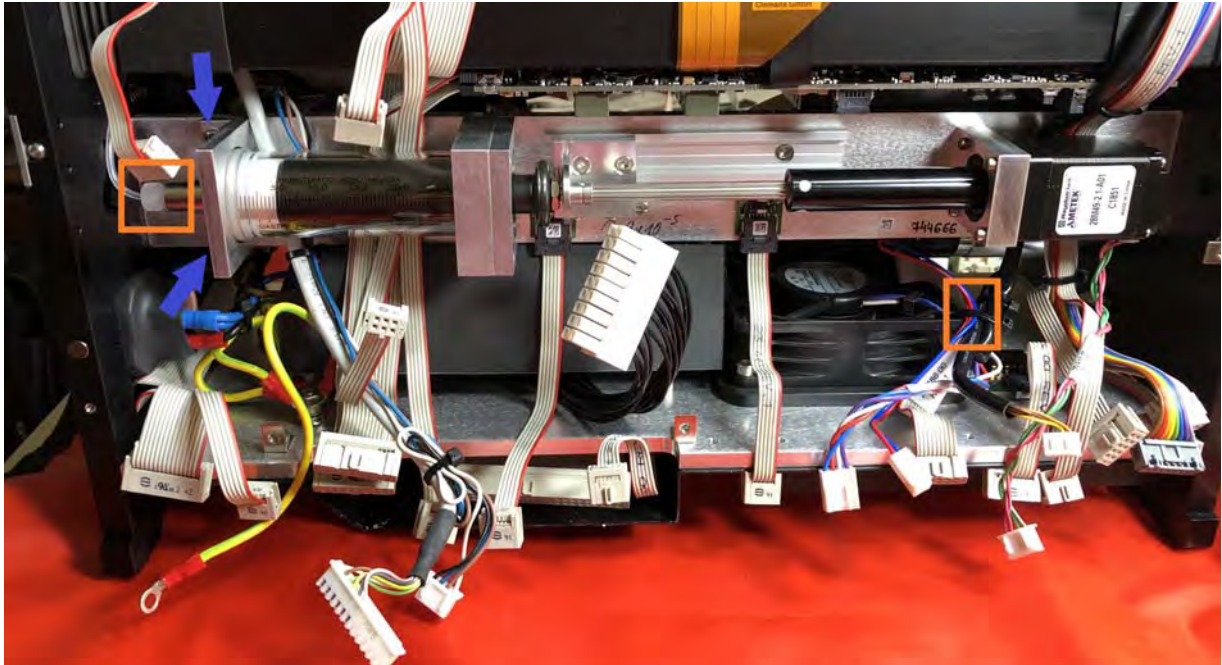
6.17 RP16 - Syringe Module

Part Required	9026387 Syringe Module
Tool(s) Required	T10 TORX tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP4 - Pneumatics Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	2 Hours
Calibration/Testing Time	2 Hours and 30 Minutes
Total Repair Time	4 Hours and 30 Minutes

Table 6.17-1 RP16 Syringe module.

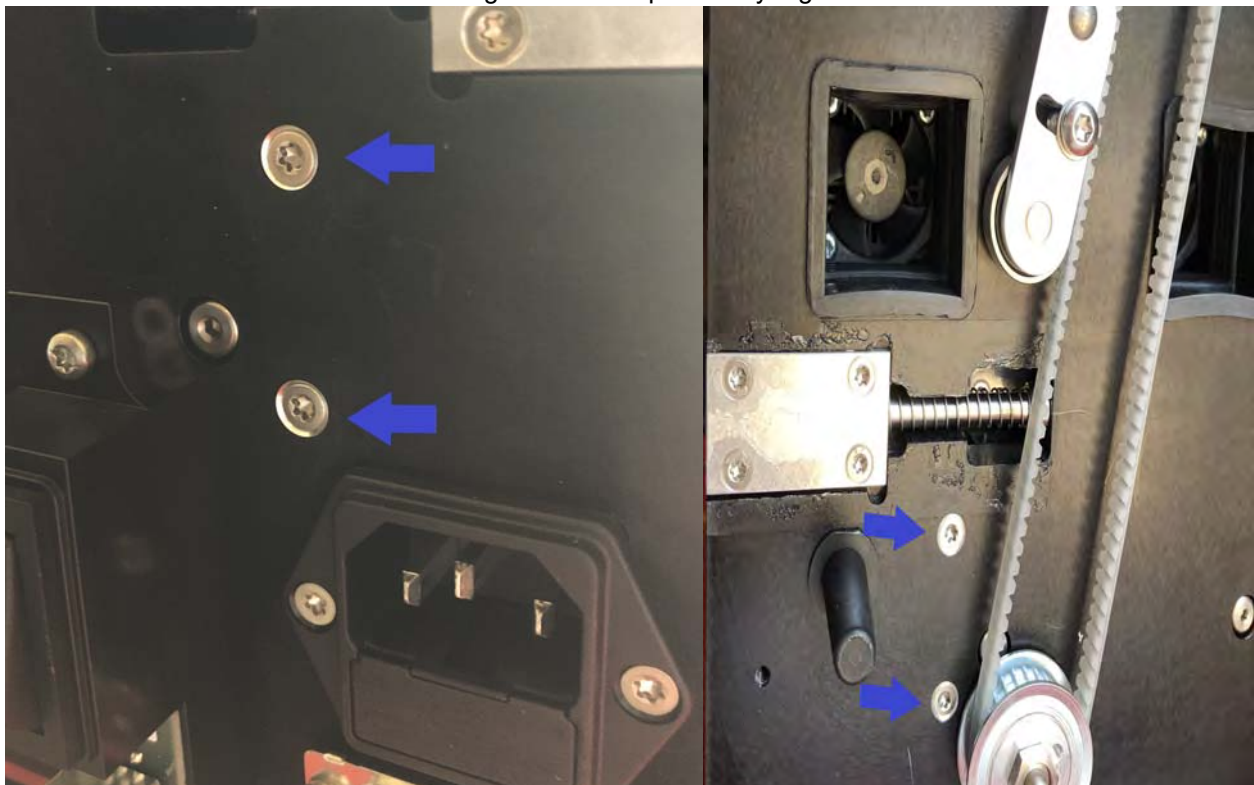
6.17.1 RP16 - Syringe Module - Removal Procedure

1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)
3. Remove the main board assembly, [see “RP10 - Main Board - Removal Procedure” on page 69.](#)
4. Remove cable tie holding the black cable to the fan.
5. Unscrew the syringe luer fitting and using a T10 TORX remove the front syringe support bracket.



Picture 6.17-2 Syringe module.

6. Move the cables from behind the syringe and motor.
7. Remove the four T10 TORX mounting screws and pull the syringe module out of the OM.



Picture 6.17-3 Syringe module mounting screws.

6.17.2 RP16 - Syringe Module - Installation Procedure

1. Mount the Syringe Module and secure it with the T10 TORX screws.
2. Slide the cables behind the syringe and syringe motor.
3. Install the syringe front support bracket with two T10 TORX screws and connect the luer fitting.
4. Install the zip tie on holding the black cable to the fan.
5. Install the Main Board, [see "RP10 - Main Board - Installation Procedure" on page 70.](#)
6. Install the module covers, [see "RP9 - AM Covers- Installation Procedure" on page 67.](#)
7. Install the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly - Installation Procedure" on page 107.](#)
8. Initialize the analyzer, [see "VP1 - System Initialization" on page 167.](#)
9. Prepare for testing, [see "VP2 - Preparation For Testing" on page 168.](#)
10. Run pneumatics test, [see "VP4 - Pneumatics Test" on page 168.](#)
11. Run ACC test, [see "VP7 - Analyzer Check Cartridge Work Flow Test" on page 169.](#)

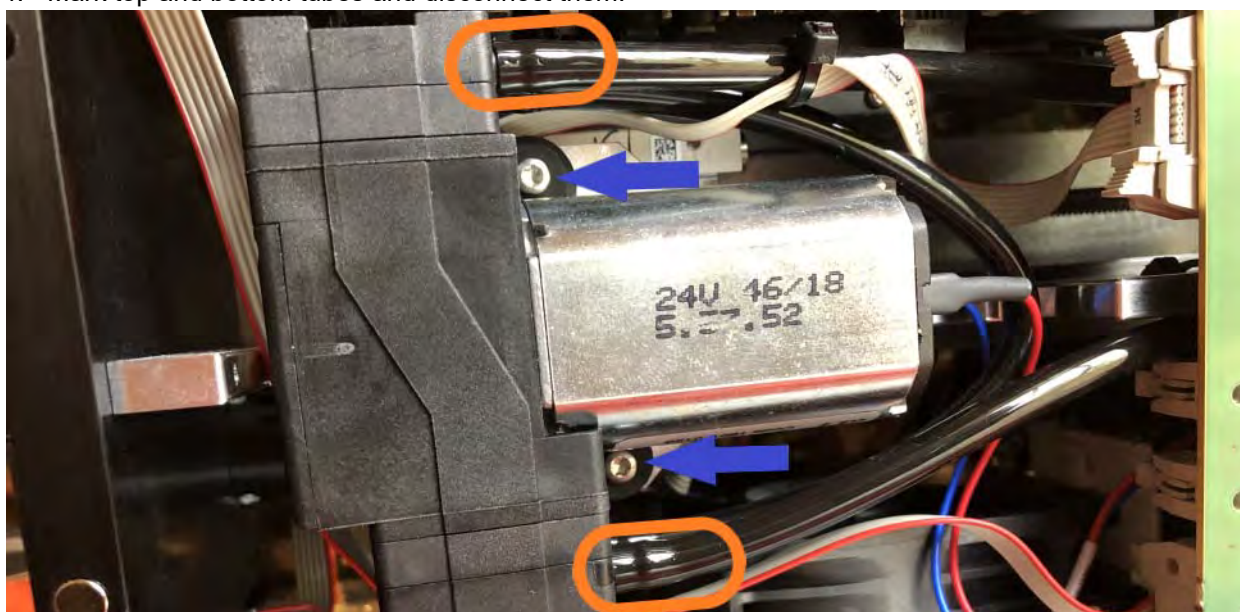
6.18 RP17 - Pressure Pump

Part Required	9026547 - Pressure Pump
Tool(s) Required	T10 Torx Tool
Calibration Required	CP1 Automated Mechanical Calibrations
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP4 - Pneumatics Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hour
Calibration/Testing Time	2 Hours and 30 Minutes
Total Repair Time	3 Hours and 30 Minutes

Table 6.18-1 RP17 Pressure pump.

6.18.1 RP17 - Pressure Pump- Removal Procedure

1. Remove the module covers, [see "RP9 - AM Covers - Removal Procedure" on page 64.](#)
2. Remove the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly- Removal Procedure" on page 106.](#)
3. Remove the Auxiliary Board, [see "RP11 - Auxiliary Board - Removal Procedure" on page 71.](#)
4. Mark top and bottom tubes and disconnect them.



Picture 6.18-2 Pressure pump screws and tubing.

5. Remove the two T10 TORX mounting screws and guide the pump out.

6.18.2 RP17 - Pressure Pump - Installation Procedure

1. Install the Pressure Pump with the two T10 TORX screws.
2. Connect the two tubes to the pump assembly.
3. Install the Auxiliary Board, [see "RP11 - Auxiliary Board - Installation Procedure" on page 72.](#)
4. Install the module covers, [see "RP9 - AM Covers- Installation Procedure" on page 67.](#)
5. Install the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly - Installation Procedure" on page 107.](#)
6. Initialize the analyzer, [see "VP1 - System Initialization" on page 167.](#)
7. Prepare for testing, [see "VP2 - Preparation For Testing" on page 168.](#)
8. Run pneumatics test, [see "VP4 - Pneumatics Test" on page 168.](#)
9. Run ACC test, [see "VP7 - Analyzer Check Cartridge Work Flow Test" on page 169.](#)

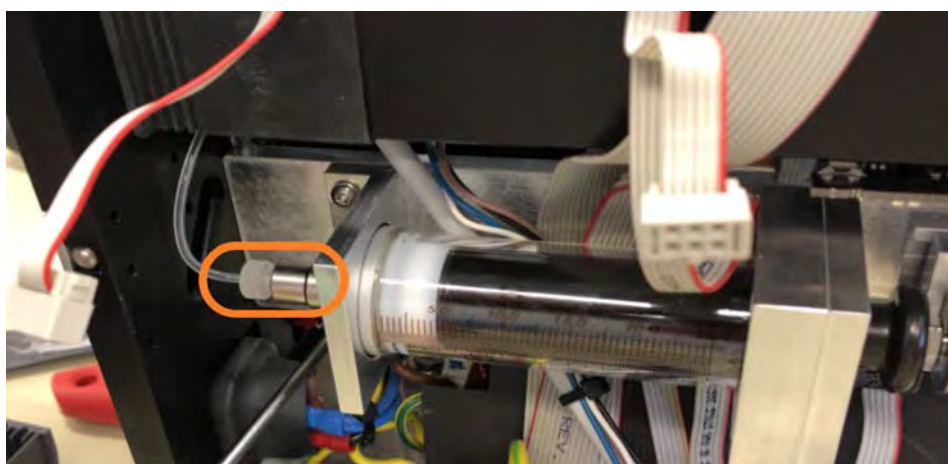
6.19 RP18 - Pneumatic Manifold

Part Required	9026548 - Pneumatic Manifold
Tool(s) Required	T10 Torx Tool
Calibration Required	CP1 Automated Mechanical Calibrations
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP4 - Pneumatics Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hour
Calibration/Testing Time	2 Hours and 30 Minutes
Total Repair Time	3 Hours and 30 Minutes

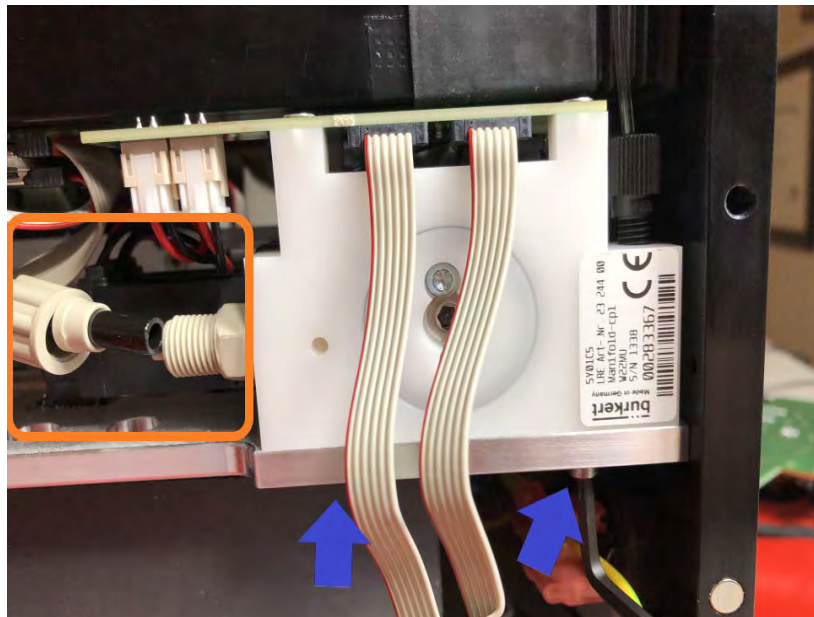
Table 6.19-1 RP18 Pneumatic manifold.

6.19.1 RP18 - Pneumatic Manifold - Removal Procedure

1. Remove the module covers, [see "RP9 - AM Covers - Removal Procedure" on page 64.](#)
2. Remove the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly- Removal Procedure" on page 106.](#)
3. Remove the Auxiliary Board, [see "RP11 - Auxiliary Board - Removal Procedure" on page 71.](#)
4. Disconnect the lure fitting on the syringe.

*Picture 6.19-2 Syringe luer fitting.*

5. Loosen the pressure fitting and disconnect the ferrule and tubing.



Picture 6.19-3 Manifold pressure fitting and screws.

6. Remove the two T10 TORX mounting screws below the mounting plate and move the manifold out.



Picture 6.19-4 Manifold pneumatic fittings.

7. Mark and disconnect the three pneumatic fittings on the rear of the assembly.

6.19.2 RP18 - Pneumatic manifold - Installation Procedure

1. Connect the three pneumatic fittings using your marks to connect to the proper orifice.
2. Slide the manifold into position and secure from the bottom with the two T10 TORX screws.
3. Connect the pressure tubing and fitting.
4. Install the Auxiliary Board, [see "RP11 - Auxiliary Board - Installation Procedure" on page 72.](#)
5. Install the module covers, [see "RP9 - AM Covers- Installation Procedure" on page 67.](#)
6. Install the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly - Installation Procedure" on page 107.](#)
7. Initialize the analyzer, [see "VP1 - System Initialization" on page 167.](#)
8. Prepare for testing, [see "VP2 - Preparation For Testing" on page 168.](#)
9. Run pneumatics test, [see "VP4 - Pneumatics Test" on page 168.](#)
10. Run ACC test, [see "VP7 - Analyzer Check Cartridge Work Flow Test" on page 169.](#)

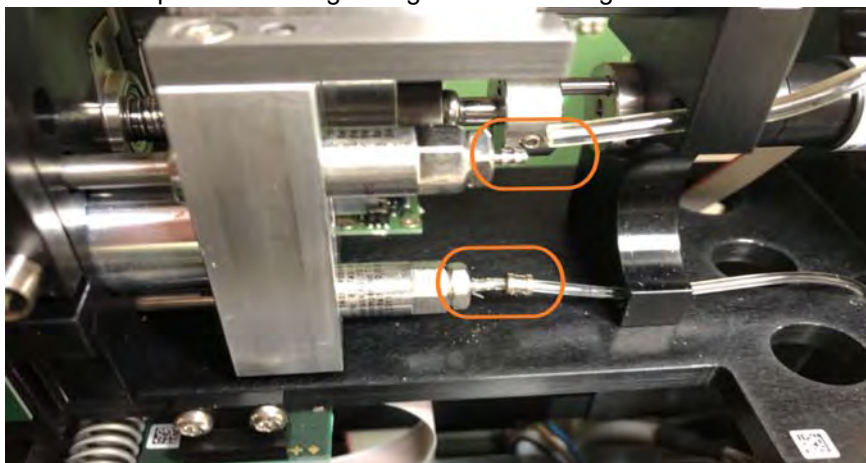
6.20 RP19 - Thermal Unit

Part Required	9026546 - Thermal Unit
Tool(s) Required	T10, T15, T20 Torx Tool, 2.0 mm Hex Tool, X-7780 Alignment Tool
Calibration Required	CP2 qPCR Optical Alignment & Thermal Unit Test , CP1 Automated Mechanical Calibrations
Testing Required	VP11 - Thermal Unit Heating Plate Alignment , VP1 - System Initialization , VP2 - Preparation For Testing , VP3 - Actuators Status Test , VP6 - Thermal Unit Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	3 Hours and 30 Minutes
Calibration/Testing Time	3 Hours
Total Repair Time	6 Hours and 30 Minutes

Table 6.20-1 RP19 Thermal unit.

6.20.1 RP19 - Thermal Unit - Removal Procedure

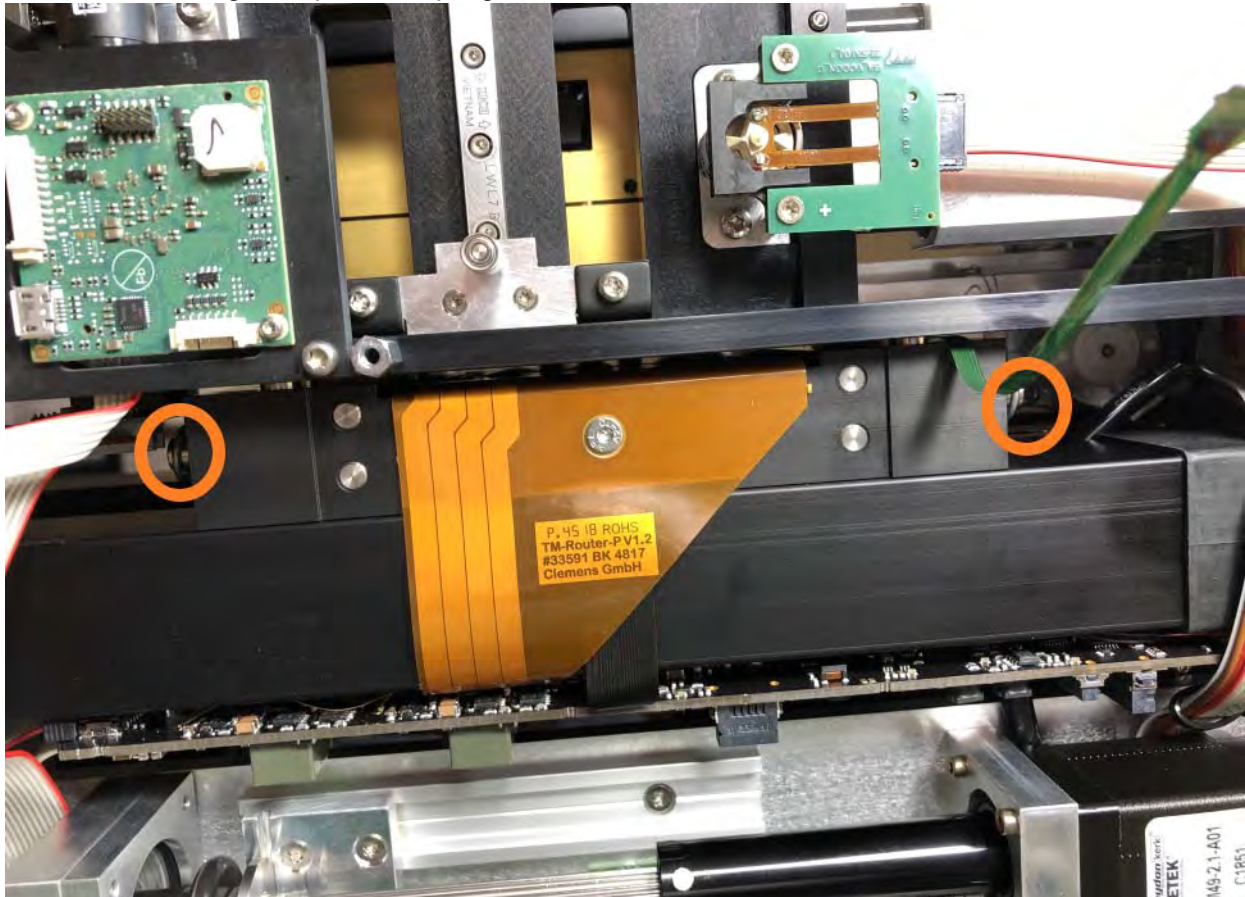
1. Remove the all module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)
3. Remove the Auxiliary Board, [see “RP11 - Auxiliary Board - Removal Procedure” on page 71.](#)
4. Remove the Main Board, [see “RP10 - Main Board - Removal Procedure” on page 69.](#)
5. Remove the top Pulley and Belt, [see “RP24 - Belt and Pulleys - Removal Procedure” on page 90.](#)
6. Remove the Cartridge Lifter Assembly, [see “RP12 - Cartridge Lifter Assembly - Removal Procedure” on page 72.](#)
7. Remove the Lid Mechanism, [see “RP14 - Lid Mechanism - Removal Procedure” on page 75.](#)
8. Remove the TC and Clamping Module, [see “RP15 - TC and Clamping Module - Removal Procedure” on page 76.](#)
9. Disconnect the vacuum and pressure tubing noting the tube routing.



Picture 6.20-2 Vacuum and pressure tubing.

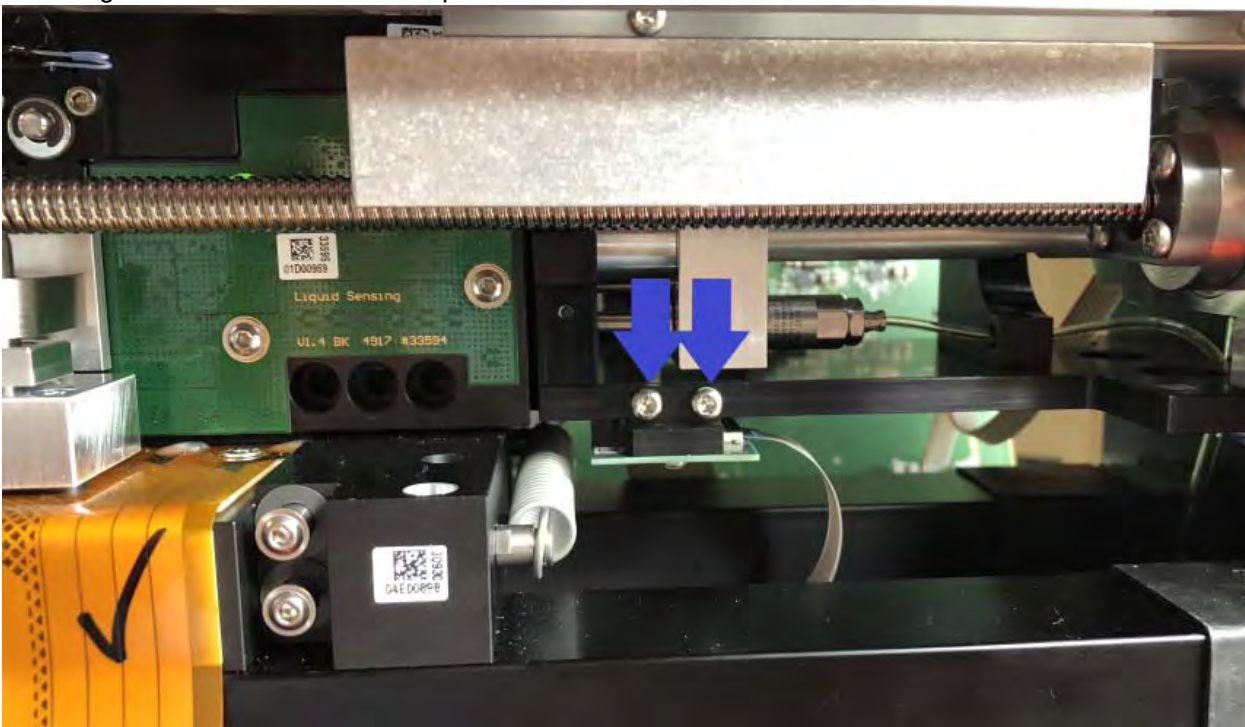
10. Remove the Syringe Module, [see “RP16 - Syringe Module - Removal Procedure” on page 79.](#)

11. Remove the large compression springs.



Picture 6.20-3 Thermal unit springs.

12. Using a T10 TORX unmount the qPCR connector board.



Picture 6.20-4 qPCR connector board.

13. Remove the four flex air tubes.



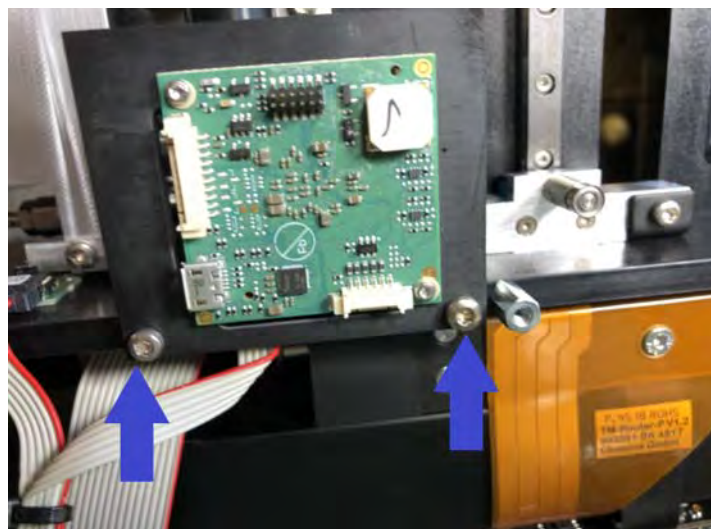
Picture 6.20-5 Thermal unit flex air tubes.

14. Remove the lifter plate and spindle by removing the two T10 TORX screws.



Picture 6.20-6 Cartridge lifter plate and spindle.

15. Remove the Voice Coil Assembly, [see "RP13 - Voice Coil - Removal Procedure" on page 74.](#)
16. Use a T10 TORX to remove the Bar Code Reader.



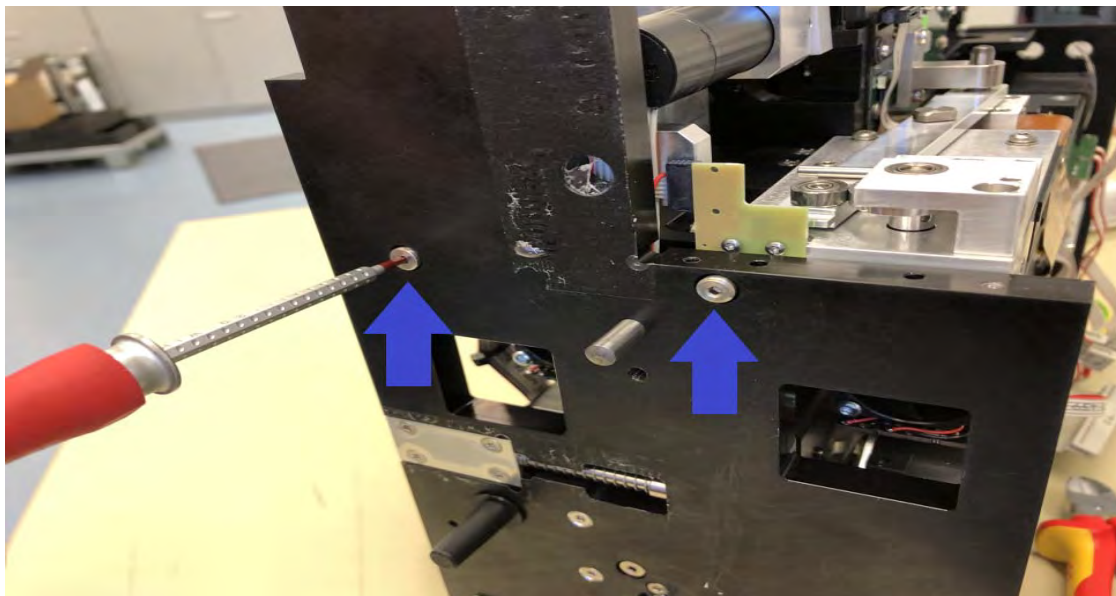
Picture 6.20-7 Bar code reader.

17. Using a 2.0mm HEX tool remove the clamping lever pin.



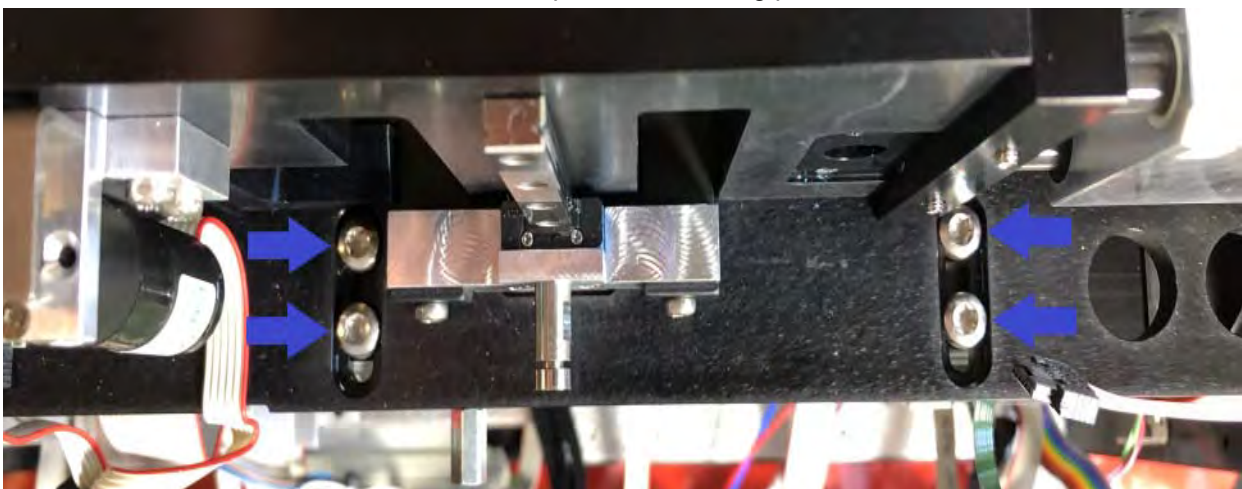
Picture 6.20-8 Clamping lever pin.

18. Remove the two 2.0 mm HEX end mounting screws on each end and lift the module out of the chassis.



Picture 6.20-9 Thermal unit mounting screws.

19. Remove the four T20 TORX screws and separate the heating plate from the Thermal Unit.



Picture 6.20-10 Thermal unit heating plate screws.

6.20.2 RP19 - Thermal Unit - Installation Procedure

1. Mount the heating plate to the heater assembly and leave the four T20 TORX screws loose.
2. Align the heating plates.
3. Install the module in the chassis with the four 2.0mm mounting screws.
4. Reinstall The Springs.
5. Install the clamping lever pin.
6. Install the bar code reader.
7. Install the voice coil, [see "RP13 - Voice Coil - Installation Procedure" on page 75.](#)
8. Install the cartridge lifter plate and spindle.
9. Install the rubber end boots.
10. Install the pQCR connector board.
11. Install the two Thermal Unit springs.
12. Reconnect the vacuum and pressure tubing.
13. Install the TC and Clamping Module, [see "RP15 - TC and Clamping Module - Installation Procedure" on page 78.](#)
14. Install the Lid Mechanism, [see "RP14 - Lid Mechanism - Installation Procedure" on page 76.](#)
15. Install the Cartridge Lifter Assembly, [see "RP12 - Cartridge Lifter Assembly - Installation Procedure" on page 74.](#)
16. Install the Belt and Pulleys, [see "RP24 - Belt and Pulleys - Installation Procedure" on page 92.](#)
17. Install the Main Board, [see "RP10 - Main Board - Installation Procedure" on page 70.](#)
18. Install the Auxiliary Board, [see "RP11 - Auxiliary Board - Installation Procedure" on page 72.](#)
19. Install the module Covers, [see "RP9 - AM Covers- Installation Procedure" on page 67.](#)
20. Install the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly - Installation Procedure" on page 107.](#)
21. Run qPCR alignment, [see "CP2 qPCR Optical Alignment & Thermal Unit Test" on page 142.](#)
22. Run mechanical calibrations, [see "CP1 Automated Mechanical Calibrations" on page 112.](#)
23. Initialize the analyzer, [see "VP1 - System Initialization" on page 167.](#)
24. Prepare for testing, [see "VP2 - Preparation For Testing" on page 168.](#)
25. Run actuators status test, [see "VP3 - Actuators Status Test" on page 168.](#)
26. Run the thermal test, [see "VP6 - Thermal Unit Test" on page 169.](#)
27. Run ACC test, [see "VP7 - Analyzer Check Cartridge Work Flow Test" on page 169.](#)

6.21 RP20 - Power Supply

Part Required	9026550 - Power Supply
Tool(s) Required	T10 TORX Tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hour
Calibration/Testing Time	1 Hour
Total Repair Time	2 Hours

Table 6.21-1 RP20 Power supply.

6.21.1 RP20 - Power Supply- Removal Procedure

1. Remove the module covers, [see "RP9 - AM Covers - Removal Procedure" on page 64.](#)
2. Remove the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly- Removal Procedure" on page 106.](#)
3. Remove the Main Board, [see "RP10 - Main Board - Removal Procedure" on page 69.](#)
4. Remove the Auxiliary Board, [see "RP11 - Auxiliary Board - Removal Procedure" on page 71.](#)

5. Push down the paper barrier and remove the four T10 TORX mounting screws, two on each side.



Picture 6.21-2 Power supply.

6. Disconnect the white line input power connector.



Picture 6.21-3 Line input power connector.

7. Remove power supply and disconnect output power supply cable.

6.21.2 RP20 - Power Supply - Installation Procedure

1. Install the output power supply cable on the board.
2. Place the paper barrier under the power supply, slide it in place and connect the input power connector.
3. Secure the power supply assembly with the four T10 TORX screws.
4. Install the Main Board, [see "RP10 - Main Board - Installation Procedure" on page 70.](#)
5. Install the Auxiliary Board, [see "RP11 - Auxiliary Board - Installation Procedure" on page 72.](#)
6. Install the module covers, [see "RP9 - AM Covers- Installation Procedure" on page 67](#)
7. Install the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly - Installation Procedure" on page 107.](#)

8. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
9. Run ACC test, [see “VP8 - Set Operational Module Serial Number” on page 169.](#)

6.22 RP21 - Module Cooling Fan

Part Required	9026551 - Cooling Fan
Tool(s) Required	T10 TORX Tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization
Replacement Time	1.0 Hours
Calibration/Testing Time	N/A
Total Repair Time	1.0 Hour

Table 6.22-1 RP21 Module Cooling fan.

6.22.1 RP21 - Module Cooling Fan - Removal Procedure

1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)
3. Remove the Auxiliary Board, [see “RP11 - Auxiliary Board - Removal Procedure” on page 71.](#)
4. Remove the Main Board, [see “RP10 - Main Board - Removal Procedure” on page 69.](#)
5. Remove the four T10 TORX mounting screws.

6.22.2 RP21 - Module Cooling Fan - Installation Procedure

1. Install the Module Cooling Fan and secure it with the Four T10 TORX screws.
2. Install the Main Board, [see “RP10 - Main Board - Installation Procedure” on page 70.](#)
3. Install the Auxiliary Board, [see “RP11 - Auxiliary Board - Installation Procedure” on page 72.](#)
4. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
5. Install the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly - Installation Procedure” on page 107.](#)
6. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)

6.23 RP22- Tilt Sensor

Part Required	9026549 - Tilt Sensor
Tool(s) Required	T10 Torx Tool
Calibration Required	CP1 Automated Mechanical Calibrations
Testing Required	VP1 - System Initialization , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	30 Minutes
Calibration/Testing Time	2 Hours
Total Repair Time	2 Hours and 30 Minutes

Table 6.23-1 RP22 Tilt sensor.

6.23.1 RP22 - Tilt Sensor- Removal Procedure

1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)

3. Remove the Main Board, [see “RP10 - Main Board - Removal Procedure” on page 69.](#)



Picture 6.23-2 Tilt sensor screws and connector.

4. Remove the two T10 TORX mounting screws and electrical connector.

6.23.2 RP22 - Tilt Sensor - Installation Procedure

1. Connect the electrical connector.
2. Mount the Tilt Sensor and secure it with the two T10 TORX screws.
3. Install the Main Board, [see “RP10 - Main Board - Installation Procedure” on page 70.](#)
4. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
5. Install the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly - Installation Procedure” on page 107.](#)
6. Run mechanical calibrations, [see “CP1 Automated Mechanical Calibrations” on page 112.](#)
7. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
8. Run ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169.](#)

6.24 RP24 - Belt and Pulleys

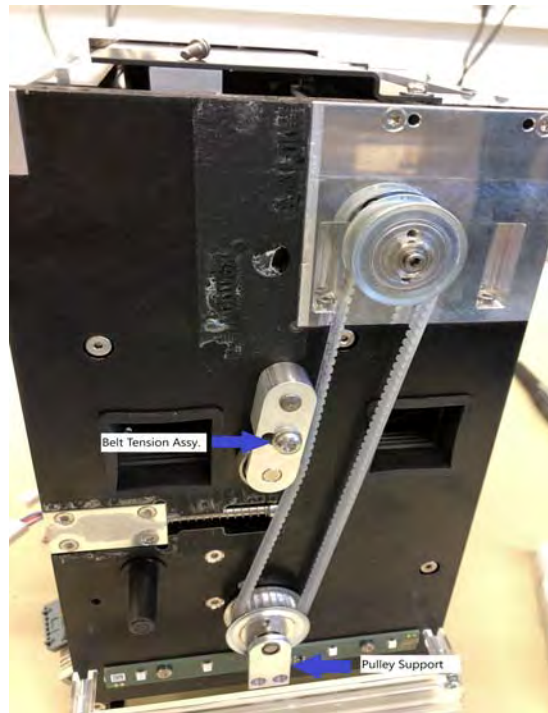
Part Required	9026561 - Cog belt
Tool(s) Required	T8, T20 TORX, Canon Plug Pliers, 10mm Socket, X-9500 Pulley Spacer Tool, Tension Meter, X-9881 Pulley Puller Tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP3 - Actuators Status Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hour
Calibration/Testing Time	2 Hour
Total Repair Time	3 Hours

Table 6.24-1 RP24 Belt and pulleys.

6.24.1 RP24 - Belt and Pulleys - Removal Procedure

1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)

3. Using T20 TORX loosen the belt tensioner.



Picture 6.24-2 Belt tensioner and pulley support.

4. Use a T8 TORX to remove the lower pulley support assembly.
5. Holding the pulleys with the canon plug pliers, use a 10mm socket to loosen the retainer nut on both pulleys but do not remove.
6. Use a T10 TORX and two screws to attach the X-9981 pulley puller tool over the retainer nut.



Picture 6.24-3 Pulley puller.

7. Use a T10 TORX slowly tighten the screws alternating between them until the pulley comes loose.
8. Remove the pulleys and belt.

6.24.2 RP24 - Belt and Pulleys - Installation Procedure

1. Slide the pulleys and belt on the shafts but do not tighten pulley retaining nuts.
2. Place the X-9500 Pulley Spacer behind the bottom pulley and push the pulley back to the spacer.



Picture 6.24-4 Pulley installation spacer.

3. With the top pulley pushed completely back and the bottom pulley flush against the spacer tool tighten the retaining nuts to 4mn using the canon plug pliers to hold each pulley.
4. Set the belt tension, [see "VP10 - Belt Tension Adjustment" on page 173.](#)
5. Install the module covers, [see "RP9 - AM Covers- Installation Procedure" on page 67.](#)
6. Install the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly - Installation Procedure" on page 107.](#)
7. Initialize the analyzer, [see "VP1 - System Initialization" on page 167.](#)
8. Prepare for testing, [see "VP2 - Preparation For Testing" on page 168.](#)
9. Run actuators status test, [see "VP3 - Actuators Status Test" on page 168.](#)
10. Run ACC test, [see "VP7 - Analyzer Check Cartridge Work Flow Test" on page 169.](#)

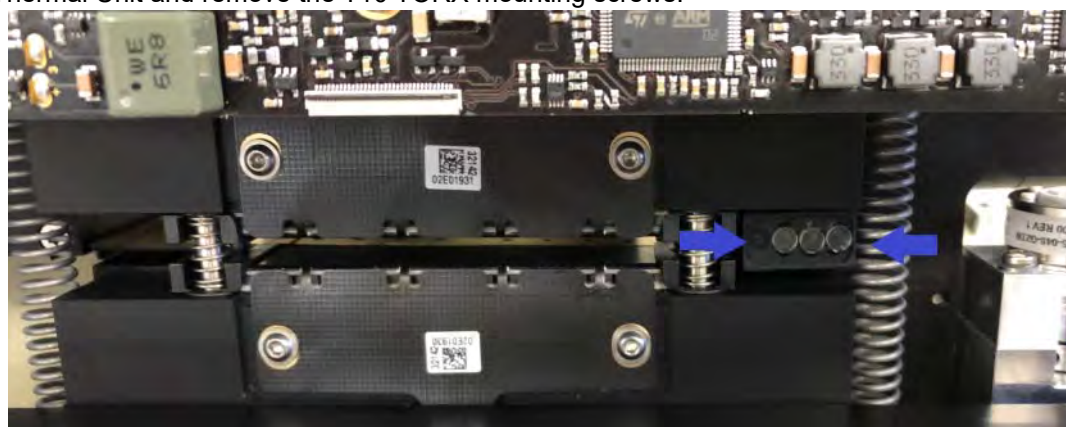
6.25 RP25- pQCR Standard

Part Required	9026533 - pQCR Standard
Tool(s) Required	T10 TORX Tool
Calibration Required	CP2 qPCR Optical Alignment & Thermal Unit Test , CP1 Automated Mechanical Calibrations
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP4 - Pneumatics Test , VP5 - qPCR Internal Standard Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	4 Hours
Calibration/Testing Time	4 Hours
Total Repair Time	8 Hours

Table 6.25-1 RP25 qPCR internal standard.

6.25.1 RP25 - qPCR Standard - Removal Procedure

1. Remove the all module covers, [see "RP9 - AM Covers - Removal Procedure" on page 64.](#)
2. Remove the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly- Removal Procedure" on page 106.](#)
3. Remove the Auxiliary Board, [see "RP11 - Auxiliary Board - Removal Procedure" on page 71.](#)
4. Remove the Main Board, [see "RP10 - Main Board - Removal Procedure" on page 69.](#)
5. Remove the top Pulley and Belt, [see "RP24 - Belt and Pulleys - Removal Procedure" on page 90.](#)
6. Remove the Cartridge Lifter Assembly, [see "RP12 - Cartridge Lifter Assembly - Removal Procedure" on page 72.](#)
7. Remove the Lid Mechanism, [see "RP14 - Lid Mechanism - Removal Procedure" on page 75.](#)
8. Remove the Transfer Cartridge, [see "RP15 - TC and Clamping Module - Removal Procedure" on page 76.](#)
9. Remove the Thermal Unit but do not remove the springs or separate the heating plate assembly from the heating unit.
10. Locate the qPCR standard on the bottom side of the Thermal Unit and remove the two T10 TORX mounting screws.
11. Invert the Thermal Unit and remove the T10 TORX mounting screws.



Picture 6.25-2 qPCR internal standard.

6.25.2 RP25 - qPCR Standard - Installation Procedure

1. Install the qPCR Standard Module securing it with two T10 TORX screws
2. Install the Thermal Unit, [see "RP19 - Thermal Unit - Installation Procedure" on page 87.](#)

3. Install the TC and Clamping Module, [see “RP15 - TC and Clamping Module - Installation Procedure” on page 78.](#)
4. Install the Lid Mechanism, [see “RP14 - Lid Mechanism - Installation Procedure” on page 76.](#)
5. Install the Cartridge Lifter Assembly, [see “RP12 - Cartridge Lifter Assembly - Installation Procedure” on page 74.](#)
6. Install the Pulley and Belt, [see “RP24 - Belt and Pulleys - Installation Procedure” on page 92.](#)
7. Install the Main Board, [see “RP10 - Main Board - Installation Procedure” on page 70.](#)
8. Install The Auxiliary Board, [see “RP11 - Auxiliary Board - Installation Procedure” on page 72.](#)
9. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
10. Install the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly - Installation Procedure” on page 107.](#)
11. Run the qPCR optical alignment, [see “CP2 qPCR Optical Alignment & Thermal Unit Test” on page 142.](#)
12. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
13. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
14. Run the pneumatics test, [see “VP4 - Pneumatics Test” on page 168.](#)
15. Run the qPCR internal standard test, [see “VP5 - qPCR Internal Standard Test” on page 168.](#)
16. Run the ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169.](#)

6.26 RP26- Main Board Technosoft Driver Modules

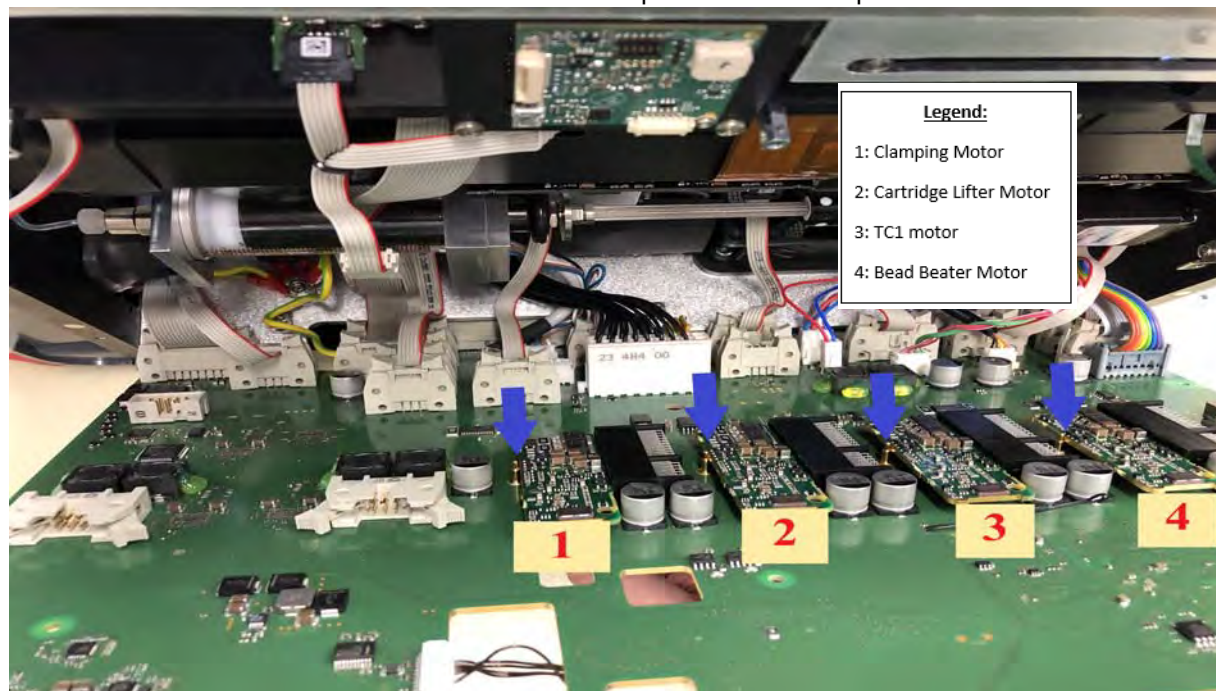
Part Required	9026474 (2 AMP) - Technosoft Driver Module
Tool(s) Required	T10 Torx Tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP3 - Actuators Status Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hours
Calibration/Testing Time	3 Hours
Total Repair Time	4 Hours

Table 6.26-1 RP26 Main board technosoft driver modules.

6.26.1 RP26 - Main Board Technosoft Driver Module - Removal Procedure

1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106](#)
3. Remove the Main Board, [see “RP10 - Main Board - Removal Procedure” on page 69.](#)

4. Locate the affected Technosoft Drive Module and press the retainer pin to release the board.



Picture 6.26-2 Main board technosoft drivers.

6.26.2 RP26 - Main Board Technosoft Driver - Installation Procedure

1. Press the retainer pin and inset the board.
2. Install the Main Board, [see "RP10 - Main Board - Installation Procedure" on page 70.](#)
3. Install the module covers, [see "RP9 - AM Covers- Installation Procedure" on page 67.](#)
4. Install the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly - Installation Procedure" on page 107.](#)
5. Initialize the analyzer, [see "VP1 - System Initialization" on page 167.](#)
6. Prepare for testing, [see "VP2 - Preparation For Testing" on page 168.](#)
7. Run actuators status test, [see "VP3 - Actuators Status Test" on page 168.](#)
8. Run ACC test, [see "VP7 - Analyzer Check Cartridge Work Flow Test" on page 169.](#)

6.27 RP27- Auxiliary Board Technosoft Driver

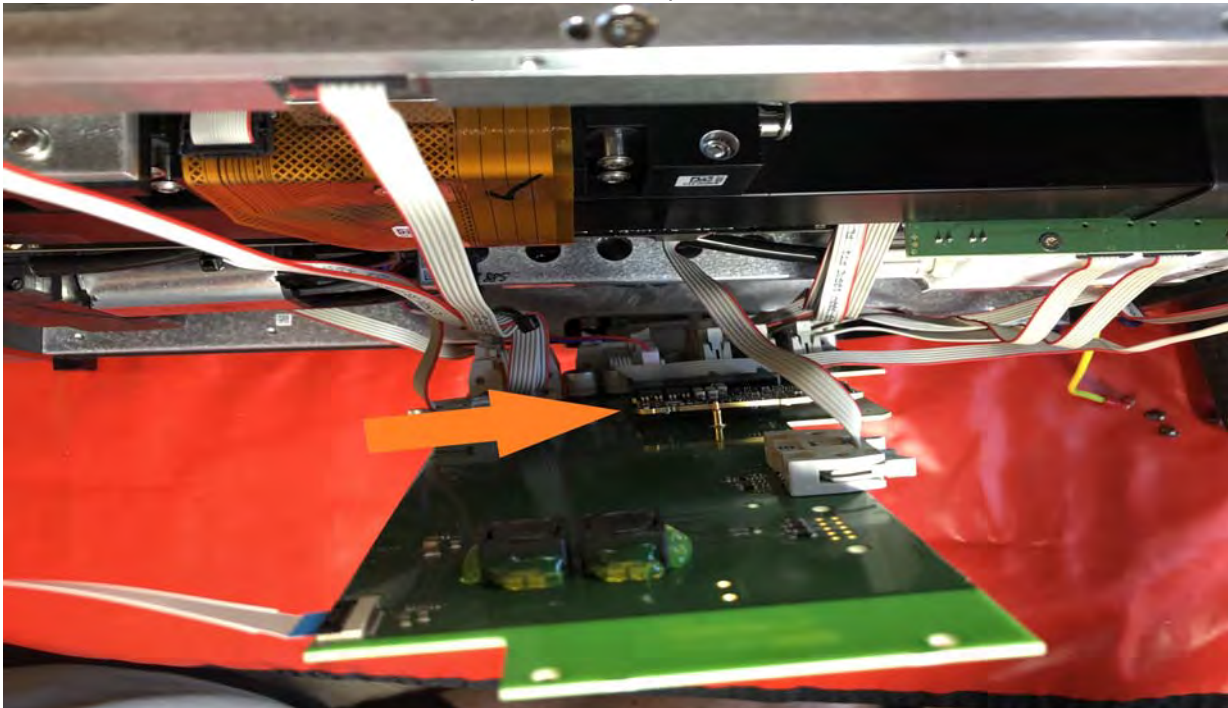
Part Required	9026475 (4 AMP) - Technosoft Driver Module
Tool(s) Required	T10 Torx Tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP3 - Actuators Status Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hours
Calibration/Testing Time	3 Hours
Total Repair Time	4 Hours

Table 6.27-1 RP27 Auxiliary board technosoft driver.

6.27.1 RP27 - Auxiliary Board Technosoft Driver - Removal Procedure

1. Remove the module covers, [see "RP9 - AM Covers - Removal Procedure" on page 64.](#)
2. Remove the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly- Removal Procedure" on page 106.](#)

3. Remove the Auxiliary Board, [see “RP11 - Auxiliary Board - Removal Procedure” on page 71.](#)
4. Locate the Technosoft Driver and depress the detent pin and remove the driver module.



Picture 6.27-2 Auxiliary board technosoft driver.

6.27.2 RP27 - Auxiliary Board Technosoft Driver - Installation Procedure

1. Press the retainer pin and inset the board.
2. Install the Auxiliary Board, [see “RP11 - Auxiliary Board - Installation Procedure” on page 72.](#)
3. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
4. Install The Filter Tray Assembly, [see “RP35 - Filter Tray Assembly - Installation Procedure” on page 107.](#)
5. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
6. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
7. Run actuators status test, [see “VP3 - Actuators Status Test” on page 168.](#)
8. Run ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169](#)

6.28 RP28 - Foil And Seal

Part Required	9026379 Foil and Seal
Tool(s) Required	T10 TORX Tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization
Replacement Time	1.0 Hours
Calibration/Testing Time	N/A
Total Repair Time	1.0 Hours

Table 6.28-1 RP-28 Foil and seal.

6.28.1 RP28 - Foil and Seal- Removal Procedure

1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106](#)
3. Locate the seal inside the plastic top cover, pull up on the seal to release the adhesive.



Picture 6.28-2 RP-28 Top cover foam seal.

4. Locate the two foil seals on the module front panel.



Picture 6.28-3 Foil seals.

5. Carefully lift a corner of the foil seal and peel it back.

6.28.2 RP28 - Foil and Seal - Installation Procedure

1. Install the seal by peeling the protective strip off the adhesive and place it in the slot in the module top cover.
2. Install the foil seal by peeling the protective strips and placing them on the identical locations.
3. Install the module covers, [see "RP9 - AM Covers- Installation Procedure" on page 67](#).
4. Install the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly - Installation Procedure" on page 107](#)

6.29 RP29 - Springs VD-080

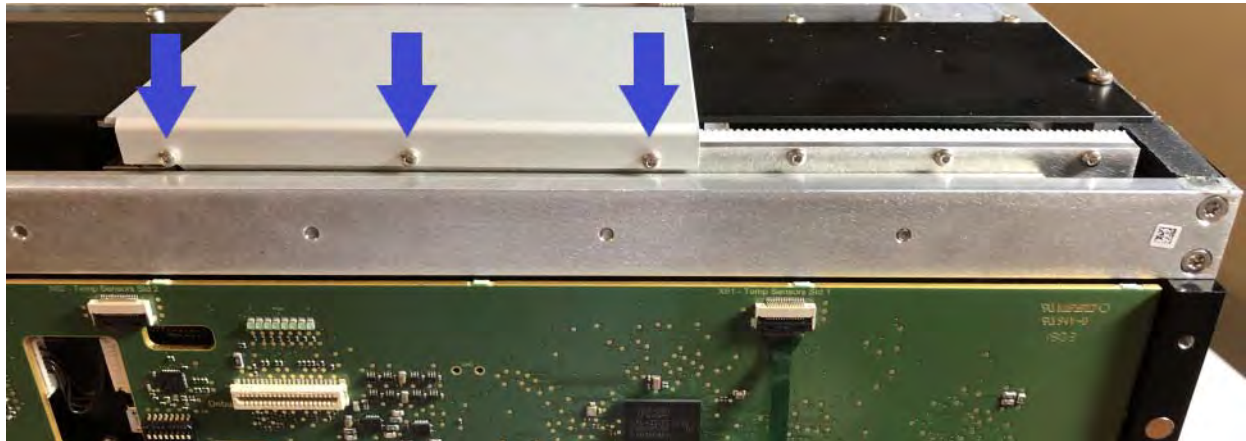
Part Required	9026381- Springs VD-080
Tool(s) Required	T10 TORX Tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP3 - Actuators Status Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hour
Calibration/Testing Time	2 Hours
Total Repair Time	3 Hours

Table 6.29-1 RP-29 Springs VD-080.

6.29.1 RP29 - Springs VD-080 - Removal Procedure

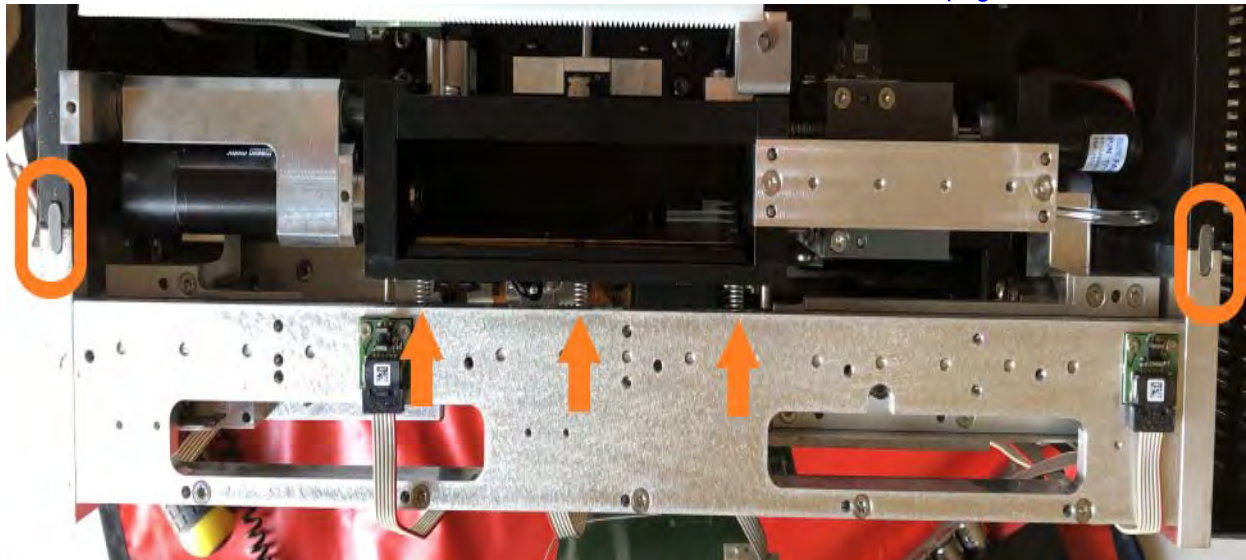
1. Remove the module covers, [see "RP9 - AM Covers - Removal Procedure" on page 64](#).

2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)
3. Remove the three T6 TORX screws and remove the door.



Picture 6.29-2 Door screws.

4. Remove the Lid Mechanism, [see “RP14 - Lid Mechanism - Removal Procedure” on page 75.](#)
5. Remove the TC and Clamping Module, [see “RP15 - TC and Clamping Module - Removal Procedure” on page 76.](#)
6. Remove the Match Pins., [see “RP30 - Match Pins - Removal Procedure” on page 100.](#)



Picture 6.29-3 Match pins and VD-80 springs.

6.29.2 RP29 - Springs VD-080 - Installation Procedure

1. Install the Transfer Cartridge, [see “RP15 - TC and Clamping Module - Installation Procedure” on page 78.](#)
2. Install the Match Pins, [see “RP30 - Match Pins - Installation Procedure” on page 100.](#)
3. Secure the four T10 TORX screws.
4. Install the Lid Mechanism, [see “RP14 - Lid Mechanism - Installation Procedure” on page 76.](#)
5. Install the door with three T6 TORX Screws.
6. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
7. Install the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly - Installation Procedure” on page 107.](#)
8. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
9. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
10. Run actuators status test, [see “VP3 - Actuators Status Test” on page 168.](#)
11. Run ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169](#)

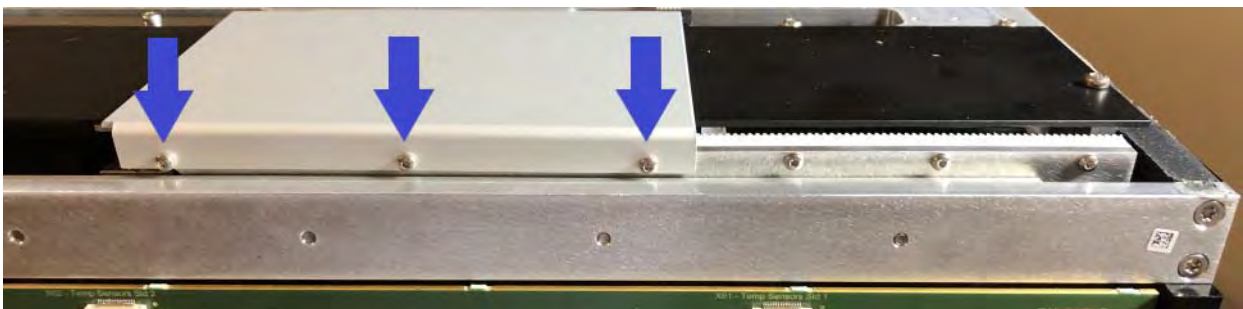
6.30 RP30 - Match Pins

Part Required	9026382 - Match Pins
Tool(s) Required	T10 TORX Tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP3 - Actuators Status Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1.0 Hours
Calibration/Testing Time	2.0 Hours
Total Repair Time	3.0 Hours

Table 6.30-1 RP-30 Match pins.

6.30.1 RP30 - Match Pins - Removal Procedure

1. Remove the module covers, [see "RP9 - AM Covers - Removal Procedure" on page 64.](#)
2. Remove the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly- Removal Procedure" on page 106.](#)
3. Remove the three T6 screws and remove the door.



Picture 6.30-2 Lid screws

4. Remove the Lid Mechanism, [see "RP14 - Lid Mechanism - Removal Procedure" on page 75.](#)
5. Remove the TC and Clamping Module, [see "RP15 - TC and Clamping Module - Removal Procedure" on page 76.](#)
6. Remove the Match Pins., [see "RP30 - Match Pins - Removal Procedure" on page 100.](#)



Picture 6.30-3 RP-30 Match pins.

6.30.2 RP30 - Match Pins - Installation Procedure

1. Install the TC and Clamping Module, [see "RP15 - TC and Clamping Module - Installation Procedure" on page 78.](#)
2. Insert the Match Pins, [see "RP30 - Match Pins - Installation Procedure" on page 100.](#)
3. Secure the four T10 TORX screws.
4. Install the Lid Mechanism, [see "RP14 - Lid Mechanism - Installation Procedure" on page 76.](#)

5. Install the door with 3 T6 TORX screws.
6. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
7. Install the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly - Installation Procedure” on page 107.](#)
8. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
9. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
10. Run actuators status test, [see “VP3 - Actuators Status Test” on page 168.](#)
11. Run ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169](#)

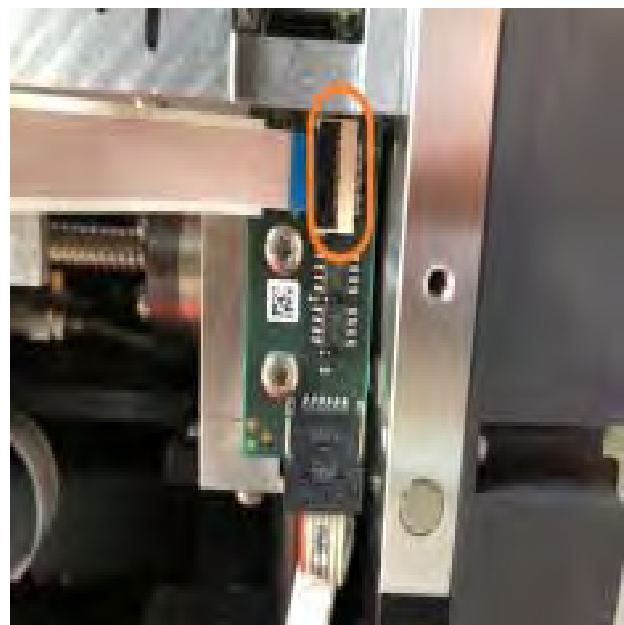
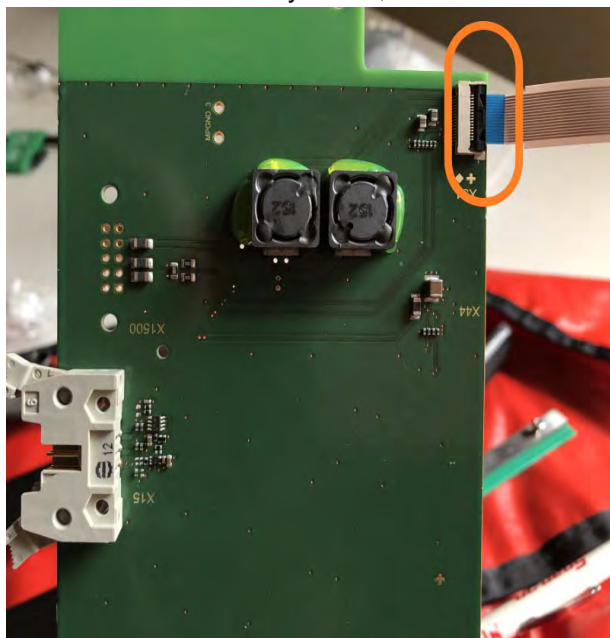
6.31 RP31 - FFC 16pol, AM

Part Required	90263831 - FFC16pol, AM
Tool(s) Required	T10 TORX Tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP3 - Actuators Status Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1.0 Hours
Calibration/Testing Time	2.0 Hours
Total Repair Time	3.0 Hours

Table 6.31-1 RP31 FFC 16pol, AM.

6.31.1 RP31 - FFC 16pol, AM - Removal Procedure

1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)
3. Remove the Auxiliary Board, [see “RP11 - Auxiliary Board - Removal Procedure” on page 71.](#)



Picture 6.31-2 RP-31 Auxiliary board flex cable, FFC 16pol.

4. Disconnect the FFC 16pol, AM from the connector.

6.31.2 RP31 - FFC 16pol, AM - Installation Procedure

1. Insert the FFC 16pol, AM into the connector on the Auxiliary Board.
2. Insert the FFC 16pol, AM into the connector on the clamping motor board.

3. Install the Auxiliary Board, [see “RP11 - Auxiliary Board - Installation Procedure” on page 72.](#)
4. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
5. Install the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly - Installation Procedure” on page 107.](#)
6. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
7. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
8. Run actuators status test, [see “VP3 - Actuators Status Test” on page 168.](#)
9. Run ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169](#)

6.32 RP32 - Tube-Diaphragm-Pump-Manifold

Part Required	9026383-Tube-Diaphragm-Pump-Manifold
Tool(s) Required	T10 TORX Tool
Calibration Required	CP1 Automated Mechanical Calibrations
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP4 - Pneumatics Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hour
Calibration/Testing Time	2 Hour
Total Repair Time	3 Hours

Table 6.32-1 RP32 Tube-diaphragm-pump-manifold.

6.32.1 RP32 - Tube-Diaphragm-Pump-Manifold - Removal Procedure

1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)
3. Remove the Auxiliary Board, [see “RP11 - Auxiliary Board - Removal Procedure” on page 71.](#)



Picture 6.32-2 Diaphragm pump tubing.

4. Disconnect the fitting on the pump and disconnect the tube from the manifold.

6.32.2 RP32 - Tube-Diaphragm-Pump-Manifold- Installation Procedure

1. Align the tube in the manifold fitting and press it on.
2. Insert the tube through the cap and ferrule and screw the cap on the pulp fitting.
3. Install the Auxiliary Board, [see “RP11 - Auxiliary Board - Installation Procedure” on page 72.](#)
4. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
5. Install the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly - Installation Procedure” on page 107.](#)
6. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
7. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
8. Run pneumatics test, [see “VP4 - Pneumatics Test” on page 168.](#)
9. Run ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169](#)

6.33 RP33 - Tube-Silencer-In

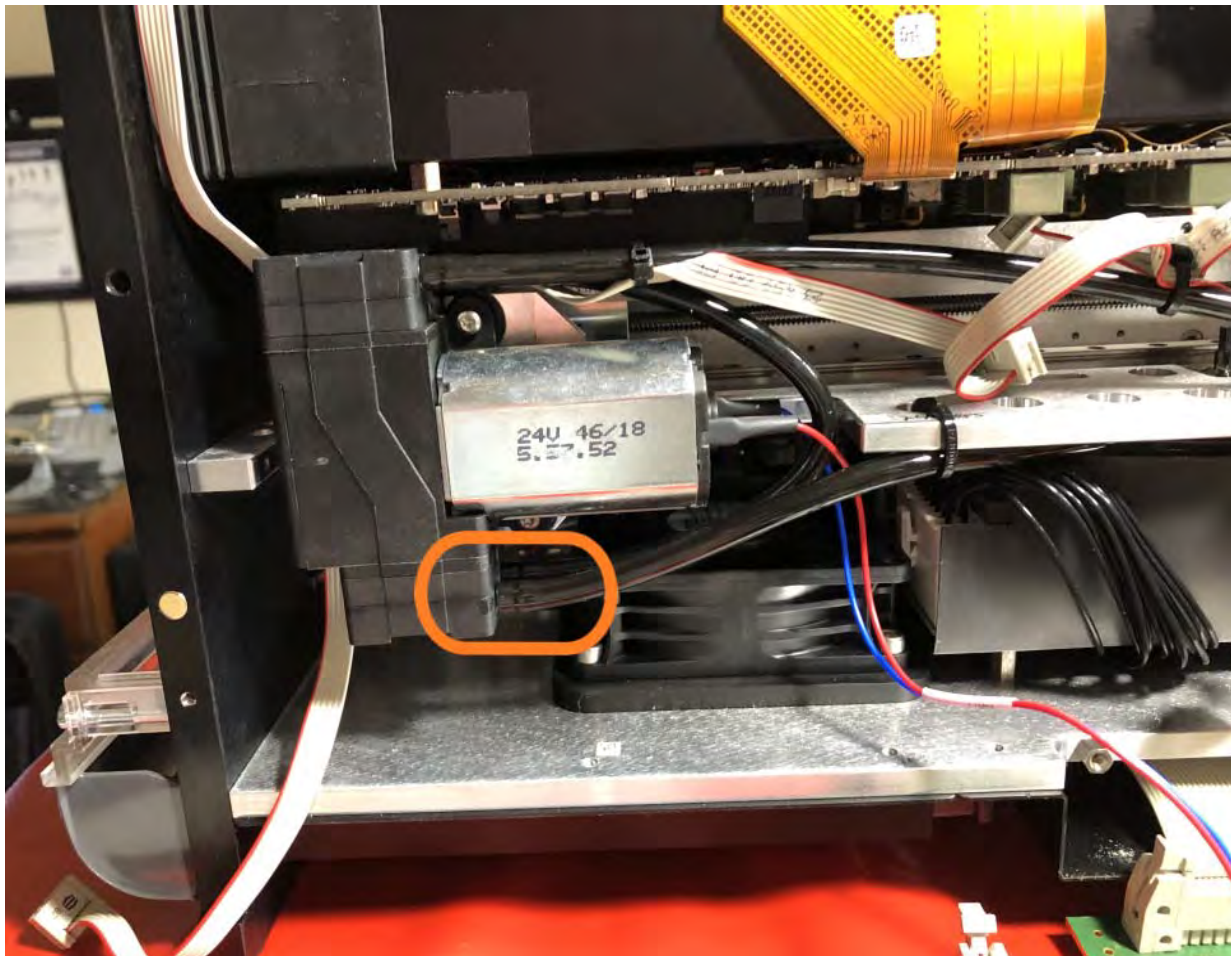
Part Required	9026385- Tube-Silencer-In
Tool(s) Required	T10 TORX Tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP4 - Pneumatics Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1.0 Hours
Calibration/Testing Time	2.0 Hours
Total Repair Time	3.0 Hours

Table 6.33-1 RP33 Tube-silencer-in.

6.33.1 RP33 - Tube-Silencer-In - Removal Procedure

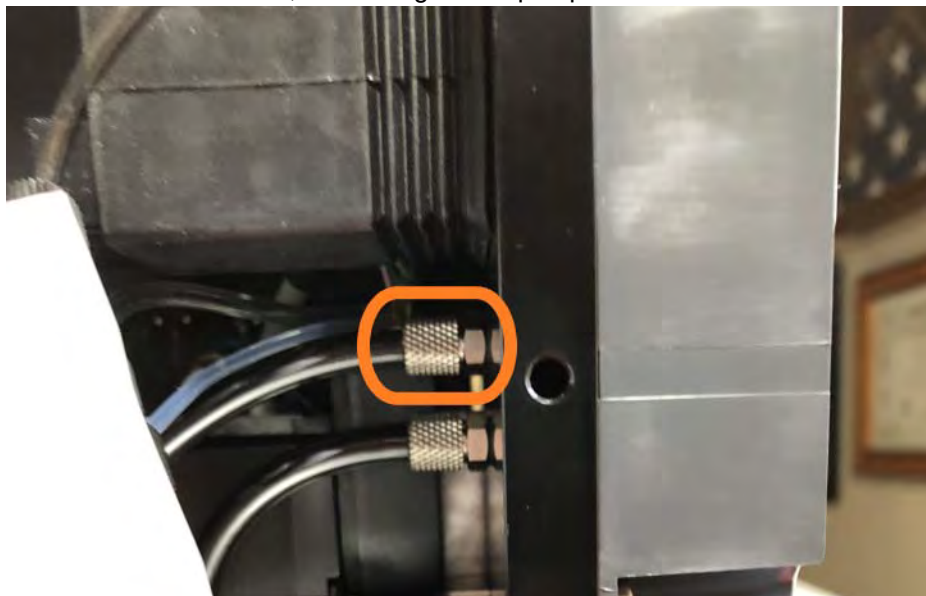
1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)
3. Remove the Auxiliary Board, [see “RP11 - Auxiliary Board - Removal Procedure” on page 71.](#)

4. Remove Pneumatic Manifold., see "RP18 - Pneumatic Manifold - Removal Procedure" on page 81.



Picture 6.33-2 Pump connection.

5. Remove the tube from the bottom, front fitting on the pump.



Picture 6.33-3 Back cover connection.

6. Follow the tubing to the back panel and unscrew the fitting to remove the tubing.

6.33.2 RP33 - Tube-Silencer-In - Installation Procedure

1. Slide the metal fitting onto the tubing and secure it on the fitting on the rear panel.

2. Feed the tubing though the slot and press it on the bottom pump fitting.
3. Install the pneumatic manifold, [see “RP18 - Pneumatic manifold - Installation Procedure” on page 82.](#)
4. Install the Auxiliary Board, [see “RP11 - Auxiliary Board - Installation Procedure” on page 72.](#)
5. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
6. Install the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly - Installation Procedure” on page 107.](#)
7. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
8. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
9. Run pneumatics test, [see “VP4 - Pneumatics Test” on page 168.](#)
10. Run ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169](#)

6.34 RP34 - Tube-Silencer-Out

Part Required	9026384- Tube-Silencer-Out
Tool(s) Required	T10 TORX Tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP4 - Pneumatics Test , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hour
Calibration/Testing Time	2 Hours
Total Repair Time	3 Hours

Table 6.34-1 RP34 Tube-silencer-in.

6.34.1 RP34 - Tube-Silencer - Out - Removal Procedure

1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the Filter Tray Assembly, [see “RP35 - Filter Tray Assembly- Removal Procedure” on page 106.](#)
3. Remove the Auxiliary Board, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
4. Remove Pneumatic Manifold, [see “RP18 - Pneumatic Manifold - Removal Procedure” on page 81.](#)
5. Find the tubing connected to the back panel lower fitting and bottom panel fitting.
6. On the bottom panel fitting press down on the white PEX out ring and remove the tubing.



Picture 6.34-2 Tube silencer out connections.

- Loosen the lower metal fitting on the back panel and remove the tubing.

6.34.2 RP34 - Tube-Silencer- Out - Installation Procedure

- Slide the metal fitting onto the tubing and secure it on the lower fitting on the rear panel.
- Feed the tubing through the slot and press it into the PEX fitting on the lower panel.
- Install the Pneumatic Manifold, [see "RP18 - Pneumatic manifold - Installation Procedure" on page 82.](#)
- Install the Auxiliary Board, [see "RP11 - Auxiliary Board - Installation Procedure" on page 72.](#)
- Install the module covers, [see "RP9 - AM Covers- Installation Procedure" on page 67.](#)
- Install the Filter Tray Assembly, [see "RP35 - Filter Tray Assembly - Installation Procedure" on page 107.](#)
- Initialize the analyzer, [see "VP1 - System Initialization" on page 167.](#)
- Prepare for testing, [see "VP2 - Preparation For Testing" on page 168.](#)
- Run pneumatics test, [see "VP4 - Pneumatics Test" on page 168.](#)
- Run ACC test, [see "VP7 - Analyzer Check Cartridge Work Flow Test" on page 169](#)

6.35 RP35 - Filter Tray Assembly

Part Required	902XXX Filter Tray Assembly
Tool(s) Required	N/A
Calibration Required	N/A
Testing Required	VP1 - System Initialization
Replacement Time	5 Minutes
Calibration/Testing Time	10 Minutes
Total Repair Time	15 Minutes

Table 6.35-1 RP35 Filter tray assembly.

6.35.1 RP35 - Filter Tray Assembly- Removal Procedure

- With your fingers feel for the tray under the front-bottom of the AM.
- Gently push upward on the tray latch and pull the tray forward.
- Remove the tray assembly and discard.



Picture 6.35-2 Filter tray assembly.

6.35.2 RP35 - Filter Tray Assembly - Installation Procedure

1. Unpack the new filter tray assembly.
2. Slide the tray into the slot under the front edge for the AM.
3. Ensure the tray is fully seated and pushed completely back in the slot to engage the sensor.
4. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)

6.36 RP36 - Motor, TC1

Part Required	9026556 - Motor, TC1
Tool(s) Required	T10 TORX Tool
Calibration Required	CP1 Automated Mechanical Calibrations
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP3 - Actuators Status Test , VP4 - Pneumatics Test , VP7 - Analyzer Check Cartridge Work Flow Test .
Replacement Time	1.0 Hours
Calibration/Testing Time	3.0 Hours
Total Repair Time	4.0 Hours

Table 6.36-1 RP36 Motor, TC1.

6.36.1 RP36 - Motor, TC1 - Removal Procedure

1. Remove the module covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the belt and lower pulley, [see “RP24 - Belt and Pulleys - Removal Procedure” on page 90.](#)
3. Remove the Main Board, [see “RP10 - Main Board - Removal Procedure” on page 69.](#)
4. Remove the Syringe Module, [see “RP16 - Syringe Module - Removal Procedure” on page 79.](#)
5. Remove the two TC1 motor mounting screws.

6.36.2 RP36 - Motor, TC1 - Installation Procedure

1. Install the TC1 motor with the two mounting screws.
2. Install the belt and lower pulley, [see “RP24 - Belt and Pulleys - Installation Procedure” on page 92.](#)
3. Install the syringe module, [see “RP16 - Syringe Module - Installation Procedure” on page 80.](#)
4. Install the Main Board, [see “RP10 - Main Board - Installation Procedure” on page 70.](#)
5. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
6. Run automated mechanical calibrations, [see “CP1 Automated Mechanical Calibrations” on page 112.](#)
7. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
8. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
9. Run the actuators status test, [see “VP3 - Actuators Status Test” on page 168.](#)
10. Run pneumatics test, [see “VP4 - Pneumatics Test” on page 168.](#)
11. Run ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169](#)

6.37 RP37 - Motor, Lifter/Lid

Part Required	9026559 - Motor, Lifter/Lid
Tool(s) Required	T10 TORX Tool
Calibration Required	N/A
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP3 - Actuators Status Test
Replacement Time	1 Hour
Calibration/Testing Time	1 Hour
Total Repair Time	2 Hours

Table 6.37-1 RP37 Motor, Lifter/Lid.

6.37.1 RP37 - Motor, Lifter/Lid - Removal Procedure

1. Remove the covers, [see "RP9 - AM Covers - Removal Procedure" on page 64.](#)
2. Remove the main board, [see "RP10 - Main Board - Removal Procedure" on page 69.](#)
3. Remove the cartridge lifter assembly, [see "RP12 - Cartridge Lifter Assembly - Removal Procedure" on page 72.](#)
4. Remove the bar code reader, [see "RP38 - Internal Bar Code Reader - Removal Procedure" on page 109.](#)
5. Remove the two T10 TORX mounting screws.



Picture 6.37-2 Lifter/Lid motor.

6.37.2 RP37 - Motor, Lifter/Lid- Installation Procedure

1. Install the motor with the two T10 TORX screws.
2. Install the internal bar code reader, [see "RP38 - Internal Bar Code Reader - Removal Procedure" on page 109.](#)
3. Install the Main Board, [see "RP10 - Main Board - Installation Procedure" on page 70.](#)
4. Install the cartridge lifter assembly, [see "RP12 - Cartridge Lifter Assembly - Removal Procedure" on page 72.](#)

5. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
6. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
7. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
8. Run the actuators status test, [see “VP3 - Actuators Status Test” on page 168.](#)

6.38 RP38 - Internal Bar Code Reader

Part Required	9026543 - Internal Bar Code Reader
Tool(s) Required	T10 TORX Tool
Calibration Required	CP1 Automated Mechanical Calibrations
Testing Required	VP1 - System Initialization , VP2 - Preparation For Testing , VP7 - Analyzer Check Cartridge Work Flow Test
Replacement Time	1 Hour
Calibration/Testing Time	1 Hour
Total Repair Time	2 Hours

Table 6.38-1 RP38 Internal bar code reader.

6.38.1 RP38 - Internal Bar Code Reader - Removal Procedure

1. Remove the covers, [see “RP9 - AM Covers - Removal Procedure” on page 64.](#)
2. Remove the main board, [see “RP10 - Main Board - Removal Procedure” on page 69.](#)
3. Remove the internal bar code reader two T10 TORX reader mounting screws.



Picture 6.38-2 Internal bar code reader.

6.38.2 RP38 - Internal Bar Code Reader- Installation Procedure

1. Install the internal bar code reader with the two T10 TORX screws.
2. Install the Main Board, [see “RP10 - Main Board - Installation Procedure” on page 70.](#)
3. Install the module covers, [see “RP9 - AM Covers- Installation Procedure” on page 67.](#)
4. Run mechanical calibrations, [see “CP1 Automated Mechanical Calibrations” on page 112.](#)
5. Initialize the analyzer, [see “VP1 - System Initialization” on page 167.](#)
6. Prepare for testing, [see “VP2 - Preparation For Testing” on page 168.](#)
7. Run ACC test, [see “VP7 - Analyzer Check Cartridge Work Flow Test” on page 169.](#)

6.39 RP39 - Power Switch, AM

Part Required	9026553 - Power Switch, AM
Tool(s) Required	T10 TORX Tool, Small flat screwdriver, Needle nosed pliers
Calibration Required	N/A
Testing Required	VP1 - System Initialization
Replacement Time	30 Minutes
Calibration/Testing Time	10 Minutes
Total Repair Time	40 Minutes

Table 6.39-1 RP39 Power switch, AM.

6.39.1 RP39 - Power Switch, AM - Removal Procedure

1. Remove the covers, [see "RP9 - AM Covers - Removal Procedure" on page 64.](#)
2. Disconnect the X72 connector from the main board, [see "RP10 - Main Board - Removal Procedure" on page 69.](#)



Picture 6.39-2 Main board X72 connector.

3. Remove the two T10 TORX screws holding the power switch/USB assembly to the analyzer.



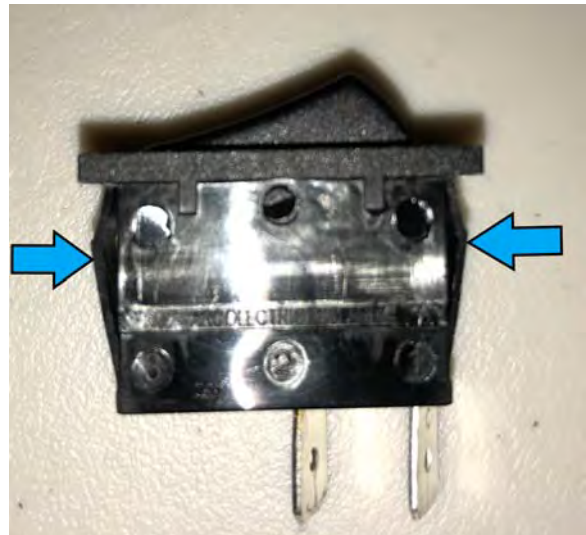
Picture 6.39-3 Power switch/USB assembly mounting screws.

4. Pull the assembly out to disconnect the two terminal connectors from the back of the switch.



Picture 6.39-4 Power switch connections.

5. Unlock the switch by using a small flat screwdriver push on the short side of the switch while pressing it out of the assembly.



Picture 6.39-5 Power switch locks.

6.39.2 RP39 - Power Switch, AM- Installation Procedure

1. Press the switch into the assembly, ensure the "0" in the top position otherwise the AM will not operate with a bridge installed.
2. Install the terminal connectors on the switch connectors in the same position.
3. Mount the switch on the back of the analyzer with the two T10 TORX screws and pull the X72 connector back toward the main board and connect it.
4. Install the Main Board, [see "RP10 - Main Board - Installation Procedure" on page 70.](#)
5. Install the module covers, [see "RP9 - AM Covers- Installation Procedure" on page 67.](#)
6. Initialize the analyzer, [see "VP1 - System Initialization" on page 167.](#)

7 Calibration Procedures (CP)

WARNING 	Electrical Hazard Disconnect the mains power cords from the power outlet before servicing. Lethal voltages inside the instrument When the instrument is connected to line power, terminals may be live, and opening covers or removing parts is likely to expose live parts.
WARNING 	Moving parts Avoid contact with moving parts during operation. If there is a problem with the QIAstat-Dx, great care should be taken if the instrument is operated with covers off.
WARNING 	Pinch and impact hazards The drive belt and drive shafts may move at any time.

7.1 Overview of Calibration Procedures

VP #	Name	Purpose
CP1	Automated Mechanical Calibration	Calibrate the motor positions, vacuum/pressure, and tilt sensors.
VP2	qPCR alignment and thermal unit test	Align the qPCR module and test the thermal unit

7.2 CP1 Automated Mechanical Calibrations

Purpose	Calibrate the motor positions, vacuum/pressure, and tilt sensors.
Tool(s) Required	USB Cable (printer), PC, 9026429 - X-8883 Force measuring fixture with X-7790 Force measuring fixture extension 9026438 - X-7293 Pressure plate, 9026440 - X-7375 Force measurement adapter X-8914 Reference cartridge 9026446 - X-8889 Pressure/Vacuum test fixture with valve control 9026435 - X-8883 Pressure/Vacuum test cartridge 9026433 - X-8737 Endtest/Leaktest cartridge 9026447 - Burster pressure sensor with X-7043 Force measuring tap
Verification Time	2 Hours 30 Minutes

Table 7.2-1 CP1 Mechanical Adjustments.

7.2.0.1 Preparation for Calibration

1. Remove the sides and top cover.
2. Connect the AM to power and to the USB port to the computer.



Picture 7.2-2 Power and CAN connection.

3. Switch the AM on and wait for it to initialize (Steady Green LED).



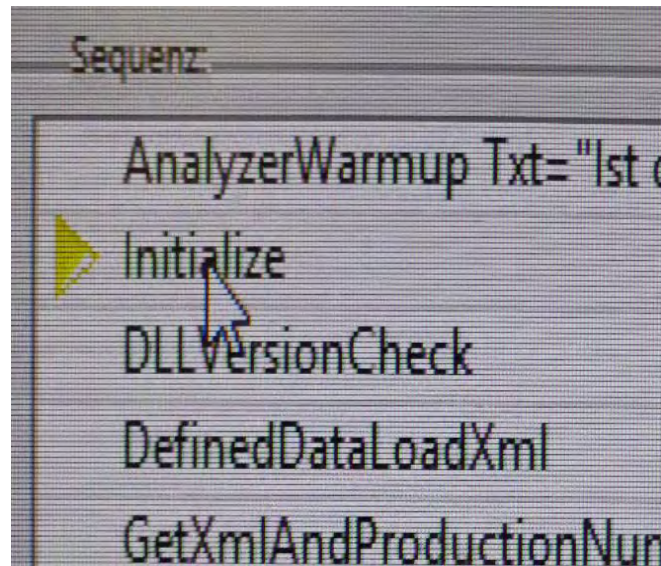
Picture 7.2-3 Power switch and status LED.

4. Open the “DiagCORECalEndMech” program.



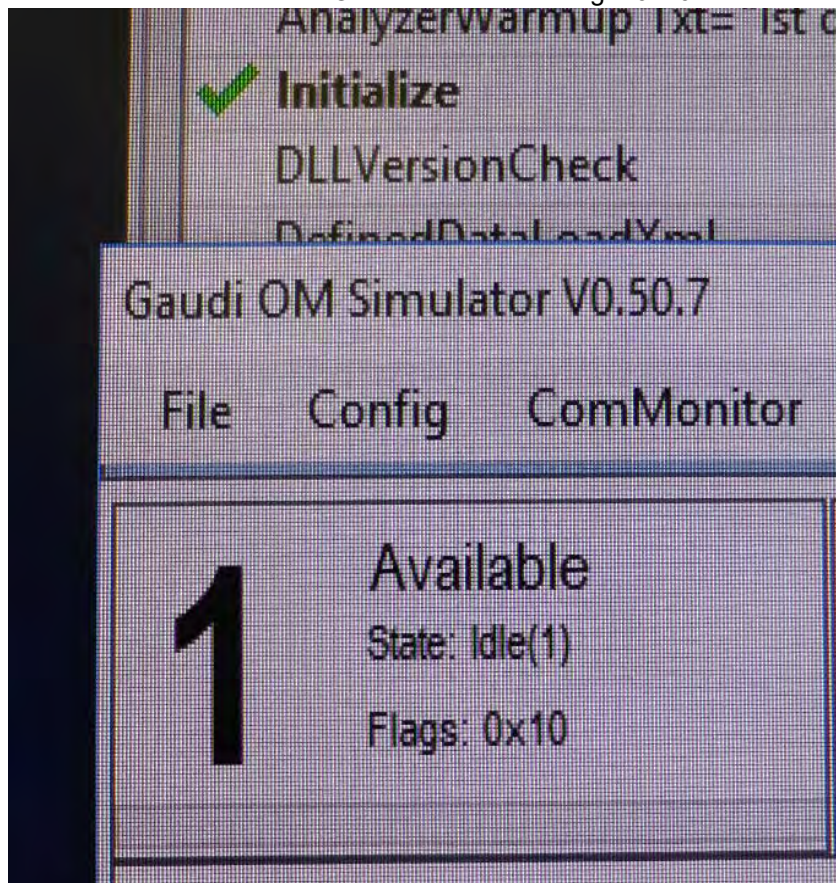
Picture 7.2-4 Automated calibration desktop icon.

5. Double click on "Initialize".



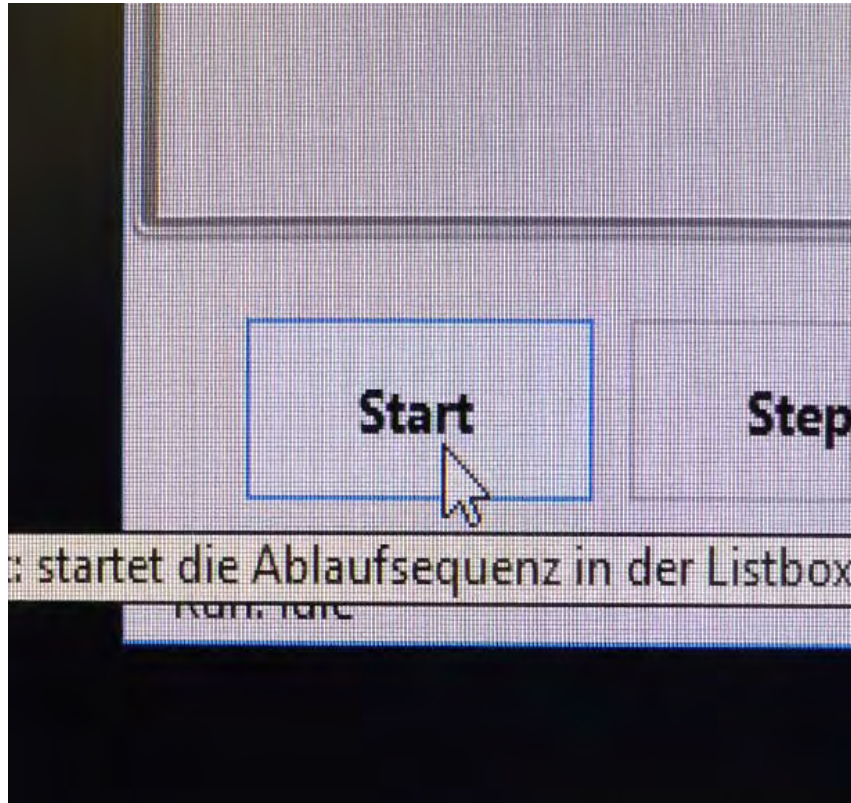
Picture 7.2-5 Initialize.

6. Wait until the instrument connects to the OM Simulator and Flags "0x10" is shown.



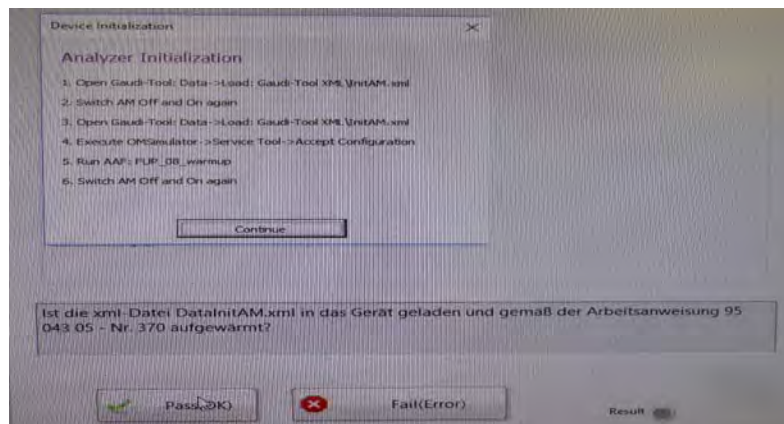
Picture 7.2-6 OM simulator status flags.

7. Click on "Start".



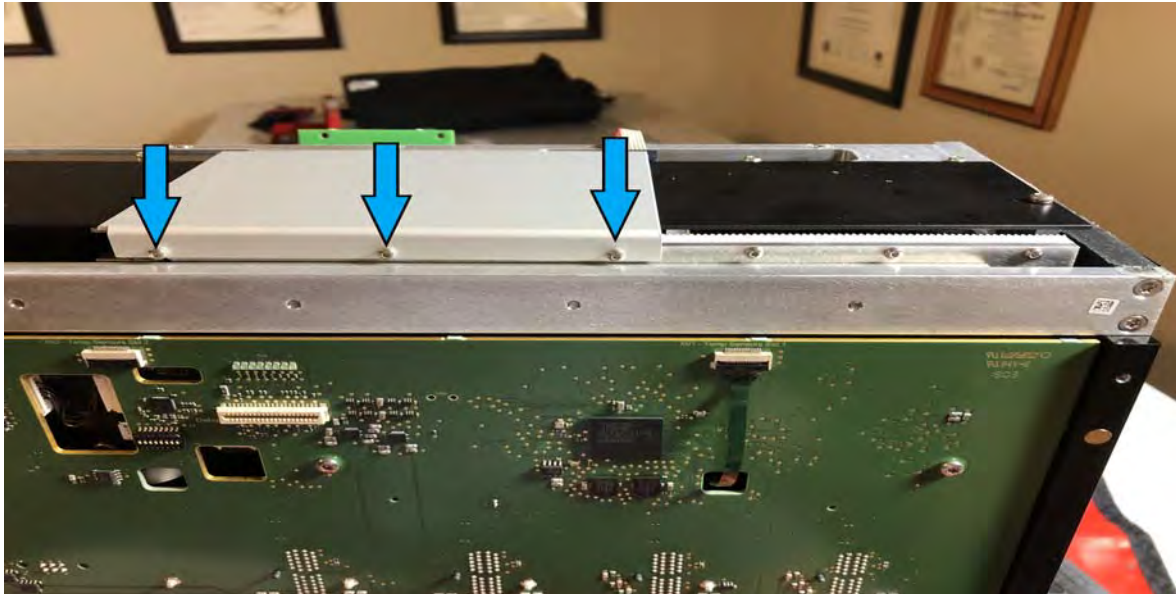
Picture 7.2-7 Start warm-up.

8. Click on "Pass" when the instrument was warmed up, approximately 14 minutes.



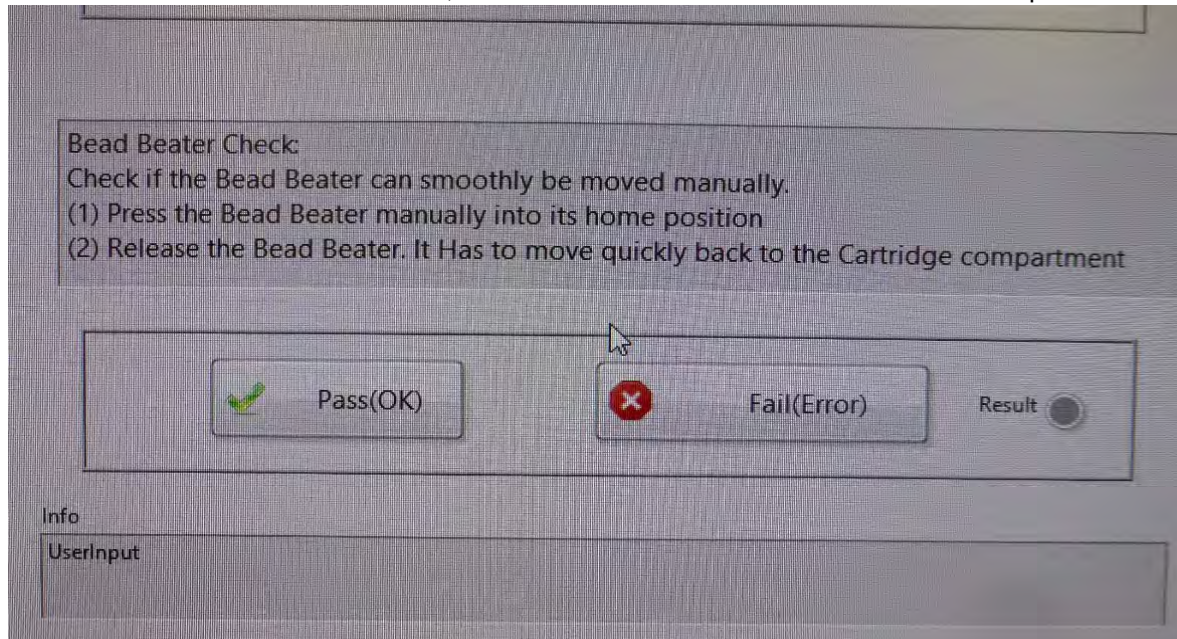
Picture 7.2-8 Pass button.

9. For this step the lid must be removed, Remove the three T6 TORX mounting screws and remove it.



Picture 7.2-9 Lid screws.

10. Check the bead beater as described, if the bead beater is movable as described then press "Pass".

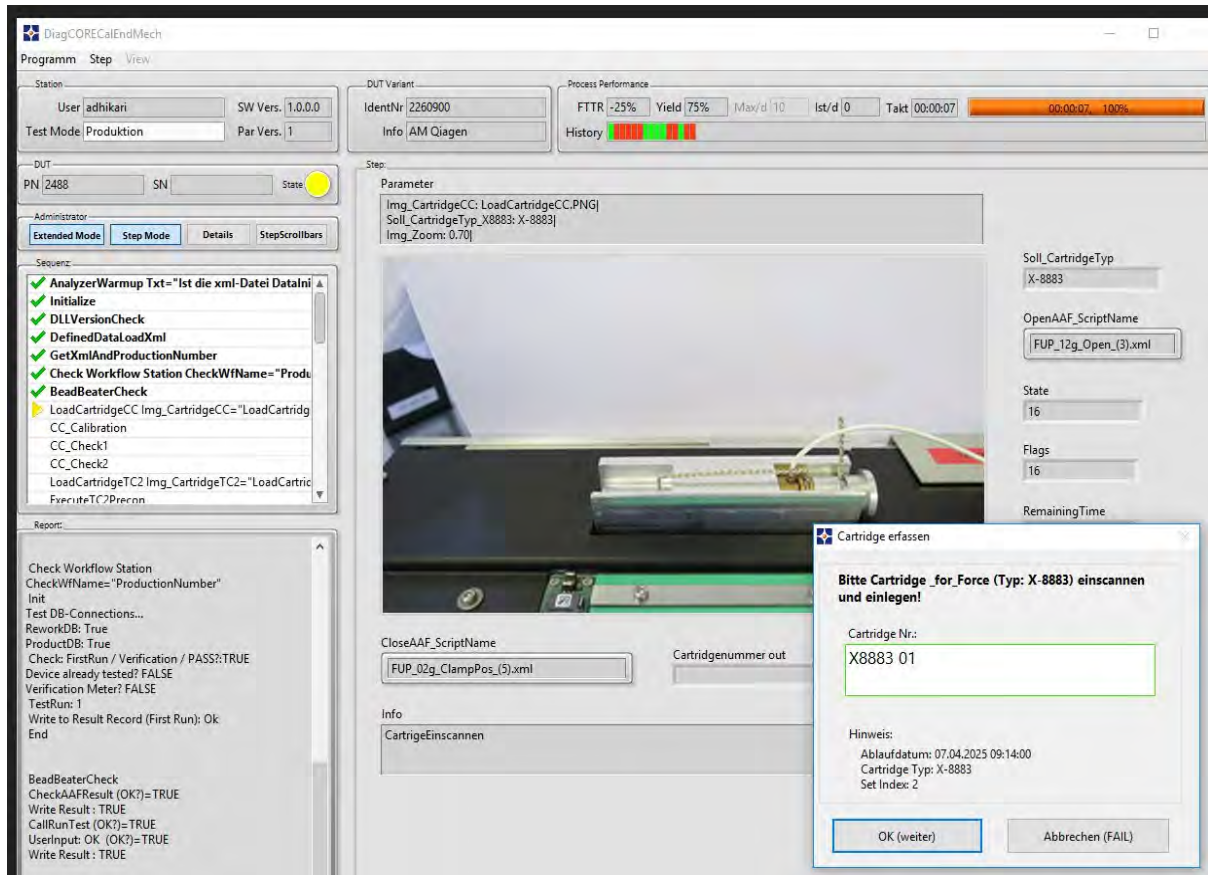


Picture 7.2-10 Bead beater pass.

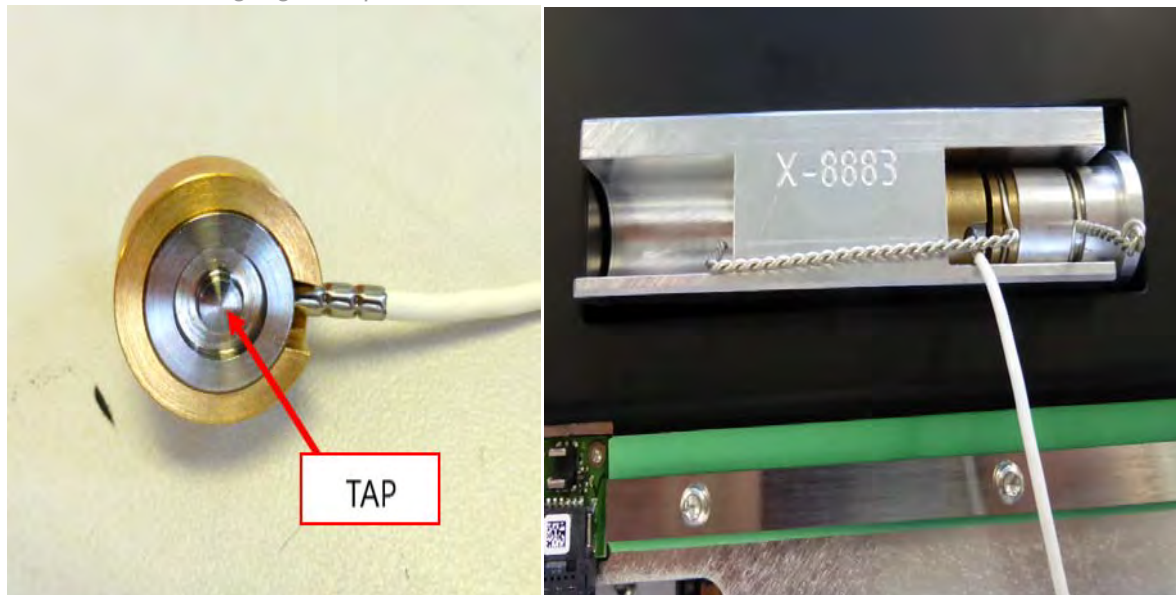
11. Reinstall the lid.

7.2.0.2 Automated Calibration Sequence

1. Setup the force gauge in the force measurement fixture as pictured below and set it in the AM cartridge slot.

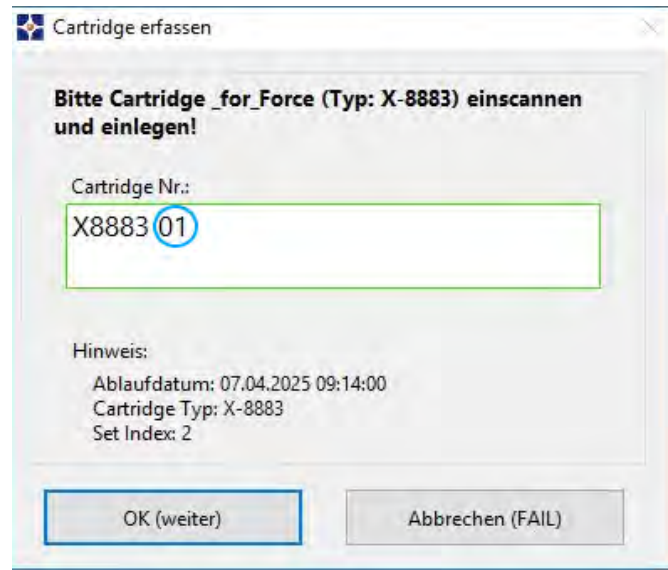


Picture 7.2-11 Force gauge setup.



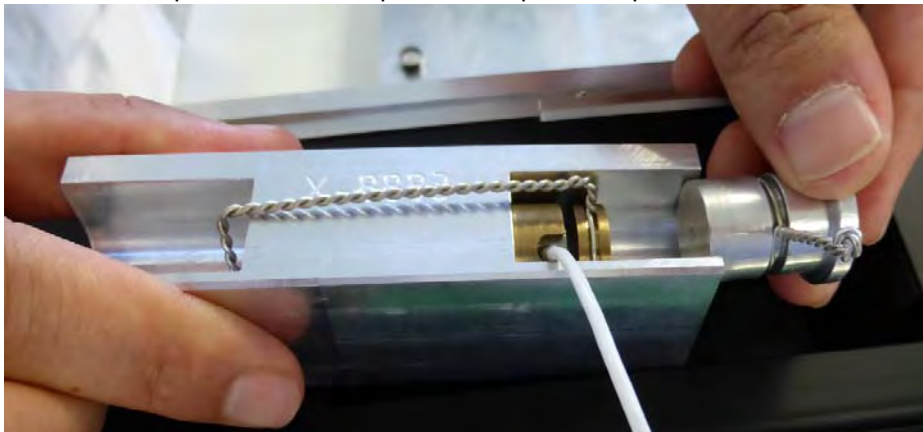
Picture 7.2-12 Force gauge setup detail

2. Insert the force measurement fixture setup in the cartridge slot, see picture 7.1.10.



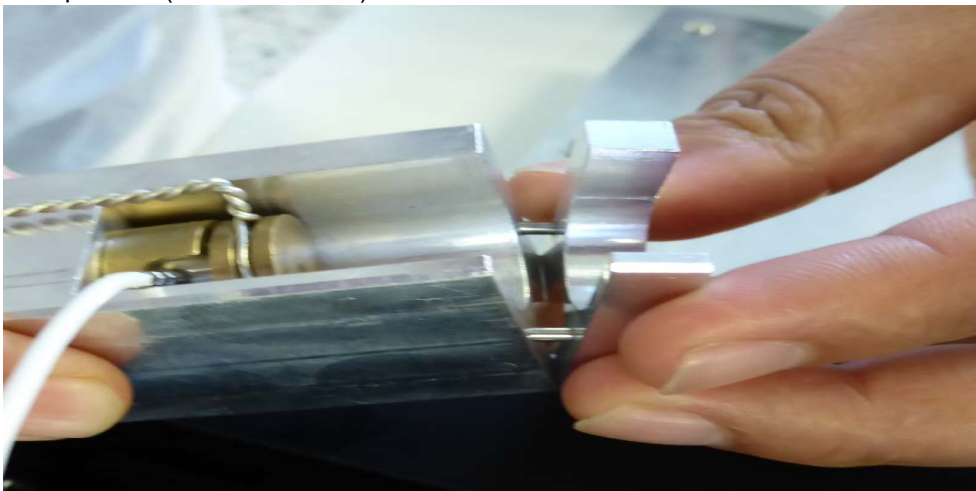
Picture 7.2-13 Check cartridge number and start calibration sequence.

3. Make sure the force measurement fixture number "X8883 XX"; check your table in the SQL database for the cartridge number.
4. Click "OK" to start the calibration sequence.
5. Once the sequence is complete remove the X8883 force measurement fixture from the analyzer and take out the X-7375 adapter. Leave the tap and force pressure plate in the fixture.

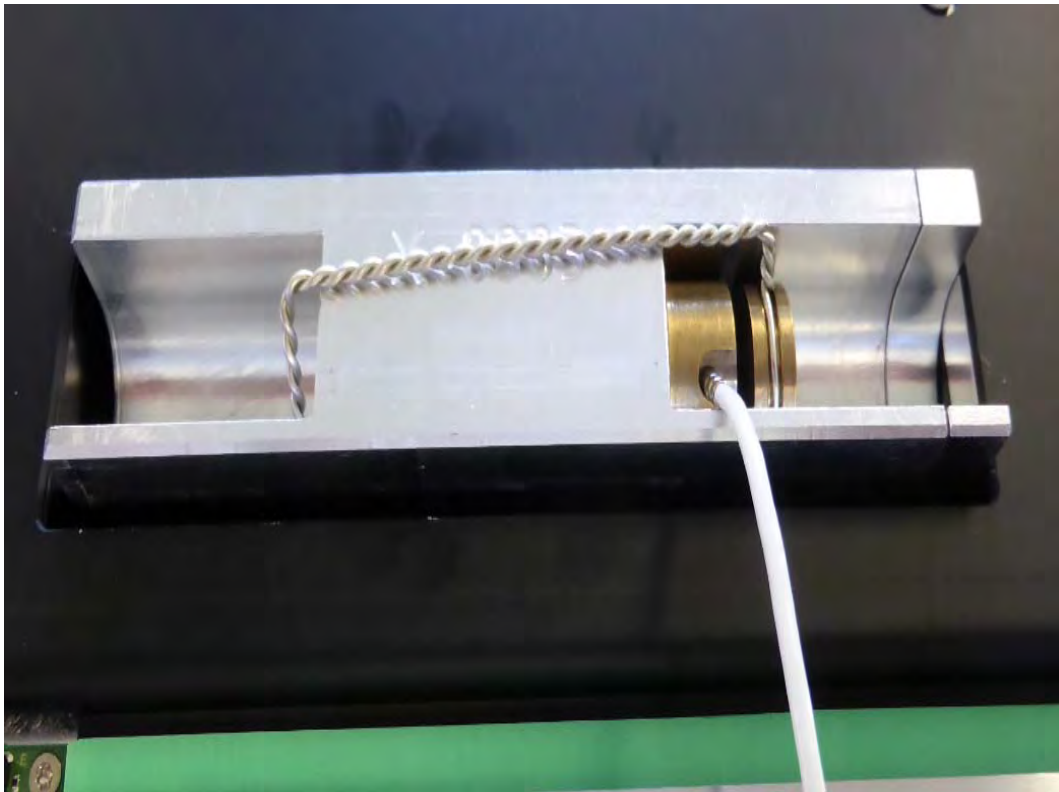


Picture 7.2-14 Remove force measurement adapter.

6. Install the X-7790 force measuring fixture extension on the end but keep the tap and the pressure plate in the same position (TC 2 calibration).

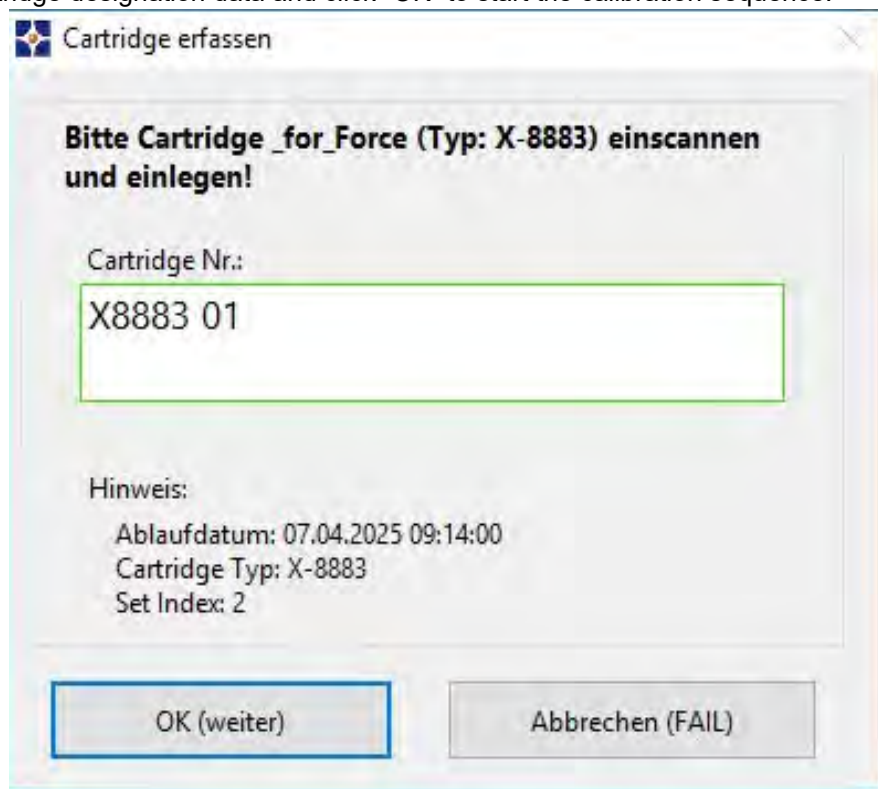


Picture 7.2-15 Force measurement fixture extension.



Picture 7.2-16 TC 2 Force measurement fixture setup.

7. Insert the TC 2 force measurement fixture in the cartridge slot; see picture 7.1.15.
8. Check cartridge designation data and click "OK" to start the calibration sequence.



Picture 7.2-17 Check cartridge number and start calibration sequence.

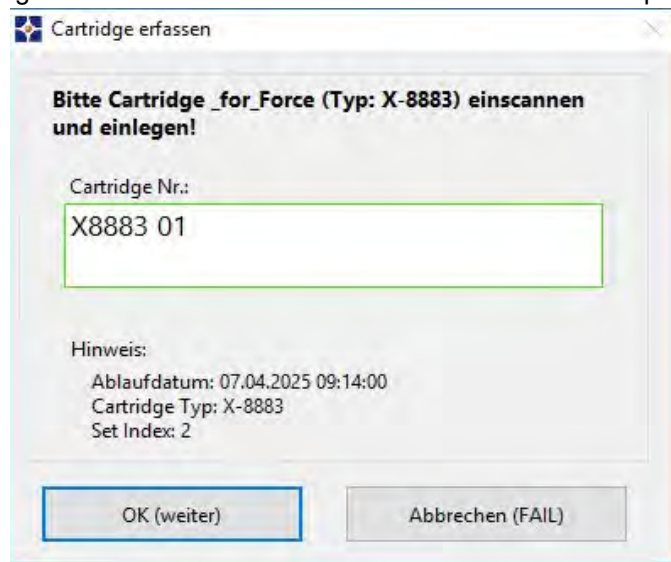
9. When the calibration is complete remove the force measurement fixture.

10. Transfer the tap and pressure plate to the other side of the calibration fixture (TC1 calibration).



Picture 7.2-18 TC 1 force fixture setup.

11. Insert the force measurement fixture in the cartridge slot, see picture 7.1-17.
12. Check cartridge designation data and click "OK" to start the calibration sequence.

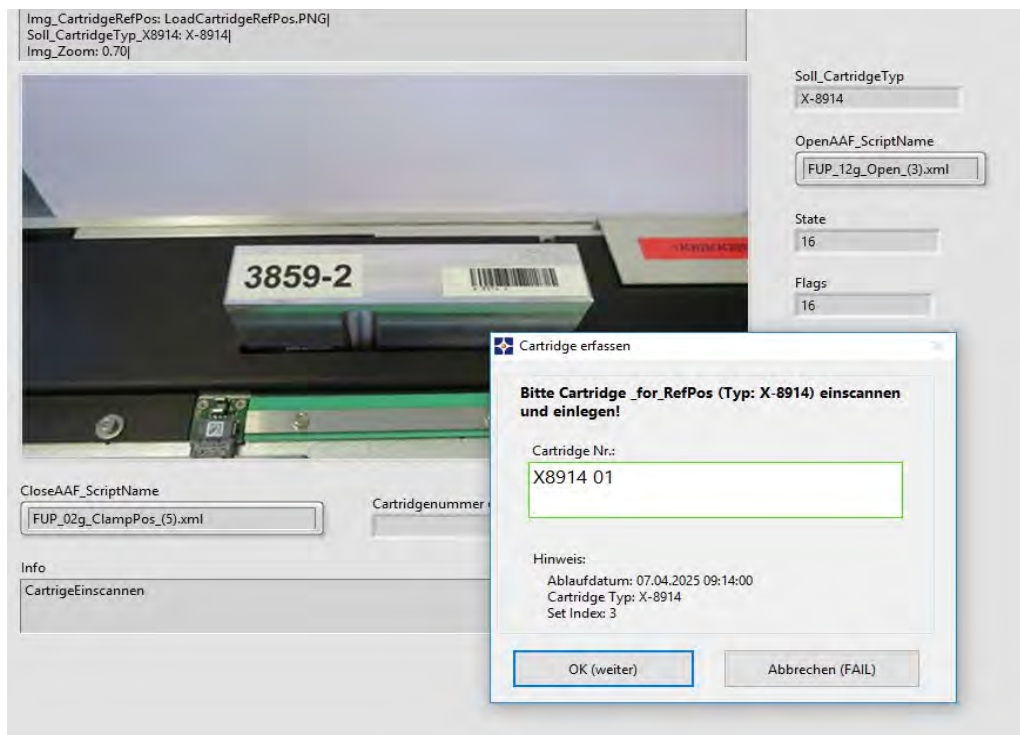


Picture 7.2-19 Check cartridge number and start calibration sequence.

13. When the sequence is complete remove the force measurement fixture x-8883.
14. Place the reference cartridge X-8914 into the cartridge slot.

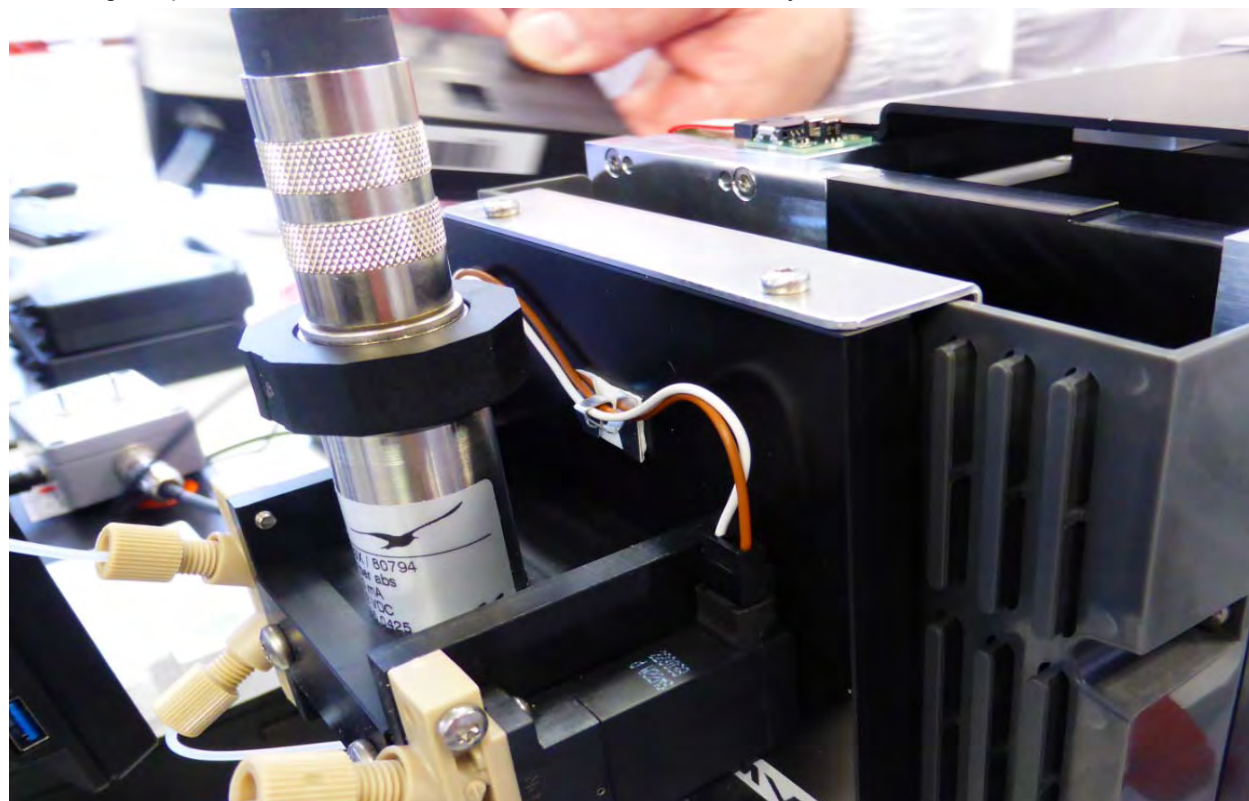


Picture 7.2-20 Insert reference cartridge in the cartridge slot.



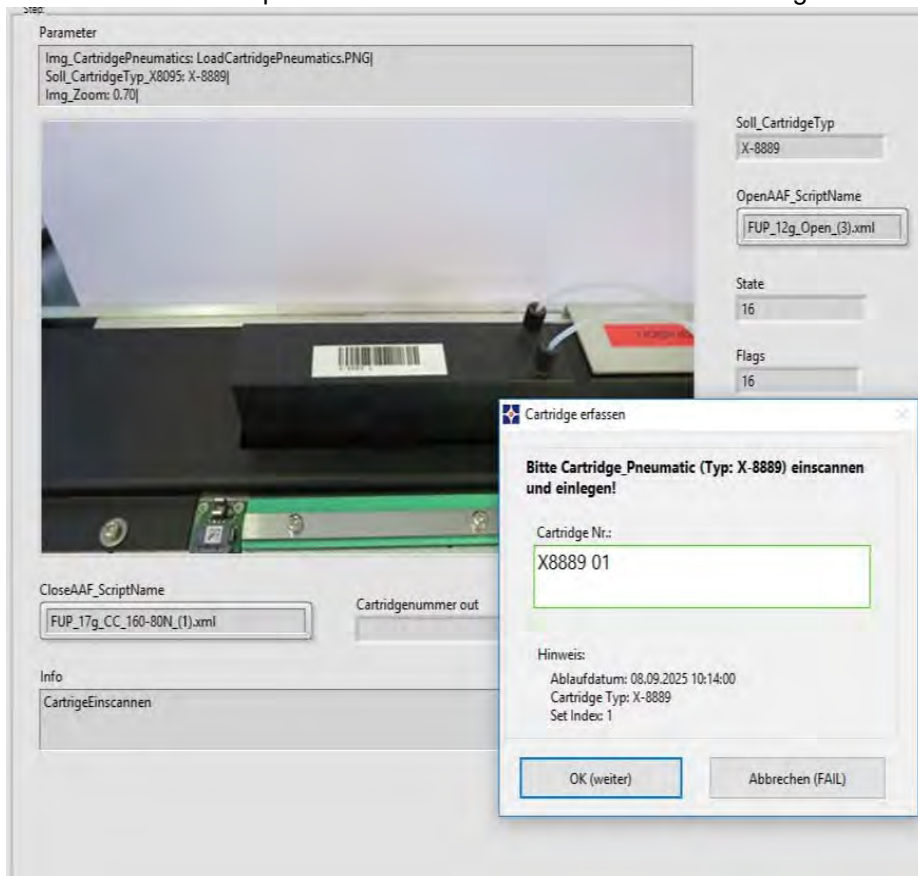
Picture 7.2-21 Enter cartridge data and click "OK" to start sequence.

15. Check cartridge designation data and click "OK" to start the calibration sequence. If the calibration fails recalibrate the failed position again.
16. When the sequence is complete remove the reference cartridge.
17. Remove the lid.
18. Hang the pressure measurement tool from the rear of the analyzer.



Picture 7.2-22 Pressure measurement tool

19. With the lid removed insert the pressure measurement fixture into the cartridge slot.



Picture 7.2-23 Insert pressure measurement fixture.

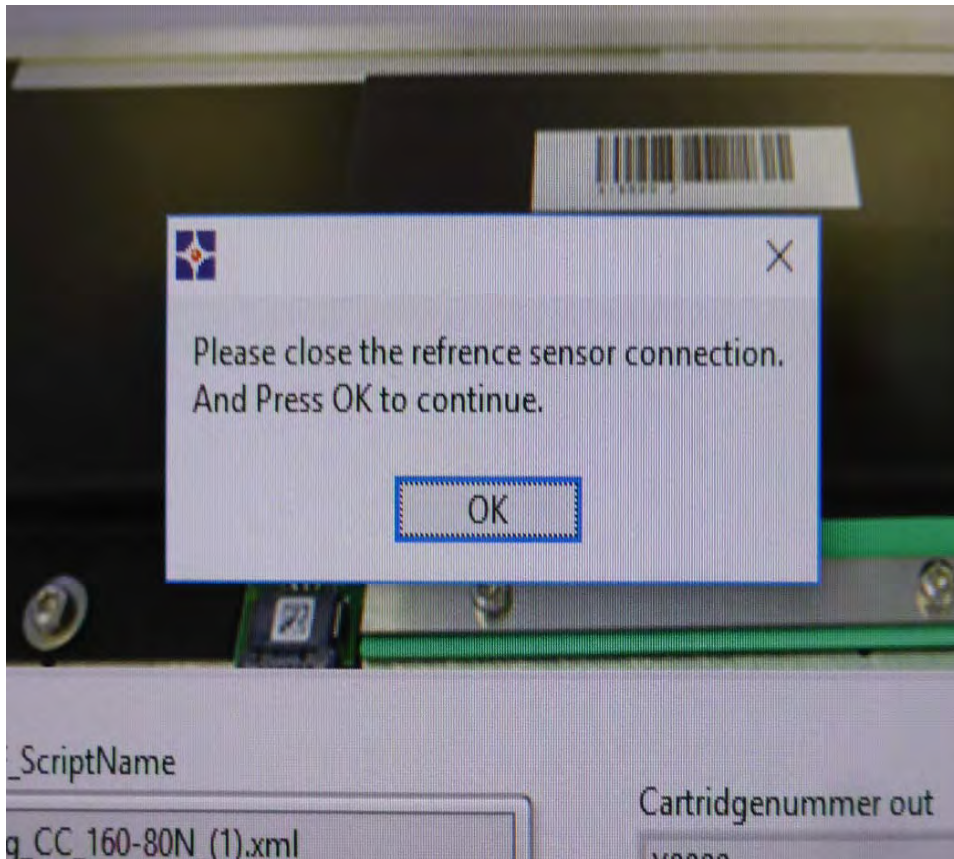
20. Check cartridge designation data and click "OK" to start the calibration sequence.

21. Close the reference sensor connection.



Picture 7.2-24 Close the reference sensor connection.

22. Click "OK". to continue.



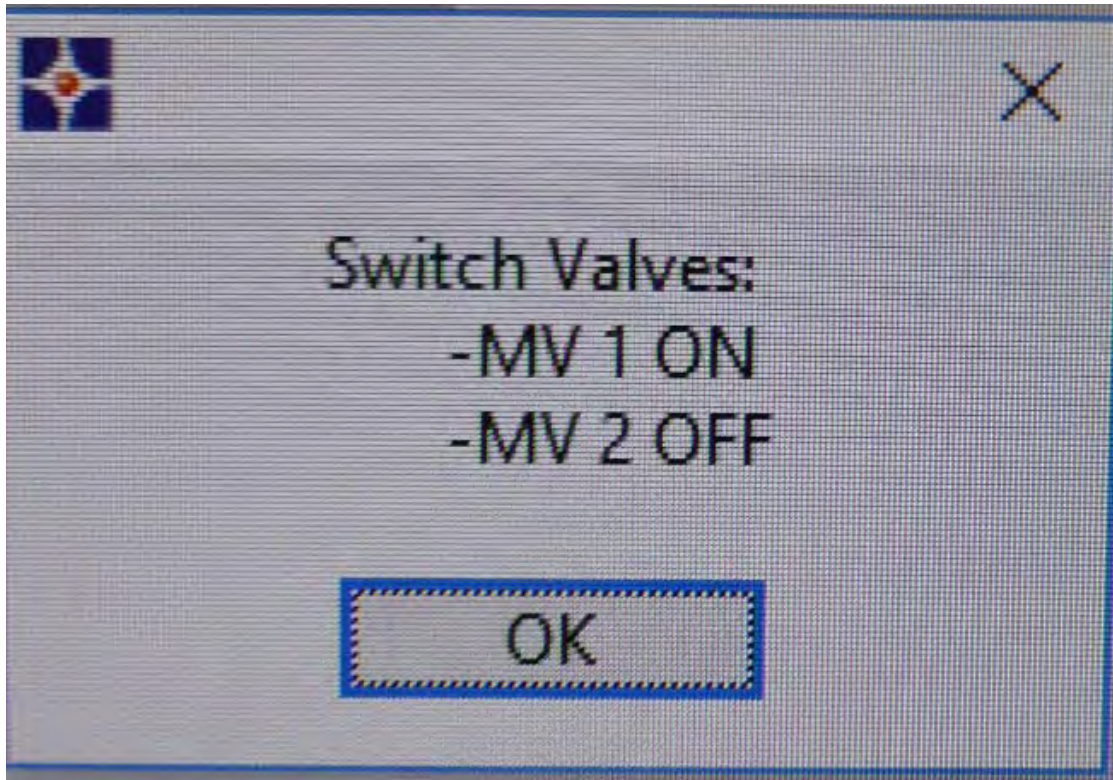
Picture 7.2-25 Reference sensor closed OK.

23. On the valve control box set "MV1" to on (EIN) and "MV2" to off (AUS).



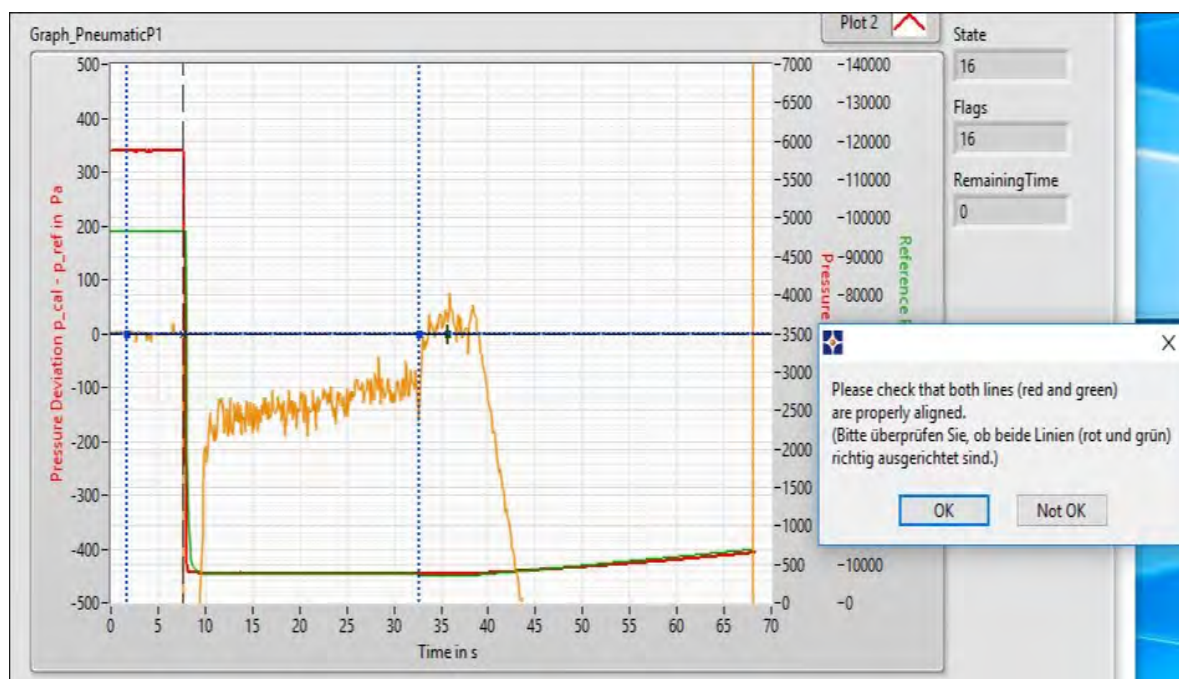
Picture 7.2-26 Valve switch box settings.

24. Then click "OK"



Picture 7.2-27 Confirm valve switches and click OK.

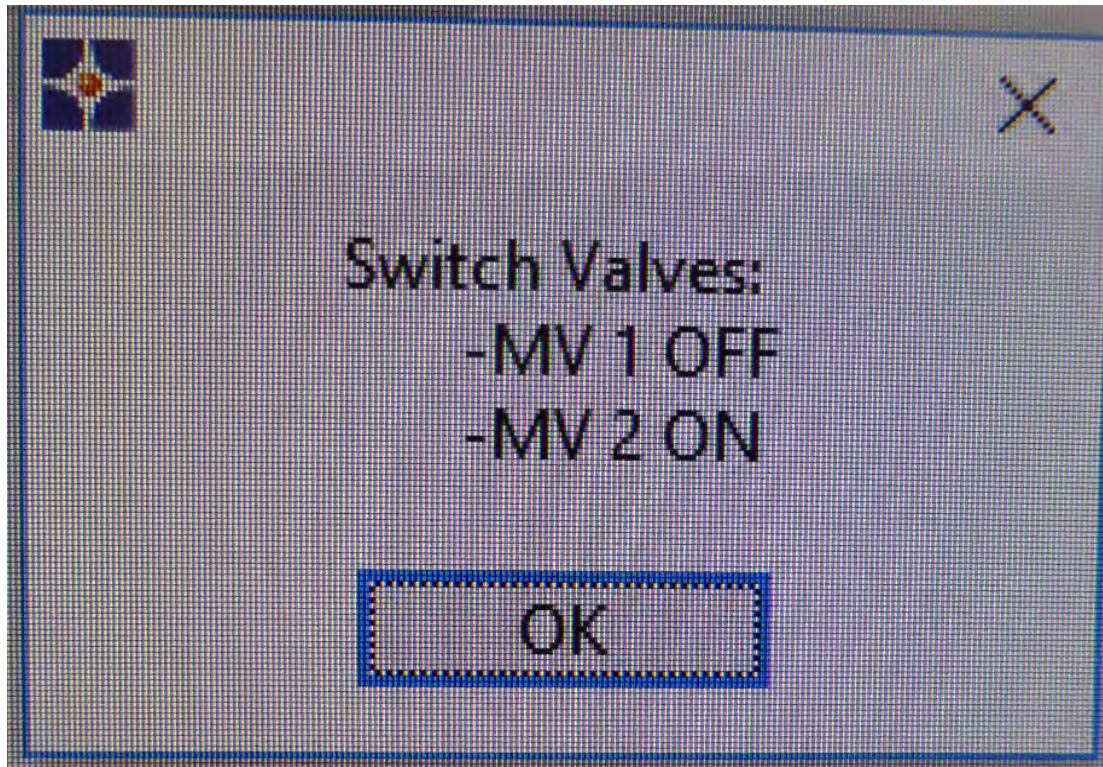
25. Observe the resulting graph and confirm the RED and GREEN vertical traces are aligned together.



Picture 7.2-28 Pressure calibration graph.

26. If the RED and GREEN traces do not align check all tubing connections on the fixture and measurement tool.
27. Repeat the calibration in the "step-mode" by selecting the calibration step "LoadcartridgePneumatic lmg_cartridgepneumatic" from the sequence list.

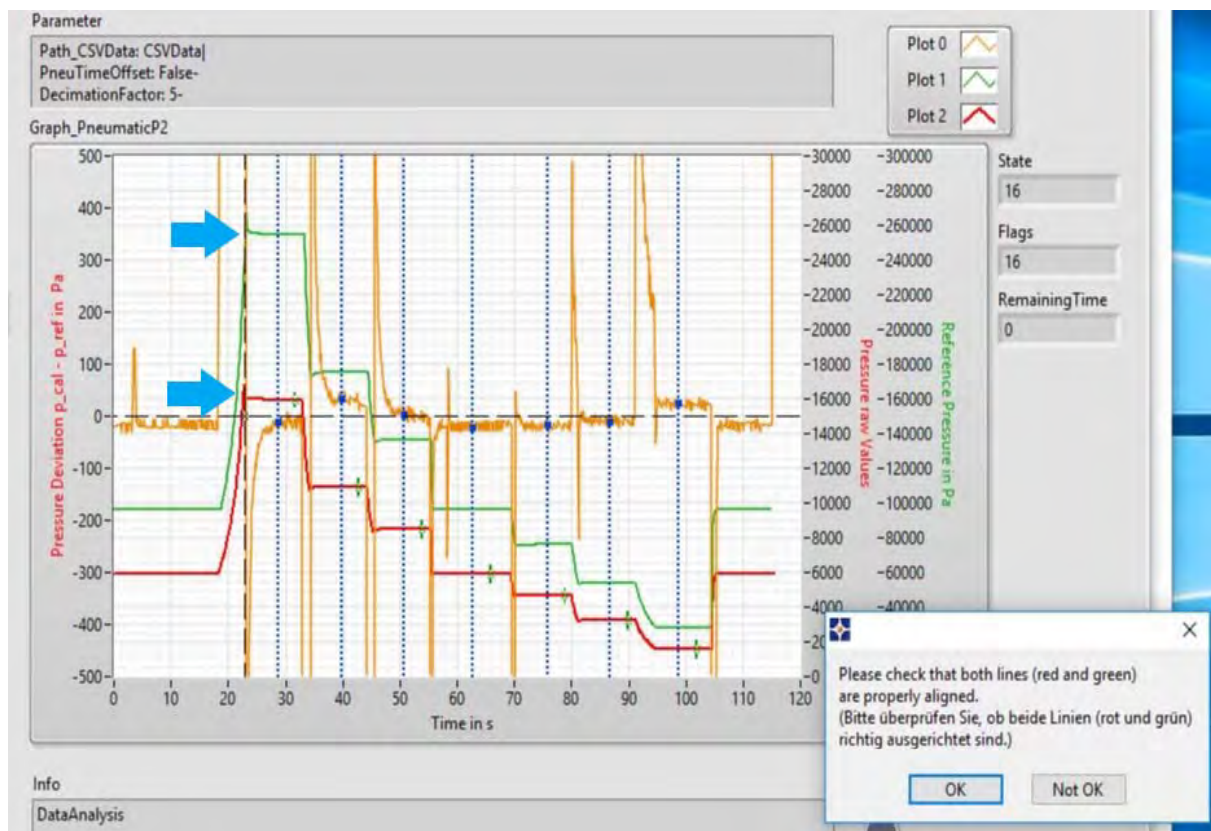
28. On the valve control box set "MV1 to off (AUS) and "MV2" to on (EIN).



Picture 7.2-29 Valve control box switch positions.

29. Click "OK" to start the calibration.

30. Observe the graph and confirm that the marked points of the RED and GREEN traces are vertically aligned on one line.



Picture 7.2-30 Pressure calibration graph.

31. If the traces are properly aligned click "OK".
32. Repeat the calibration in the "step-mode" by selecting the calibration step "LoadcartridgePneumatic lmg_cartridgepneumatic" from the sequence list.
33. Remove the pressure measurement fixture from the cartridge slot.
34. On the valve control box switch both valves, MV1 and MV2 to off (AUS) position.
35. Loosen the reference sensor connection by 3 turns.



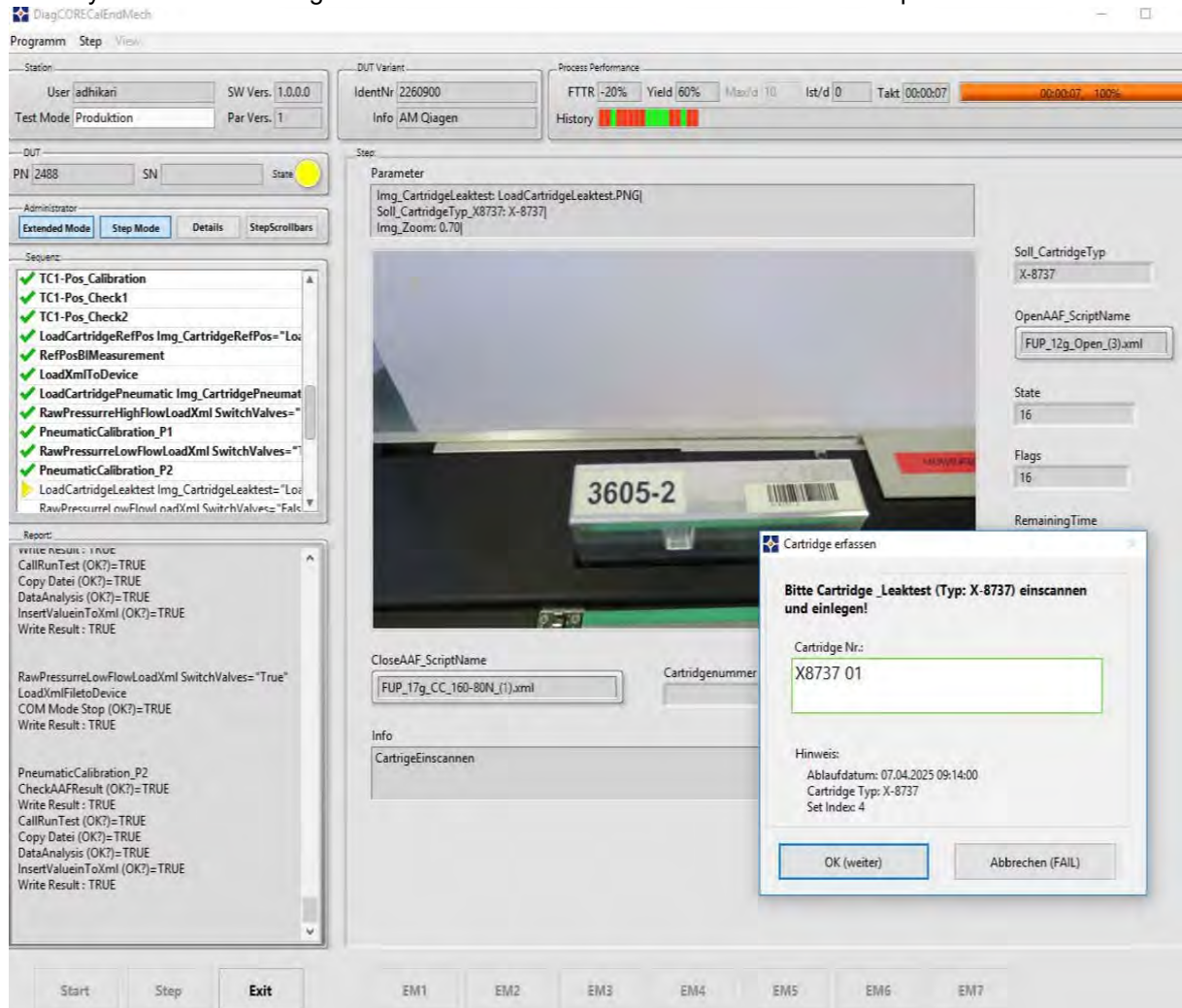
Picture 7.2-31 Loosen reference sensor connection.

36. Remove the pressure measurement fixture from the cartridge slot.
37. Install the lid.
38. Insert the Endtest/Leaktest X-8737 cartridge into the cartridge slot.



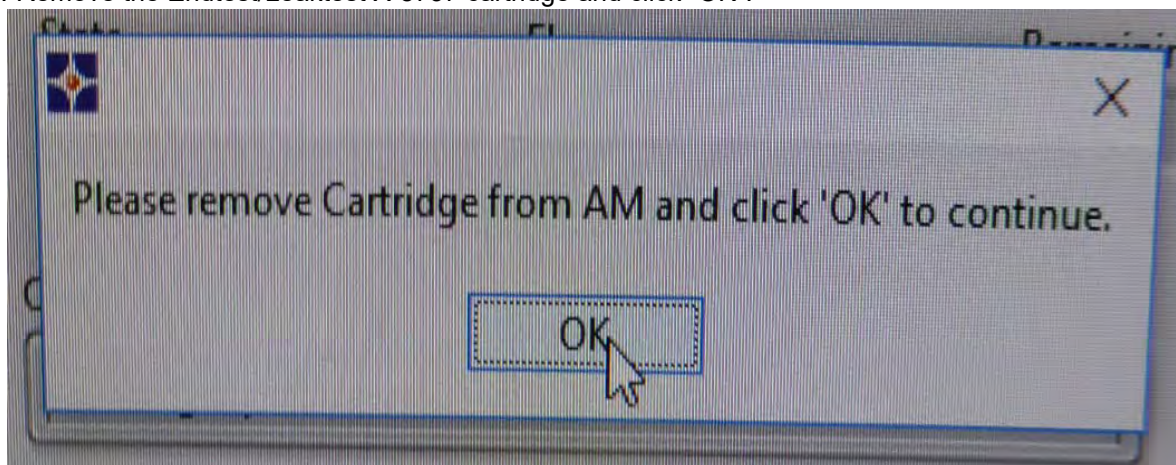
Picture 7.2-32 Insert leak test cartridge in the cartridge slot.

39. Verify the correct cartridge number and click “OK” to start the calibration sequence.



Picture 7.2-33 Verify cartridge number and click OK.

40. Remove the Endtest/Leaktest X-8737 cartridge and click “OK”.



Picture 7.2-34 End test/Leak test calibration complete.

41. Remove the pressure measurement tool from the back of the AM.

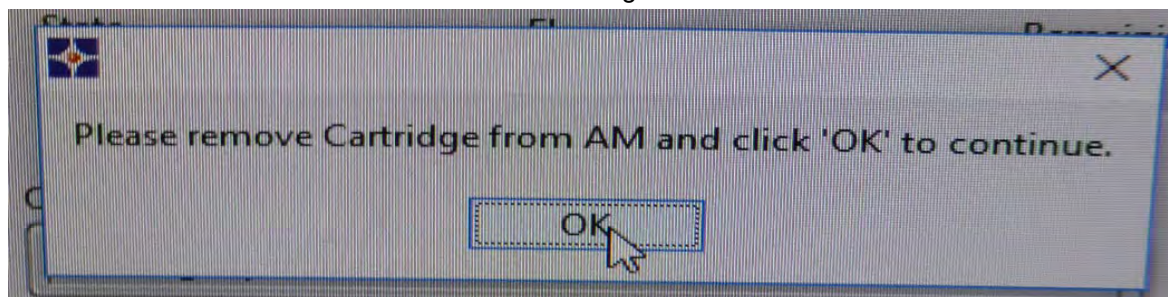
42. To start the cartridge detection determine if the thermal unit was replaced. If not click “Continue”; if the thermal unit was replaced go to the next step.

43. Locate the CD_Cartridge_Threshold value that was determined on CP3 step 60; enter it in the Cartridge detection Threshold then click "Continue".

Table 7.2-35 Cartridge detection threshold CP2 step 60

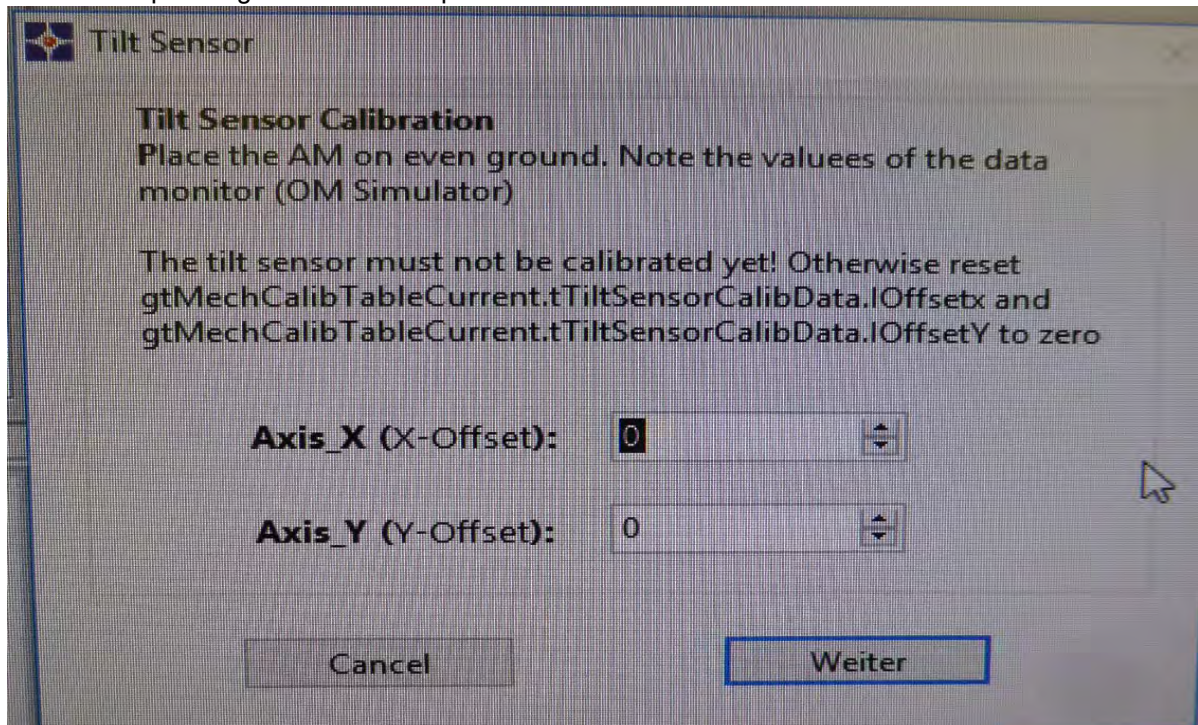
Table 7.2-36 Cartridge detection setup.

44. Enter the threshold in the box and click "continue" to start the calibration.
 45. When the calibration has finished remove the cartridge and click "Continue".



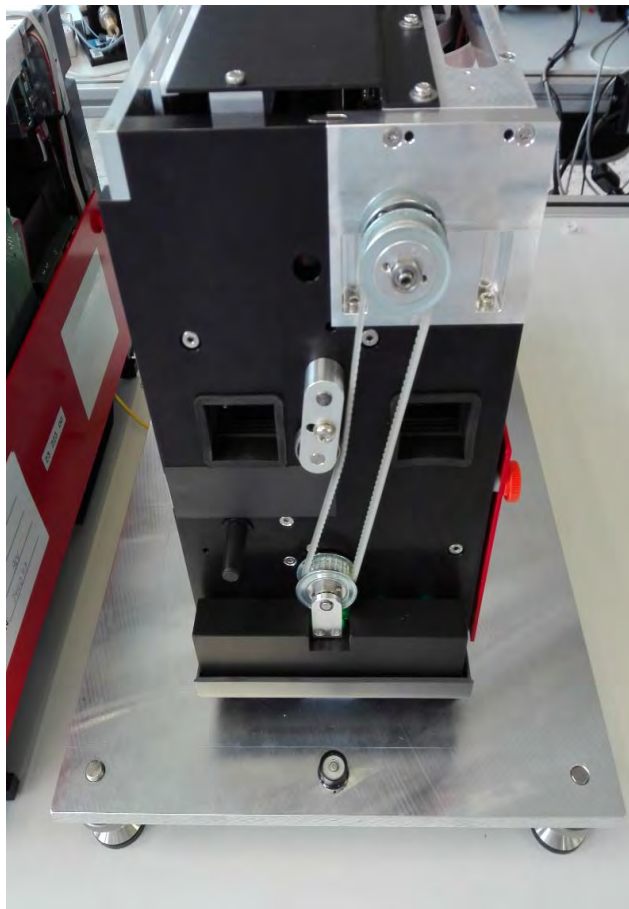
Picture 7.2-37 Calibration step complete.

46. To start the tilt sensor calibration determine if the sensor was replaced, if not click “Weiter (Continue)”. If it was replaced go to the next step.



Picture 7.2-38 Tilt sensor calibration.

47. Ensure the AM is on a level surface.



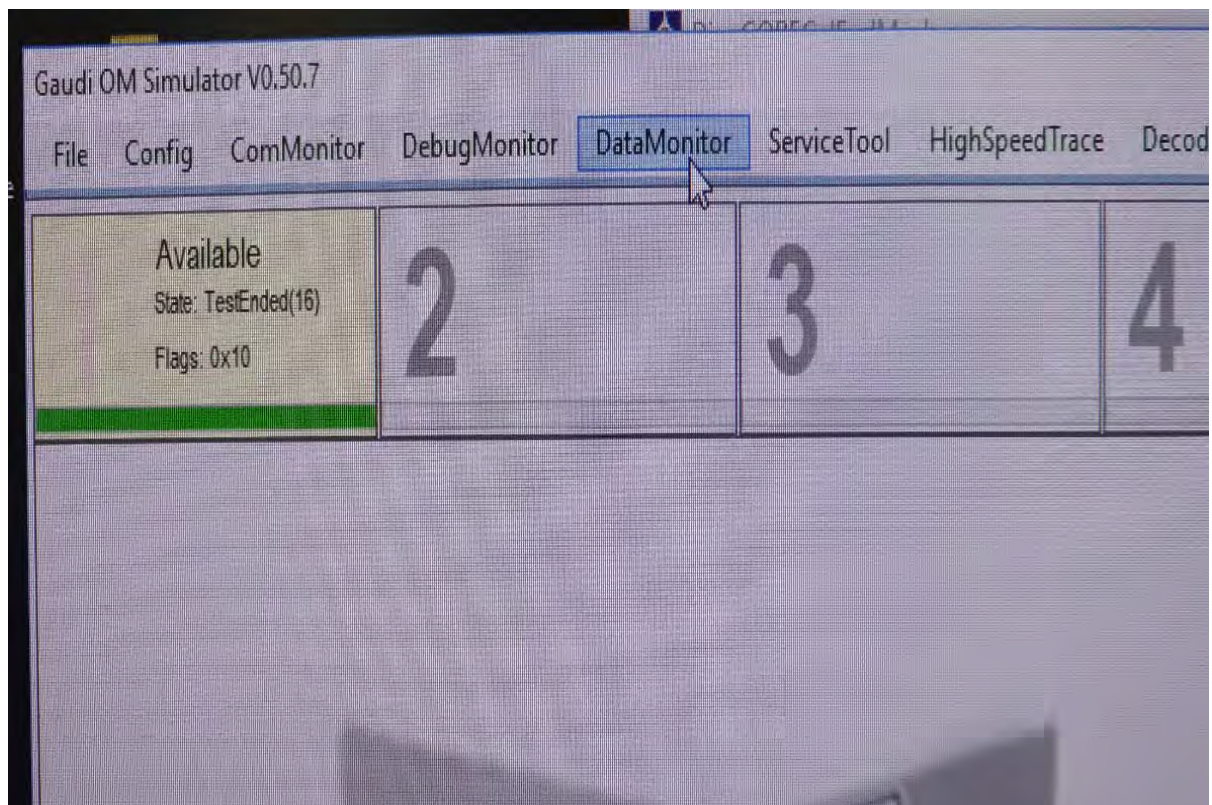
Picture 7.2-39 AM on a level surface.

48. If not already open, open the GAUDI OM Simulator.



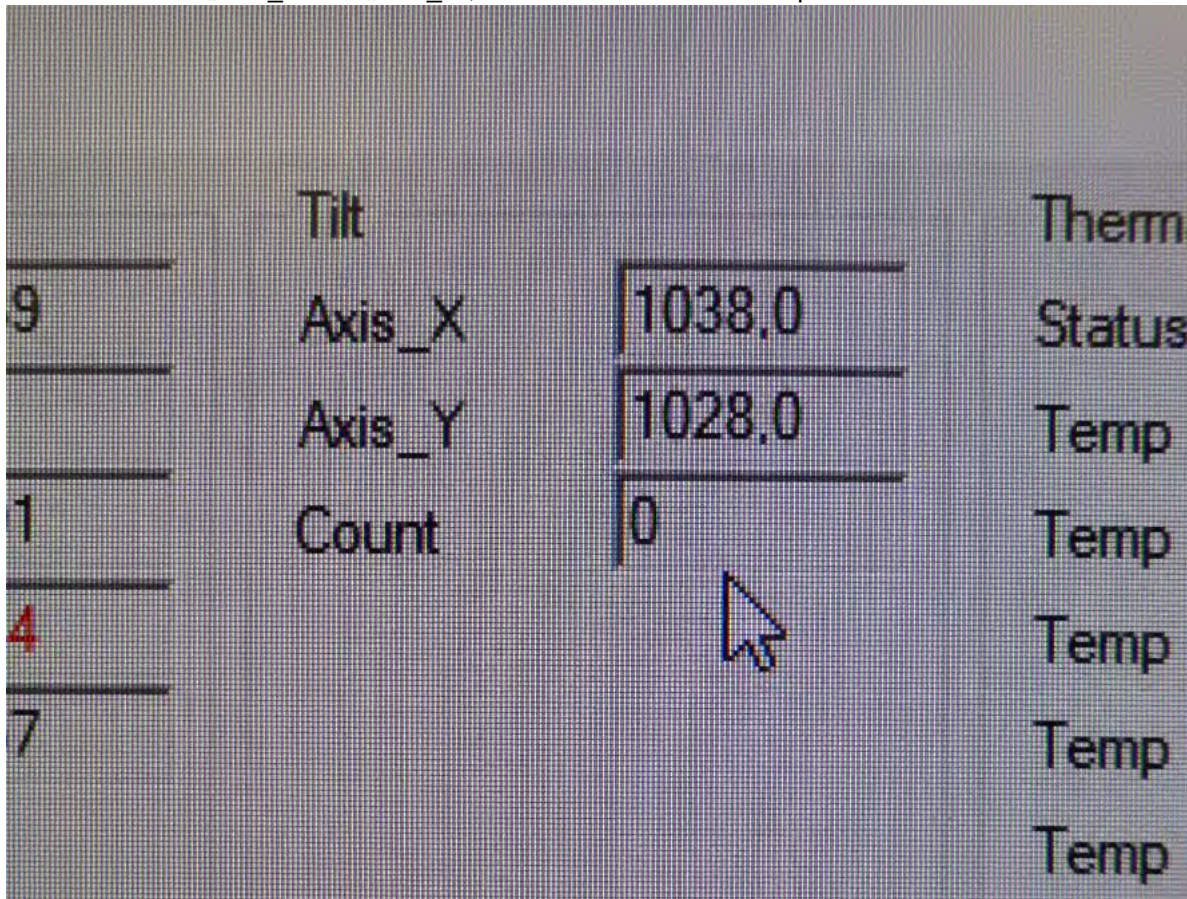
Picture 7.2-40 GAUDI OM Simulator icon.

49. Click on DATAMonitor in the OM Simulator.



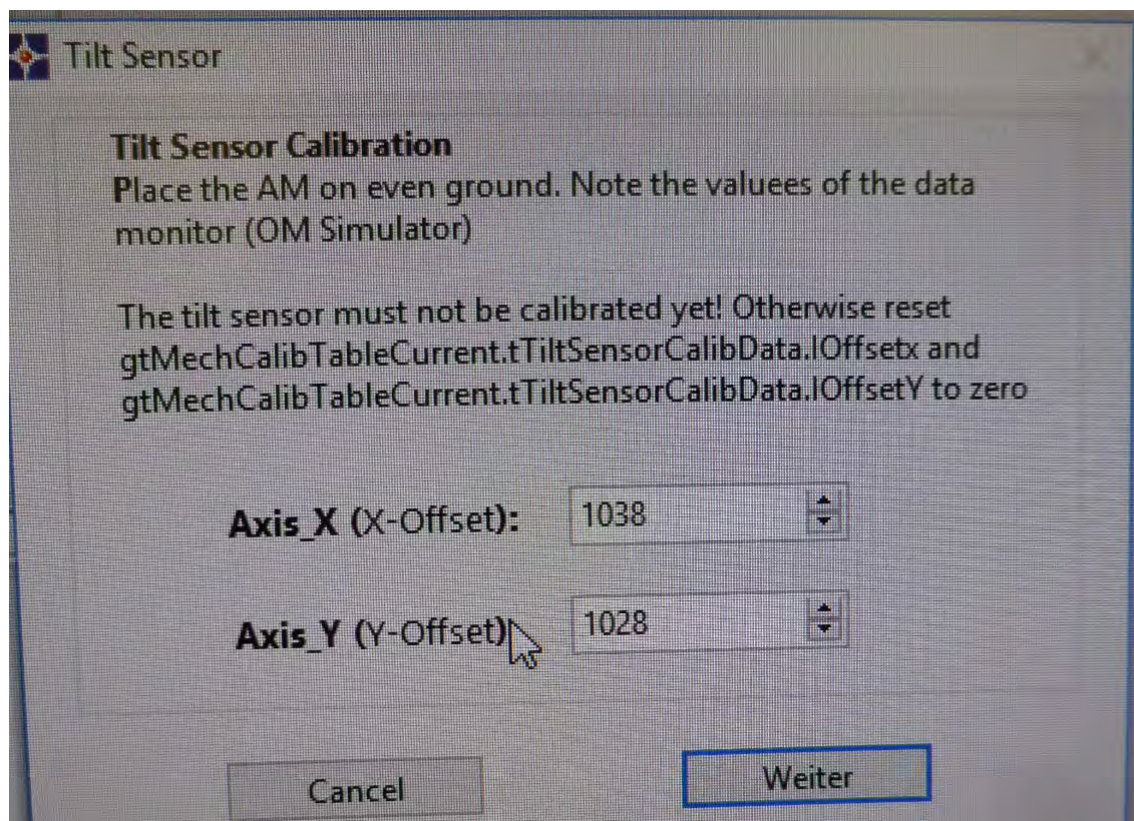
Picture 7.2-41 GAUDI OM Simulator DataMonitor.

50. Note the values „Axis_X“ and „Axis_Y“, insert the values in to the open Mechanical Calibration window.



Picture 7.2-42 Data Monitor axis values.

51. Place the axis values in the tilt sensor calibration window.



Picture 7.2-43 Tilt sensor calibration window.

52. Click Weiter (Continue).
 53. Close the MeasurementDataMonitor and click "OK".

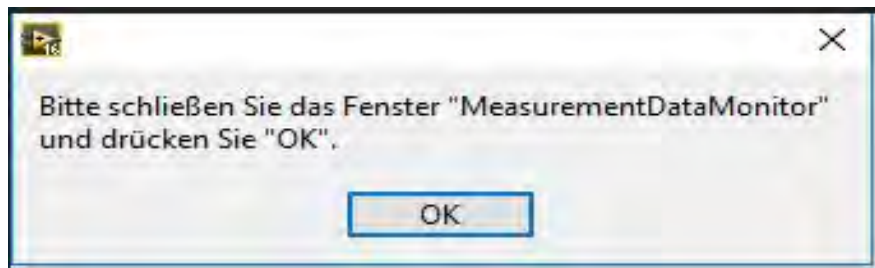
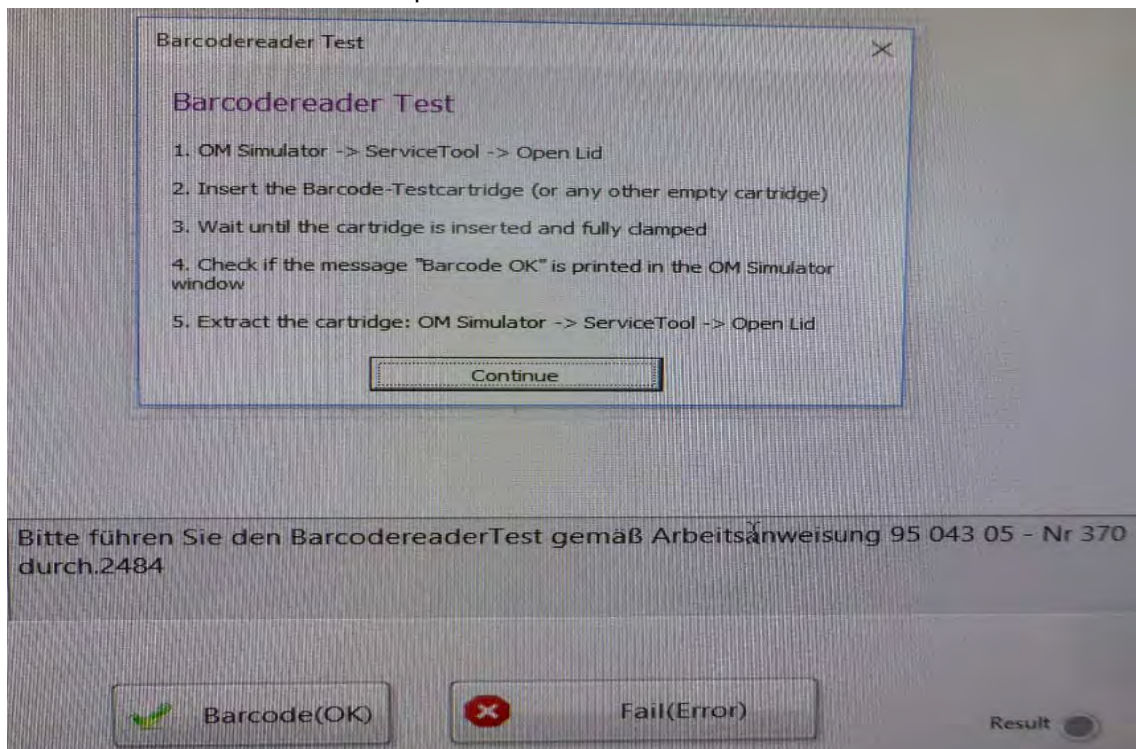


Table 7.2-44 Close MeasurementDataMonitor.

54. Follow the bar code reader test sequence.



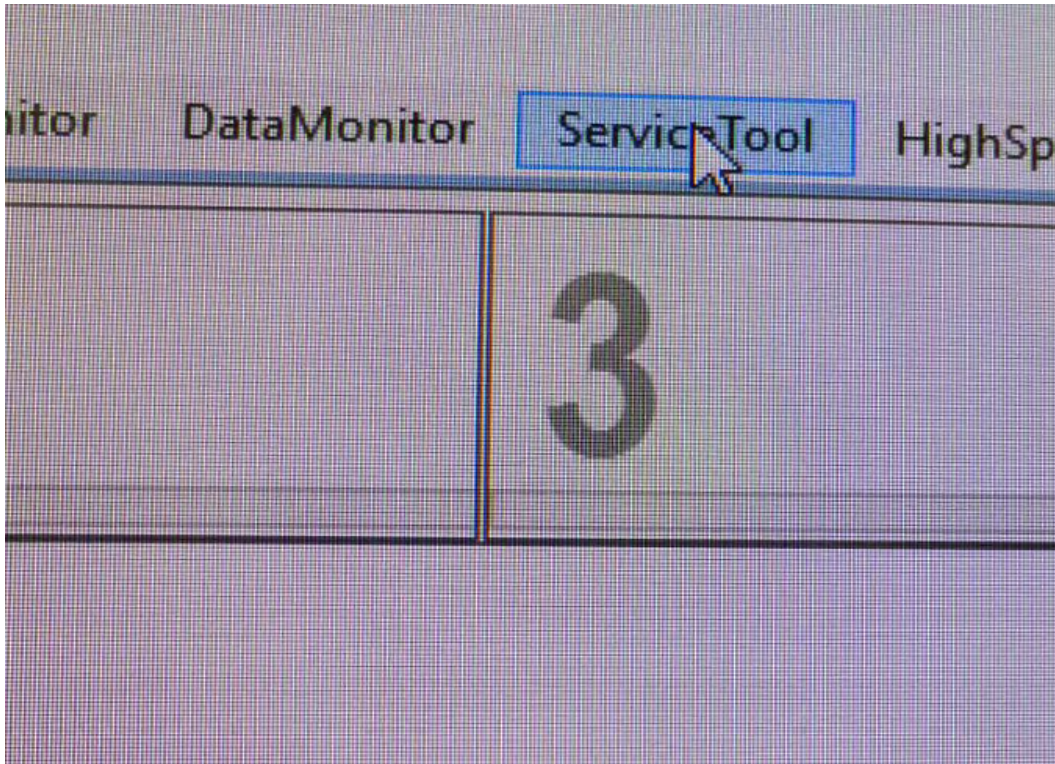
Picture 7.2-45 Bar code reader test script.

55. If not already open, open the GAUDI OM Simulator.



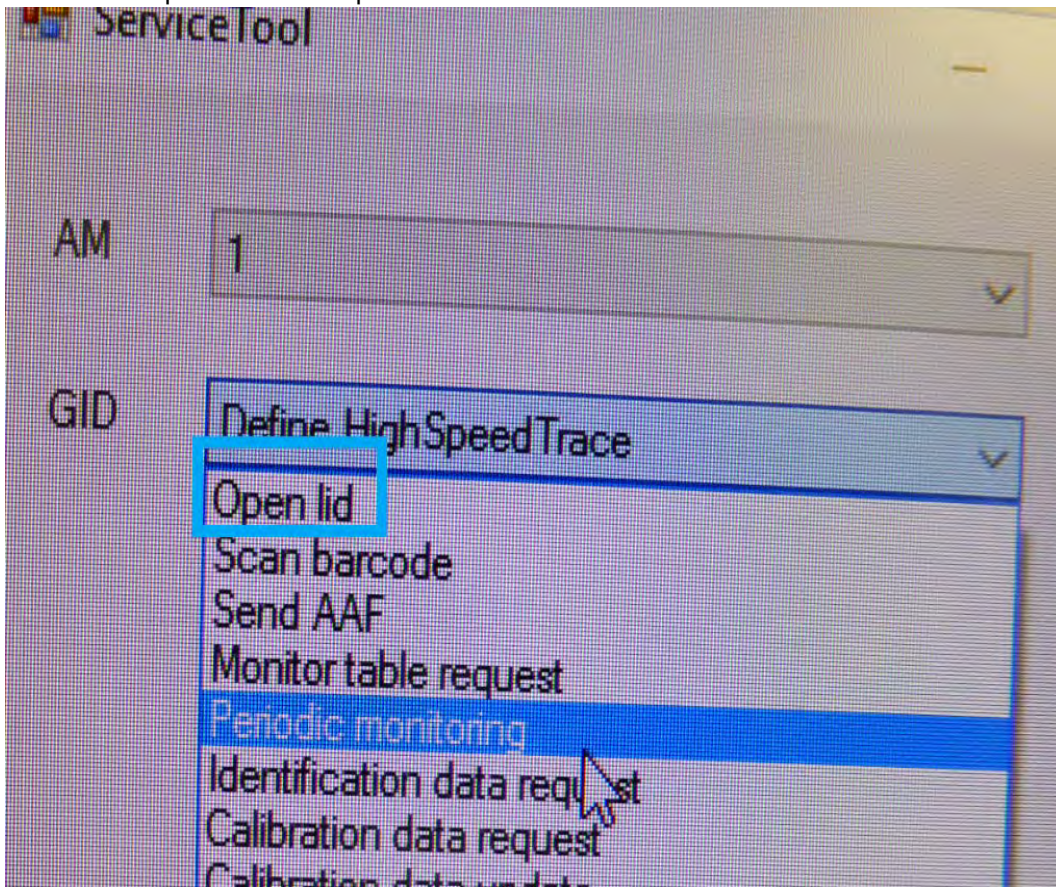
Picture 7.2-46 GAUDI OM Simulator Icon.

56. On the GAUDI OM Simulator open the Service Tool.



Picture 7.2-47 Open ONM simulator service tool.

57. From the GID dropdown Select "open Lid".



Picture 7.2-48 OM simulator service tool menu.

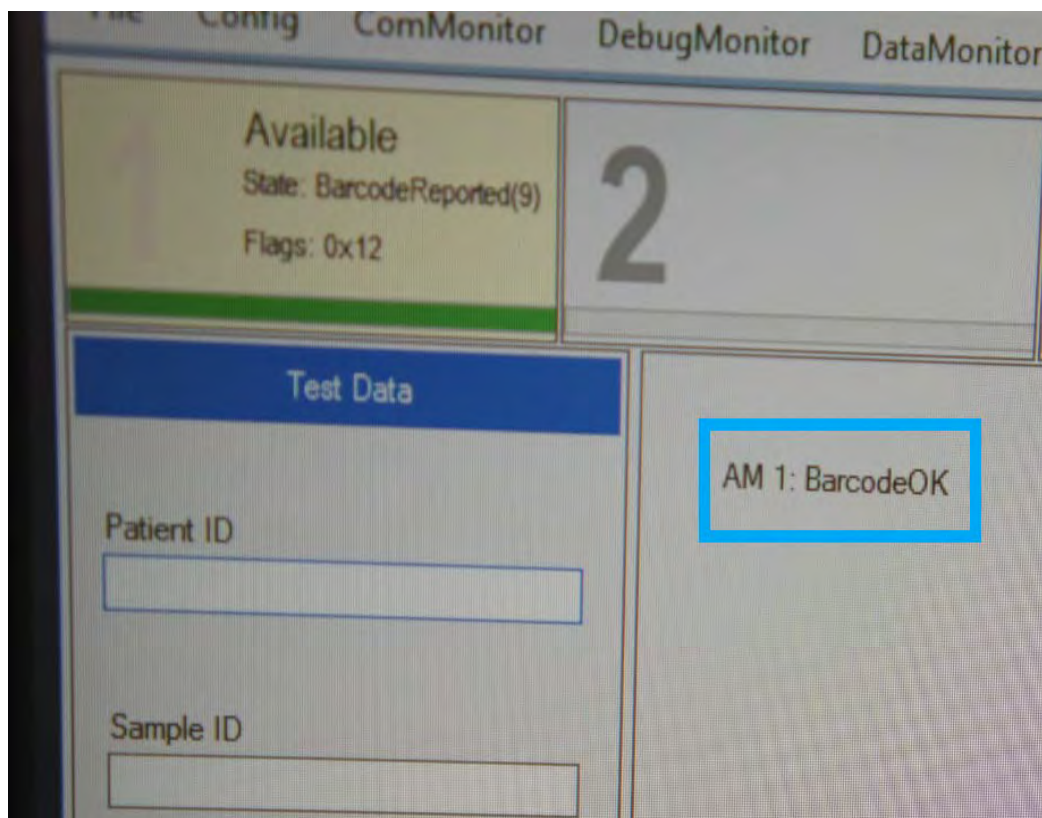
58. Click "OK" to send.

59. Place any cartridge with a bar code in the cartridge slot.



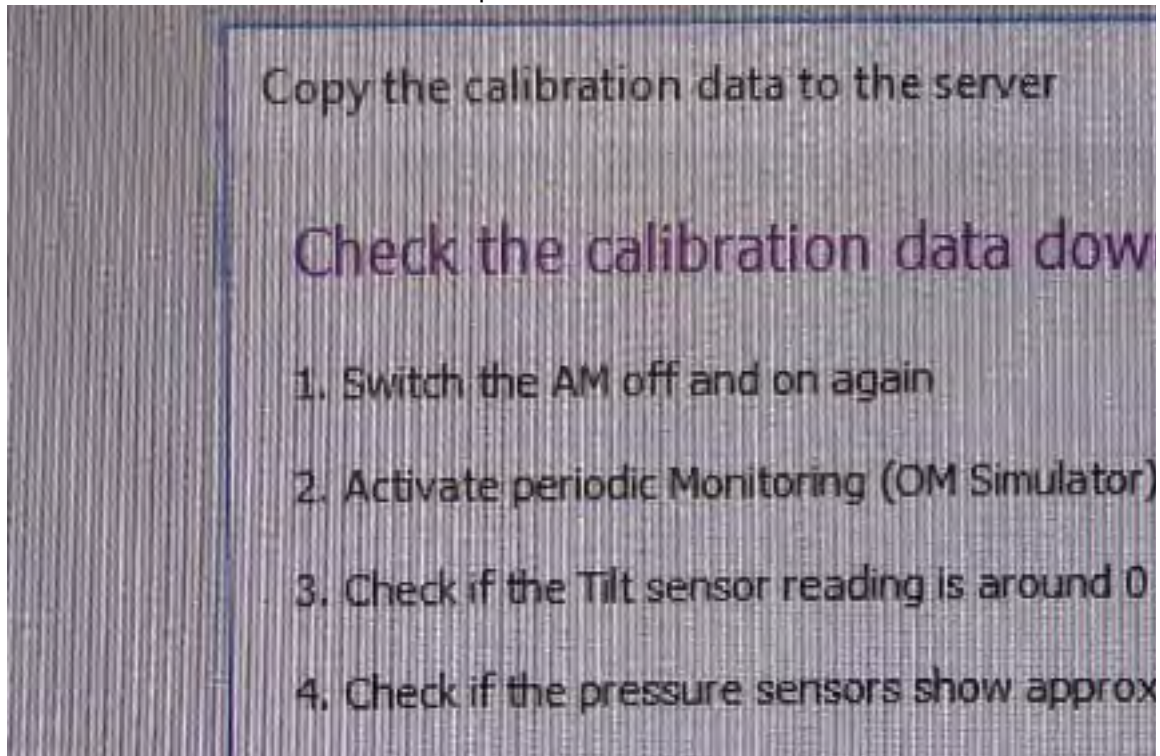
Picture 7.2-49 Insert cartridge in slot.

60. Check for the "BarcodeOK" message.



Picture 7.2-50 Bar code "OK".

61. On the OM simulator GID dropdown menu select "open Lid".
62. Remove the cartridge.
63. To check the calibration download data follow the script. Leave this window open; the result of the calibration checks will determine the final pass/fail.



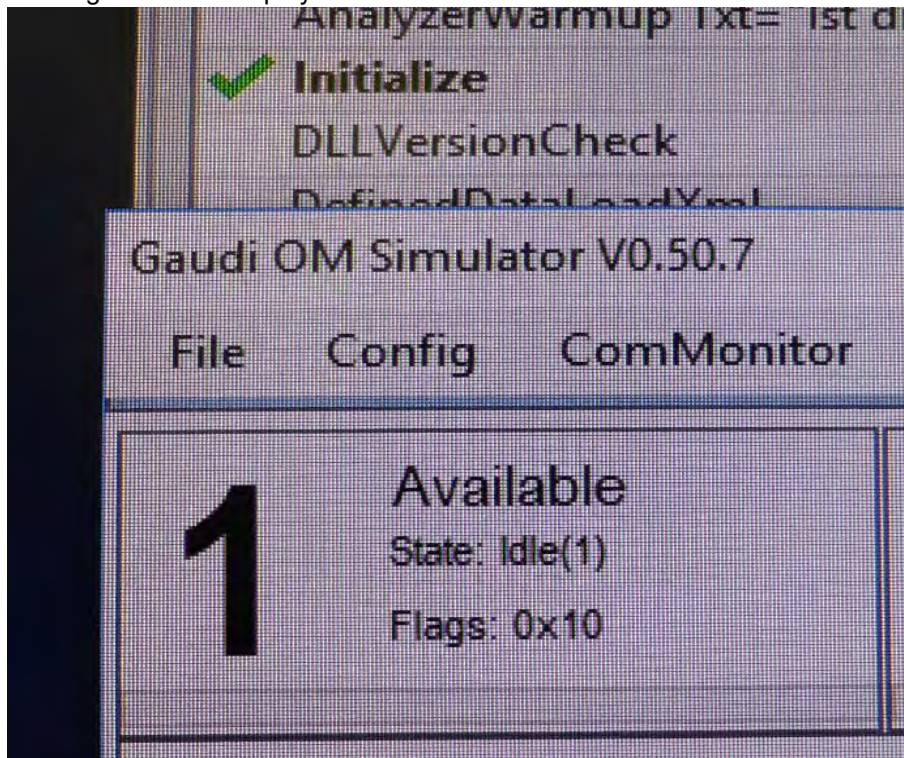
Picture 7.2-51 Calibration check script and result.

64. Set the AM power switch to "off" and then back to "on".
65. Open the GAUDI OM Simulator.



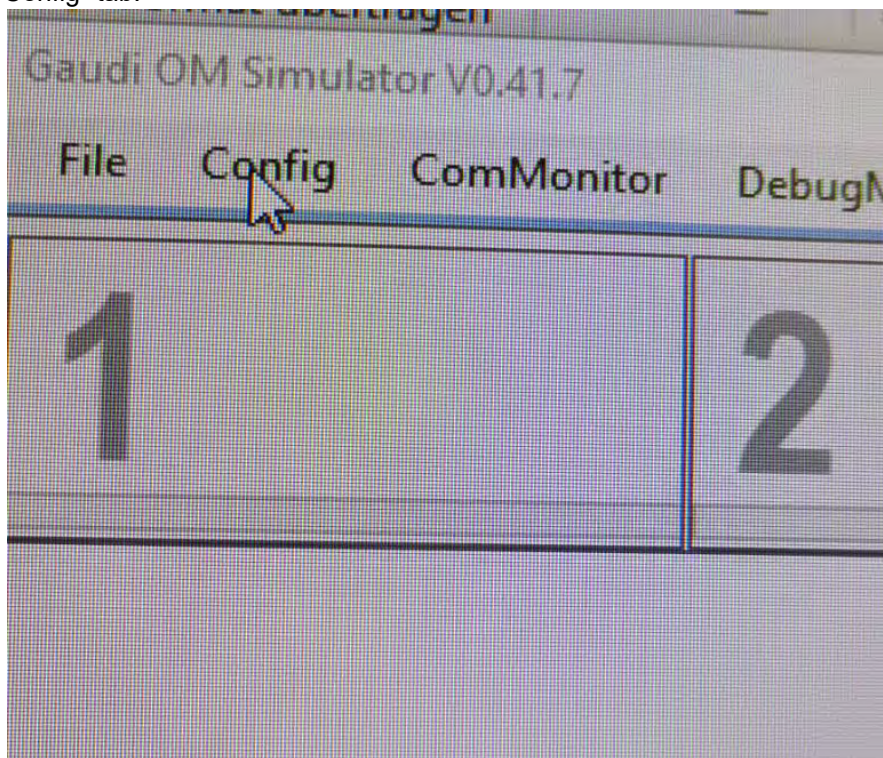
Picture 7.2-52 GAUDI OM Simulator ICON

66. Wait until the “Flags: 0x10” is displayed.



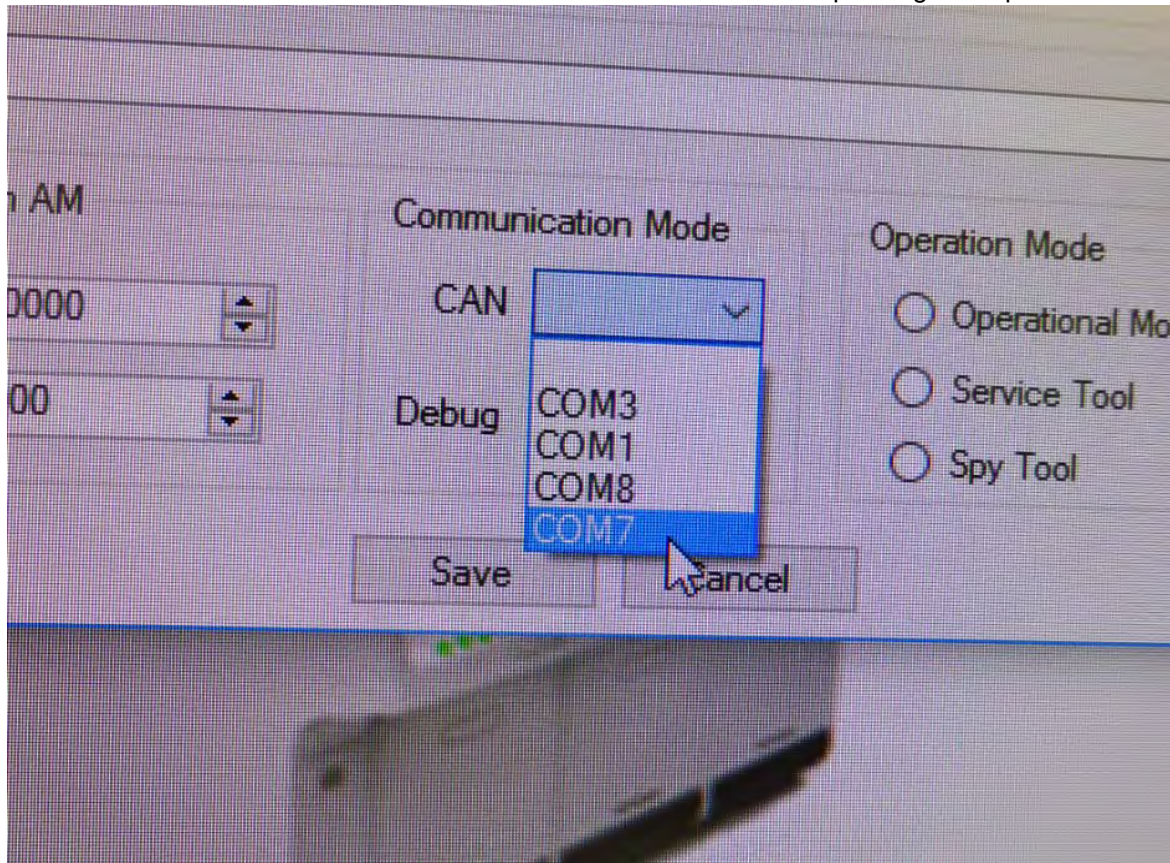
Picture 7.2-53 GAUDI OM Simulator Flags 0x10.

67. Click on the “Config” tab.



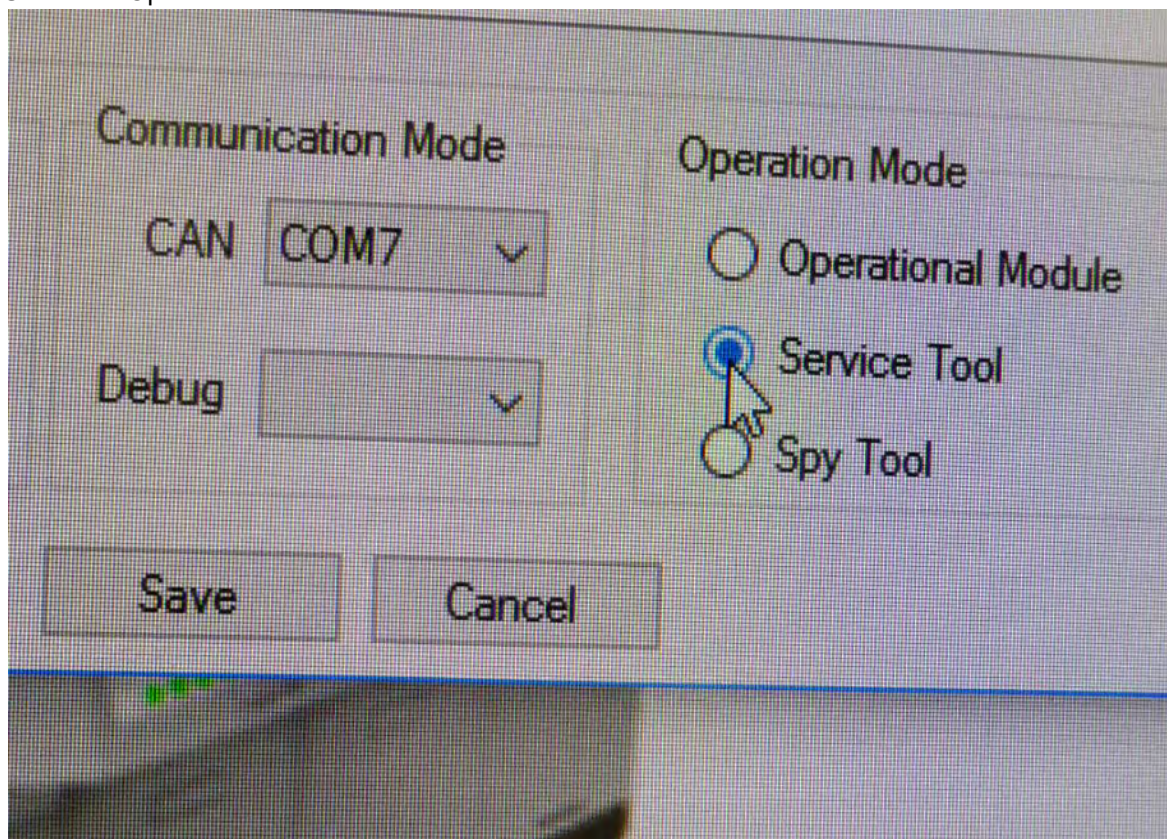
Picture 7.2-54 GAUDI OM Simulator Config tab.

68. Check the Communication Mode is set to "CAN" and set to the corresponding COM port.



Picture 7.2-55 COM port settings.

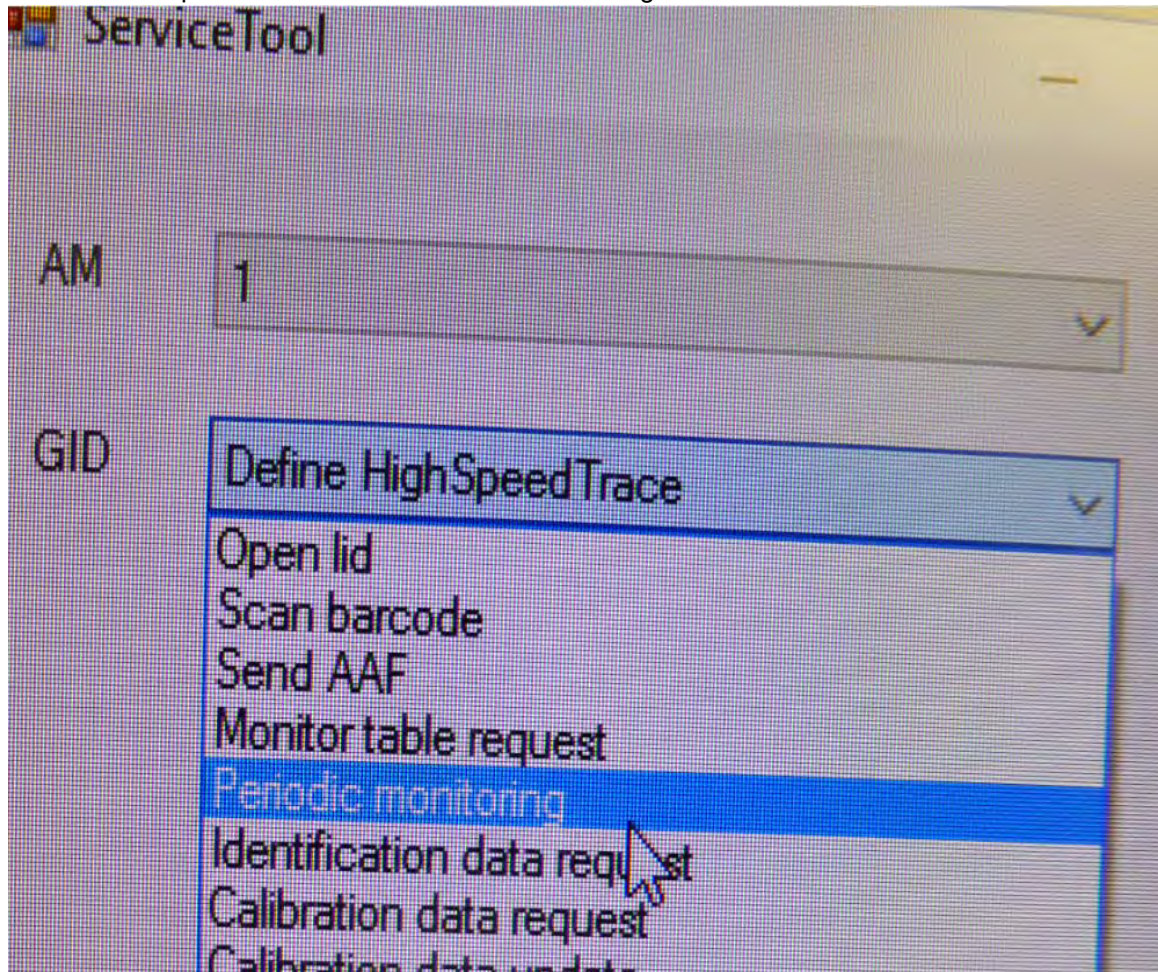
69. Under the Operation Mode select the "Service Tool" button.



Picture 7.2-56 GAUDI ON Simulator config Operation Mode.

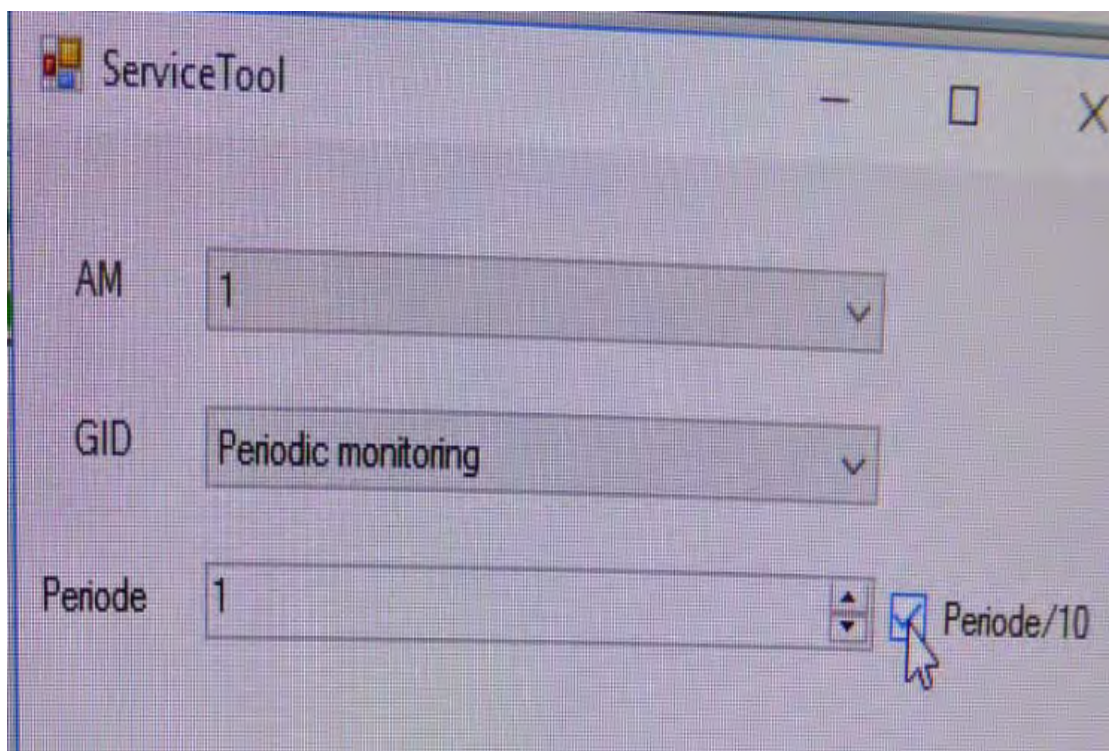
70. Click "Save".

71. In the GID dropdown menu select "Periodic Monitoring".



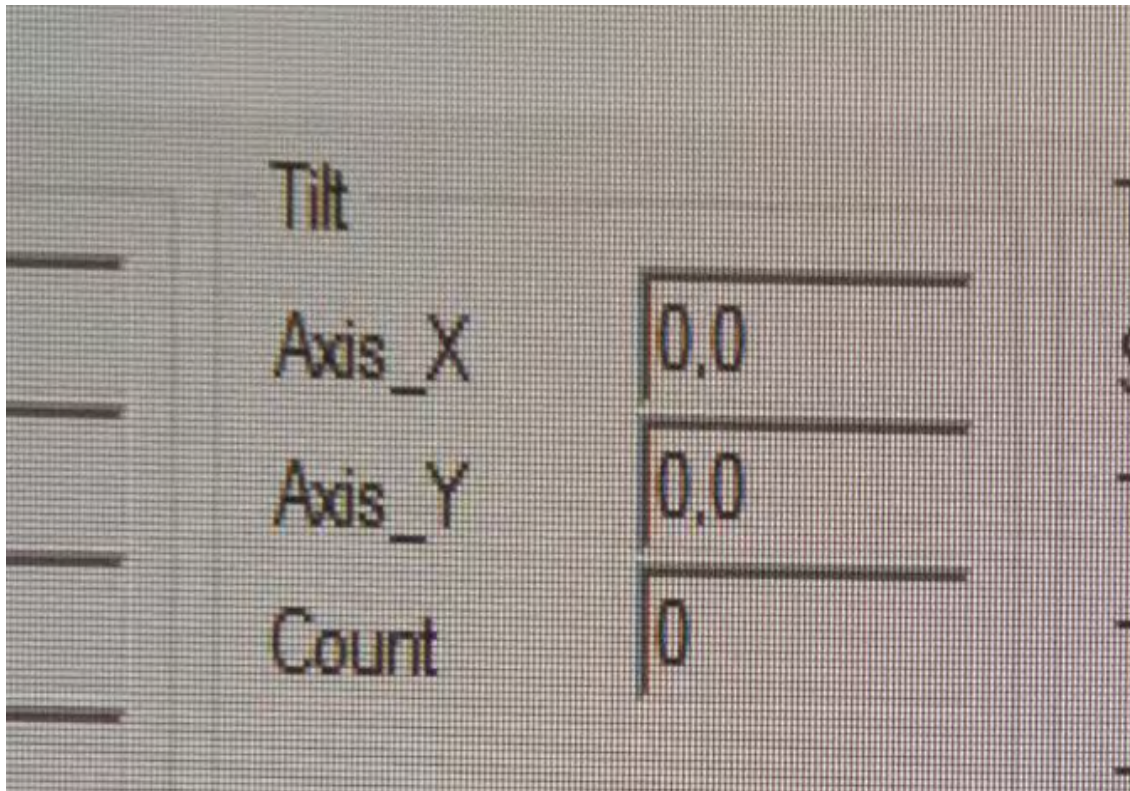
Picture 7.2-57 GAUDI OM Simulator GID dropdown menu.

72. Select Period 1 and check the Period/10 box.



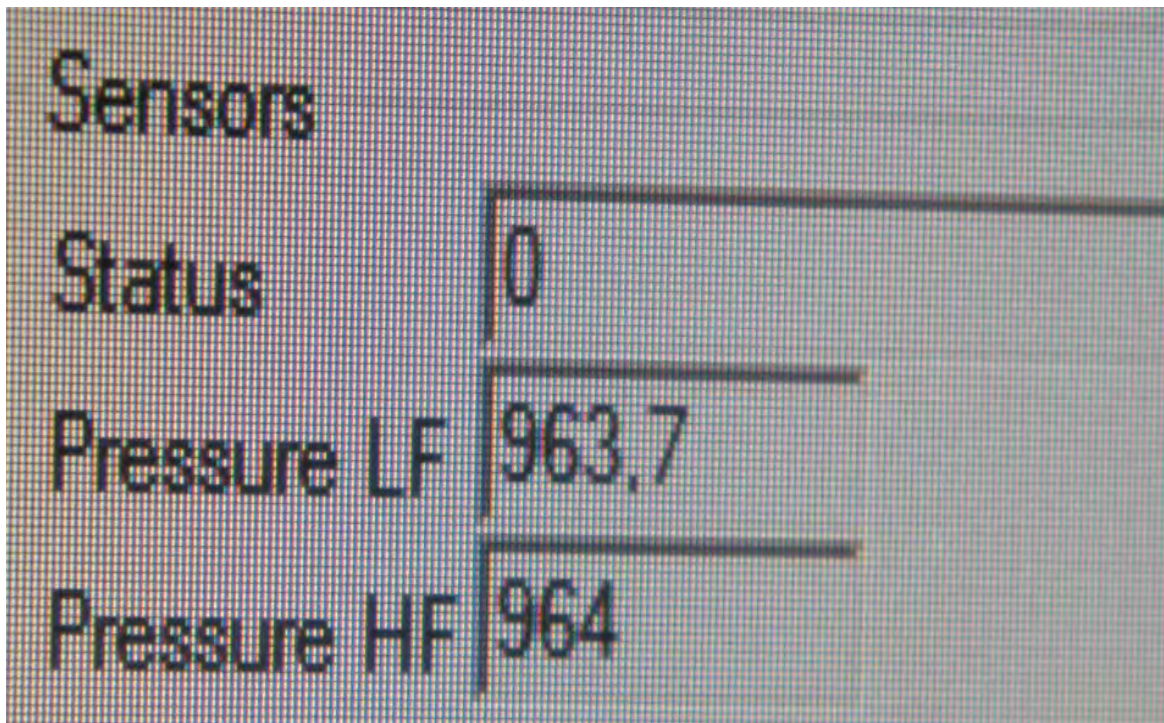
Picture 7.2-58 GAUDI OM Simulator set Periodic Monitoring.

73. Ensure the Tilt Sensor X and Y axis settings are 0 +/- 0.3.



Picture 7.2-59 Tilt Sensor final calibration settings.

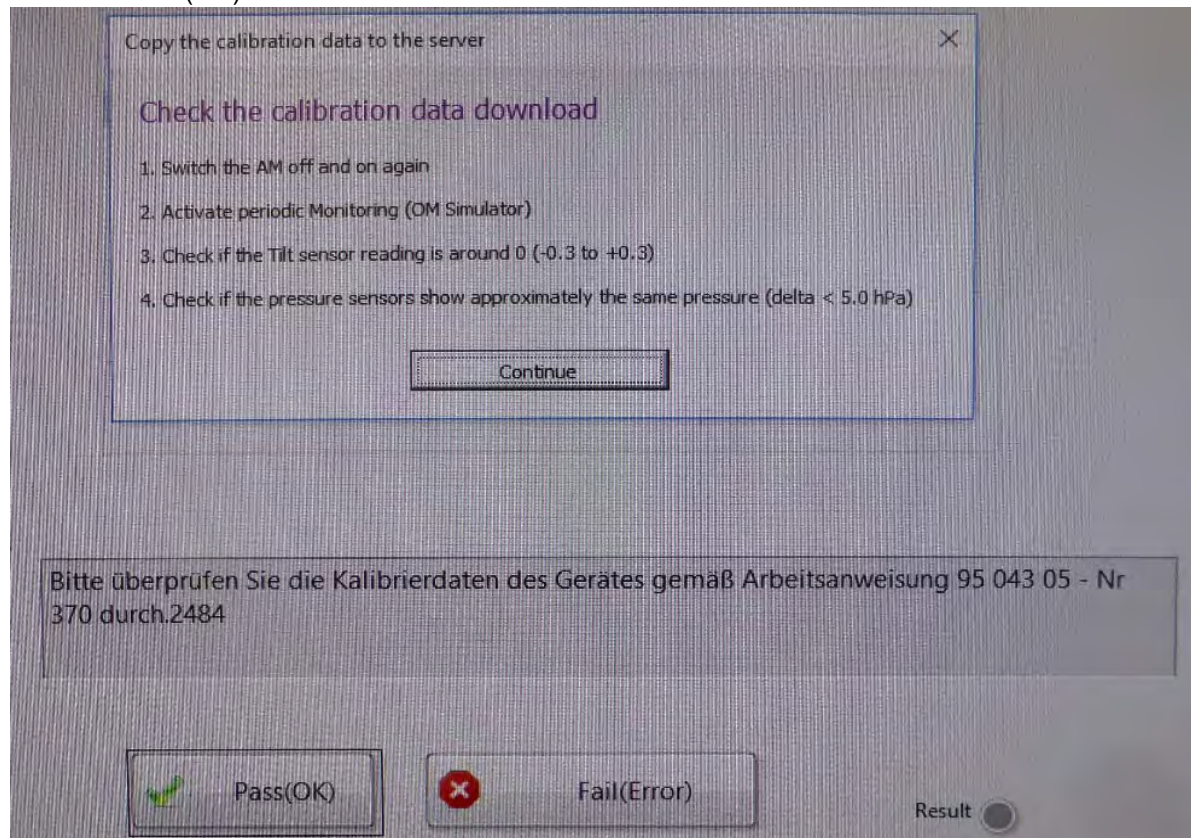
74. Ensure the delta for the values for the Pressure LF and Pressure HF is less than 5 hPa. Example:
Pressure HF – Perssure LF = 964 hPa – 963,7 hPa = 0,3 hPa => 0,3 hPa < 5 hPa => OK



Picture 7.2-60 Pressure sensors calibration.

75. Close the GAUDI OM Simulator.

76. Click "PASS"(OK).



Picture 7.2-61 Calibration check results PASS/FAIL.

77. Calibration complete.

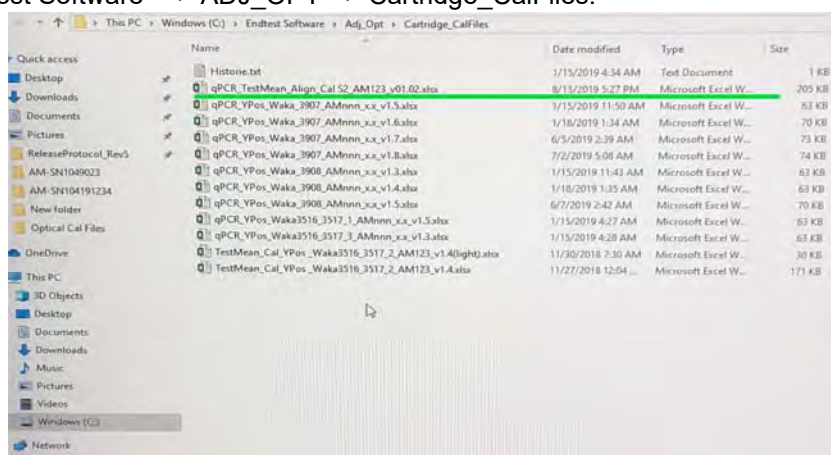
7.2.1 CP2 Optical Alignment Cartridge Registration

Purpose	Register the cartridges in the SQL database.
Tool(s) Required	Optical Alignment Cartridge Calibration File.
Calibration/Testing Time	30 Minutes

Table 7.2-62 CP2 Register Optical Cartridges.

INFORMATION 	The Optical Alignment cartridges must be loaded into the SQL database prior to the test. This step can be skipped if they have been loaded previously.
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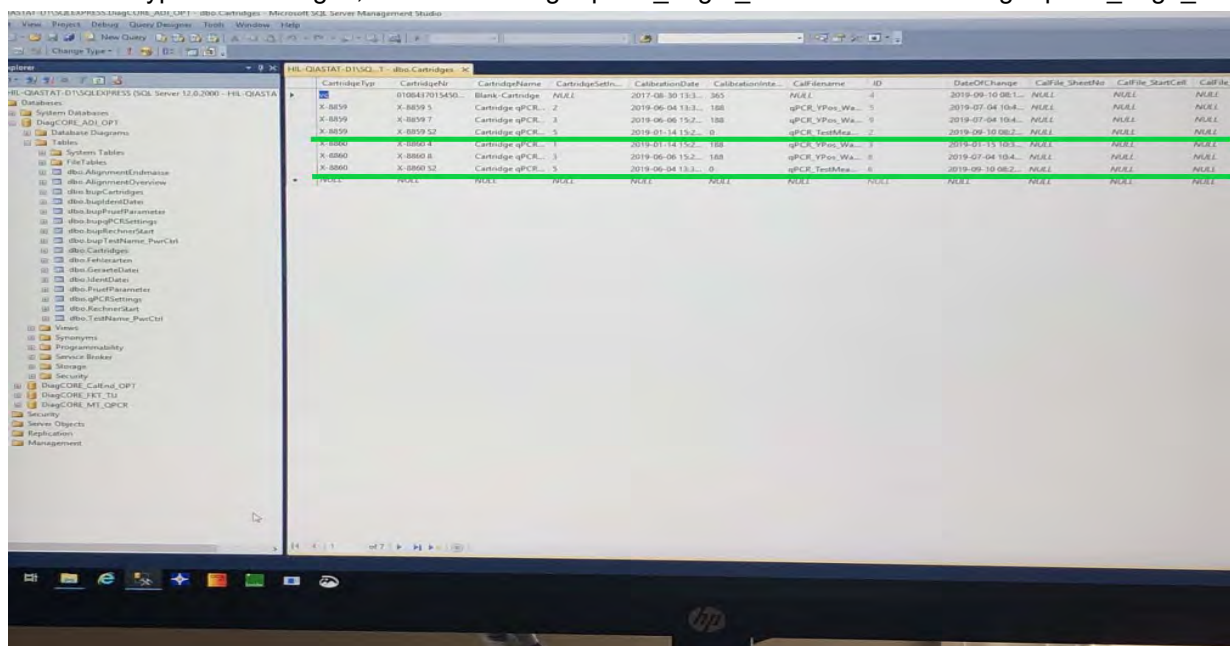
1. Ensure you have the current calibration data for the cartridges from LRE.
2. Log onto the repair station computer.
3. Got to "This PC"--> C: drive.
4. Open "End Test Software" --> ADJ_OPT --> Cartridge_CalFiles.



Picture 7.2-63 Calibration file location.

5. Drop the Excel "qPCR_TestMean_Align_CAL S2 AM123" file to this folder; and then copy the file name; highlight and press Cntrl and C.
6. Click on the WINDOWS Icon and type "SQL" and select "Microsoft SQL Service Management Studio".
7. In the popup window the server name should be "HIL-QISTAT-D(X)\SQLEXPRESS" press Connect.
8. Click on "Databases" --> "Tables" --> "File Tables".

9. Right click on “dbo.Cartridges” and select “Edit All Rows” and the cartridge registration will open, there are two types of cartridges; X-8859-Cartridge qPCR_Align_2 and X-8860-Cartridge qPCR_Align_1.



Picture 7.2-64 Cartridge registration.

INFORMATION



The Cartridge Number is specific to that cartridge by the bar code on the cartridge. The RRC in GTN has been assigned as set 2, so the cartridge number will be X-88XX S2

10. Paste the name of the calibration file name in the “CAI Filename” block of both cartridges.
11. Add a space and “S2” to the “CartridgeNr”.
12. Change the “Cartridgesetting” to “5” for both cartridges.
13. Set the “Calibrationinterval” to “0”.

7.2.2 CP2 qPCR Optical Alignment & Thermal Unit Test

Purpose	Align the qPCR Module and Thermal Unit Test.
Tool(s) Required	Optical Alignment Cartridges; 9026427 - X-8860 Align 1; 9026426 - X-8859 Align 2 9026442 - Power Distributor fixture TU module 9026445 - Control unit TC1-TC2-CC-TU, 9026444 - QPCR Optic Alignment Communication Adapter, Dark Cloth 3ft X 3ft, Ceramic gauges.
Calibration/Testing Time	2.0 Hours.

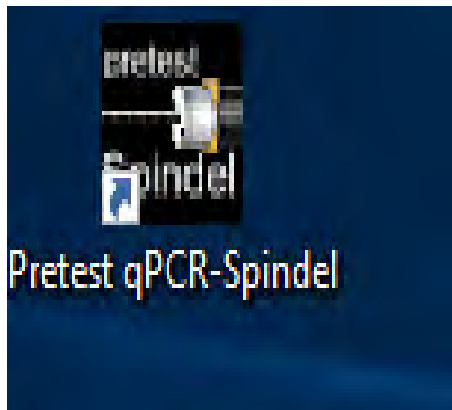
Table 7.2-65 CP2 qPCR module alignment

INFORMATION



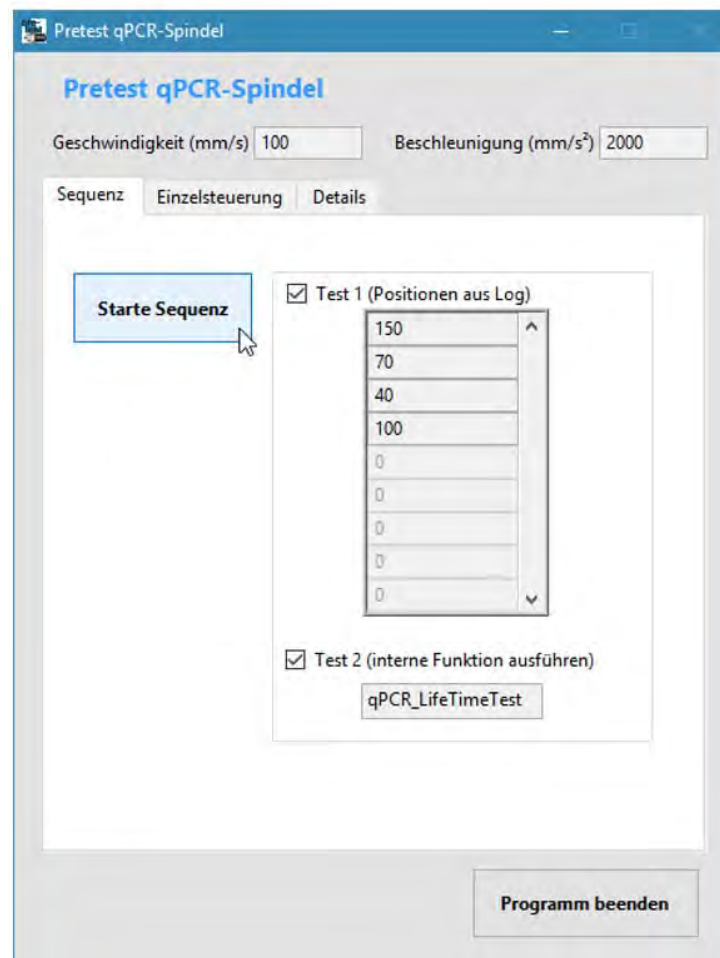
The Optical Alignment cartridges must be loaded into the SQL database prior to the test.

1. If installed remove the cartridge lifter assembly.
2. Log onto the repair station computer.
3. Open "Pretest qPCR Spindle" to perform a pre-test of the spindle.



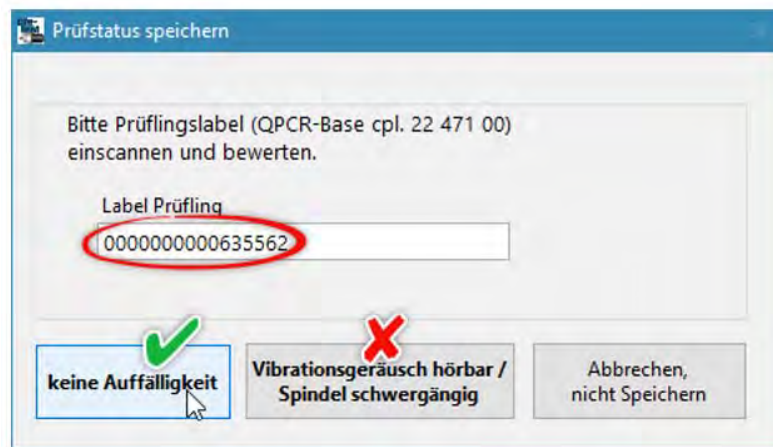
Picture 7.2-66 Pretest qPCR spindle icon.

4. The qPCR motor moves to the "home position" and the buttons are activated.
5. Listen carefully to the sound of the motor detect any vibration or unusual noise which could indicate a faulty spindle.



Picture 7.2-67 Pre-test qPCR spindle screen.

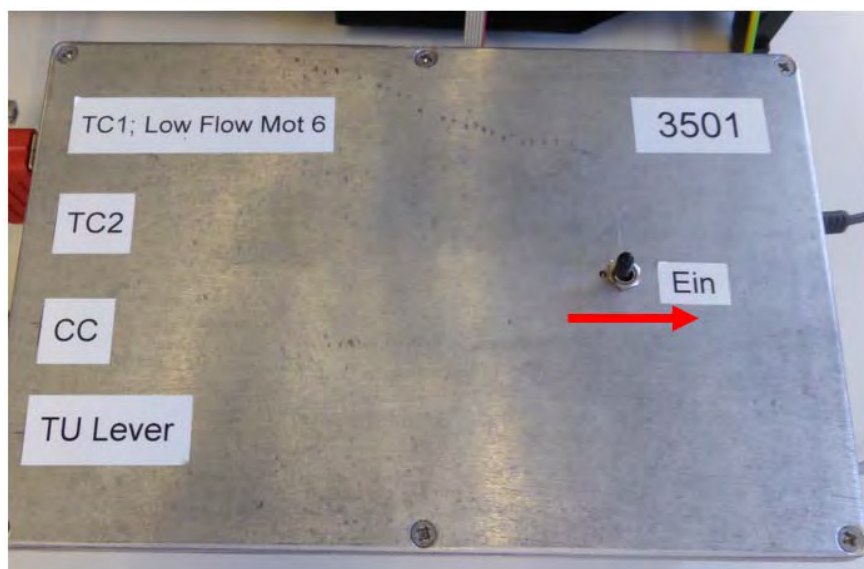
6. To start the sequence click "Starte Sequenze".
7. Test Evaluation: No vibration or noise = PASS; Vibration or noise/stiff spindle = Fail.
8. To repeat the test do not select PASS or FAIL, Select Cancel, Do Not Save.

*Picture 7.2-68 Test Result Screen.*

9. Open DiagCORE_ADJ_OPT.

*Picture 7.2-69 DiagCORE_ADJ_OPT icon.*

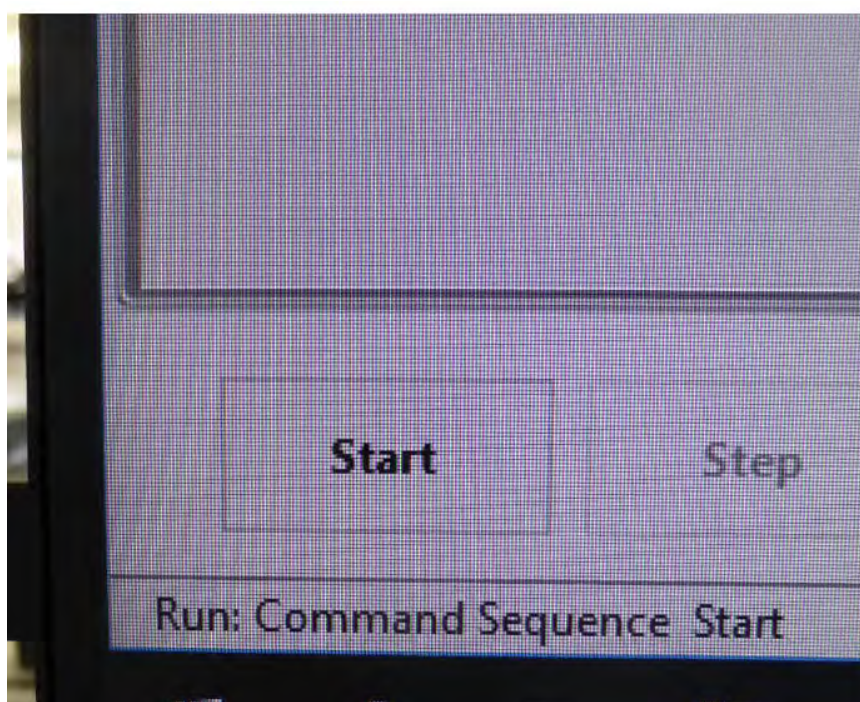
10. Switch on the control unit TC1-TC2-CC-TU (EIN).



Picture 7.2-70 Control Unit TC1-TC2-CC-TU.

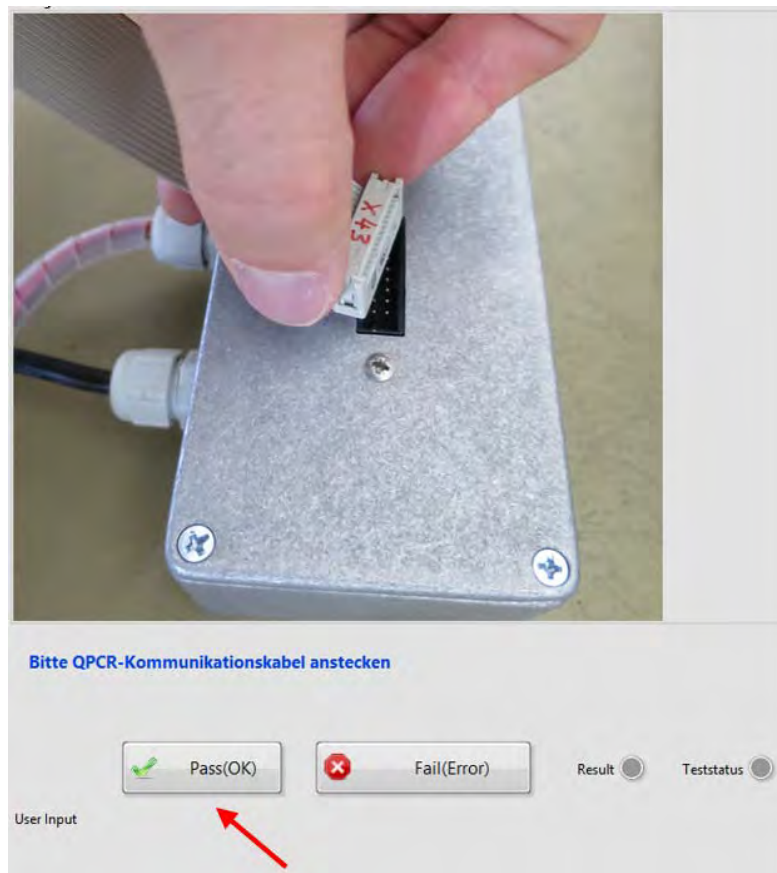
11. AXIS 1: qPCR Motor TC1; AXIS 3: Cartridge Clamping; AXIS 4: TU Lever.

12. Click "Start".



Picture 7.2-71 Click start.

13. Connect the qPCR communication cable (X43) to the qPCR Optical Alignment Communication & Power Adapter.



Picture 7.2-72 qPCR optical alignment communication & power adapter.

14. On the screen click "PASS (OK)".
15. Connect qPCR motor (X38), Encoder (X45), and Sensor to AXIS 1.

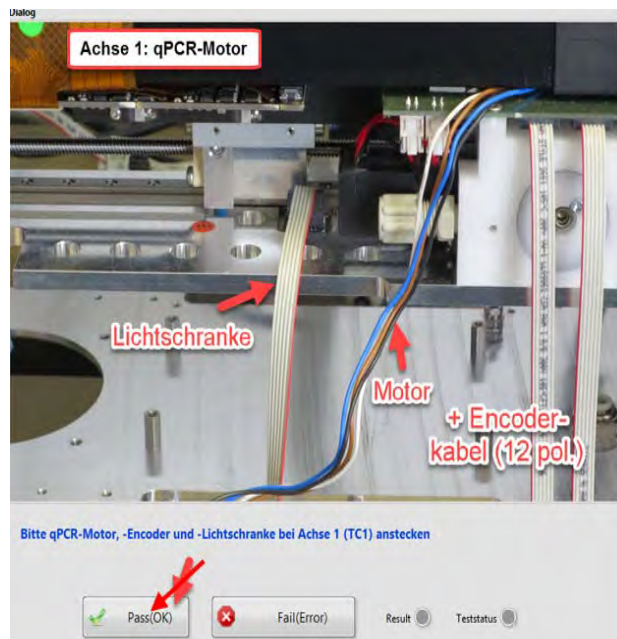
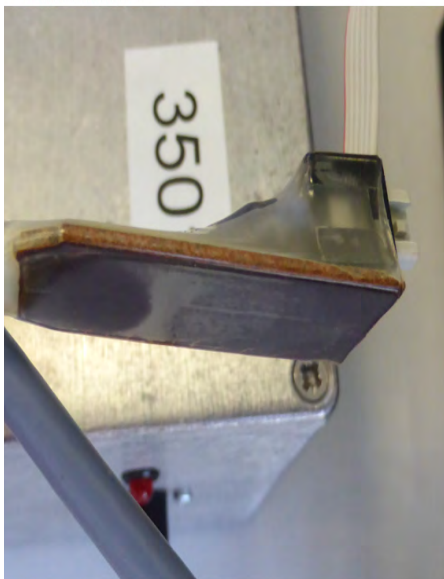
16. Click "Pass(OK)".
(4 polig, X38)



qPCR-Encoder-cabel (12 polig, X45)



R-lightbarrier (6 polig)



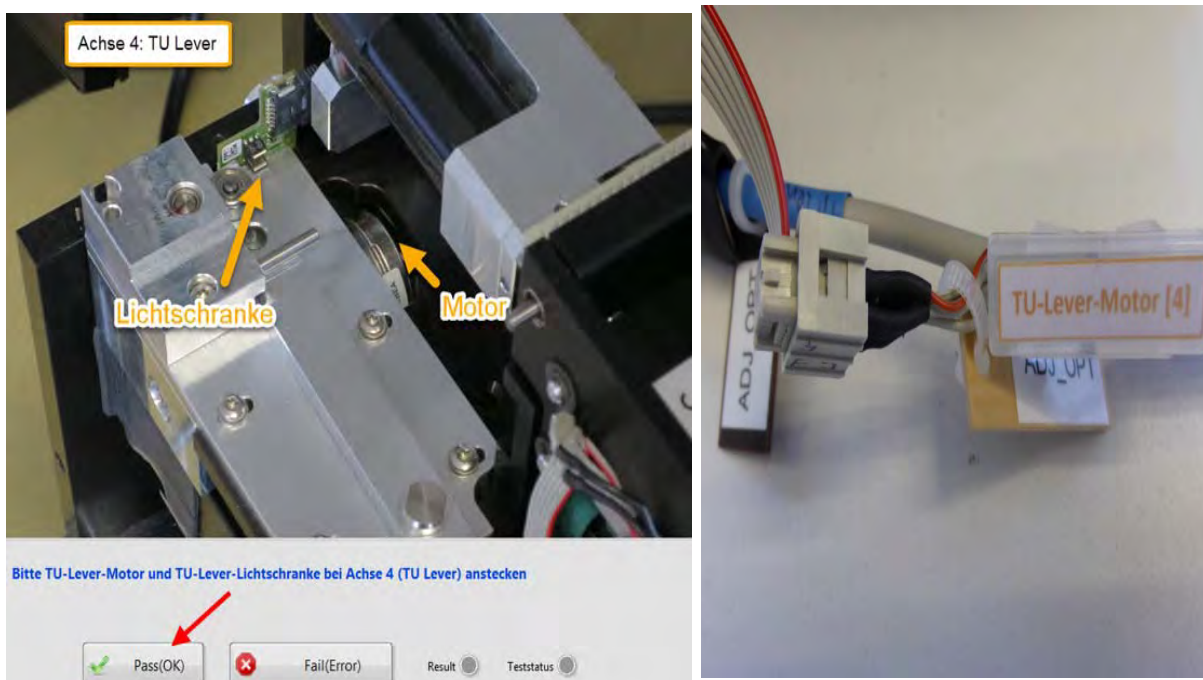
Picture 7.2-73 qPCR cable to adapter connections.

17. Plug the Clamping Motor and Clamping Motor Sensor to AXIS 3.



Picture 7.2-74 Cartridge clamping to adapter connections.

18. Connect the TU Lever motor and TU Lever Sensor to AXIS 4.



Picture 7.2-75 TU motor and sensor to adapter connection.

19. Scan the QR code on the TU Lever the click "Pass(OK)".



Picture 7.2-76 TU QC code.

20. Scan the qPCR Module QR code.



Picture 7.2-77 qPCR Module QR code.

21. Find the current standard on the left side of the TU, and enter it into the block via the drop down menu. then click "OK" (Example 1.8 is shown).

Endmaß Einstellung

Endmaß auswählen
1,8 (Standard) ▼

Bitte gewähltes Endmaß einbauen.
Bitte erst mit "OK" bestätigen, wenn dieser Arbeitsschritt korrekt durchgeführt wurde.

OK (weiter) **abbrechen (FAIL)**

Justage-Durchlauf Nr. 1, maximal 3 Durchläufe erlaubt. **Freipruefung?** ●

Endmaß Historie

Endmaße (aktuelle Session)	0 mm	0 mm	0 mm
Endmaße (letzter Durchgang - DB)	0 mm	0 mm	0 mm

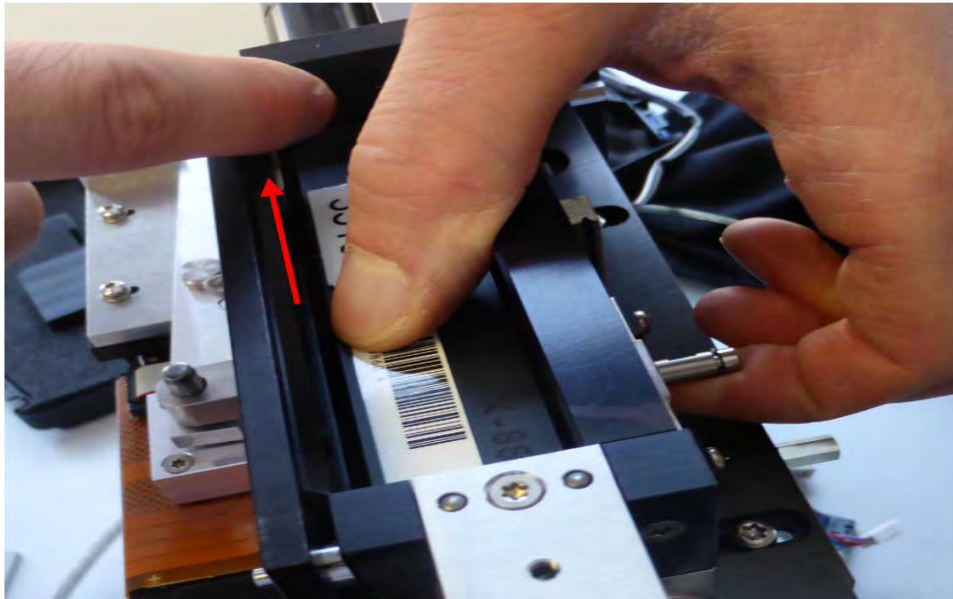
Picture 7.2-78 qPCR current calibration.

22. Complete the steps in the picture. If the Clamping Module is installed omit steps 1 and 2.



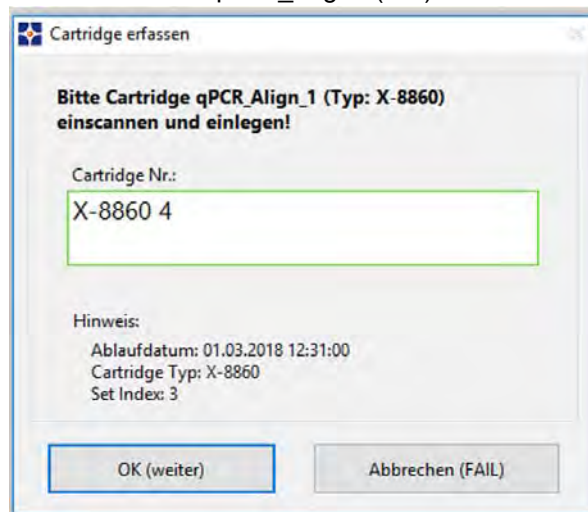
Picture 7.2-79 Conditional steps.

23. Insert the cartridge (Test Mean qPCR_Align1 (Cal) X-8860), the lifter must be raised a little and the cartridge pushed to the left (in the direction of the arrow).



Picture 7.2-80 Insert cartridge X-8860

24. Scan the top bar code on the Test Mean qPCR_Align1 (Cal) X-8860 cartridge then click "OK".



Picture 7.2-81 Scan bar code.

25. Cover the AM with the dark cloth.



Picture 7.2-82 Cover AM for Testing.

26. Complete the steps in the picture. If the Clamping Module is installed omit steps 1 and 2.



Picture 7.2-83 Conditional steps.

27. Insert the cartridge (Test Mean qPCR_Align2 (Cal) X-8859), the lifter must be lifted slightly and the cartridge pushed to the left (in the direction of the arrow).



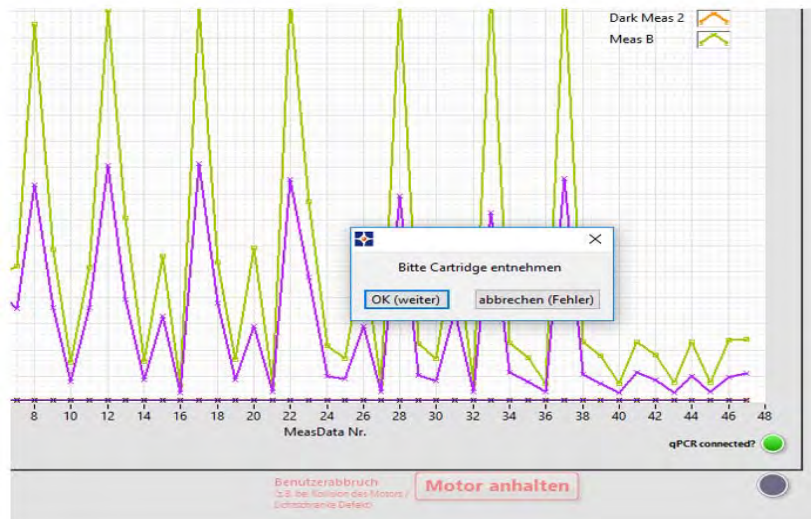
Picture 7.2-84 Insert cartridge X-8859.

28. Cover the AM with the dark cloth.



Picture 7.2-85 Cover AM for Testing.

29. Click “Eject Cartridge” and “OK”.

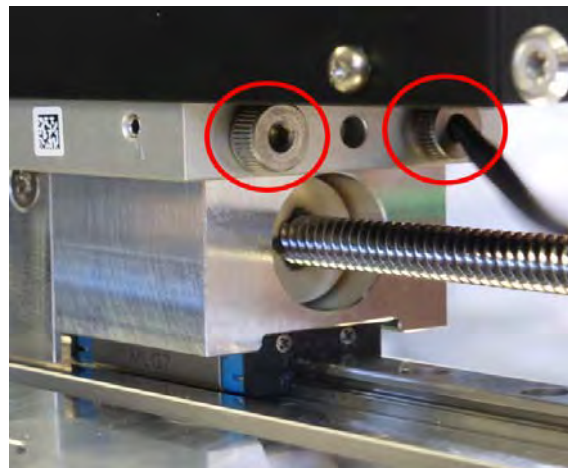
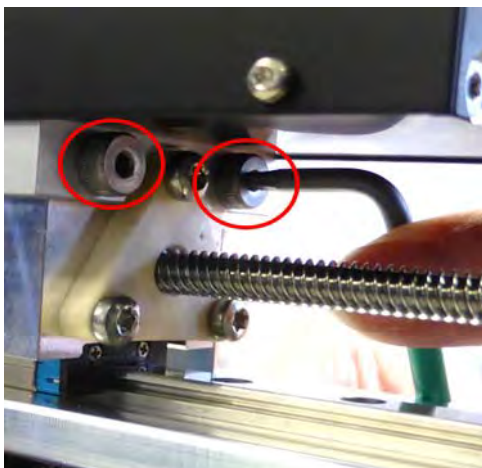


Picture 7.2-86 qPCR Graph and select box.

30. If the adjustment is unsuccessful, the optics must be readjusted with the suggested final dimension (1.7 mm in this case). To do this, perform steps 31 to 38 and re-test. If the test passes click “OK” and continue to step 39.

Picture 7.2-87 qPCR alignment results

31. Loosen the 4 mounting screws..



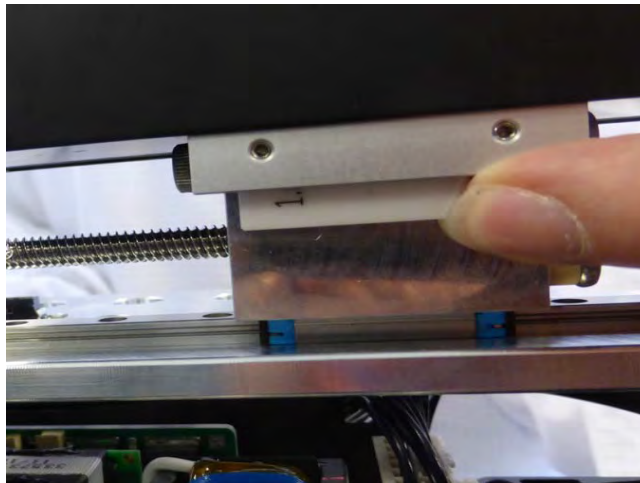
Picture 7.2-88 qPCR mounting screws.

32. Place a cloth over the cooling fan to prevent the ceramic gauge from entering the fan.
33. Slightly loosen the two 1.3mm alignment set screws on the opposite side from the ceramic block.



Picture 7.2-89 Loosen Alignment set screws.

34. Push the appropriate ceramic gauge between the QPCR adapter IGUS and the QPCR adjustment block.



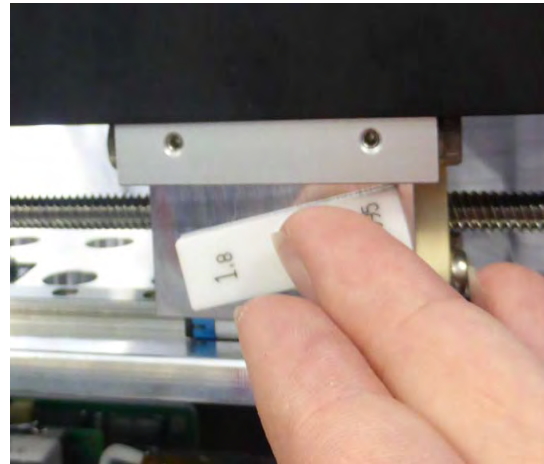
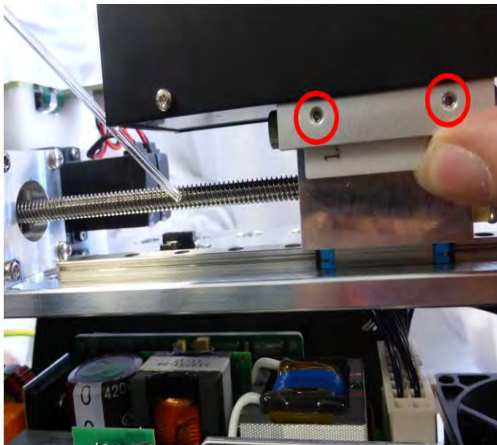
Picture 7.2-90 qPCR gauge.

35. Tighten the two 1.3mm alignment set screws on the opposite side from the ceramic block.



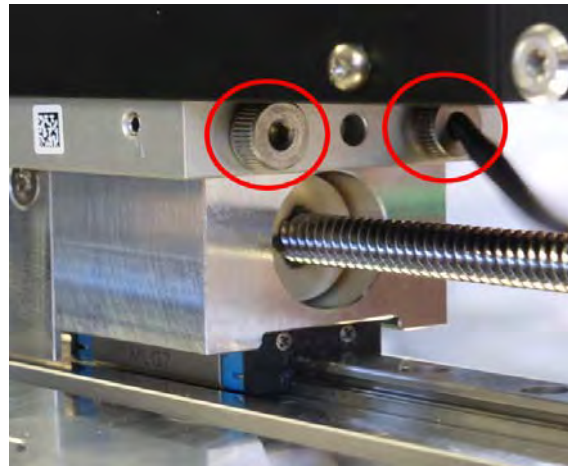
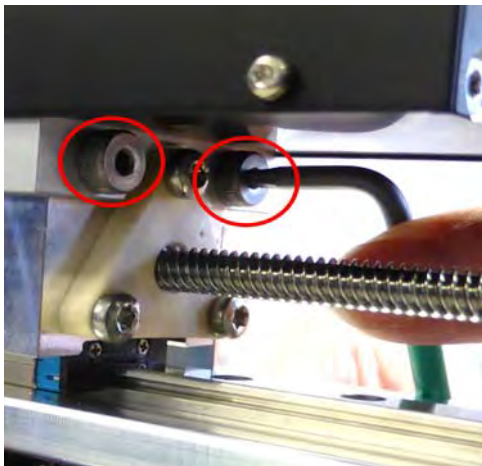
Picture 7.2-91 qPCR Alignment screws.

36. Tighten the two setscrews (SW 1.3 mm) on the side of the gauge block. Thus, the ceramic gauge is free and can be removed.



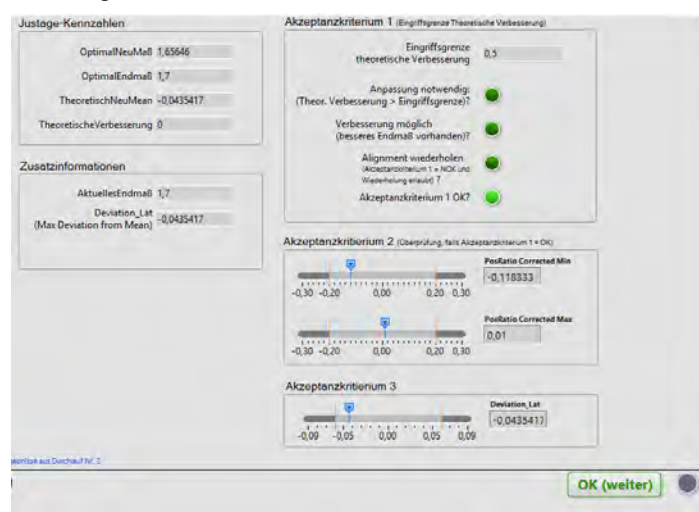
Picture 7.2-92 Remove the ceramic gauge.

37. Tighten the 4 qPCR mounting screws.



Picture 7.2-93 qPCR mounting screws.

38. If "OK" is highlighted the alignment was successful. Click "OK"



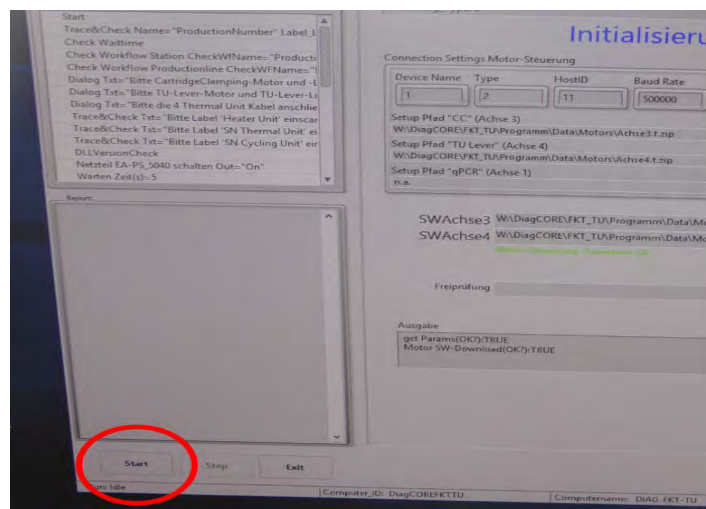
Picture 7.2-94 qPCR alignment successful.

39. Open the „DiagCORE_FKT_TU“ software.



Picture 7.2-95 Diagcore icon.

40. Click “Start”.



Picture 7.2-96 Diagcore start.

41. Scan the QR Code on the TU Lever and click “OK”.



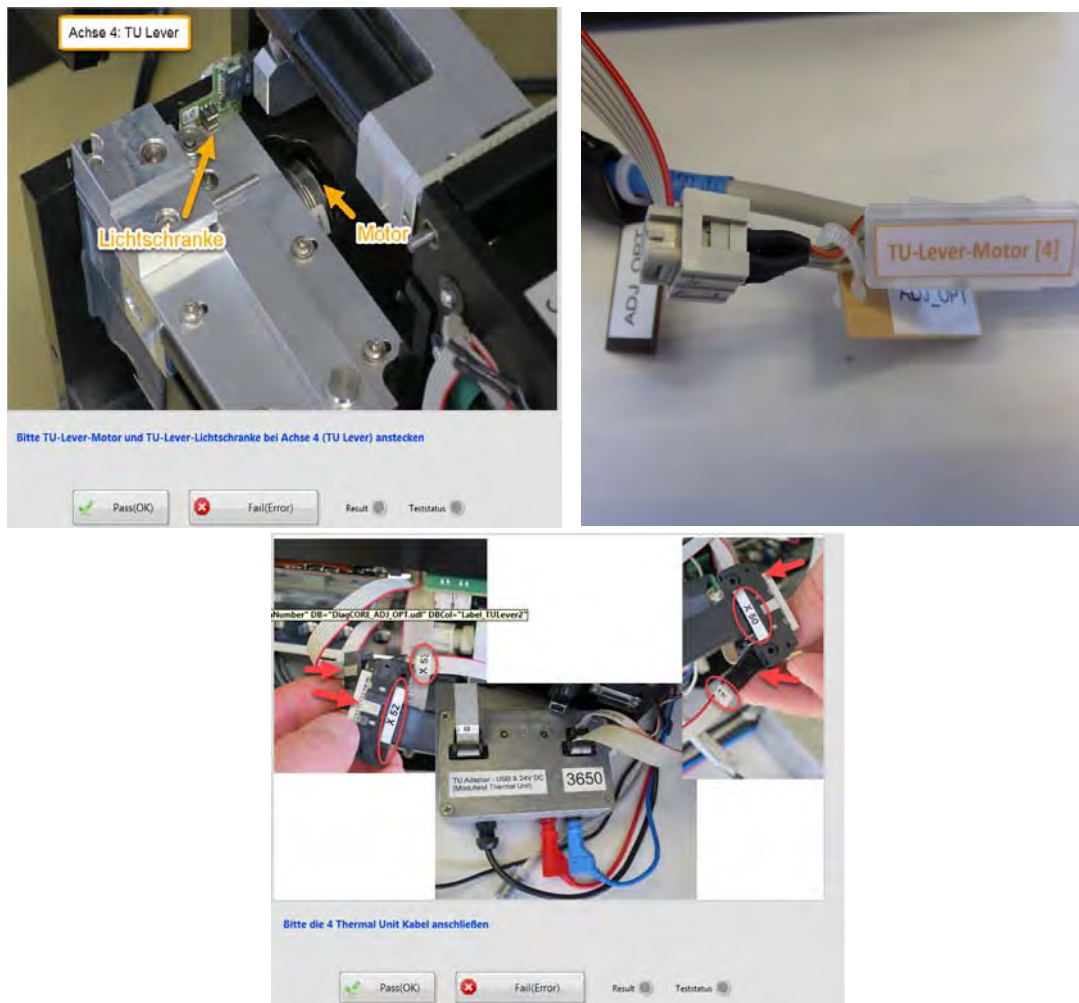
Picture 7.2-97 Thermal unit lever QR code.

42. Connect the Cartridge Clamping motor and sensor to axis 3 and click "OK".



Picture 7.2-98 Clamping module test connection.

43. Connect TU-Lever motor and TU-Lever sensor to axis 4 and click "OK".



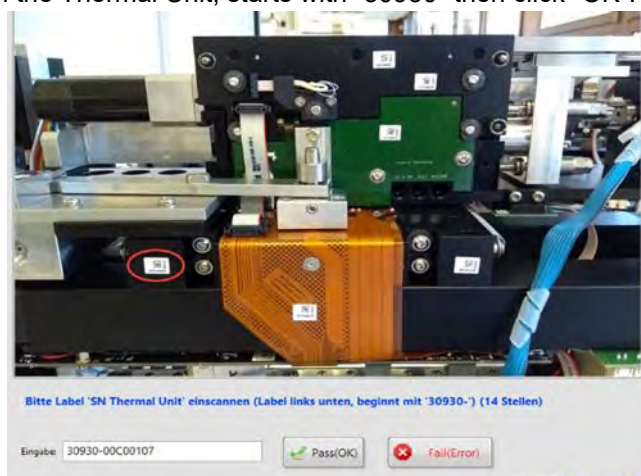
Picture 7.2-99 Thermal unit lever test connection.

44. Scan the QR code label on the Heating Unit, starts with “30961” then click “OK”.



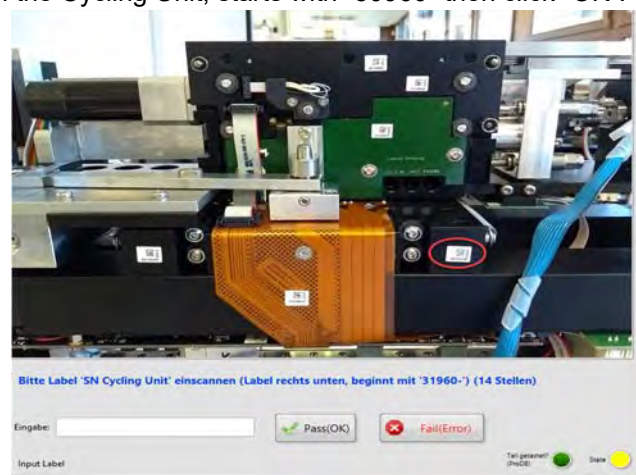
Picture 7.2-100 Heating unit QR code.

45. Scan the QR code on the Thermal Unit, starts with “30930” then click “OK”.



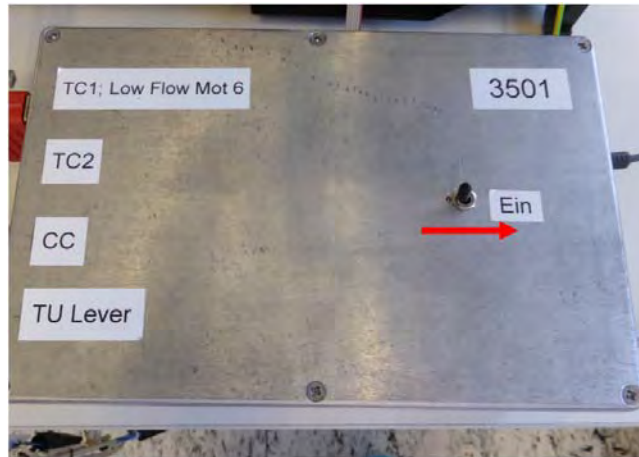
Picture 7.2-101 Thermal unit QR code.

46. Scan the QR code on the Cycling Unit, starts with “30960” then click “OK”.



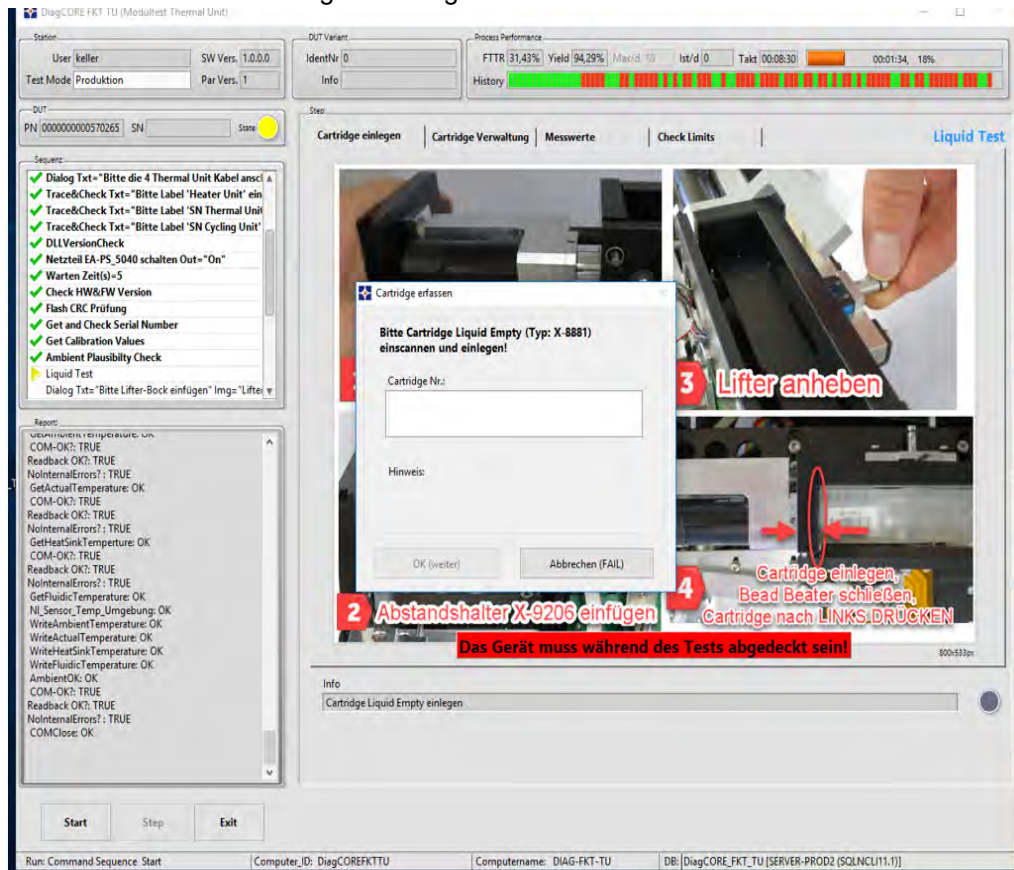
Picture 7.2-102 Cycling unit QR code.

47. Switch on the Controller TC1-TC2-CC-TU Lever.



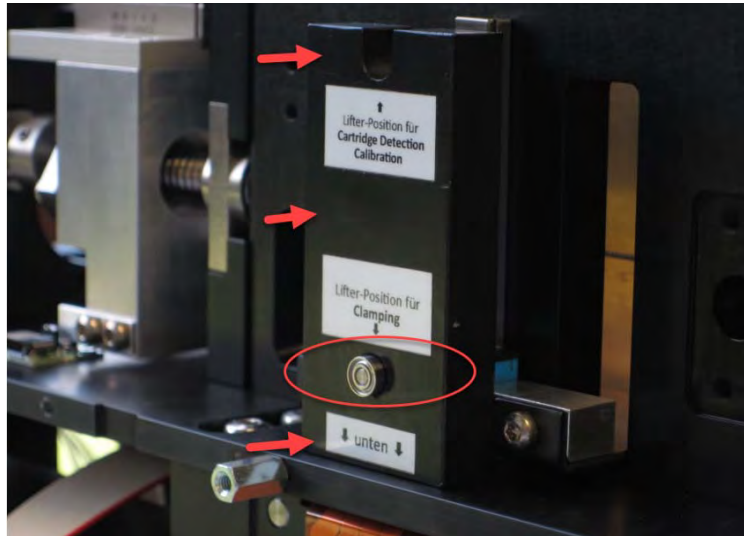
Picture 7.2-103 TC1-TC2-CC-TU lever controller.

48. In the Diagcore menu scan the Test Mean Liquid Empty X-8881 bar code and click "OK"; then slightly lift the lifter and insert the cartridge ensuring it is seated as shown.



Picture 7.2-104 Liquid empty test setup.

49. Insert the lifter block to keep the cartridge in the correct position.



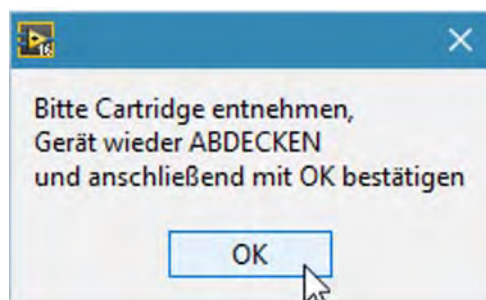
Picture 7.2-105 Lifter Block Installation.

50. Cover the AM with the dark cloth.



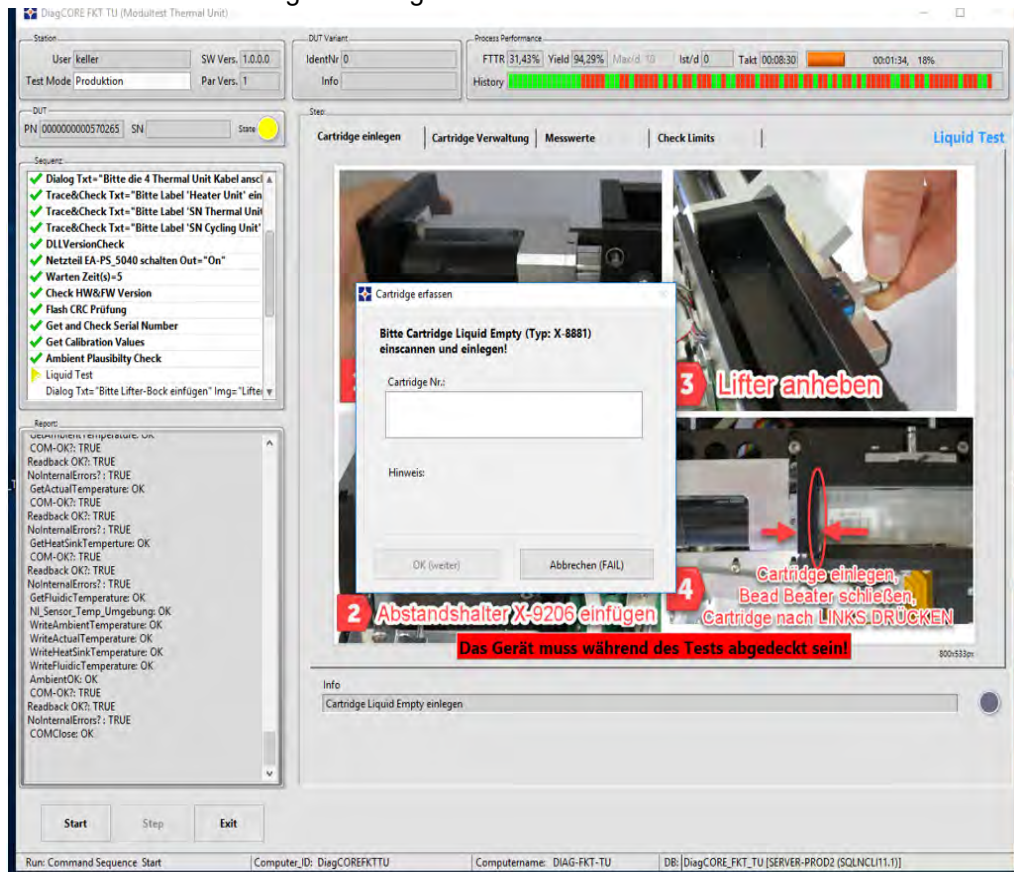
Picture 7.2-106 Cover AM for testing.

51. When the dialog box appears, the Liquid Test is complete, remove the Liquid Filled Cartridge and confirm with OK.



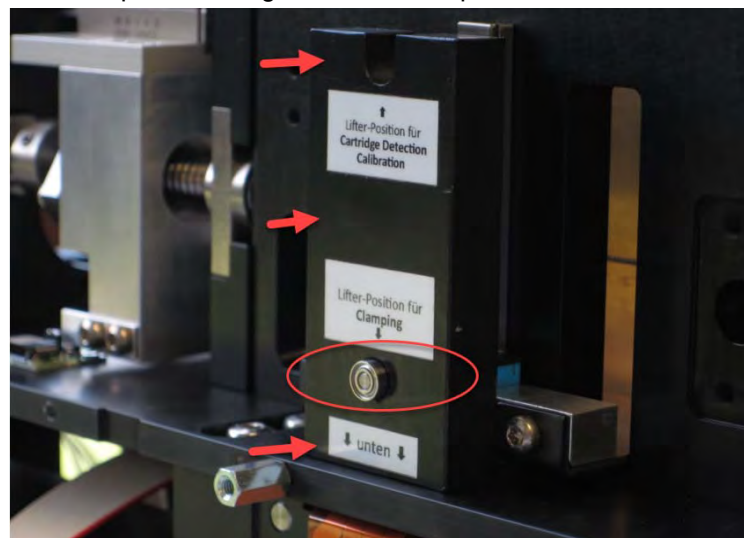
Picture 7.2-107 Liquid test complete.

52. In the Diagcore menu scan the Test Mean Liquid Filled X-8886 bar code and click "OK"; then slightly lift the lifter and insert the cartridge ensuring it is seated as shown.



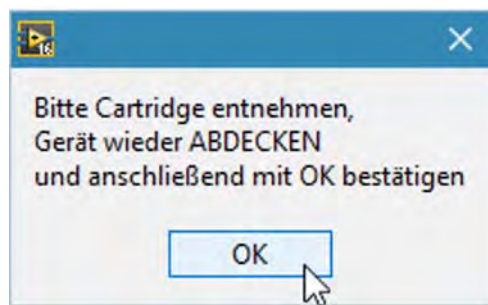
Picture 7.2-108 Liquid filled test setup.

53. Insert the lifter block to keep the cartridge in the correct position.



Picture 7.2-109 Lifter block installation.

54. When the dialog box appears, the Liquid Test is complete, remove the Liquid Filled Cartridge and confirm with OK.



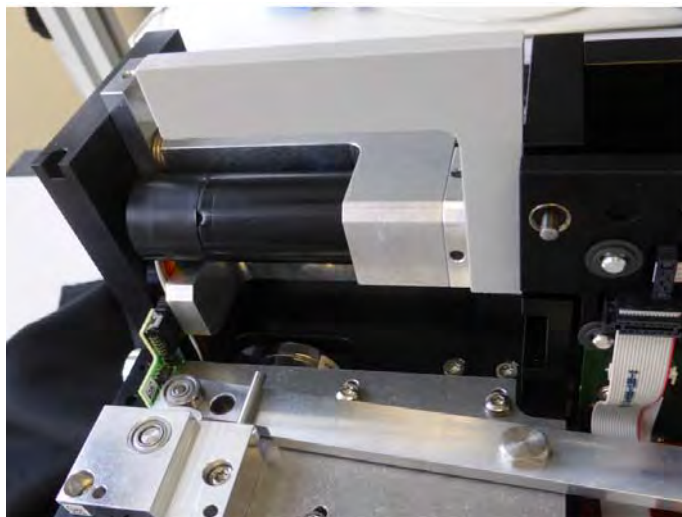
Picture 7.2-110 Liquid test complete.

55. Mount the cartridge lifter block as shown and click "OK".



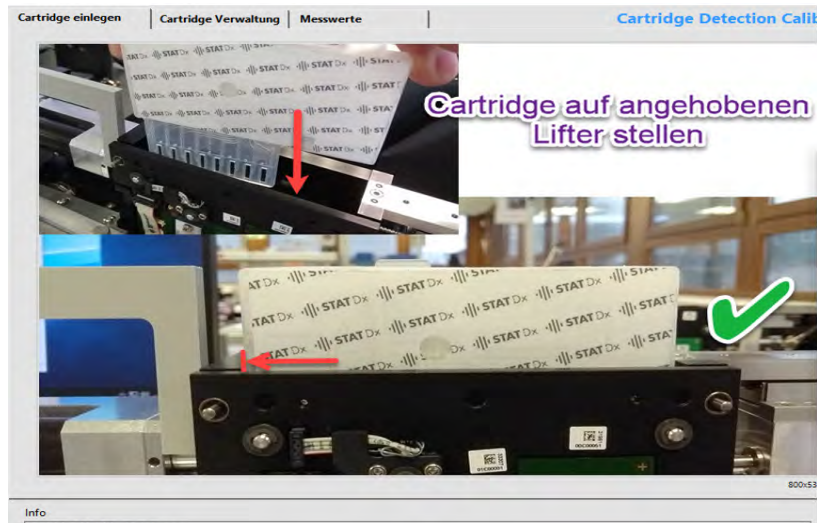
Picture 7.2-111 Lifter block installation.

56. Insert the X-9206 tool.



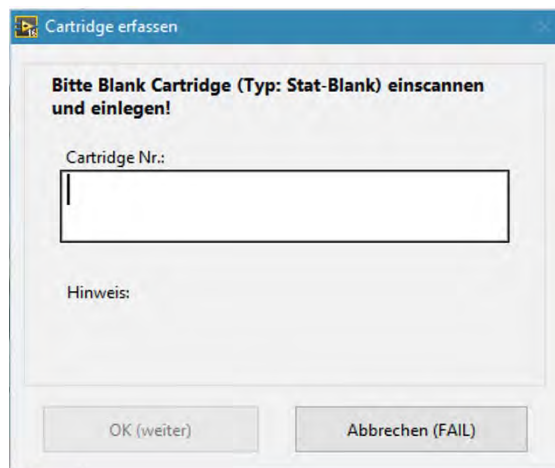
Picture 7.2-112 X-9206 Tool installed.

57. Place any blank cartridge on the lifter.



Picture 7.2-113 Blank cartridge on the lifter.

58. Scan the cartridge bar code and click "OK"



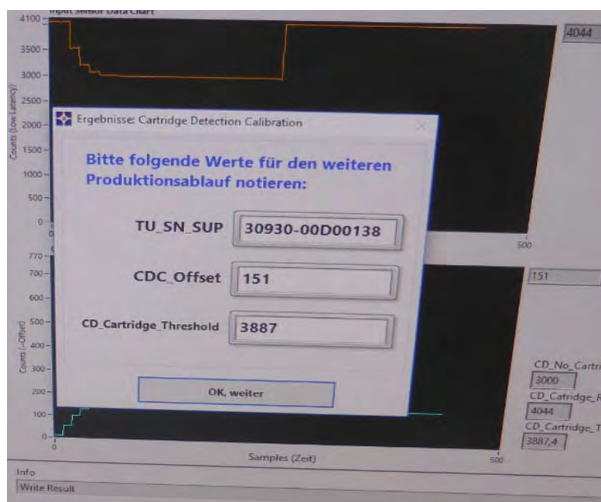
Picture 7.2-114 Scan blank cartridge.

59. Cover the AM with the dark cloth.



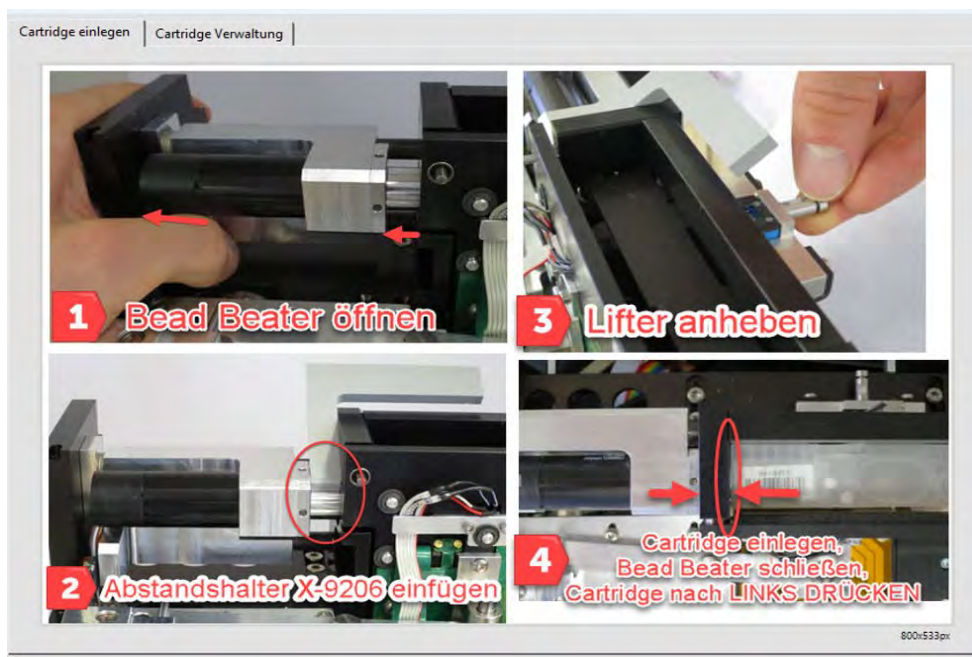
Picture 7.2-115 Cover AM for testing.

60. After testing use the snipping tool to print the Cartridge Detection Calibration data. Enclose the printout with the module for the Release Test. Click "OK"



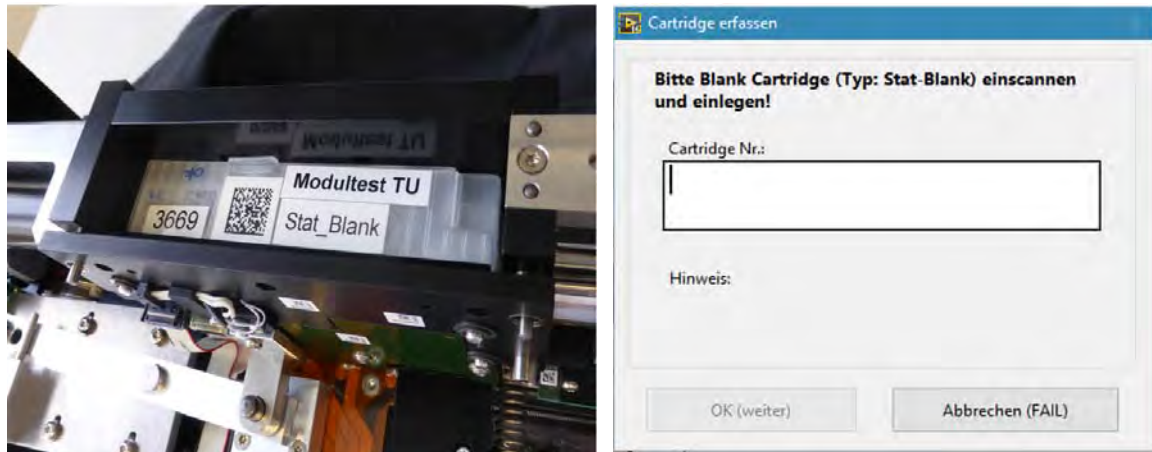
Picture 7.2-116 Cartridge detection calibration data.

61. Complete the steps as pictured below; if a Clamping Module is installed omit steps 1 and 2 in the picture.
 62. Raise the lifter and insert a blank cartridge, lower it down into position and ensure it is aligned to the left as in step 4 of the picture.



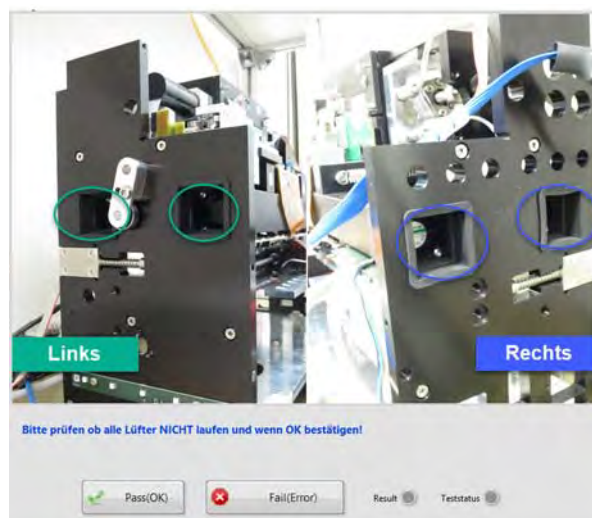
Picture 7.2-117 Cartridge alignment.

63. Verify cartridge alignment, scan the cartridge bar code and click "OK".



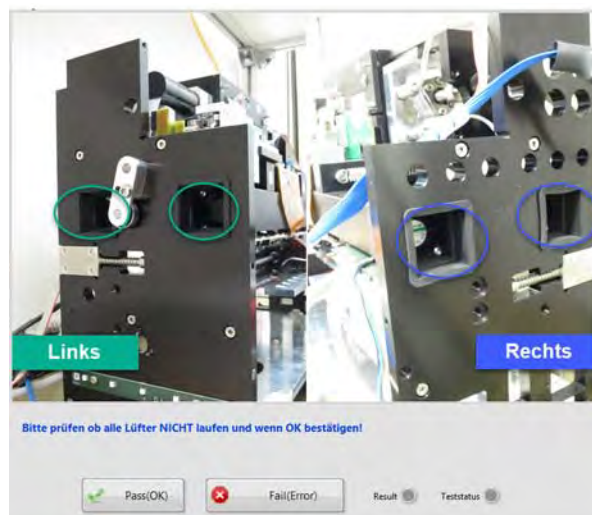
Picture 7.2-118 Cartridge alignment and bar code scan entry.

64. Make sure the 4 Thermal Unit fans are NOT running by feeling for airflow at the ports. Click "OK".



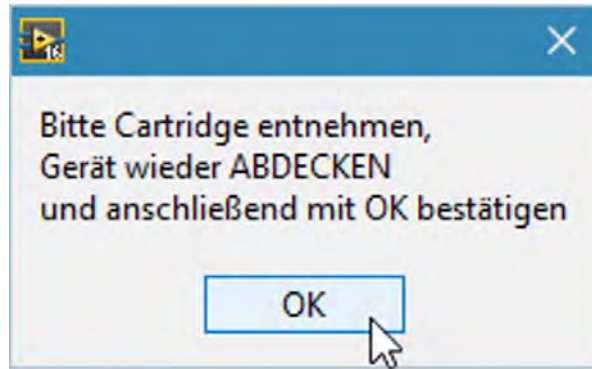
Picture 7.2-119 Thermal unit check fans.

65. Make sure the 4 Thermal Unit fans are ARE running by feeling for airflow at the ports. Click "OK".



Picture 7.2-120 Thermal unit check fans.

66. Remove the cartridge.



Picture 7.2-121 Remove cartridge.

67. Test is Completed.

8 Verification Procedures (VP)

8.1 Overview of Verification Procedures

VP #	Name	Purpose
VP1	System Initialization	Bring the analyzer to the initialized state
VP2	Preparation For Testing	Prepare the AM for testing
VP3	Actuators Status	Functional Test of AM Actuators
VP4	Pneumatics	Pneumatics subsystem Functional Test
VP5	qPCR Internal Standard	Tests the qPCR Internal Standard
VP6	Thermal Unit	Checks heating/cooling
VP7	Analyzer Check Cartridge Work flow Test	Overall functional test of the Analytical Module
VP8	Set Operational Module serial Number	Set serial number when the Mamba Board is replaced
VP9	Lid Mechanism Alignment	Align cartridge slot
VP10	Belt Tension Adjustment	Set belt tension
VP11	Thermal Unit Plate Alignment	Align thermal plates
VP12	Operational Module Functional Test	Verify Operational Module functions
VP13	Release Test	Analytical module final test
VP14	Infield Service Tool Test	Functional Test of the AM

Table 8.1-1 Verification Procedure Overview.

8.2 Analytical Module Verifications

8.2.1 VP1 - System Initialization

Purpose	Bring the analyzer to the initialized state
Tool(s) Required	N/A
Verification Time	5 Minutes

Table 8.2-1 VP1 System initialization.

1. Ensure the Operational Module and the Analytical Module(s) are properly connected, bridged, and the power cord is connected to the AM and a power source.
2. Place the AM power switch to "ON" and press the power button on the front of the OM.
3. When the self test and initialization is completed the LEDs on the AM will turn "Green".

8.2.2 VP2 - Preparation For Testing

Purpose	Prepare the Analytical Module(s) for Testing
Tool(s) Required	9026465-CAN Connection Cable, 9026467-CAN to USB adapter
Verification Time	5 Minutes.

Table 8.2-2 VP2 Preparation for testing.

1. Ensure there are no cartridges in the AM.
2. Connect the CAN to USB adapter to one of the 15 pin ports on the AM and to the USB port on the computer.
3. Start the Infield Service Tool on the computer.
4. Select CAN to USB and Infield Mode, then OK.
5. Move the AM power switch to ON position and when the AM has linked to the computer the AM serial number will appear in the status field.

8.2.3 VP3 - Actuators Status Test

Purpose	Functional Test of the AM actuators
Tool(s) Required	9026464- Tool-Cartridge Service, AM-3D Printed Cartridge
Verification Time	10 Minutes.

Table 8.2-3 VP3 Actuators status test.

1. Start the Actuators Status Test.
2. Insert the 3D printed cartridge when prompted and press "OK".
3. After the test is complete remove the cartridge and press "OK".

8.2.4 VP4 - Pneumatics Test

Purpose	Pneumatics Leakage and Functional Test
Tool(s) Required	9026464- Tool-Cartridge Service, AM-3D Printed Cartridge
Verification Time	25 Minutes.

Table 8.2-4 VP4 Pneumatics test.

1. Start the Pneumatics Test, the analyzer will ask for the 3D printed cartridge, Insert the cartridge when prompted and press "OK".
2. After the test is complete remove the 3D printed cartridge and press "OK".

8.2.5 VP5 - qPCR Internal Standard Test

Purpose	Tests the validity of the qPCR internal Standard
Tool(s) Required	None
Verification Time	5 Minutes.

Table 8.2-5 VP5 qPCR internal standard.

1. Start the qPCR Internal Standard Test.
2. After the test press "OK".

8.2.6 VP6 - Thermal Unit Test

Purpose	Tests the Thermal Unit heating.
Tool(s) Required	9026464- Tool-Cartridge Service, AM-3D Printed Cartridge
Verification Time	15 Minutes.

Table 8.2-6 VP6 Thermal unit test.

1. Start the Thermal Unit, the analyzer will ask for the Leakage Cartridge (9026464, Tool-Cartridge Service, AM, QSTAT). Insert it in the cartridge slot when prompted.
2. When the test is complete remove the test cartridge and press "OK".

8.2.7 VP7 - Analyzer Check Cartridge Work Flow Test

Purpose	Overall Functional Test of the Analytical Module
Tool(s) Required	9026466, Analyzer Check Cartridge (ACC)
Verification Time	90 Minutes.

Table 8.2-7 VP7 Analyzer Check Cartridge Test.

1. Once the analyzer has initialized load the file "[ADF_ACC_V06.asy](#)" file (from GPS Intranet) on the OM from the USB drive.
2. Select "RUN" to start the test.
3. Choose a patient ID e.g. "ACC_09/05/2019_AM1".
4. Scan the ACC Cartridge bar code on the OM scanner.
5. When the door opens insert the cartridge.
6. Repeat steps 2 through 5 for all AM's.
7. When the test is complete the results will be "PASSES" on the display.

8.2.8 VP8 - Set Operational Module Serial Number

Purpose	Set the Operational Module serial Number when the Mamba Board is replaced
Tool(s) Required	Laptop Computer, Ethernet Cable, WinSCP FTP Client Software
Verification Time	15 Minutes

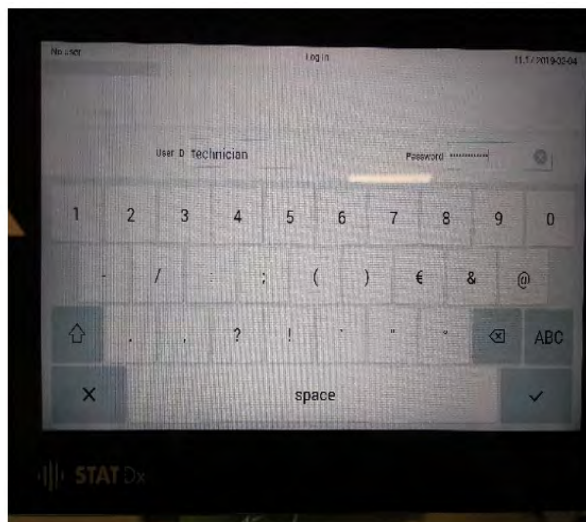
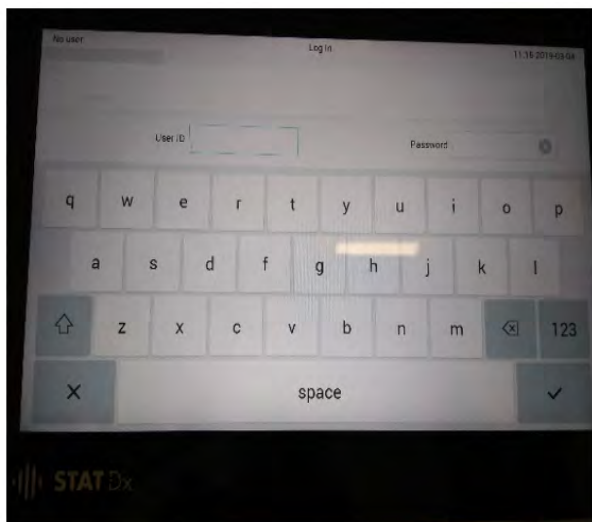
Table 8.2-8 VP8 Set Operational Module Serial Number.

8.2.8.1 Connect the Analytical Module (AM) to the Operational Module (OM)

1. Connect the Ethernet cable to the OM and to the network router.
2. Set the power switch on the AM to "ON".
3. Press the power switch on the front of the OM and wait until the OM is ready.

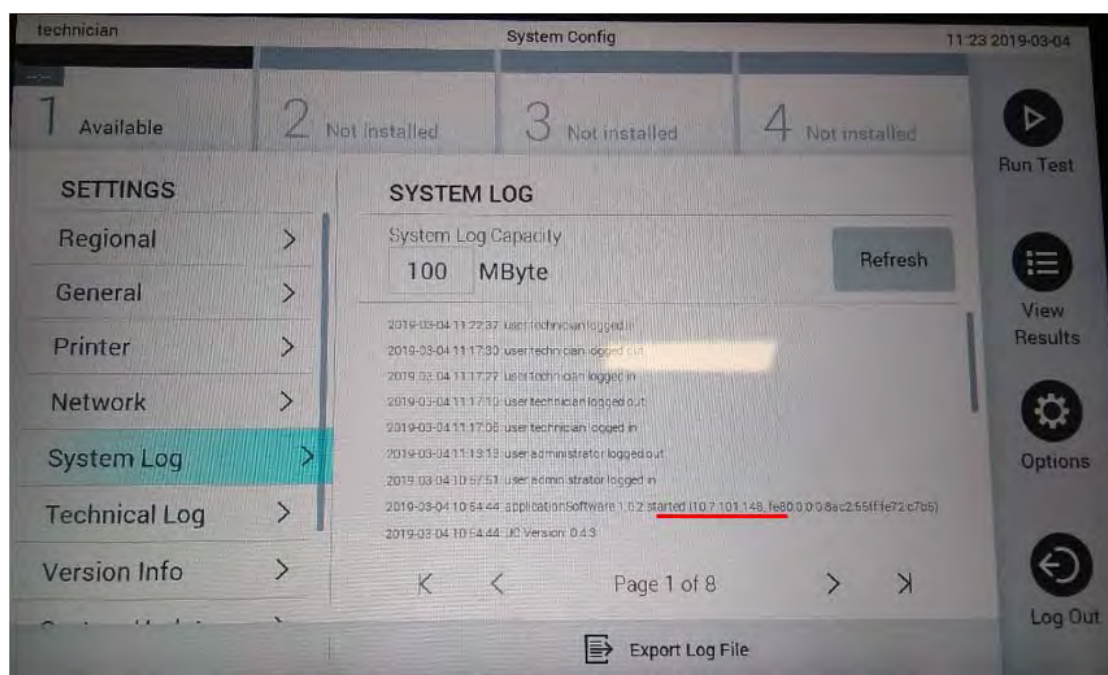
8.2.8.2 Login and Get the IP Address.

1. Login: User ID: Insert USB with the service tool and use your QIAsymphony PIN.



Picture 8.2-9 Login Screen on the OM.

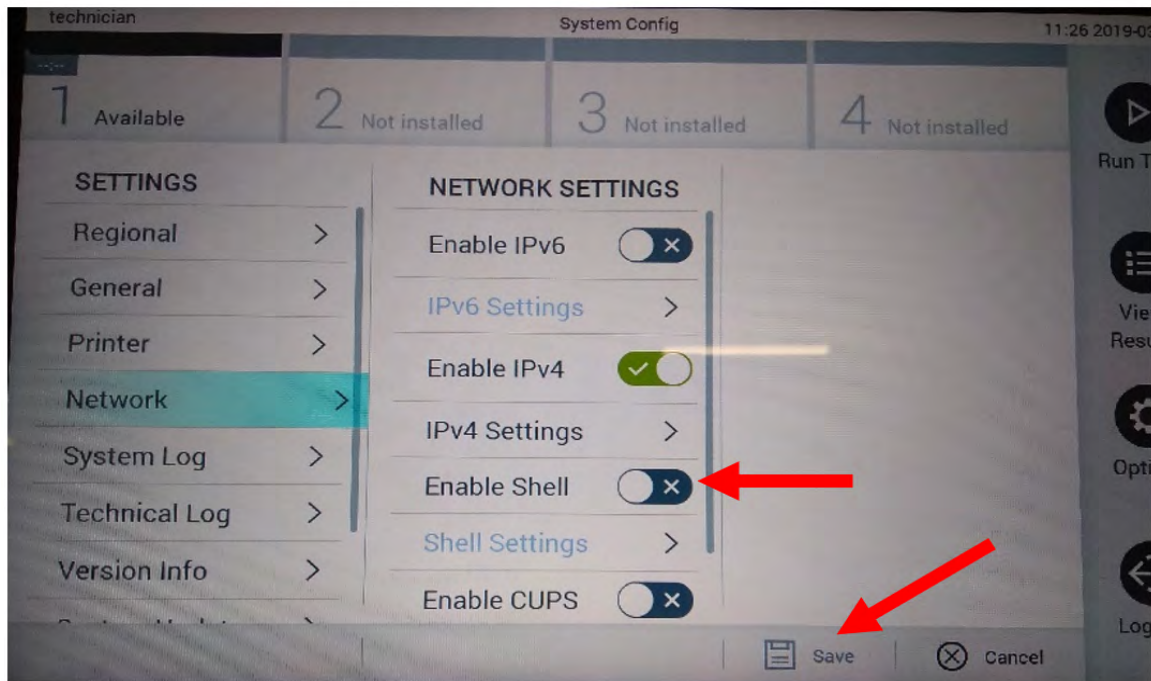
2. Get the current IP Address: Options -->System Config -->System Log. Note the IP Address.



Picture 8.2-10 Obtain IP Address.

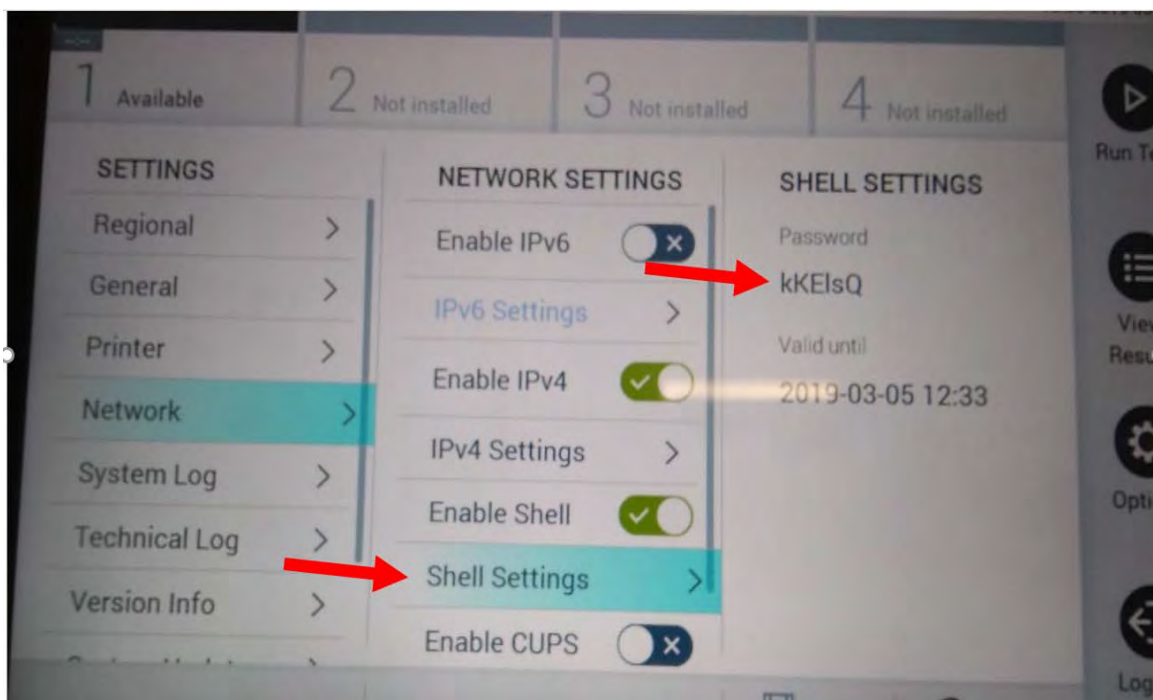
8.2.8.3 Enable shell and get the shell password.

1. Options --> System Config --> Network --> Enable Shell --> Save.



Picture 8.2-11 Enable Shell.

2. Select "Shell Settings. Note the password". NOTE: Valid for 24 hours only.

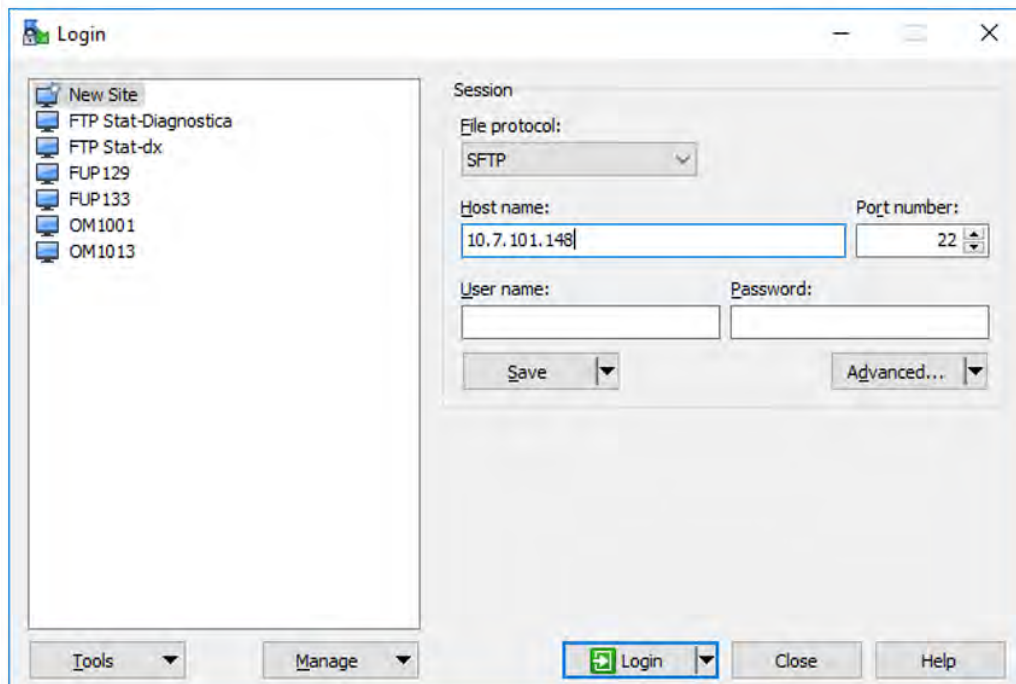


Picture 8.2-12 Obtain Password.

8.2.8.4 Edit OM "omdevice.xml" file using FTP Client software.

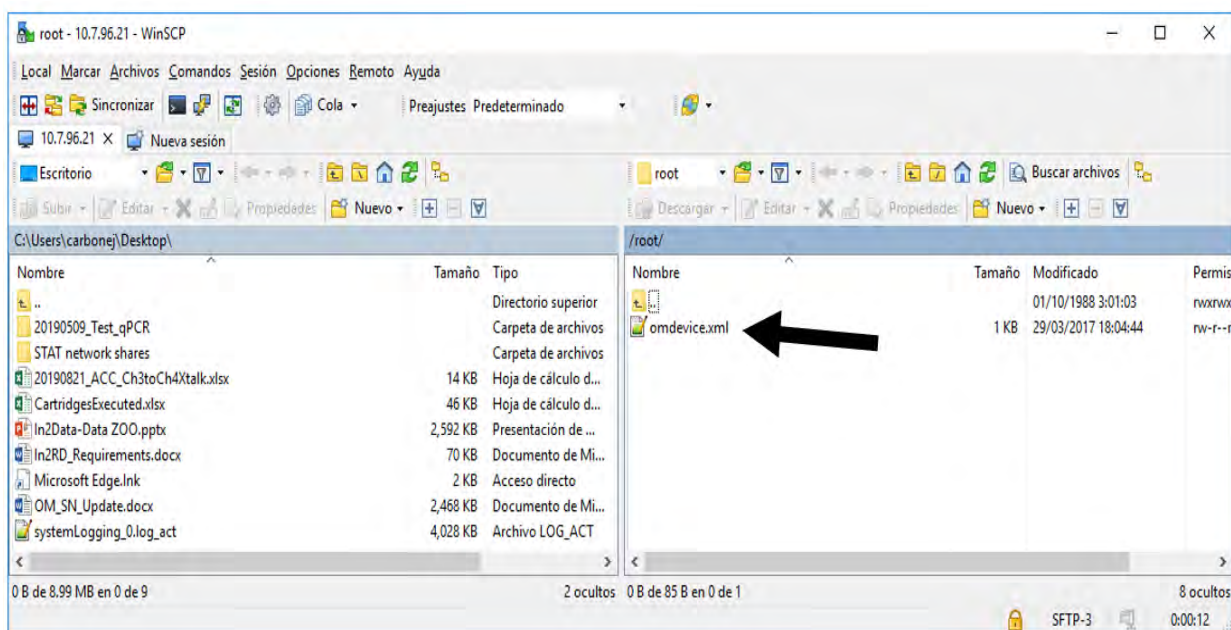
1. From a computer using a FTP client connect to the OM.

2. Host Name: IP address noted previously; User Name: root; Password: Shell Setting Password.



Picture 8.2-13 Enter Host Name.

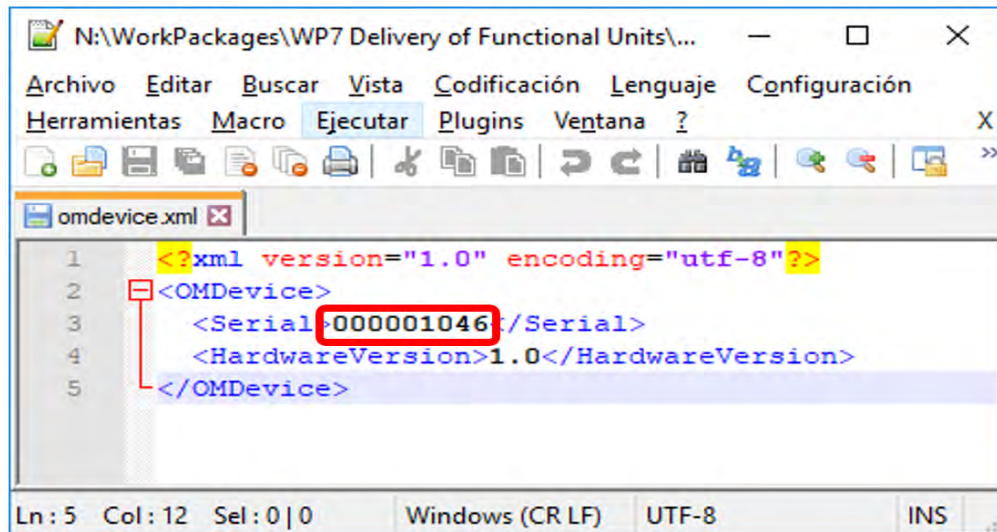
3. Click "Login"
4. From WinSCP edit move the "omdevice.xml" to the OM path (/root/) and save it.
5. Right click and open with Notepad++.



Picture 8.2-14 VP8 Save file and open.

6. Right click and open with Notepad.

7. Edit the serial number to match the serial number on the OM.



Picture 8.2-15 VP-07 Edit Serial Number.

8. Close WinSCP.

8.2.9 VP9 - Lid Mechanism Alignment

Purpose	Lid Mechanism Alignment
Tool(s) Required	#10 TORX Driver, X-9900 Alignment Tool
Verification Time	10 Minutes.

Table 8.2-16 VP9 Lid mechanism alignment.

1. Install the Lid Mechanism but leave screws loose.
2. Use the X-9900 alignment tool to properly position the top cover and tighten screws in place.



Picture 8.2-17 Adjust lid mechanism.

8.2.10 VP10 - Belt Tension Adjustment

Purpose	Adjust tension of the drive belt
Tool(s) Required	#20 TORX and Infrared Belt Tension Meter
Verification Time	10 Minutes

Table 8.2-18 VP10 Belt tension adjustment.

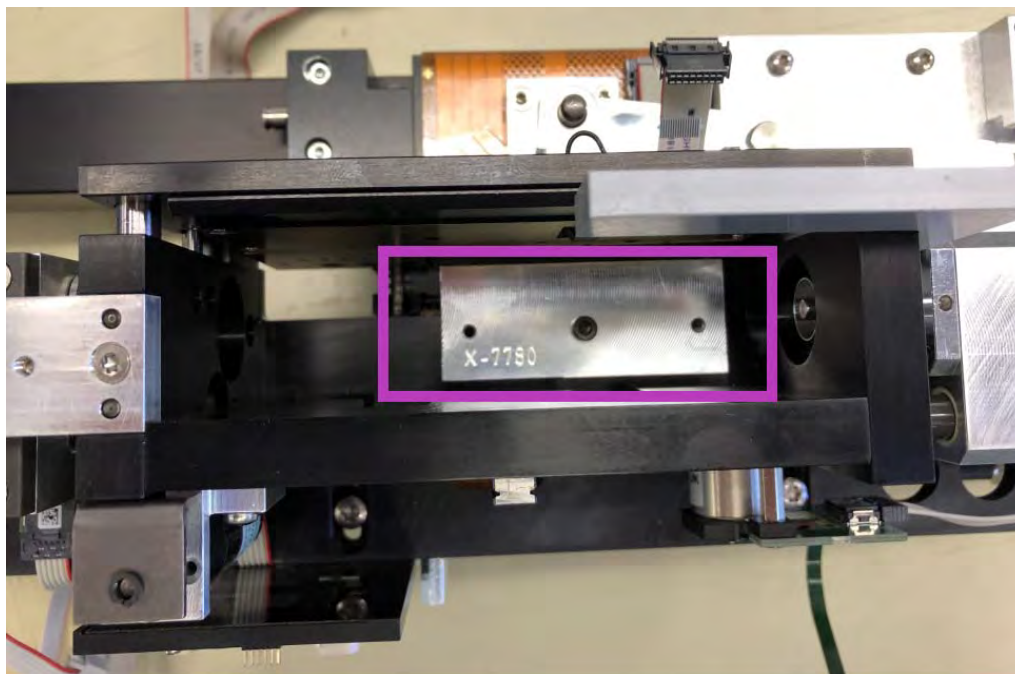
1. With the belt and pulleys installed apply pressure light pressure on the belt with the belt tensioner.
2. Place the belt tension meter sensor approximately 3/4" from the belt and gently "snap" the belt.
3. The belt tension should be between 65Hz and 75Hz, if not adjust the belt tensioner-increase pressure for higher frequency; less pressure for lowering the frequency.
4. Tighten the belt tensioner screw.

8.2.11 VP11 - Thermal Unit Heating Plate Alignment

Purpose	Set the gap in the heating plates
Tool(s) Required	T20 TORX, X-7780 Alignment Tool
Verification Time	15 Minutes

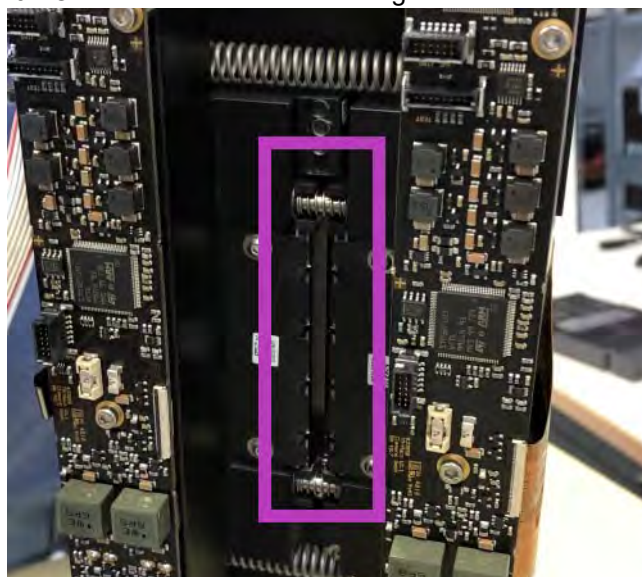
Table 8.2-19 VP11 Thermal unit heating plate alignment

1. With the 4 heating plate mounting screws loose, insert the X-7780 alignment tool in the cartridge well and secure it with the middle screw.



Picture 8.2-20 Thermal Unit heating plate alignment tool placement.

- Carefully press together the heater assembly until the plates are snug against the sides of the alignment tool. Tighten the four T20 TORX screws. Remove the alignment tool.



Picture 8.2-21 Thermal Unit heating plates alignment.

8.2.12 VP12 - Operational Module Functional Check

Purpose	Tests the function of the Touchscreen, USB Port and Bar Code Reader
Tool(s) Required	USB drive, Unused Assay Cartridge
Verification Time	5 Minutes

Table 8.2-22 VP12 Operational module functional check.

- Have the OM mounted to the AM, bridge and power cord installed.
- Set the AM power switch to "ON" and press the OM power switch. Wait for analyzer to initialize.
- Manipulate the touch screen by pressing selections and watching the displayed response to input.
- Insert a USB drive into the OM USB ports one at a time, Export the Technical Log to the USB drive inserted at each port. Options --> System Config --> Technical Log --> Export
- Scan an unused assay cartridge, it can only be scanned ONCE!

8.2.13 VP13 - Release Test

Purpose	Final Testing of the Analytical Module
Tool(s) Required	USB to CAN Adapter, Cartridge Test Fixture WAKA 3603, Blank Assay cartridge, Modified blank cartridge WAKA 3760, AM to OM interconnect Cable
Verification Time	2 Hours 30 Minutes.

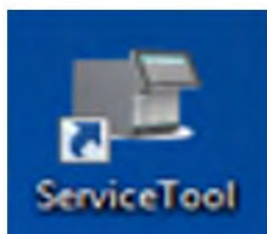
Table 8.2-23 VP13 Release test.

1. Connect the USB to CAN cable and the power cable to the AM.



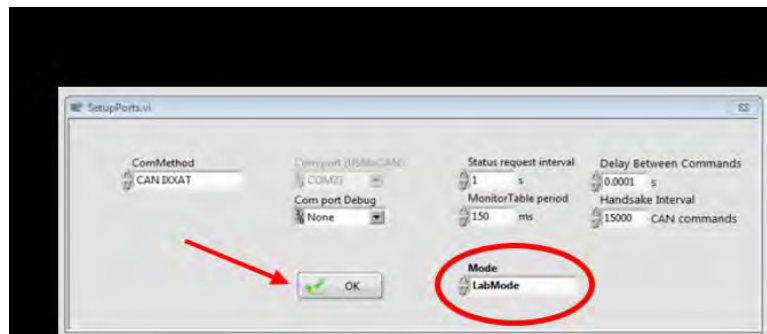
Picture 8.2-24 Power and CAN connection.

2. Turn on the power and wait for the LEDs on the front to turn green.
3. Open the “Service Tool” software.



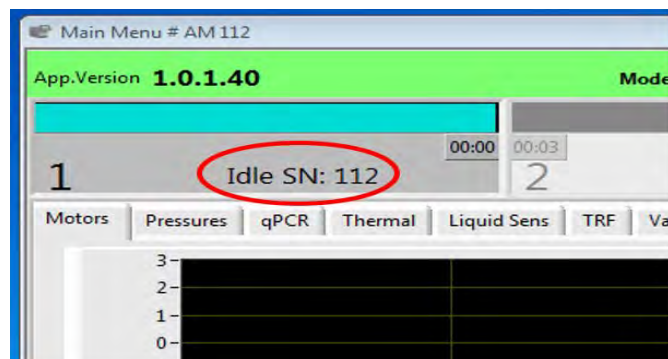
Picture 8.2-25 Service tool icon.

4. Once it opens set the mode to “LabMode” and Click “OK”.



Picture 8.2-26 LabMode.

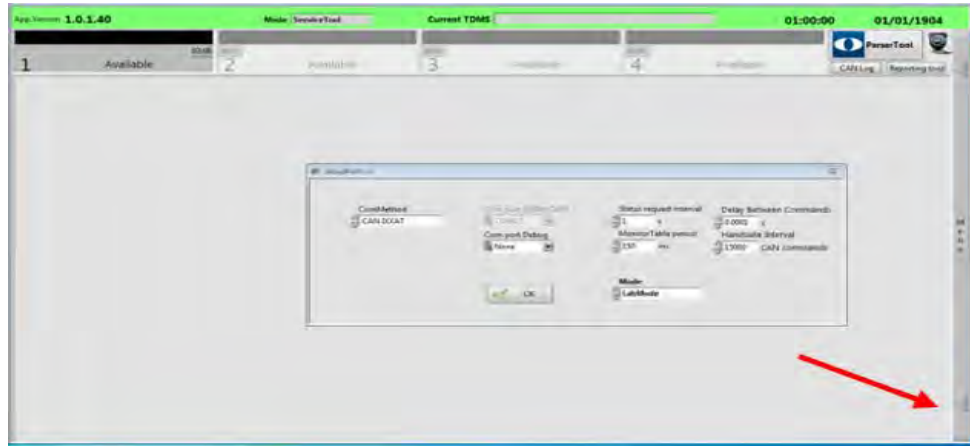
5. With until the “Idle SN” reads the AM serial number.



Picture 8.2-27 Main Menu Idle SN.

6. If the “Idle SN” is “0” complete the following steps:

- Open the Menu Bar".



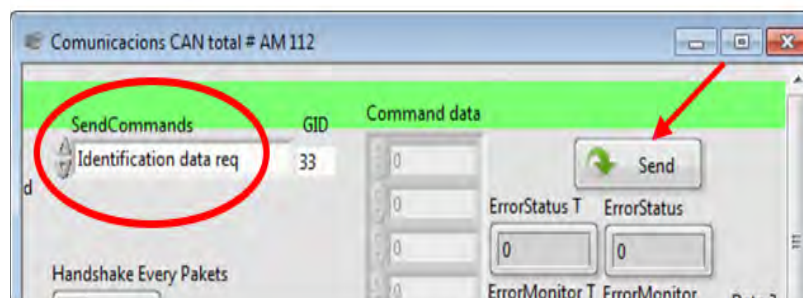
Picture 8.2-28 Menu bar.

- Click "Direct CAN".



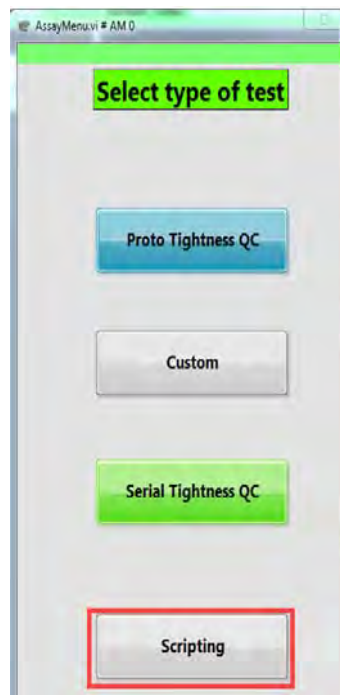
Picture 8.2-29 Menu Direct CAN.

- Set "Send Commands" to "Identification data req" and click "Send".



Picture 8.2-30 Identification data request.

7. In "Type of Test" select "Scripting".



Picture 8.2-31 Type of test select scripting.

8. Zero the X-9246 test fixture by placing the flat side of the X-9801 against the plunger and pushing it completely against the stop plate. Press the "Zero" button on the gauge and hold until the meter reads "0".



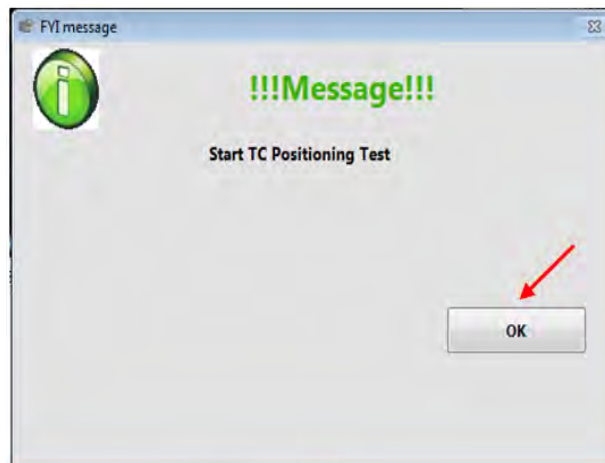
Picture 8.2-32 Zero Test fixture gauge.

9. Enter "tc01" in "cartridge SN" and click "Confirm"



Picture 8.2-33 Set cartridge serial number.

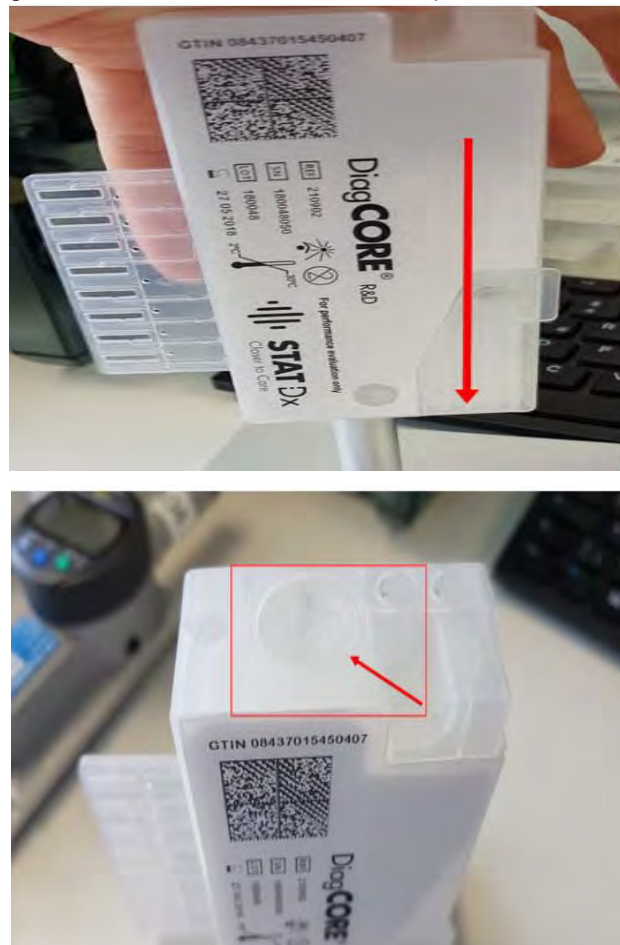
10. Open the file "TC-Positioning-v2.prg" in the following-path C:-ReleaseProtocol-ReleaseProtocol vx.x (always use the latest version).
11. Click "OK"



Picture 8.2-34 Start TC positioning test.

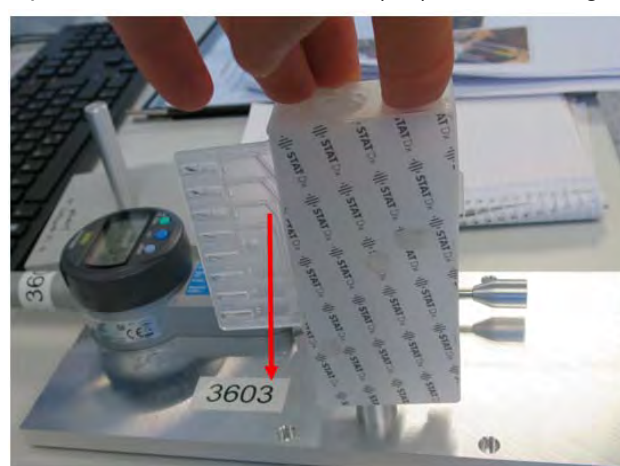
12. Make sure there is no cartridge in the AM and click "OK".
13. Click "OK" when Message = "Push TC to 0 and measure the initial TC position of the Cartridge" appears.

14. Place the blank cartridge on the on the X-9246 test fixture pestle.



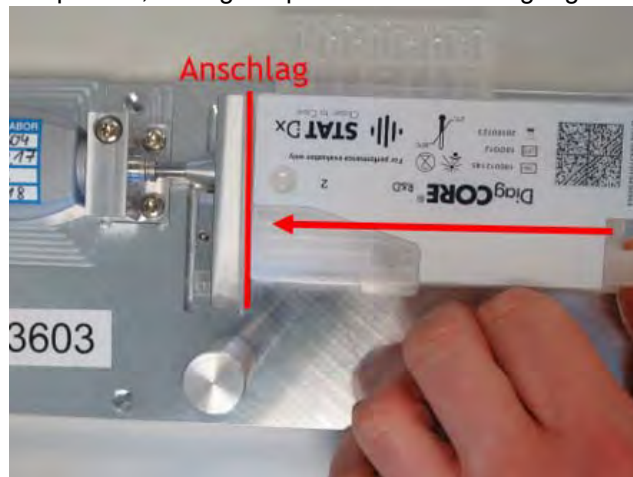
Picture 8.2-35 Plunger set top.

15. Insert the Blank Cartridge for TC Positioning Test on the pestle of the X-9246 test fixture and press down until it stops. This pushes the transfer chamber (TC) of the cartridge to the zero position.



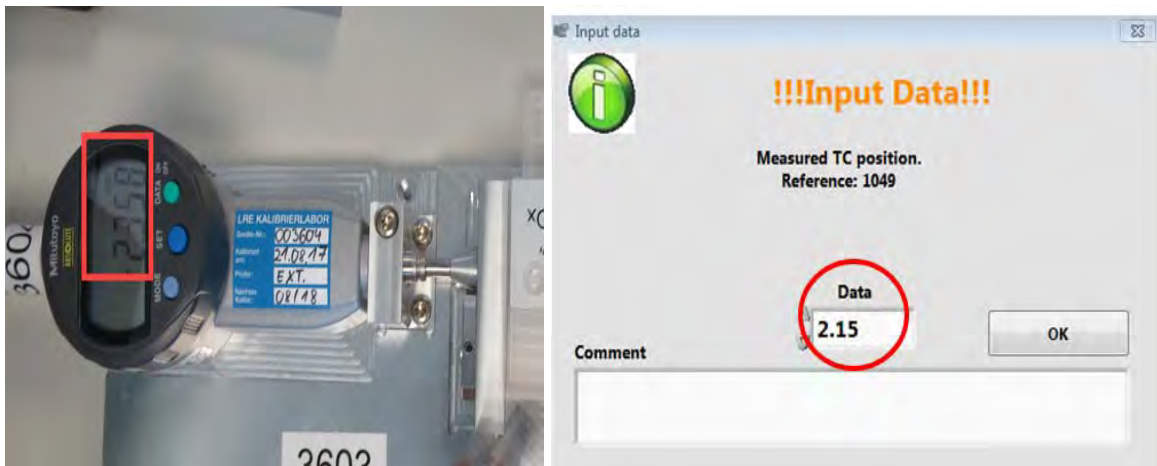
Picture 8.2-36 Cartridge pushed to the "0" position.

16. Push the blank cartridge for TC Positioning Test on the X-9246 test fixture. Ensure it is seated completely against the stop block, driving the pusher towards the gauge.



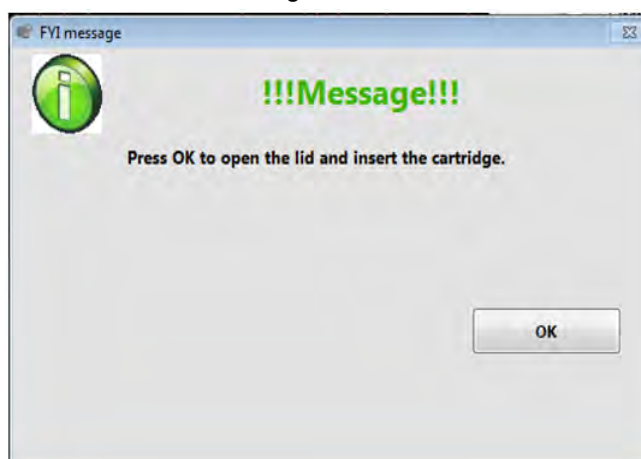
Picture 8.2-37 Push cartridge to stop.

17. Read the measured value from and enter it in the “Data” field, two decimal places, ignoring the (-) negative symbol. Click “OK”



Picture 8.2-38 TC Measurement data.

18. Click “OK to open the lid and insert the cartridge.



Picture 8.2-39 Open the lid and insert cartridge.

19. The following five steps must be performed equally for all measurements of the TC position (reference: 1050, 1055, 1051, 1052, 1053, 1054, 1056, 1057 and 1058).

20. Push the Blank Cartridge for TC Positioning Test on the X-9246 test fixture against the pusher of the length measuring device to the stop.



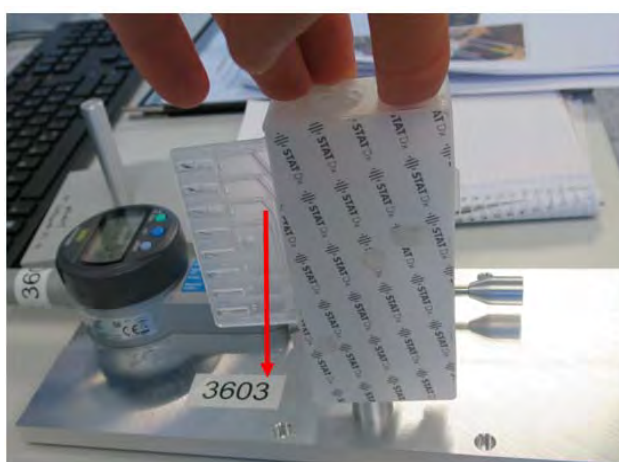
Picture 8.2-40 Push cartridge to the stop.

21. Read the measured value from and enter it in the “Data” field, two decimal places, ignoring the (-) negative symbol. Click “OK”



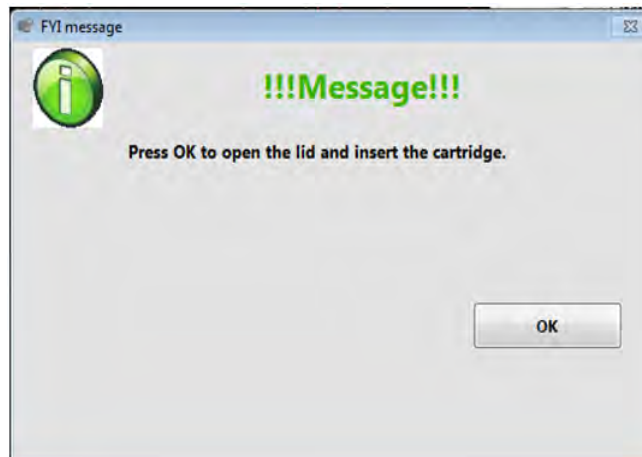
Picture 8.2-41 TC Measurement Data.

22. Insert the Blank Cartridge for TC Positioning Test on the pestle of the X-9246 test fixture and press down until it stops. This pushes the transfer chamber (TC) of the cartridge to the zero position.



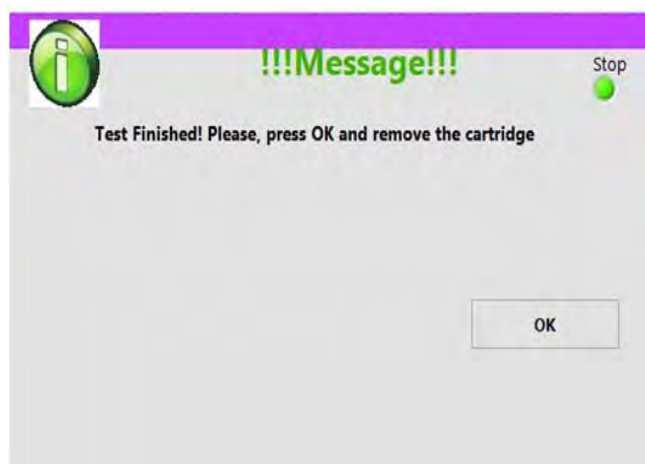
Picture 8.2-42 Cartridge pushed to the “0” position.

23. Click "OK" to open the lid and insert the cartridge.



Picture 8.2-43 Open the lid and insert cartridge.

24. After the last measurement remove the cartridge and click "OK"



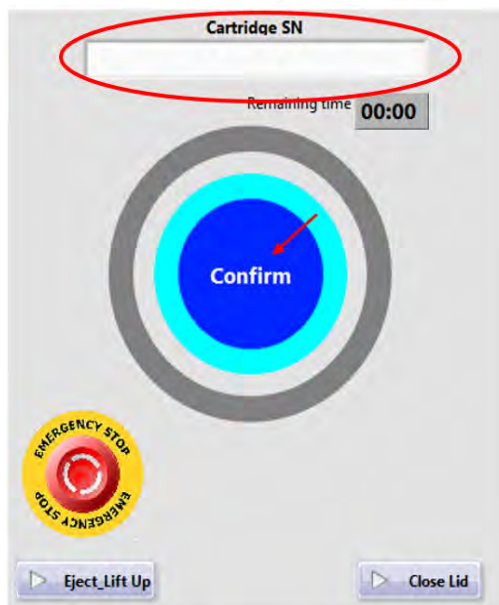
Picture 8.2-44 Test complete.

25. Open the following Excel file in the Summary Test Tab "D:\ReleaseProtocol" ReleaseProtocol vx.x" ReleaseProtocolReport.xlsm and enter if the pass test or failed.

COVER SHEET		TEST SUMMARY	
AM Serial Number:	1062	Test	PASS/FAIL
AM Hardware Version:	1.0	TC Positioning Test	PASS
Firmware/Bootware Version:	0.6.21.0	Pneumatics Functional Test	PASS
OM Serial Number:		Actuators Functional Test	PASS
OM Hardware Version:	1.0	qPCR Test	PASS
Application Software Version:		Thermal Module Test	PASS
QPCR Serial Number	572467	Liquid Sensor Test	PASS
QPCR Firmware Version:	0.6.2.0	Workflow Test	PASS
Thermal Unit Serial Number	30930-00000146		
SVC Firmware Version	1.80.0.0		
TRF Serial Number	34170160		
TRF Firmware Version	2.20.66.0		
Service Tool Version	1.0.2.55		
TC Positioning TDMS Filename	20180316090015_FU1062_TC_Positioning		
Pneumatics TDMS Filename	20180316092240_FU1062_Pneumatics		
Actuators TDMS Filename	20180316102016_FU1062_Actuators		
QPCR TDMS Filename	20180316113746_FU1062_qPCR_Test		
Thermal TDMS Filename	20180316105718_FU1062_Thermal		
Liquid Sensors TDMS Filename	20180316111003_FU1062_LiquidSensors_Test		
Workflow Test TDMS Filename	20180319072817_FU1062_180040008		
Comments:			
Ref. NC/CAPA:			

Picture 8.2-45 Test Summary Data Sheet.

26. In the “Cartridge SN” enter “pneum01” and click “Confirm”.



Picture 8.2-46 Set Cartridge SN to pneum01.

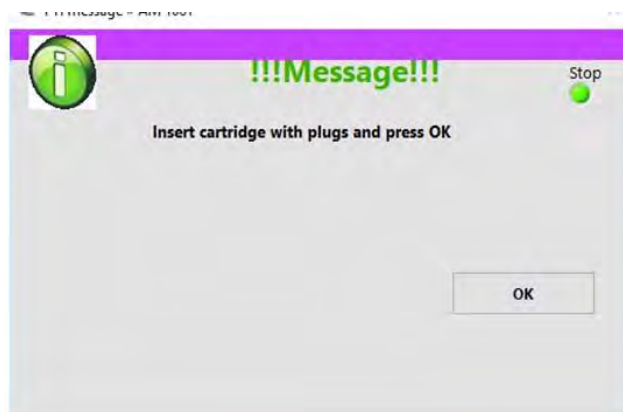
27. Open the Pneumatics-v2.prg file in the following path “C:-ReleaseProtocol-ReleaseProtocol vx.x” (always use the latest version).

28. Use modified blank cartridge WAKA 3760 with inserts or 3D printed cartridge. (see picture).



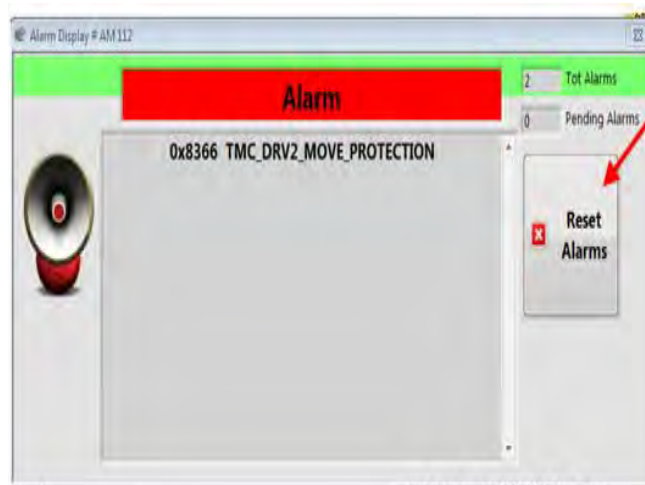
Picture 8.2-47 WAKA 3760 cartridge and 3D printed cartridge.

29. Insert the cartridge and press “OK”



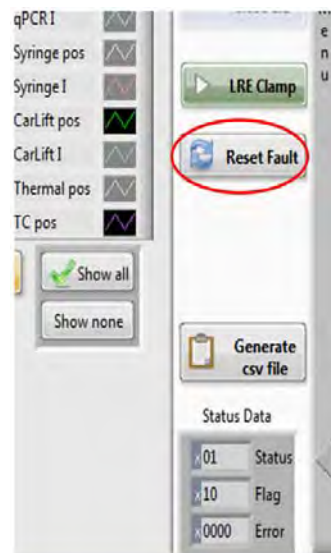
Picture 8.2-48 Insert and click OK.

30. A deliberate alarm is triggered during the test step (too high pressure) at the end of the test. Click on "Reset Alarms" to clear.



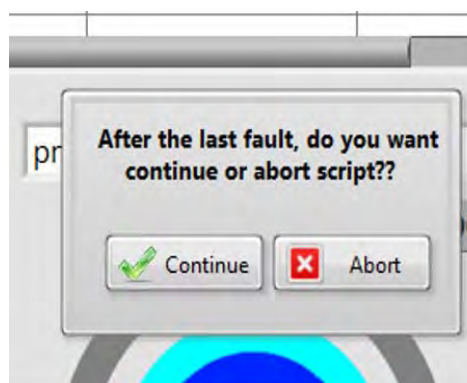
Picture 8.2-49 Reset alarm.

31. Reset the Fault.



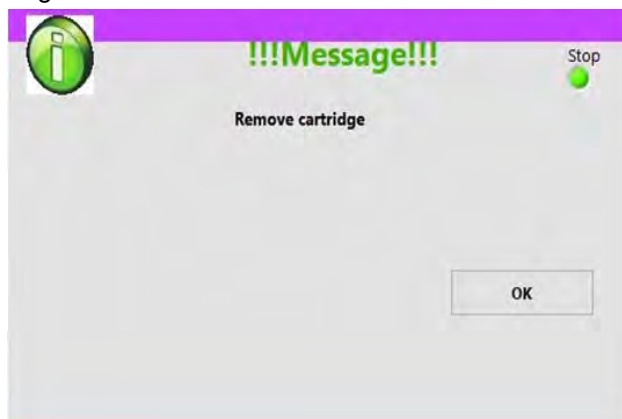
Picture 8.2-50 Reset fault.

32. Click "Continue"



Picture 8.2-51 Continue script select.

33. Remove the blank cartridge and click "OK"



Picture 8.2-52 Remove test cartridge.

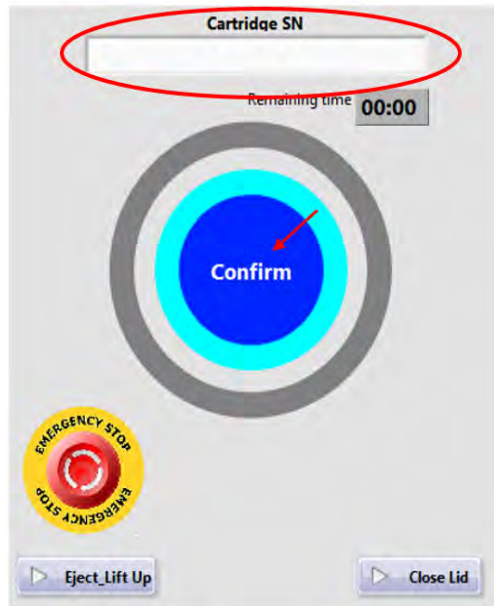
34. Open the following Excel file "C:\ReleaseProtocol\ ReleaseProtocol vx.x" ReleaseProtocolReport.xlsm and in the SHEET SUMMARY_Test and enter test passed or failed (always use the latest version).

COVER SHEET		TEST SUMMARY	
AM Serial Number:	1062	Test	PASS/FAIL
AM Hardware Version:	1.0	TC Positioning Test	PASS
Firmware/Software Version:	0.6.21.0	Pneumatics Functional Test	PASS
OM Serial Number:		Actuators Functional Test	PASS
OM Hardware Version:	1.0	QPCR Test	PASS
Application Software Version:		Thermal Module Test	PASS
QPCR Serial Number	572467	Liquid Sensor Test	PASS
QPCR Firmware Version	0.6.2.0	Workflow Test	PASS
Thermal Unit Serial Number:	30930-00D00146		
SVC Firmware Version:	1.80.0.0		
TRF Serial Number	34170160		
TRF Firmware Version	2.20.66.0		
Service Tool Version	1.0.2.55		
TC_Positioning TDMS Filename	20180316090015_FU1062_TC_Positioning		
Pneumatics TDMS Filename	20180316092240_FU1062_Pneumatics		
Actuators TDMS Filename	20180316102016_FU1062_Actuators		
QPCR TDMS Filename	20180316113746_FU1062_qPCR_Test		
Thermal TDMS Filename	20180316105718_FU1062_Thermal		
Liquid Sensors TDMS Filename	20180316111003_FU1062_LiquidSensors_Test		
Workflow Test TDMS Filename	20180319072817_FU1062_180040008		
Comments:			
Ref. NC/CAPA:			

SUMMARY_Test | TC_Positioning | Pneumatics | Actuators_Functional | QPCR | Thermal_Module | Liquid_Sensors | Header_TC_Pos

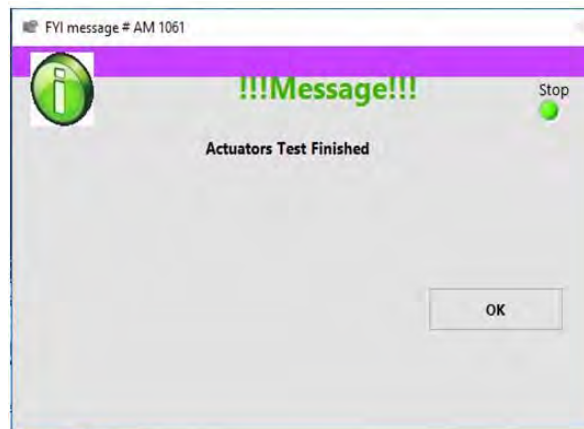
Picture 8.2-53 Test Summary Data Sheet.

35. In the "Cartridge SN" enter ""ac01". and click "Confirm".



Picture 8.2-54 Set cartridge serial number.

36. Open the Actuators-v2.prg file in the following path "C:-ReleaseProtocol-ReleaseProtocol vx.x" (always use the latest version).
37. Click "OK" to start the Actuators Test.
38. Ensure no cartridge is in the AM.
39. Press "OK" and insert the 3D printed cartridge.
40. After the test remove the cartridge and click "OK".



Picture 8.2-55 Actuators test complete

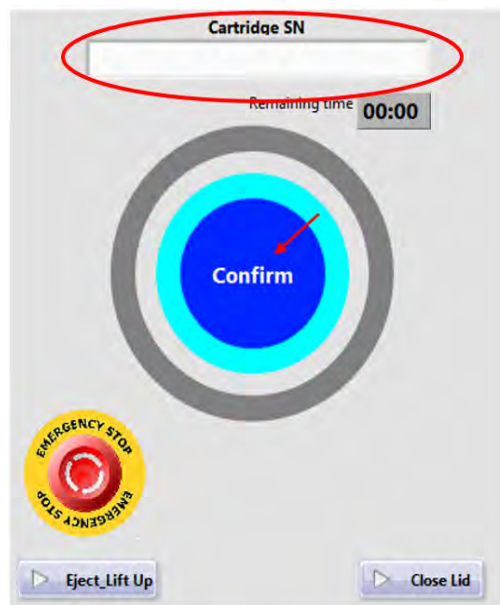
41. Open the following Excel file "C:\ReleaseProtocol\ReleaseProtocol vx.x\ReleaseProtocolReport.xlsm" and in the SHEET SUMMARY_Test enter if passed or failed (always use the latest version).

COVER SHEET		TEST SUMMARY	
AM Serial Number:	1062	Test	PASS/FAIL
AM Hardware Version:	1.0	TC Positioning Test	PASS
Firmware/Software Version:	0.6.21.0	Pneumatics Functional Test	PASS
OM Serial Number:		Actuators Functional Test	PASS
OM Hardware Version:	1.0	QPCR Test	PASS
Application Software Version:		Thermal Module Test	PASS
QPCR Serial Number	572467	Liquid Sensor Test	PASS
QPCR Firmware Version	0.6.2.0	Workflow Test	PASS
Thermal Unit Serial Number	30930-00D00146		
SVC Firmware Version	1.80.0.0		
TRF Serial Number	34170160		
TRF Firmware Version	2.20.66.0		
Service Tool Version	1.0.2.55		
TC Positioning TDMS Filename	20180316090015_FU1062_TC_Positioning		
Pneumatics TDMS Filename	20180316092240_FU1062_Pneumatics		
Actuators TDMS Filename	20180316102016_FU1062_Actuators		
QPCR TDMS Filename	20180316113746_FU1062_qPCR_Test		
Thermal TDMS Filename	20180316105718_FU1062_Thermal		
Liquid Sensors TDMS Filename	20180316111003_FU1062_LiquidSensors_Test		
Workflow Test TDMS Filename	20180319072817_FU1062_180040008		
Comments:			
Ref. NC/CAPA:			

SUMMARY_Test TC_Positioning Pneumatics Actuators_Functional QPCR Thermal_Module Liquid_Sensors Header_TC_Pos

Picture 8.2-56 Test Summary Data Sheet.

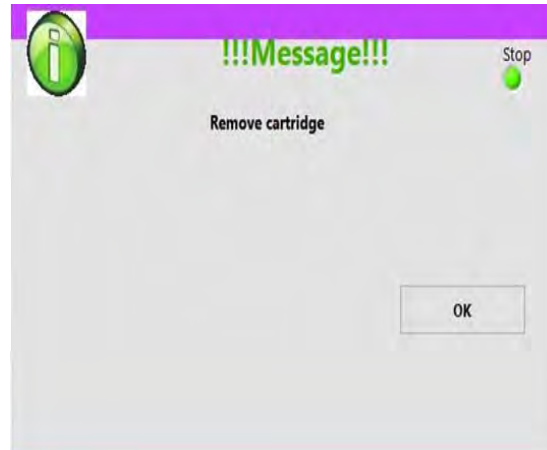
42. In the "Cartridge SN" enter ""tm01". and click "Confirm".



Picture 8.2-57 Set cartridge serial number.

43. Open the thermal v2.prg file in the following path "C:\ReleaseProtocol-ReleaseProtocol vx.x" (always use the latest version).
 44. Insert the 3D printed cartridge.

45. When the test is complete remove the test cartridge.



Picture 8.2-58 Thermal test complete.

46. Open the following Excel file "C:\ReleaseProtocol\ ReleaseProtocol vx.x" ReleaseProtocolReport.xlsmand in the SHEET SUMMARY_TEST and check if the test passed or failed. pass test (always use the latest version).

COVER SHEET

AM Serial Number:	1062
AM Hardware Version:	1.0
Firmware/Bootware Version:	0.6.21.0
OM Serial Number:	
OM Hardware Version:	1.0
Application Software Version:	
QPCR Serial Number	572467
QPCR Firmware Version	0.6.2.0
Thermal Unit Serial Number	30930-00000146
SVC Firmware Version	1.80.0.0
TRF Serial Number	34170160
TRF Firmware Version	2.20.66.0
Service Tool Version	1.0.2.55
TC_Positioning TDMS Filename	20180316090015_FU1062_TC_Positioning
Pneumatics TDMS Filename	20180316092240_FU1062_Pneumatics
Actuators TDMS Filename	20180316102016_FU1062_Actuators
QPCR TDMS Filename	20180316113746_FU1062_qPCR_Test
Thermal TDMS Filename	20180316105718_FU1062_Thermal
Liquid Sensors TDMS Filename	20180316111003_FU1062_LiquidSensors_Test
Workflow Test TDMS Filename	20180319072817_FU1062_180040008
Comments:	
Ref. NC/CAPA:	

TEST SUMMARY

Test	PASS/FAIL
TC_Positioning Test	PASS
Pneumatics Functional Test	PASS
Actuators Functional Test	PASS
QPCR Test	PASS
Thermal Module Test	PASS
Liquid Sensor Test	PASS
Workflow Test	PASS

SUMMARY Test

TC_Positioning

Pneumatics

Actuators_Functional

QPCR

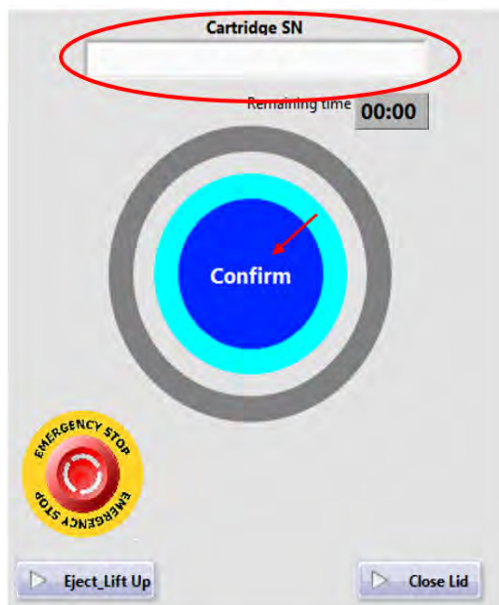
Thermal_Module

Liquid_Sensors

Header_TC_Pos

Picture 8.2-59 Test Summary data sheet.

47. In the “Cartridge SN” enter “”liquid01.prg” and click “Confirm”.



Picture 8.2-60 Set cartridge serial number.

48. Open the File "LiquidSensors_Test-v2" in the following path "C:-ReleaseProtocol-ReleaseProtocol vx.x(always use the latest version).

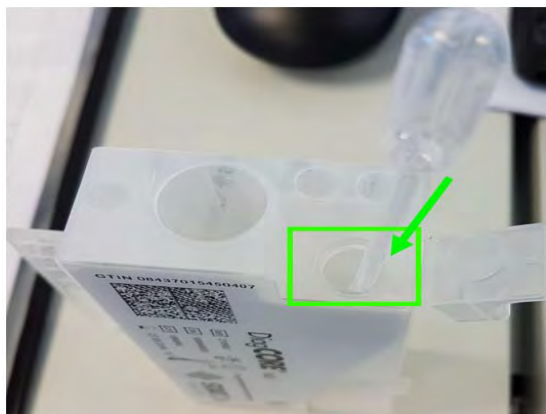
49. Open the “Swab Port” of a Blank Cartridge, this will be used as a Liquid Sensor Test Cartridge.



Picture 8.2-61 Liquid Sensor Test Cartridge.

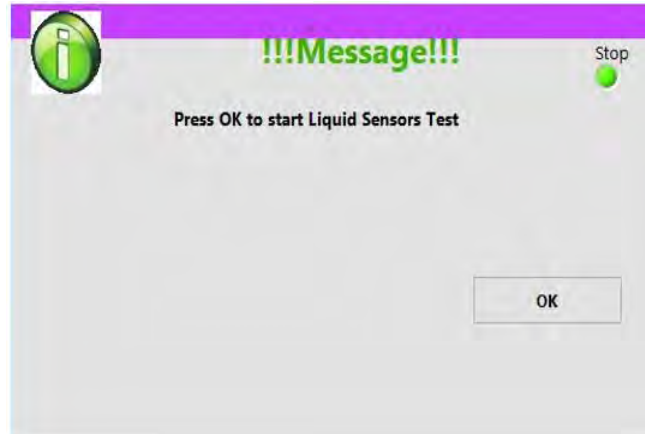
50. Fill a pipette with 300ul of water (used as a buffer).

51. Place the 300ul of water in the “Swab Port” and close.



Picture 8.2-62 Cartridge swab port.

52. Click "OK"



Picture 8.2-63 Start liquid sensors test.

53. Insert the cartridge and click "OK"



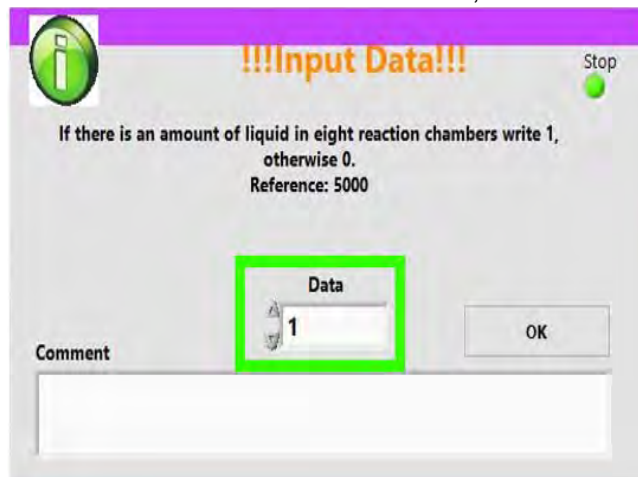
Picture 8.2-64 Insert Cartridge.

54. Remove Blank Cartridge and ensure that the eight cartridge reaction chambers are filled with approximately with the same level of water.



Picture 8.2-65 Check cartridge liquid distribution.

55. In the box enter a "1" if the water is in all 8 reaction chambers; "0" if not.



Picture 8.2-66 Set water level condition.

56. Click on "OK" and insert the Water Filled Blank Cartridge back in the AM to perform the condensation test.



Picture 8.2-67 Insert Cartridge for the condensation test.

57. Click "OK".



Picture 8.2-68 Remove the cartridge to check for condensation.

58. Check the front and back of the reaction chambers to ensure that the water evenly condensed (no larger air spaces).



Picture 8.2-69 Evenly distributed condensation.

59. If condensation is evenly distributed enter “1”, if not enter “0”.

!!!Input Data!!!

Stop

If both sides of the reaction chambers are equally condensated write 1, otherwise 0.
Reference: 5001

Data

1

OK

Comment

Picture 8.2-70 Enter condensation test result.

60. The Test is complete, discard the test cartridge and click “OK”.

!!!Message!!!

Stop

Liquid Sensors Test finished.

OK

Picture 8.2-71 Liquid sensors test complete.

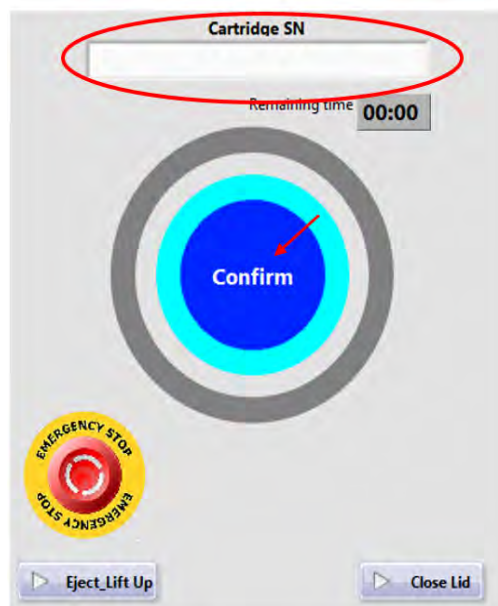
61. Open the following Excel file "C:\ReleaseProtocol\ReleaseProtocol vx.x\ReleaseProtocolReport.xlsm" and in the SHEET SUMMARY_Test and enter test passed or failed (always use the latest version).

COVER SHEET		TEST SUMMARY	
AM Serial Number:	1062	Test	PASS/FAIL
AM Hardware Version:	1.0	TC Positioning Test	PASS
Firmware/Bootware Version:	0.6.21.0	Pneumatics Functional Test	PASS
OM Serial Number:		Actuators Functional Test	PASS
OM Hardware Version:	1.0	QPCR Test	PASS
Application Software Version:		Thermal Module Test	PASS
QPCR Serial Number	572467	Liquid Sensor Test	PASS
QPCR Firmware Version	0.6.2.0	Workflow Test	PASS
Thermal Unit Serial Number	30930-00D00146		
SVC Firmware Version	1.80.0.0		
TRF Serial Number	34170160		
TRF Firmware Version	2.20.66.0		
Service Tool Version	1.0.2.55		
TC_Positioning TDMS Filename	20180316090015_FU1062_TC_Positioning		
Pneumatics TDMS Filename	20180316092240_FU1062_Pneumatics		
Actuators TDMS Filename	20180316102016_FU1062_Actuators		
QPCR TDMS Filename	20180316113746_FU1062_qPCR_Test		
Thermal TDMS Filename	20180316105718_FU1062_Thermal		
Liquid Sensors TDMS Filename	20180316111003_FU1062_LiquidSensors_Test		
Workflow Test TDMS Filename	20180319072817_FU1062_180040008		
Comments:			
Ref. NC/CAPA:			

SUMMARY_Test TC_Positioning Pneumatics Actuators_Functional QPCR Thermal_Module Liquid_Sensors Header_TC_Pos

62. Test Summary Data Sheet.

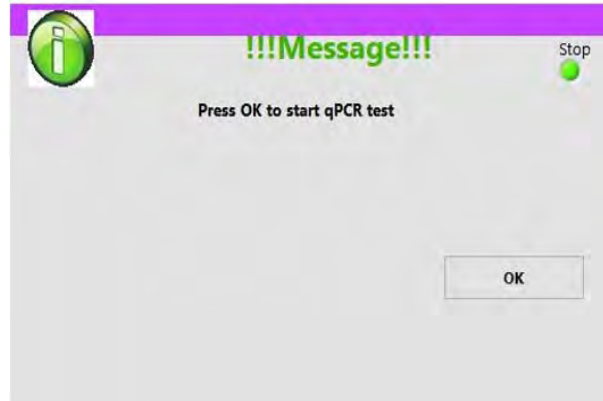
63. In the "Cartridge SN" enter ""qPCR01" and click "Confirm".



Picture 8.2-72 Set cartridge serial number.

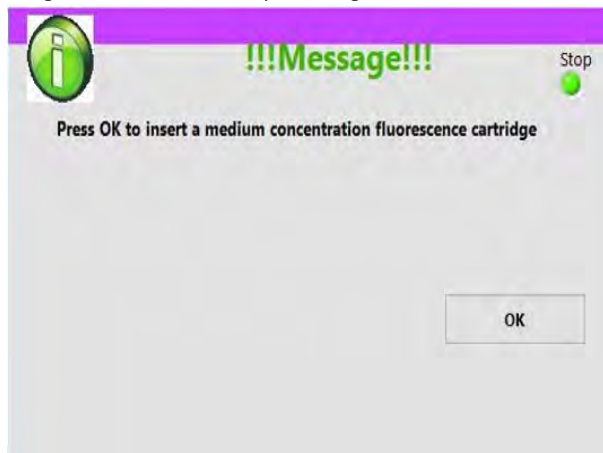
64. Open the qPCR_Test-v2.prg file in the following path "C:-ReleaseProtocol-ReleaseProtocol vx.x(always use the latest version).

65. Click "OK".



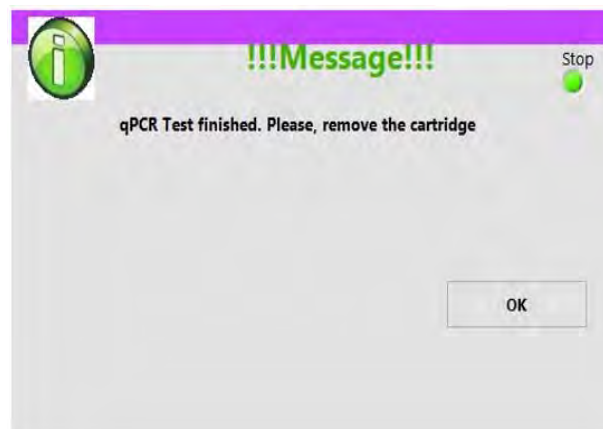
Picture 8.2-73 Start qPCR test.

66. Insert Fluorophore Cartridge for QPCR test. (Cartridge is stored in the refrigerator). and click "OK".



Picture 8.2-74 Insert medium fluorescence cartridge.

67. Remove Fluorophore Cartridge for QPCR test and click on "OK" (Return cartridge back to the refrigerator correctly oriented).



Picture 8.2-75 Remove qPCR test cartridge.

68. Open the following Excel file "C:\ReleaseProtocol\ReleaseProtocol vx.x\ReleaseProtocolReport.xlsm" and in the SHEET SUMMARY_Test and enter test passed or failed (always use the latest version).

COVER SHEET		TEST SUMMARY	
AM Serial Number:	1062	Test	PASS/FAIL
AM Hardware Version:	1.0	TC Positioning Test	PASS
Firmware/Bootware Version:	0.6.21.0	Pneumatics Functional Test	PASS
OM Serial Number:		Actuators Functional Test	PASS
OM Hardware Version:	1.0	QPCR Test	PASS
Application Software Version:		Thermal Module Test	PASS
QPCR Serial Number	572467	Liquid Sensor Test	PASS
QPCR Firmware Version	0.6.2.0	Workflow Test	PASS
Thermal Unit Serial Number	30830-00D00146		
SVC Firmware Version	1.80.0.0		
TRF Serial Number	34170160		
TRF Firmware Version	2.20.66.0		
Service Tool Version	1.0.2.55		
TC Positioning TDMS Filename	20180316090015_FU1062_TC_Positioning		
Pneumatics TDMS Filename	20180316092240_FU1062_Pneumatics		
Actuators TDMS Filename	20180316102016_FU1062_Actuators		
QPCR TDMS Filename	20180316113746_FU1062_qPCR_Test		
Thermal TDMS Filename	20180316105718_FU1062_Thermal		
Liquid Sensors TDMS Filename	20180316111003_FU1062_LiquidSensors_Test		
Workflow Test TDMS Filename	20180319072817_FU1062_180040008		
Comments:			
Ref. NC/CAPA:			

SUMMARY_Test TC_Positioning Pneumatics Actuators_Functional QPCR Thermal_Module Liquid_Sensors Header_TC_Pos

69. Test Summary data sheet.

70. Log out of the "Service Tool".

71. Place the AM power switch to "OFF" (0).

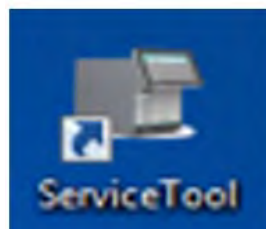
72. Connect the Analytical and Operational Modules using WAKA 3761 cable or equivalent.



Picture 8.2-76 AM to OM interconnect cable.

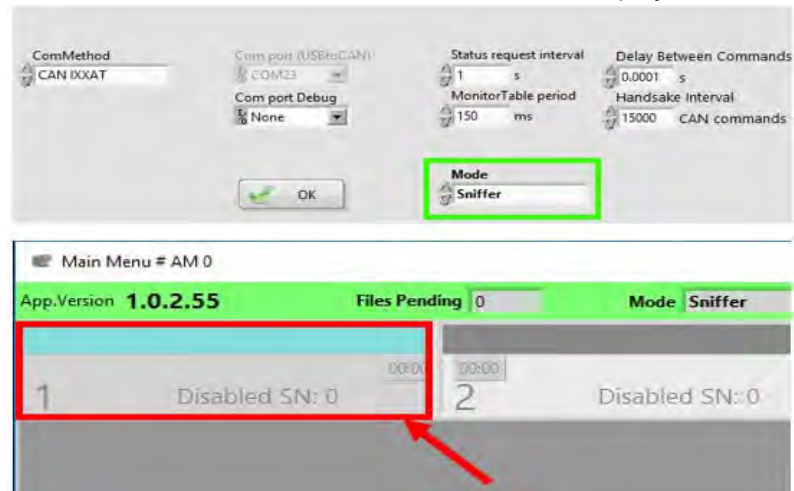
73. Set the AM power switch to "ON" and on the OM press the power switch.

74. Open the "Service Tool" software.



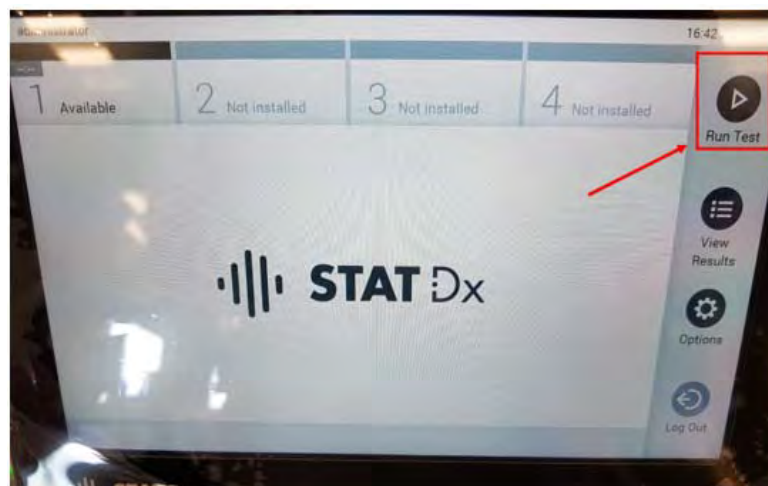
Picture 8.2-77 Service tool icon.

75. Set the mode to "Sniffer" and then wait until the serial number is displayed.



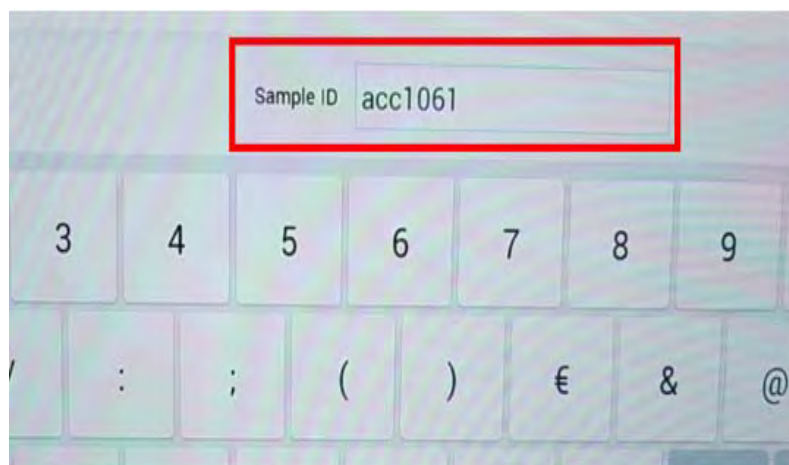
Picture 8.2-78 Enable sniffer mode.

76. On the OM touch screen select "Run Test".



Picture 8.2-79 OM touch screen entry.

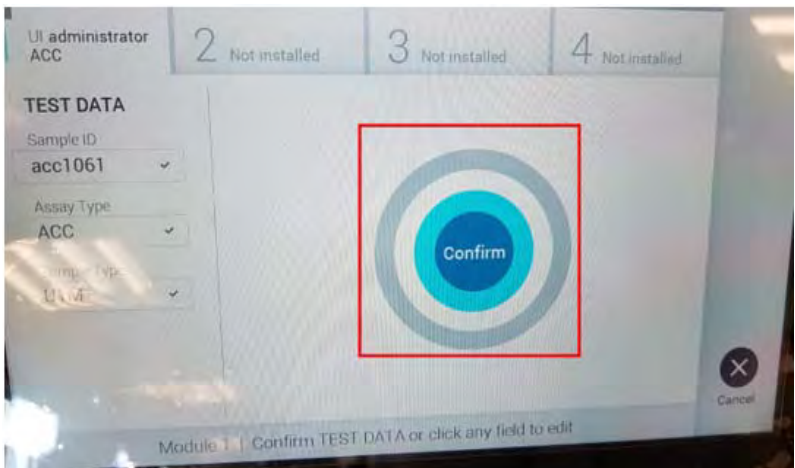
77. Enter in the text field "acc" + serial number of the AM (e.g. "acc1061"). The serial number is visible in the "Service Tool".



Picture 8.2-80 OM Touch screen entry.

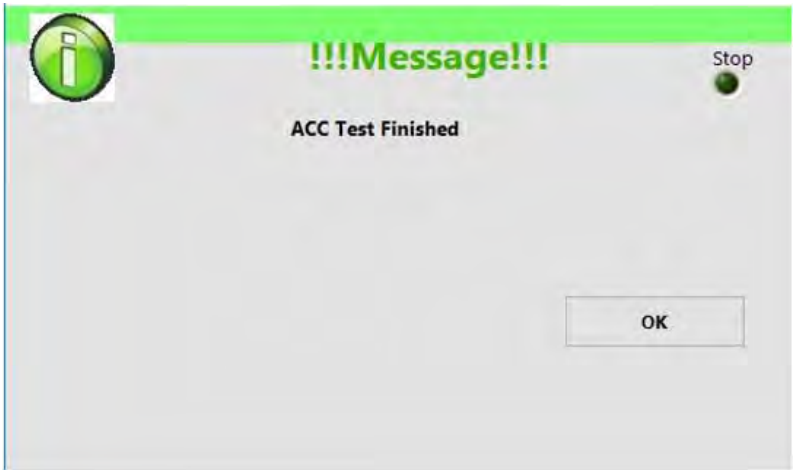
78. Scan the XXXX bar code using the Operational Module.

79. Click on "Confirm" at the OM and insert Analyzer Check Cartridge 50 021 00.



Picture 8.2-81 OM touch screen entry.

80. Click on "OK" and click on "Test Complete" on the OM touch screen and select Remove Cartridge.



Picture 8.2-82 Test complete.

81. Open the following Excel file "C:\ReleaseProtocol\ ReleaseProtocol vx.x" ReleaseProtocolReport.xlsm and in the SHEET SUMMARY_Test and enter test passed or failed (always use the latest version).

COVER SHEET		TEST SUMMARY	
AM Serial Number:	1062	Test	PASS/FAIL
AM Hardware Version:	1.0	TC Positioning Test	PASS
Firmware/Bootware Version:	0.6.21.0	Pneumatics Functional Test	PASS
OM Serial Number:		Actuators Functional Test	PASS
OM Hardware Version:	1.0	QPCR Test	PASS
Application Software Version:		Thermal Module Test	PASS
QPCR Serial Number	572467	Liquid Sensor Test	PASS
QPCR Firmware Version	0.6.2.0	Workflow Test	PASS
Thermal Unit Serial Number	30930-00D00146		
SVC Firmware Version	1.80.0.0		
TRF Serial Number	34170160		
TRF Firmware Version	2.20.66.0		
Service Tool Version	1.0.2.55		
TC_Positioning TDMS Filename	20180316090015_FU1062_TC_Positioning		
Pneumatics TDMS Filename	20180316092240_FU1062_Pneumatics		
Actuators TDMS Filename	20180316102016_FU1062_Actuators		
QPCR TDMS Filename	20180316113746_FU1062_qPCR_Test		
Thermal TDMS Filename	20180316105718_FU1062_Thermal		
Liquid Sensors TDMS Filename	20180316111003_FU1062_LiquidSensors_Test		
Workflow Test TDMS Filename	20180319072817_FU1062_180040008		
Comments:			
Ref. NC/CAPA:			

SUMMARY_Test TC_Positioning Pneumatics Actuators_Functional QPCR Thermal_Module Liquid_Sensors Header_TC_Pos

82. Test summary data sheet.

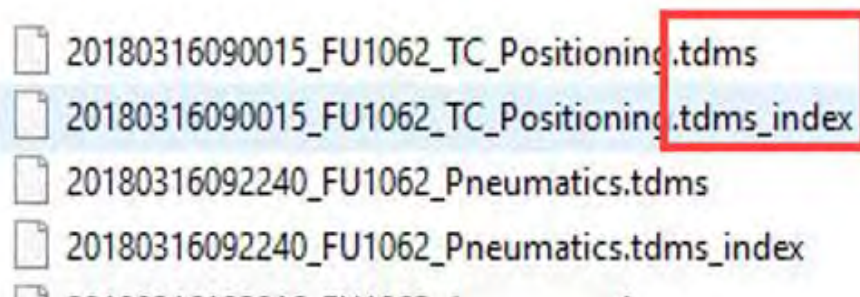
83. Open Excel Log and find the listed result documents in the SUMMA-RY_TEST sheet in the following path (each ending in .tdms and .tdms_index): C:\Users\mschneider\Documents\LabVIEW Data\26568-ServiceTool\Results (Attention: Path may vary depending on the logged-on user).

TC Positioning TDMS Filename	20180316090015_FU1062_TC_Positioning
Pneumatics TDMS Filename	20180316092240_FU1062_Pneumatics
Actuators TDMS Filename	20180316102016_FU1062_Actuators
QPCR TDMS Filename	20180316113746_FU1062_qPCR_Test
Thermal TDMS Filename	20180316105718_FU1062_Thermal
Liquid Sensors TDMS Filename	20180316111003_FU1062_LiquidSensors_Test
Workflow Test TDMS Filename	20180319072817_FU1062_180040008

Dateinamen

Picture 8.2-83 List of results documents.

84. Copy files to the following directory: C:\ReleaseProtocol\ready\AM-SNXXXX Re-lease Protocol Vx.x\TDMS. The folders have to be created manually by the user (always use the latest version).



Picture 8.2-84 Copy results.

85. Copy Excel document from path C:\ReleaseProtocol\ReleaseProtocol v1.4 to the directory C:\ReleaseProtocol\ready\AM-SNXXXX Re-lease Protocol Vx.x\ copy and copy the entire folder to the serial number performed and paste it in the designated shared location.
86. Print Excel log (= form99 009 18) and fill in or sign the marked fields. This completed cover page must be placed in the box with the AM.

Comments:	
Ref. NC/CAPA:	
This unit has met all required specifications for Release	
YES / NO	
Tester :	Date:

Picture 8.2-85 Customer report.

87. Testing Completed.

8.3 VP-14 Infield Service Tool

Purpose	Functional Test for the Analytical Module
Tool(s) Required	PC with Windows 10 with the Infield Service Tool and IXXAT driver loaded; 9026467 USB to Can Adapter, AM, QSTAT; 9026465 CAN Connection Cable, AM, QSTAT; 9026464 Tool-Cartridge Service, AM, QSTAT(3D Printed Cartridge)
Verification Time	90 Minutes.

Table 8.3-1 Infield Service Test

8.3.1 Start the Infield Service Tool

NOTE: 	The first five tests will be performed on EACH AM individually, the subsequent ACC Workflow test is performed with all AM's connected via the AM bridges.	
---	---	--

1. Power off the instrument, if not already powered off.

Action	Duration
Start Infield Service Tool	5 minutes

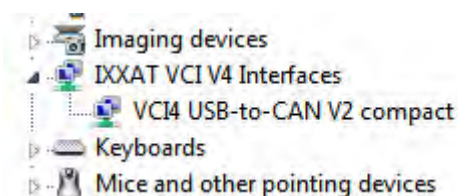
Table 8.3-2 Start the Infield Service Tool

2. Remove the power cord from the back of the instrument.
3. Remove the OM-AM bridge from the back of the first AM.



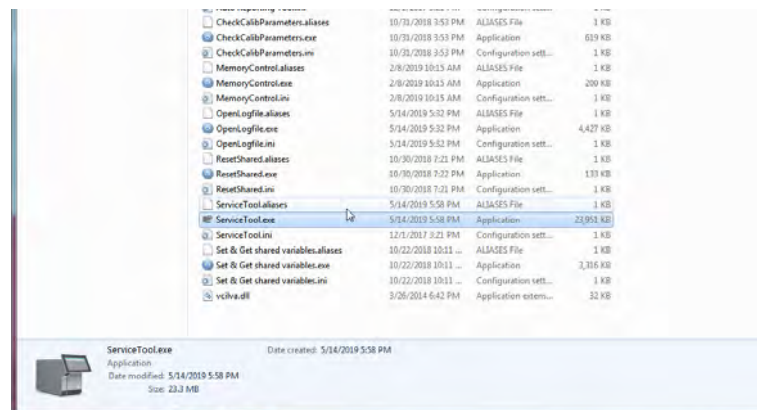
Picture. 8.3-1 AM CAN bus connection.

4. Connect the CAN to USB Adapter to one of the 15 Pin ports of the first AM.
5. Connect the CAN to USB cable to a USB port on the FSE laptop
6. The IXXAAT US device should be shown in the device manager on the FSE laptop. To access the device manager: Right click on the "My PC" icon; click "Manage"; and click "Device Manager" folder.



Picture. 8.3-2 Device manager.

7. Connect the power cord to this AM and place the AM power switch to 1 position (on).
8. Start the Infield Service Tool by selecting the desktop icon or the Service Tool executable found in the DiagCORE-Functional Unit folder or desktop icon.



Picture. 8.3-3 Infield service tool executable location.

9. This should open the “QIAstat Infield Service Tool” main window.



Picture 8.3-3 Infield service tool main window.

10. Select the first five tests: Visual Inspection, Actuators Status test, Pneumatics, qPCR Internal Standard, and Thermal Unit test by clicking the check boxes.



Picture 8.3-4 Service tool selected tests.

8.3.2 Start the Visual Inspection Test

Action	Duration
Visual Inspection Test	5 minutes

Table 8.3-5 Visual inspection.

1. From in the QIAstat Infield Service Tool, ensure the check boxes next to the first five tests are checked and the AM to be tested is selected.



Picture 8.3-6 Infield service tool visual Inspection.

2. Click the Run test button to start the test.
3. A prompt to save the result will be displayed. Save the Result Report file with the default name.
4. Acknowledge the welcome message.



Picture 8.3-7 Start the Vvsual inspection.

5. After clicking on OK, wait for the instrument to complete the first portion of the test. The instrument will move to a cleaning position.

6. Check and clean the TC1 Pusher part, as described in the prompt, if necessary clean with 70% isopropyl alcohol and swab.



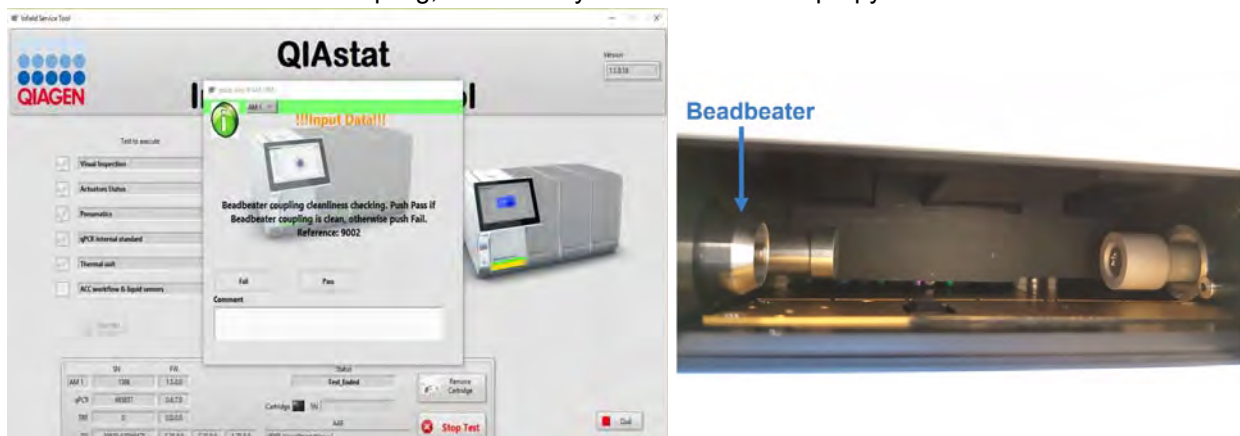
Picture 8.3-8 Select pass or fail.

7. Confirm cleanliness by pressing pass.
8. Check the TC2 Pusher as described in the prompt, if necessary clean with 70% isopropyl alcohol and swab.



Picture 8.3-9 Select pass or fail.

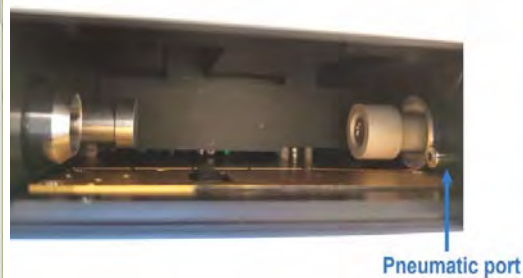
9. Confirm cleanliness by clicking PASS.
10. Check the Bead Beater coupling, if necessary clean with 70% isopropyl alcohol and swab.



Picture 8.3-10 Select pass or fail.

11. Confirm cleanliness by clicking PASS.

12. Check pneumatic ports, if necessary clean with 70% isopropyl alcohol and swab.



Picture 8.3-11 Select pass or fail.

13. Confirm cleanliness by clicking PASS.

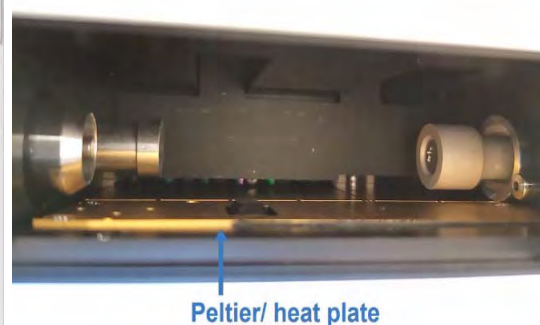
14. Check the Cartridge Housing (cartridge slot area), if necessary clean with 70% isopropyl Alcohol and swab.



Picture 8.3-12 Select pass or fail.

15. Confirm cleanliness by clicking PASS.

16. Check the peltier heat plate, if necessary clean with 70% Isopropyl Alcohol and swab.



Picture 8.3-13 Select pass or fail.

17. Confirm cleanliness by clicking PASS.

18. Check the filter tray cleanliness according to the prompt in the software (changed previously).



Picture 8.3-14 Select pass or fail.

19. Confirm filter cleanliness by clicking PASS.

20. The next prompt will indicate to listen for qPCR noise.

20.1. The sound should be smooth and consistent; no friction, scratching or squeaking sounds.



Picture 8.3-15 qPCR Motion test start.



Picture 8.3-16 qPCR motion test start pass/fail.

21. Confirm the qPCR sound was smooth and consistent, by clicking PASS, if not the AM has failed and must be repaired.

22. Wait for the software to generate report files, and for the instrument to perform some movements. This will take around a minute.

8.3.3 Actuators Status Test

Action	Duration
Actuators Status Test	10 min

Table 8.3-17 Actuators status test.

1. The Actuators Status test will open automatically, select OK to begin.
2. A prompt to save the result will be displayed. Save the result report file with the default name.



Picture 8.3-18 Actuators status test.

3. The software will display a prompt in which to start the Actuators Status test and insert the 3D printed Tool- Cartridge Service (9026464).



Picture 8.3-19 Run actuators status test.

4. Click OK for the actuators Sstatus test to start. Wait for the cartridge door to open, and gently place the cartridge in the correct orientation (the label should be on the left side).
 - 4.1.The cartridge will be automatically lowered into position, and the cartridge door closed.
 - 4.2.The test will start and progress automatically.

5. Once the test has completed, the software will prompt for the removal of the cartridge.



Picture 8.3-20 Remove the cartridge.

6. The software will generate the final reports, and a prompt will appear that the test has finished.

8.3.4 Pneumatics Function Test

Action	Duration
Pneumatics Functional Test	25 min

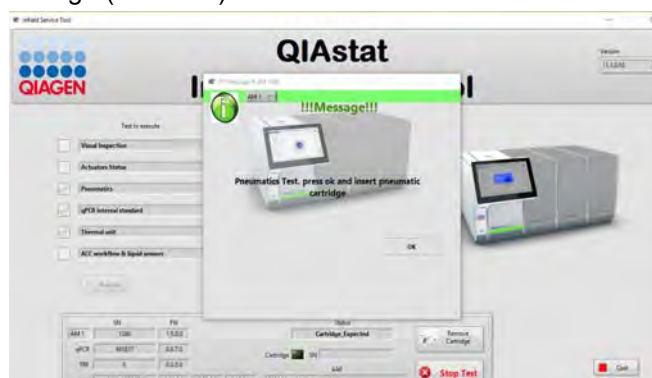
Table 8.3-21 Pneumatics functional test.

1. The pneumatics test will open automatically, select OK to begin.



Picture 8.3-22 Pneumatics functional test.

- A prompt to save the result will be displayed. Save the Result Report file with the default name.
- The software will display a prompt to insert a pneumatic cartridge. This is the 3D printed Tool Service Cartridge (9026464)



Picture 8.3-23 Start test and insert 3D printed cartridge.

- Click OK on the prompt, and wait for the cartridge door to open. Once the cartridge door is open, insert the cartridge.
 - The instrument will automatically lower the cartridge and close the cartridge door.

- 4.2. The Pneumatics test will automatically start running.
- 5. Wait for the Pneumatics test to complete.

INFORMATION


During the test, the cartridge will be lifted up and down a few times. Do not remove the cartridge until the test is finished.

- 6. When the test finishes, a message will be displayed to click OK. When the door is opened and the cartridge has been completely lifted, remove the cartridge from the instrument.



Picture 8.3-24 Remove the cartridge.

- 7. Click OK to exit the test and go to the next test.

8.3.5 qPCR Internal Standard Test

Action	Duration
qPCR Internal Standard Test	5 min

Tab. 8.3-25 qPCR internal standard test.

- 1. The qPCR internal standard test will open automatically, select OK to begin.



Picture 8.3-26 qPCR standard test.

- 2. A prompt to save the result will be displayed. Save the Result Report file with the default name.
- 3. The qPCR internal standard test will run; no FSE interaction is required during this test.
- 4. Acknowledge completion of the test completed prompt.
- 5. After the test completes, the test status should indicate a pass.

8.3.6 Thermal Unit Test

Action	Duration
Thermal unit Test	15 min

Tab. 8.3-27 Thermal unit test

1. The Thermal unit test will start automatically, select OK to begin.



Picture. 8.3-4 Thermal unit test.

2. A prompt to save the result will be displayed. Save the result report file with the default name.
3. The software will prompt to confirm starting the thermal unit test and insert the 3D printed cartridge.



Picture 8.3-28 Test prompt and 3D printed cartridge.

4. Click OK on the prompt, and wait for the cartridge door to open. Once the cartridge door is open, insert the cartridge in the correct orientation.
 - 4.1. The instrument will automatically lower the cartridge and close the cartridge door.
 - 4.2. The thermal unit test will automatically start running.
5. Once the test is complete, the cartridge will be lifted out of the instrument, and presented for removal. Remove the cartridge from the instrument.

INFORMATION 	The instrument performs heating and cooling cycles during this test. The cartridge might be warm afterwards; this is normal.
---	--

6. Press OK to finalize the test.
7. If there are additional AM's connected repeat steps 2.9.1 to 2.9.6 on all additional AM's by clicking on the AM to be tested on the Infield Service Tool. There is no reason to connect the CAN bus adapter to each AM individually.
8. END of TEST. To run the ACC Workflow test see ["VP7 - Analyzer Check Cartridge Work Flow Test" on page 169.](#)

9 Annual Inspection

9.1 Introduction

The objectives of the QIAstat-Dx Annual Inspection procedure are:

- Ensure the reliability and functionality of the system
- Perform a complete system inspection to ensure maximum system performance
- Upgrade or update the system according to the upgrade policies stated in the relevant instrument specific ISI notes
- The implementation and/or renewal of service contracts

This document is the guideline for performing the annual Annual Inspection for the QIAstat-Dx. To keep this document as compact as possible we refer to other service documentation whenever applicable.

All **QIAstat-Dx documents** relevant for service can be downloaded from Master Control.

9.2 Terms

In the Annual Inspection procedure in chapter 9.5 we use the term *inspection*. An inspection includes the following tasks:

- Look for damages
- Look for corrosion
- Look for dirt, precipitations from buffers, etc.

9.3 Requirements

All preventative maintenance (PM) procedures outlined in this guideline shall only be performed by QIAGEN authorized personnel.

INFORMATION 	Ensure all required material is on-site before starting the PM.
---	--

9.3.1 Protective material

In order to prevent possible exposure to biological hazards, make sure to utilize appropriate hand, eye, and body protection.

Material	Comment
Eye protection	Safety glasses
Hand protection	Gloves (rubber, nitrile, etc.)
Body protection	Lab coat

Table 9.3-1 Protective material

9.3.2 Useful Documents

Document	Comment
HB-2636-001_MDx_UM_QIAstat-Dx_Analyzer_SW	QIAstat-Dx User Manual
PMVR-5000-0031	PM Documentation Form

Tab. 9.3-2

9.3.3 Software

Software	Comment
QIAstat-Dx Infield Service Tool	Installed During Training/GPS Intranet
IXXAT CAN to USB Driver	Installed During Training/GPS Intranet
StatCAN2USB Driver	Installed During Training/GPS Intranet
ADF_ACC_V06.asy	From GPS Intranet, for the ACC workflow test

Tab. 9.3-3

9.3.4 Tools

Tools	Comment
9026465, CAN Connection Cable, QSTAT	
9026467, USB to CAN Adapter, QSTAT	
9026464, Tool-Cartridge Service, AM, QSTAT	3D Printed Cartridge for Leakage Test
9026466, Analyzer Check Cartridge, AM QSTAT	

Table 9.3-4 Tools

9.3.5 Parts

SAP	Part Name	Comment
9026189	Air Filter Tray, AM, QSTAT	

Table 9.3-5 Parts

9.3.6 Other

Material	Comment
CCD Cleaning Swab	for qPCR cleaning
Tissue, Lint Free	Suitable for cleaning optics
Mild Detergent or isopropanol	For cleaning
USB Drive	For software update (if required)

Table 9.3-6 Parts, Other

9.4 Annual Inspection schedule

The approximate duration of the Annual Inspection procedure is 6 hours. This time may vary, depending on the status and performance of the QIAstat-Dx.

Subject	Task	Time
Check ISI's	Check Sales Force for ISI's that need to be completed	0.1 hours
Cleaning	Clean the instrument	0.5 hours
Testing/ Verification	Perform function tests and adjustments	4.0 hours
Report	Fill in the PM visit record and sign off	0.5 hours
TOTAL		5.1 hours

9.5 Annual Inspection procedure

9.5.1 See PM-5000-0009 Annual Inspection Guideline

10 Software and Software Tools

This Chapter is divided into different sections: User Software and Service Tools.

10.1 QIAstat-Dx User Software

10.1.1 General Information

The QIAstat-Dx is shipped with the latest version of the User Software.

The proprietary user software runs in the QIAstat-Dx Operational Module (OM) and is accessed using an integrated touchscreen.

The software is updated using a USB drive connected to a USB port on the analyzer.

10.1.2 Installation and Update Analyzer Software

Installation, update, or re-installation of the QIAstat-Dx software can be performed by the user or service technician.

INFORMATION 	Before updating/installing the analyzer software perform a system backup. Unless the backup is requested by the customer this backup will be deleted immediately after service.
---	--

1. Perform a system backup to a USB drive, insert a USB drive on any USB port and on the main menu:
 - Press Options --> System Config --> Make Backup
2. To install/update software insert the USB drive with the software on any USB port and on the main menu:
 - Press Options --> System Config --> System Update --> Proceed

10.2 Software for Service and Troubleshooting

10.2.1 General Information

The Analytical Module (AM) service software, Infield Service Tool, is resident on a laptop computer for field service technicians and repair center technicians. Connection to the AM is made via the CAN bus connector on the back of the module using a CAN to USB adapter and CAN bus adapter to AM cable.

The service software is for use in the repair center only. Connection to the AM is made via the CAN bus connector on the back of the module to the repair center computer using a CAN to USB adapter and CAN bus adapter; additional connection to AM is made using a USB printer cable.

10.2.2 Text editor

- Notepad++ available from <http://notepad-plus-plus.org/>
- jEdit available from <http://www.jedit.org/>

10.2.3 Analyze log files

The system log contains system status information and error codes logged with the date/time of occurrence and are accessed through the main screen:

- Options --> system Config --> System Log

INFORMATION

The System log files can be downloaded to a USB drive for archiving and troubleshooting.

10.2.4 QIAstat-Dx Instrument Service Software.

The service software includes the Infield Service tool loaded on a laptop for use during Annual Maintenance and IQ/OQ activities. The repair center may also utilize this software for troubleshooting.

- Infield Service Tool is an automated testing protocol that is used to determine and validate the overall analyzer AM condition and status. This software is used for the Annual and IQ/OQ to validate the analyzer. See Chapter 9, Annual Maintenance procedures for guidance running this test.

Alignments and calibrations are completed via software loaded on the test station computer and can not be utilized in the field. See Chapter 7, Calibrations and Chapter 8, Verifications for guidance running this software.

- GAUDI Service Tool.
- Optical Alignment and Thermal Unit Module Test.
- AM Mechanical Final Test.
- AM Optical Final Test.
- AM Release Test.
- AM Mechanical Calibrations.
- Setting the OM serial number.

10.2.5 Install System Update

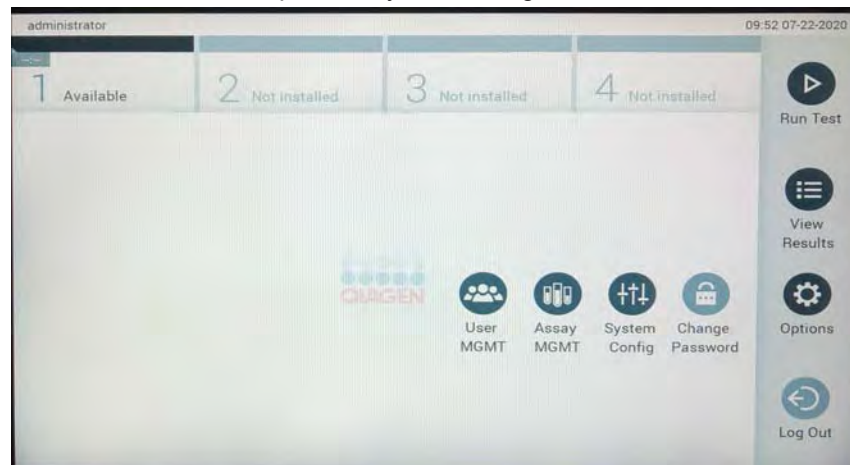
1. Perform a system backup if the analyzer is not a new install.
2. Go to the GPS folder and download the new software files.



Picture 10.2-1 GPS file location

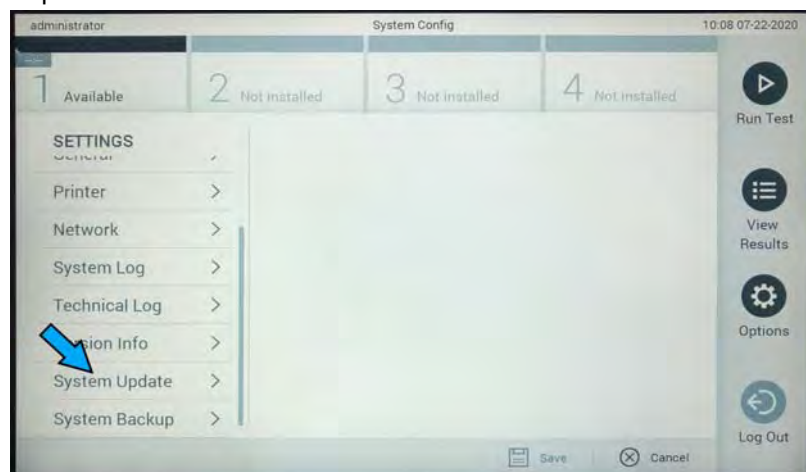
3. On you laptop format the USB drive to "FAT 32" file system.
4. The Unzip the file to the root directory of the USB drive.

5. After the analyzer is initialized insert the USB drive into the analyzer USB port.
6. Login as administrator and select: options> system config.



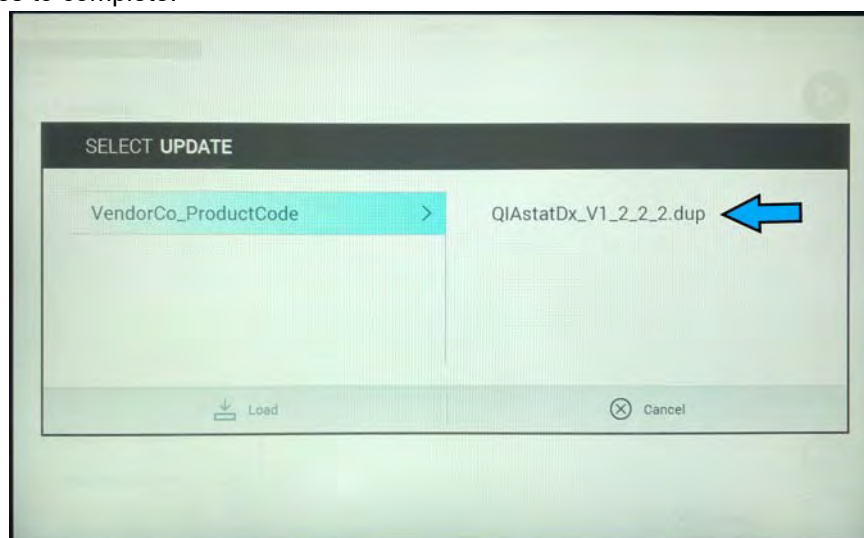
Picture 10.2-2 System main screen.

7. Select "System Update".



Picture 10.2-3 Options screen.

8. When the update screen appears select the update file and begin the update. The process can take up to 30 minutes to complete.



Picture 10.2-4 System update file

11 Troubleshooting

11.1 Most frequent errors and recommended actions

Error code*	Brief error description	Reason	Error occurs at times		Error occurs various times consecutively	
			Action	Contact level 2 technical support	Action	Contact level 2 technical support
0x0003	Barcode reading failed	Barcode reading error or cartridge Datamatrix code poor printing quality or damaged	Use another cartridge		Change cartridge lot, if it still occurs, replace AM	X
0x0004	Communication error OM to AM	Checksum error during file transfer of Assay Software (ADF) from OM to AM	Retry test, same cartridge can be used		Possible hardware failure, unit replacement	X
0x0005	Communication error OM to AM	Assay software parsing error	Retry test, same cartridge can be used		Possible hardware failure, unit replacement	X
0x04F0	Cartridge already used	Used cartridges are not allowed to run more than once	Use another cartridge		Possible hardware failure, unit replacement	X

Error code*	Brief error description	Reason	Error occurs at times		Error occurs various times consecutively	
			Action	Contact level 2 technical support	Action	Contact level 2 technical support
0x1010001	Test failed called by Result Call algorithm	The result call algorithm considered the test result as fail. The most common cause for this error is that the Internal Control (IC) and no other target have amplified and therefore the result is not valid. This could be due to a failure of the fluidic workflow during the cartridge execution or to the presence of inhibitors in the PCR. Other possible cases may be the presence of noise in the raw data, chamber under/over dosing, etc.	Try running the same sample in another cartridge. If the problem persists try to run a different sample in a new cartridge.		If the error persists for different samples and the incidence is above 2%, scale up to technical level 2 support	X
0x1000000-0x1000FFF	Fluidic protocol execution error	Errors in this range are triggered by the failure in one of the checkpoints in the fluidic protocol execution inside of the cartridge. A possible cause for fluidic failure could be that the input sample is not correct (different kind of sample, volume out of range, presence of debris, etc.)	Retry test on another cartridge. Check with the user if the sample used was correct for the intended assay. Check with a negative sample.		Retry with another lot of cartridge. Possible hardware failure, unit replacement	X
0x1008000-0x1008FFF (except the errors below)	Instrument hardware problem	Failure of a subsystem of the analyzer	Restart unit. Use another cartridge		Possible hardware failure, unit replacement	X

Error code*	Brief error description	Reason	Error occurs at times		Error occurs various times consecutively	
			Action	Contact level 2 technical support	Action	Contact level 2 technical support
0x1008024	Air Filter related error	Missing filter drawer / bottom filter drawer is open	Check if the filter is placed and/or close the filter drawer		Possible hardware failure, unit replacement	X
0x100800a	Tilt error	The analyzer is not lying on a flat surface. The instrument has to lay in a horizontal surface in order to perform a correct execution of the fluidic protocol.	Place the analyzer in an even surface.		Possible hardware failure, unit replacement	X

Table 11.1-1 Frequent Errors and actions.

11.2 System

11.2.1 Error Overview

From	To	
0x0000	0xFFFF	Operational Module Errors
0xA000000	0xA000FFF	Cartridge Execution Errors
0xA0[0-6]1000	0xA0[0-6]1FFF	Result Call Algorithm Errors
0xA008000	0xA008FFF	Analytical Module Errors

Table 11.2-1 Error overview

11.2.2 Troubleshooting Operational Module errors (0x0000 to 0xFFFF)

Error code	Title	Description
0x101	DiskSpaceWarningLevelReached	Free disc space reached warning level.
0x102	DiskSpaceCriticalLevelReached	Free disc space reached critical level.
0x103	ExportFailed	Export failed!
0x104	FileCanNotCreate	Failed to create file
0x105	FileExists	File already exists
0x106	FileNotExists	File does not exist
0x107	InvalidFileName	Invalid file name
0x205	SerialPortOpenError	SerialPort open error.
0x206	SerialPortCloseError	SerialPort close error.

Error code	Title	Description
0x208	Usb2CanMessageNotAcknowledged	Usb2Can Message not Acknowledged.
0x20A	SerialPortReadError	SerialPort Read Error
0x20C	SerialPortWriteError	SerialPort Write Error
0x20F	OMAMComNotInitialized	OMAMCom not initialized
0x210	AnalyticalModuleNotInitialized	AnalyticalModule not initialized
0x211	GidDataFieldLengthError	Gid Data Length Error
0x212	SerialFrameError	SerialPort Frame Error
0x213	SerialFrameLengthError	SerialPort Frame Length Error
0x301	AssayReadError	Assay file not readable.
0x302	AssayInvalidXmlFile	Assay file invalid.
0x303	AssayInvalidDasVersion	Assay requires version {1}
0x304	AssayAlraedyInDb	Assay already imported.
0x305	AssayImportFailed	Importing failed.
0x401	AssayNotPresent	Assay not available.
0x402	AssayNotActive	Assay not active.
0x403	AssayNoUserRightsToBeExecuted	User has no right to execute assay .
0x404	AssayNewerDasVersionToBeExecuted	Assay requires version {1}.
0x405	AssayNewerAmVersionToBeExecuted	Assay {1} requires version {2}.
0x406	AssayNewerVersionRequired	A newer version of the assay is required.
0x420	OpenLidTimeoutError	Cartridge not inserted in .
0x424	EjectCartridgeTooHot	Eject not possible
0x31	OmBarcodeError	Failed to scan barcode.
0x33	DifferentBarcodeError	Different Barcode.
0x90	ProcessingModuleInvalidError	The processing module is not valid.
0xE0	ResultRawDataEvaluationError	The evaluation of the raw data failed
0xE1	MissingResultRawData	Test ended without or with too less raw result data.
0xF0	CartridgeAlreadyUsedError	Cartridge already used.
0xF1	CartridgeExpired	Cartridge expired.
0x100	TestCanceledByUser	The test process has been canceled by user {1}.
0x110	BarcodeDataCrcError	Transmitting barcode failed
0x111	BarcodeDataLengthError	Transmitting barcode failed
0x112	LoggingDataCrcError	Transmitting logging data failed
0x113	LoggingDataLengthError	Transmitting logging data failed
0x116	IdentificationDataCrcError	Invalid identification data
0x117	IdentificationDataLengthError	Invalid identification data
0x118	ResultDataCrcError	Transmitting test raw data failed
0x119	ResultDataLengthError	Transmitting test raw data failed
0x11A	CalibrationDataCrcError	Invalid calibration data
0x11B	CalibrationDataLengthError	Invalid calibration data
0x11C	CalibrationParametersCrcError	Calibration Parameters Crc Error
0x11D	CalibrationParametersLengthError	Calibration Parameters Length Error
0x11E	CalibrationRequired	Calibration of required in {1} days.
0x11F	MaintenanceRequired	Maintenance of required in {1} days.

Error code	Title	Description
0x120	RejectOlderTestResult	Test record rejected - test start time is older than 90 minutes.
0x121	TestResultDataLost	Test result data lost.
0x122	NoFreeModuleAvailable	No free module available.
0x0001	AssayInvalidCRC	Assay invalid CRC
0x0002	UserInvalidCRC	User data invalid CRC
0x0003	UserProfileInvalidCRC	User profile data invalid CRC
0x0004	TestRecordInvalidCRC	Test record invalid CRC
0x0005	DatabaseNotFound	Database not found.
0x0006	DatabaseIncompatible	Database is not compatible.
0x001	PMError_setInitializationData	Initialization of PM failed with error
0x002	PMError_setRawData	Process Module failed by setRawData with error
0x003	PMError_checkResults	Process Module failed by checkResults with error
0x004	PMError_getTestResults	Process Module failed by getTestResults with error
0x005	PMError_getAmplificationPlot	Process Module failed by getAmplificationPlot with error
0x006	PMError_getMeltingPlot	Process Module failed by getMeltingPlot with error
0x1101	PowerFailOnLastShutdown	Unexpected power off. Data may be lost
0x01	AmOpenLidError	OpenLidError
0x02	AmCloseLidError	Error by closing lid.
0x03	AmBarcodeError	Barcode reading failed.
0x04	AmAafCrcError	Downloading test failed
0x05	AmAafParseError	AAF parse error
0x09	AmTestFault	Test Fault
0x0A	AmHomingError	Homing Error
0x10	AmFault	Fault
0x11	AmFwCrcError	FW Crc Error
0x12	AmFwFlashingError	Flashing Error
0x13	AmAafTooLong	AAF too long
0xF000	AmError	AM Error
0xF001	WrongAmError	Unexpected AM found
0xF002	AmUnexpectedBehavior	Unexpected behavior of .
0xF003	AmFwUpdateError	Error during firmware update.
0x1000	AuthenticationFailed3Times	Authentication failed 3 times. Please wait 1 minute to try again.
0x1001	BackupCreatedWithNewerSoftware	Backup created with newer software.
0x1002	RestoreNotPossible	Restore not possible.
0x1003	InvalidUpdateFile	Invalid update file.
0x1004	UpdateCanceled	Update canceled.
0x1005	UpdateFailed	Update failed!
0x1006	IllegalServiceMode	Illegal ServiceMode!

Table 11.2-2 AM troubleshooting matrix.

11.2.3 Troubleshooting cartridge execution errors (0xA000000-0xA000FFF)

11.2.3.1 Explanation of Error Codes

The last three digits of the error code in hexadecimal have to be converted to a decimal number.

The decimal error code has the format: ABBCD, where A defines the code category, B the error location, C the class of the error and D the type of error. Each error class has his own table of maximum 9 error types. This allows a creation maximal 891 error codes (99 x 9) without considering the error location (B).

Code	Number	Explanation
A	0	AAF workflow related checkpoints
BB	00 to 99	Location of the checkpoint failure which will be the number of the step variable of each function
C	0 to 9	Class of failure / error
D	0 to 9	Type of failure / error

Table 11.2-3 Cartridge error matrix

11.2.3.2 Category A definitions

A - Category	
0	AAF workflow related checkpoints
1	Reserved 1
2	Reserved 2
3	QC related checkpoints

Table 11.2-4 Category A

11.2.3.3 Error Class and Error Type

The ErrorClass and the ErrorType is defined at the position of interest in the code.

11.2.3.4 11.2.3.3.1 Error Class

The Error Class and the Error Type is defined at the position of interest in the code. All classes of errors are listed in following table:

C – Class of error		
B	Class	Description
0	Clamping and timeout	All errors related to the clamping procedures
1	Pressure	All errors related to pressure checkpoint failures
2	Temperature	All errors related to temperature checkpoint failures
3	qPCR	All errors related to qPCR measurements
4	qPCR other	All errors related to qPCR measurements at 95 °C and other qPCR measurements
5	Bead Beating	All errors related to the bead beating procedure
6	RC dosing correction	All errors related to reaction chamber dosing correction procedure
7	DCC dosing	All errors related to drychem chamber dosing
8	RC dosing	All errors related to reaction chamber dosing procedure
9	RC dispensing	All errors related to reaction chamber dispensing procedure

Table 11.2-5 Error class.

11.2.3.3.2 Error Type

D – Type – Class 0: Clamping and timeout	
D	Type
0	NA
1	Syringe timeout
2	Liquid detected in port 1 of the cartridge
3	TC clamping error
4	BB clamping error (current too high)
5	ATM pressure correction factor is off-limits
6	No sample type defined
7	NA
8	NA
9	NA

Table 11.2-6 Class 0: clamping and timeout.

D – Type – Class 1: Pressure	
D	Type
0	Out of range
1	Value not reached
2	Value is surpassed
3	Zero pressure not reached
4	Lysate membrane clogging
5	Wash buffer 1 membrane clogging
6	Wash buffer 2 membrane clogging
7	Elate membrane clogging
8	Pressure of pump not reached
9	Zero pressure not reached after purging of membrane

Table 11.2-7 Class 1: Pressure.

D – Type – Class 3: qPCR	
D	Type
0	Out of range of mean of all TL and TR
1	Out of range TL 1
2	Out of range TL 2
3	Out of range TL 3
4	Out of range TL 4
5	Out of range TR 1
6	Out of range TR 2
7	Out of range TR 3
8	Out of range TR 4
9	Reading time exceeds 14 seconds

Table 11.2-8 Class 3: qPCR.

D – Type – Class 4: qPCR other	
D	Type
0	Temperature 95 is not reached
1	Internal Standard (IS1Ch0) measurement is out of range (> 33 % than calibration value). Maintenance needed.
2	Internal Standard (IS2Ch1) measurement is out of range (> 33 % than calibration value). Maintenance needed.
3	Internal Standard (IS2Ch2) measurement is out of range (> 33 % than calibration value). Maintenance needed.
4	Internal Standard (IS2Ch3) measurement is out of range (> 33 % than calibration value). Maintenance needed.
5	Internal Standard (IS4Ch4) measurement is out of range (> 33 % than calibration value). Maintenance needed.
6	Internal Standard (IS4Ch5) measurement is out of range (> 33 % than calibration value). Maintenance needed.
7	Dark current of CH0 is out of range (>110% than calibration value). Maintenance needed.
8	Dark current of CH1 is out of range (>110% than calibration value). Maintenance needed.
9	Dark current of CH2 is out of range (>110% than calibration value). Maintenance needed.

Table 11.2-9 Class 4: qPCR.

D – Type – Class 5: Bead beating	
D	Type
0	BB speed not reached
1	BB speed is too high
2	Beat Beater motor current too high
3	Dark current of CH3 is out of range (>110 % than calibration value). Maintenance needed.
4	Dark current of CH4 is out of range (>110 % than calibration value). Maintenance needed.
5	Dark current of CH5 is out of range (>110 % than calibration value). Maintenance needed.
6	NA
7	NA
8	NA
9	NA

Table 11.2-10 Class 5: bead beating.

D – Type – Class 6: RC dosing correction	
D	Type
0	RC 8 dosing pressure not reached
1	Pressure too high in RC8 due to missing liquid or a too big air bubble
2	Bubble removal procedure did not succeed twice
3	Pressure condition fail for dosing RC7
4	Pressure not reached in DCC to remove air bubble
5	NA
6	NA
7	NA
8	NA
9	NA

Table 11.2-11 Class 6: RC dosing correction.

D – Type – Class 7: DCC dosing	
D	Type
0	Dosing pressure not reached in maximum dosing time
1	Dosing pressure is too high (out of range)
2	DC has a high leakage rate after dosing
3	TC pressure release fail after dosing
4	Pressure not reached in TC before DCC mixing
5	Value not reached in DCC mixing when going to lower pressures
6	Value not reached in DCC mixing when going to higher pressures
7	Pressure not reached in DC before DCC mixing
8	Minimum pressure not reached when recovering liquid to TC after DCC mixing
9	No liquid could be dosed to the DCC

Table 11.2-12 Class 7: dosing.

Table 11.2-13

D – Type – Class 8: RC dosing	
D	Type
0	Pressure equilibrium at 0 mbar not reached in RC1
1	RC 1 dosing failure
2	RC 2 dosing failure
3	RC 3 dosing failure
4	RC 4 dosing failure
5	RC 5 dosing failure
6	RC 6 dosing failure
7	RC 7 dosing failure
8	RC 8 dosing failure
9	Liquid detected in RC after RC dosing discard

Table 11.2-14 Class 8: RC dosing.

D – Type – Class 9: RC dispensing	
D	Type
0	Pressure not reached in Press1 as a dispensing condition for RC2
1	RC 1 dispensing failure
2	RC 2 dispensing failure
3	RC 3 dispensing failure
4	RC 4 dispensing failure
5	RC 5 dispensing failure
6	RC 6 dispensing failure
7	RC 7 dispensing failure
8	RC 8 dispensing failure
9	Lower pressure not reached and dispensing condition for next RC is not given

Table 11.2-15 Class 8 RC dispensing.

11.2.3.5 Analytical Module errors

Error code	Title	Description
8201	GENERAL_ERROR_WRONG_TEST_ID	
8001	GENERAL_ERROR_WRONG_COMMAND	
8002	GENERAL_ERROR_NO_RESULT_AVAILABLE	
8003	GENERAL_ERROR_INIT_NOT_FINISHED	
8004	GENERAL_ERROR_FLASH_READ	
8005	GENERAL_ERROR_FLASH_WRITE	
8006	GENERAL_ERROR_NO_DATA_TRANSFER	
8007	GENERAL_ERROR_TEMPERATURE_TOO_LOW	
8008	GENERAL_ERROR_TEMPERATURE_TOO_HIGH	
8009	GENERAL_ERROR_MEAS_TEMPERATURE_TOO_HIGH	
800A	GENERAL_ERROR_TILT	
800B	GENERAL_ERROR_NO_SERVICE_MODE	
800C	GENERAL_ERROR_FW_UPDATE_NEEDED	
800D	GENERAL_ERROR_SELFTEST_TIMEOUT	
800E	GENERAL_ERROR_BARCODE_TASK_TIMEOUT	
800F	GENERAL_ERROR_BARCODE_READER	
8010	GENERAL_ERROR_PARSE_TASK_TIMEOUT	
8011	GENERAL_ERROR_MEAS_TASK_TIMEOUT	
8012	GENERAL_ERROR_NO_VALID_CAN_ADDRESS	
8013	GENERAL_ERROR_NO_HST_DATA_DEFINED	
8014	GENERAL_ERROR_TOO_MUCH_HST_DATA_DEFINED	
8015	GENERAL_ERROR_POSTCODE	
8016	GENERAL_ERROR_INVALID_RESULT_TYPE	
8017	GENERAL_ERROR_OPEN_LID_NOT_ALLOWED	
8018	GENERAL_ERROR_CARTRIDGE_NOT_PROPER_LOADED	
8019	GENERAL_ERROR_NO_FW_DOWNLOADED	
801A	GENERAL_ERROR_WRONG_DEVICE_SELECTED	
801B	GENERAL_ERROR_PARAMETER_MISMATCH	
801C	GENERAL_ERROR_INVALID_START_STEP	
801D	GENERAL_ERROR_NO_RESET_FAULT_ON_DB_ERROR	
801E	GENERAL_ERROR_WRONG_SEQUENCE_NUMBER	
801F	GENERAL_ERROR_QPCR_INTERNAL_STANDARDS	
8020	GENERAL_ERROR_TRF_INTERNAL_STANDARD	
8021	GENERAL_ERROR_TU_INIT_HEATING	
8022	GENERAL_ERROR_PRESSURE_SENSORS_DIFFERENCE	
8023	GENERAL_ERROR_UNEXPECTED_STATE_CAUSED_ERROR_CLOSING_LID	
8024	GENERAL_ERROR_MISSING_FILTER	

8025	GENERAL_ERROR_QPCR_DARK_MEASUREME NT	
8026	GENERAL_ERROR_QPCR_DARK_MEAS_DEVIA TION	
8027	GENERAL_ERROR_TRF_NOT_CONFIGURED	
0x807F	GENERAL_ERROR_MAX_ERROR	
0x8080	MEAS_ERROR_ALL_DAMAGED	
8081	MEAS_ERROR_AAF_FAILURE	
0x80FF	MEAS_ERROR_MAX_ERROR	
0x8100	MOT10_VC_LOOKUP_TABLE_RANGE_ERROR	
8101	MOT10_VC_LOOKUP_TABLE_INDEX_ERROR	
8102	MOT10_VC_FREQUENCY_RANGE_ERROR	
8103	MOT10_VC_AMPLITUDE_RANGE_ERROR	
8104	MOT10_VC_OFFSET_RANGE_ERROR	
8105	MOT10_VC_AMPLITUDE_OFFSET_RANGE_ER ROR	
8106	MOT10_VC_NO_CURRENT_ERROR	
8107	MOT10_VC_CURRENT_HIGH_ERROR	
0x813F	MOT10_VC_ERROR_MAX_ERROR	
0x8140	PUMP_NO_CURRENT_ERROR	
8141	PUMP_WRONG_CURRENT_ERROR	
0x817F	PUMP_ERROR_MAX_ERROR	
0x8180	HW_ERROR_DC_MOT_DRIVER	
8181	HW_ERROR_ACTUATOR	
0x81FF	HW_ERROR_MAX_ERROR	
0x8200	SENSOR_ERROR_WRONG_SENSOR	
8201	SENSOR_ERROR_TEMPERATURE_NTC_AIR_IN NER_SIDE	
8202	SENSOR_ERROR_TEMPERATURE_INT	
8203	SENSOR_ERROR_TEMPERATURE_LM73_1_M B_TOP	
8204	SENSOR_ERROR_TEMPERATURE_LM73_2_M B_BOTTOM	
8205	SENSOR_ERROR_TEMPERATURE_EXT	
8206	SENSOR_ERROR_TEMPERATURE_ADT7420_1 _QPCR_STD	
8207	SENSOR_ERROR_TEMPERATURE_ADT7420_2 _TRF_STD	
8208	SENSOR_ERROR_NO_VALID_TEMPERATURE	
8209	SENSOR_ERROR_PRESSURE_INT	
820A	SENSOR_ERROR_PRESSURE_EXT	
820B	SENSOR_ERROR_PRESSURE_TOO_HIGH	
0x822F	SENSOR_ERROR_MAX_ERROR	
0x8230	MOTOR_NUMBER_RANGE_ERROR	
8231	MOTOR_NO_HOMING_ERROR	
8232	MOTOR_NO_HOMING_ERROR_MOT5	
8233	MOTOR_NO_HOMING_ERROR_MOT6	
8234	MOTOR_NO_HOMING_ERROR_MOT8	
8235	MOTOR_POSITION_RANGE_ERROR	

8236	MOTOR_5_POSITION_NOT_REACHED_ERROR	
8237	MOTOR_6_POSITION_NOT_REACHED_ERROR	
8238	MOTOR_8_POSITION_NOT_REACHED_ERROR	
0x8250	STEPPER_FREQ_RANGE_ERROR	
8251	STEPPER_NOT_A_STEPPER_ERROR	
8252	STEPPER_TUCLAMP_MODE_ERROR	
8253	STEPPER_TUCLAMP_POSITION_RANGE_ERROR	
8254	STEPPER_TUCLAMP_MICROSTEP_ERROR	
8255	STEPPER_TUCLAMP_MOVE_PROTECT_ERROR	
0x82A0	DCMOT_NUMBER_RANGE_ERROR	
82A1	DCMOT_PWM_RANGE_ERROR	
82A2	DCMOT_FREQUENCY_RANGE_ERROR	
82A3	DCMOT_CURRENT_LIMIT_RANGE_ERROR	
0x82FF	MOTOR_ERROR_MAX_ERROR	
8300	TMC_DRIVES_UNKNOWN_ID	At least one Technosoft drive with an unknown axis ID has been detected in the CAN network
8301	TMC_DRV1_NO_RESPONSE	No CAN-response received from Technosoft drive with axis ID 1
8302	TMC_DRV2_NO_RESPONSE	No CAN-response received from Technosoft drive with axis ID 2
8303	TMC_DRV3_NO_RESPONSE	No CAN-response received from Technosoft drive with axis ID 3
8304	TMC_DRV4_NO_RESPONSE	No CAN-response received from Technosoft drive with axis ID 4
8305	TMC_DRV7_NO_RESPONSE	No CAN-response received from Technosoft drive with axis ID 7
8306	TMC_DRV1_AMBIGUOUS_ID	Two or more Technosoft drives with identical axis ID 1 have been detected in the CAN network
8307	TMC_DRV2_AMBIGUOUS_ID	Two or more Technosoft drives with identical axis ID 2 have been detected in the CAN network
8308	TMC_DRV3_AMBIGUOUS_ID	Two or more Technosoft drives with identical axis ID 3 have been detected in the CAN network
8309	TMC_DRV4_AMBIGUOUS_ID	Two or more Technosoft drives with identical axis ID 4 have been detected in the CAN network
830A	TMC_DRV7_AMBIGUOUS_ID	Two or more Technosoft drives with identical axis ID 7 have been detected in the CAN network
830B	TMC_DRV1_INVALID_CONFIG	Invalid configuration of Technosoft drive with axis ID 1 (Technosoft drive HW needs to be replaced)

830C	TMC_DRV2_INVALID_CONFIG	Invalid configuration of Technosoft drive with axis ID 2 (Technosoft drive HW needs to be replaced)
830D	TMC_DRV3_INVALID_CONFIG	Invalid configuration of Technosoft drive with axis ID 3 (Technosoft drive HW needs to be replaced)
830E	TMC_DRV4_INVALID_CONFIG	Invalid configuration of Technosoft drive with axis ID 4 (Technosoft drive HW needs to be replaced)
830F	TMC_DRV7_INVALID_CONFIG	Invalid configuration of Technosoft drive with axis ID 7 (Technosoft drive HW needs to be replaced)
8310	TMC_DRV1_UPDATE_ERROR	Update or verify of Technosoft drive software has failed
8311	TMC_DRV2_UPDATE_ERROR	Update or verify of Technosoft drive software has failed
8312	TMC_DRV3_UPDATE_ERROR	Update or verify of Technosoft drive software has failed
8313	TMC_DRV4_UPDATE_ERROR	Update or verify of Technosoft drive software has failed
8314	TMC_DRV7_UPDATE_ERROR	Update or verify of Technosoft drive software has failed
8315	TMC_DRV1_INVALID_SETUP	Technosoft drive with axis ID 1 has invalid setup data (bit MER.2 is set or gtSetupTableCurrent.tMotorSetup[0] is invalid)
8316	TMC_DRV2_INVALID_SETUP	Technosoft drive with axis ID 2 has invalid setup data (bit MER.2 is set or gtSetupTableCurrent.tMotorSetup[1] is invalid)
8317	TMC_DRV3_INVALID_SETUP	Technosoft drive with axis ID 3 has invalid setup data (bit MER.2 is set or gtSetupTableCurrent.tMotorSetup[2] is invalid)
8318	TMC_DRV4_INVALID_SETUP	Technosoft drive with axis ID 4 has invalid setup data (bit MER.2 is set or gtSetupTableCurrent.tMotorSetup[3] is invalid)
8319	TMC_DRV7_INVALID_SETUP	Technosoft drive with axis ID 7 has invalid setup data (bit MER.2 is set or gtSetupTableCurrent.tMotorSetup[4] is invalid)
831A	TMC_DRV1_FAULT_SET	Technosoft drive with axis ID 1 has Fault (bit SRH.15)
831B	TMC_DRV2_FAULT_SET	Technosoft drive with axis ID 2 has Fault (bit SRH.15)

831C	TMC_DRV3_FAULT_SET	Technosoft drive with axis ID 3 has Fault (bit SRH.15)
831D	TMC_DRV4_FAULT_SET	Technosoft drive with axis ID 4 has Fault (bit SRH.15)
831E	TMC_DRV7_FAULT_SET	Technosoft drive with axis ID 7 has Fault (bit SRH.15)
831F	TMC_DRV1_FAULT_SHORT_CIRCUIT	Technosoft drive with axis ID 1 has Fault-status "Short-circuit" (bit MER.1)
8320	TMC_DRV2_FAULT_SHORT_CIRCUIT	Technosoft drive with axis ID 2 has Fault-status "Short-circuit" (bit MER.1)
8321	TMC_DRV3_FAULT_SHORT_CIRCUIT	Technosoft drive with axis ID 3 has Fault-status "Short-circuit" (bit MER.1)
8322	TMC_DRV4_FAULT_SHORT_CIRCUIT	Technosoft drive with axis ID 4 has Fault-status "Short-circuit" (bit MER.1)
8323	TMC_DRV7_FAULT_SHORT_CIRCUIT	Technosoft drive with axis ID 7 has Fault-status "Short-circuit" (bit MER.1)
8324	TMC_DRV1_FAULT_CONTROL_ERROR	Technosoft drive with axis ID 1 has Fault-status "Control error" (bit MER.3)
8325	TMC_DRV2_FAULT_CONTROL_ERROR	Technosoft drive with axis ID 2 has Fault-status "Control error" (bit MER.3)
8326	TMC_DRV3_FAULT_CONTROL_ERROR	Technosoft drive with axis ID 3 has Fault-status "Control error" (bit MER.3)
8327	TMC_DRV4_FAULT_CONTROL_ERROR	Technosoft drive with axis ID 4 has Fault-status "Control error" (bit MER.3)
8328	TMC_DRV7_FAULT_CONTROL_ERROR	Technosoft drive with axis ID 7 has Fault-status "Control error" (bit MER.3)
8329	TMC_DRV1_FAULT_POS_WRAPAROUND	Technosoft drive with axis ID 1 has Fault-status "Position wraparound" (bit MER.5)
832A	TMC_DRV2_FAULT_POS_WRAPAROUND	Technosoft drive with axis ID 2 has Fault-status "Position wraparound" (bit MER.5)
832B	TMC_DRV3_FAULT_POS_WRAPAROUND	Technosoft drive with axis ID 3 has Fault-status "Position wraparound" (bit MER.5)
832C	TMC_DRV4_FAULT_POS_WRAPAROUND	Technosoft drive with axis ID 4 has Fault-status "Position wraparound" (bit MER.5)
832D	TMC_DRV7_FAULT_POS_WRAPAROUND	Technosoft drive with axis ID 7 has Fault-status "Position wraparound" (bit MER.5)
832E	TMC_DRV1_FAULT_LSP_LSN_ACTIVE	Technosoft drive with axis ID 1 has Fault-status "Limit switches LSP and LSN both active" (bits MER.6)
832F	TMC_DRV2_FAULT_LSP_LSN_ACTIVE	Technosoft drive with axis ID 2 has Fault-status "Limit switches LSP and LSN both active" (bits MER.6)
8330	TMC_DRV3_FAULT_LSP_LSN_ACTIVE	Technosoft drive with axis ID 3 has Fault-status "Limit switches LSP and LSN both active" (bits MER.6)

8331	TMC_DRV4_FAULT_LSP_LSN_ACTIVE	Technosoft drive with axis ID 4 has Fault-status "Limit switches LSP and LSN both active" (bits MER.6
8332	TMC_DRV7_FAULT_LSP_LSN_ACTIVE	Technosoft drive with axis ID 7 has Fault-status "Limit switches LSP and LSN both active" (bits MER.6
8333	TMC_DRV1_FAULT_OVER_CURRENT	Technosoft drive with axis ID 1 has Fault-status "Over current" (bit MER.8)
8334	TMC_DRV2_FAULT_OVER_CURRENT	Technosoft drive with axis ID 2 has Fault-status "Over current" (bit MER.8)
8335	TMC_DRV3_FAULT_OVER_CURRENT	Technosoft drive with axis ID 3 has Fault-status "Over current" (bit MER.8)
8336	TMC_DRV4_FAULT_OVER_CURRENT	Technosoft drive with axis ID 4 has Fault-status "Over current" (bit MER.8)
8337	TMC_DRV7_FAULT_OVER_CURRENT	Technosoft drive with axis ID 7 has Fault-status "Over current" (bit MER.8)
8338	TMC_DRV1_FAULT_I2T	Technosoft drive with axis ID 1 has Fault-status "I2t" (bit MER.9)
8339	TMC_DRV2_FAULT_I2T	Technosoft drive with axis ID 2 has Fault-status "I2t" (bit MER.9)
833A	TMC_DRV3_FAULT_I2T	Technosoft drive with axis ID 3 has Fault-status "I2t" (bit MER.9)
833B	TMC_DRV4_FAULT_I2T	Technosoft drive with axis ID 4 has Fault-status "I2t" (bit MER.9)
833C	TMC_DRV7_FAULT_I2T	Technosoft drive with axis ID 7 has Fault-status "I2t" (bit MER.9)
833D	TMC_DRV1_FAULT_OVERTEMP_MOTOR	Technosoft drive with axis ID 1 has Fault-status "Over temperature motor" (bit MER.10)
833E	TMC_DRV2_FAULT_OVERTEMP_MOTOR	Technosoft drive with axis ID 2 has Fault-status "Over temperature motor" (bit MER.10)
833F	TMC_DRV3_FAULT_OVERTEMP_MOTOR	Technosoft drive with axis ID 3 has Fault-status "Over temperature motor" (bit MER.10)
8340	TMC_DRV4_FAULT_OVERTEMP_MOTOR	Technosoft drive with axis ID 4 has Fault-status "Over temperature motor" (bit MER.10)
8341	TMC_DRV7_FAULT_OVERTEMP_MOTOR	Technosoft drive with axis ID 7 has Fault-status "Over temperature motor" (bit MER.10)
8342	TMC_DRV1_FAULT_OVERTEMP_DRIVE	Technosoft drive with axis ID 1 has Fault-status "Over temperature drive" (bit MER.11)
8343	TMC_DRV2_FAULT_OVERTEMP_DRIVE	Technosoft drive with axis ID 2 has Fault-status "Over temperature drive" (bit MER.11)

8344	TMC_DRV3_FAULT_OVERTEMP_DRIVE	Technosoft drive with axis ID 3 has Fault-status "Over temperature drive" (bit MER.11)
8345	TMC_DRV4_FAULT_OVERTEMP_DRIVE	Technosoft drive with axis ID 4 has Fault-status "Over temperature drive" (bit MER.11)
8346	TMC_DRV7_FAULT_OVERTEMP_DRIVE	Technosoft drive with axis ID 7 has Fault-status "Over temperature drive" (bit MER.11)
8347	TMC_DRV1_FAULT_OVER_VOLTAGE	Technosoft drive with axis ID 1 has Fault-status "Over voltage" (bit MER.12)
8348	TMC_DRV2_FAULT_OVER_VOLTAGE	Technosoft drive with axis ID 2 has Fault-status "Over voltage" (bit MER.12)
8349	TMC_DRV3_FAULT_OVER_VOLTAGE	Technosoft drive with axis ID 3 has Fault-status "Over voltage" (bit MER.12)
834A	TMC_DRV4_FAULT_OVER_VOLTAGE	Technosoft drive with axis ID 4 has Fault-status "Over voltage" (bit MER.12)
834B	TMC_DRV7_FAULT_OVER_VOLTAGE	Technosoft drive with axis ID 7 has Fault-status "Over voltage" (bit MER.12)
834C	TMC_DRV1_FAULT_UNDER_VOLTAGE	Technosoft drive with axis ID 1 has Fault-status "Under voltage" (bit MER.13)
834D	TMC_DRV2_FAULT_UNDER_VOLTAGE	Technosoft drive with axis ID 2 has Fault-status "Under voltage" (bit MER.13)
834E	TMC_DRV3_FAULT_UNDER_VOLTAGE	Technosoft drive with axis ID 3 has Fault-status "Under voltage" (bit MER.13)
834F	TMC_DRV4_FAULT_UNDER_VOLTAGE	Technosoft drive with axis ID 4 has Fault-status "Under voltage" (bit MER.13)
8350	TMC_DRV7_FAULT_UNDER_VOLTAGE	Technosoft drive with axis ID 7 has Fault-status "Under voltage" (bit MER.13)
8351	TMC_DRV1_FAULT_COMMAND_ERROR	Technosoft drive with axis ID 1 has Fault-status "Command error" (bit MER.14)
8352	TMC_DRV2_FAULT_COMMAND_ERROR	Technosoft drive with axis ID 2 has Fault-status "Command error" (bit MER.14)
8353	TMC_DRV3_FAULT_COMMAND_ERROR	Technosoft drive with axis ID 3 has Fault-status "Command error" (bit MER.14)
8354	TMC_DRV4_FAULT_COMMAND_ERROR	Technosoft drive with axis ID 4 has Fault-status "Command error" (bit MER.14)
8355	TMC_DRV7_FAULT_COMMAND_ERROR	Technosoft drive with axis ID 7 has Fault-status "Command error" (bit MER.14)
8356	TMC_DRV1_FAULT_ENABLE_INACTIVE	Technosoft drive with axis ID 1 has Fault-status "Enable input inactive" (bit MER.15)
8357	TMC_DRV2_FAULT_ENABLE_INACTIVE	Technosoft drive with axis ID 2 has Fault-status "Enable input inactive" (bit MER.15)
8358	TMC_DRV3_FAULT_ENABLE_INACTIVE	Technosoft drive with axis ID 3 has Fault-status "Enable input inactive" (bit MER.15)
8359	TMC_DRV4_FAULT_ENABLE_INACTIVE	Technosoft drive with axis ID 4 has Fault-status "Enable input inactive" (bit MER.15)
835A	TMC_DRV7_FAULT_ENABLE_INACTIVE	Technosoft drive with axis ID 7 has Fault-status "Enable input inactive" (bit MER.15)

835B	TMC_DRV1_HOMING_ERROR	Homing of motor with axis ID 1 has failed (TC1 Actuation)
835C	TMC_DRV2_HOMING_ERROR	Homing of motor with axis ID 2 has failed (TC2 Actuation)
835D	TMC_DRV3_HOMING_ERROR	Homing of motor with axis ID 3 has failed (Centering and Clamping)
835E	TMC_DRV4_INIT_ERROR	Initialization of motor with axis ID 4 has failed (Bead beater)
835F	TMC_DRV7_HOMING_ERROR	Homing of motor with axis ID 7 has failed (Cartridge Lift)
8360	TMC_DRV1_TIMEOUT_ERROR	Movement timeout of motor with axis ID 1 (TC1 Actuation)
8361	TMC_DRV2_TIMEOUT_ERROR	Movement timeout of motor with axis ID 2 (TC2 Actuation)
8362	TMC_DRV3_TIMEOUT_ERROR	Movement timeout of motor with axis ID 3 (Centering and Clamping)
8363	TMC_DRV4_TIMEOUT_ERROR	Movement timeout of motor with axis ID 4 (Bead beater)
8364	TMC_DRV7_TIMEOUT_ERROR	Movement timeout of motor with axis ID 7 (Cartridge Lift)
8365	TMC_DRV1_MOVE_PROTECTION	Move protection for motor with axis ID 1 (TC1 Actuation)
8366	TMC_DRV2_MOVE_PROTECTION	Move protection for motor with axis ID 2 (TC2 Actuation)
8367	TMC_DRV3_MOVE_PROTECTION	Move protection for motor with axis ID 3 (Centering and Clamping)
8368	TMC_DRV4_MOVE_PROTECTION	Move protection for motor with axis ID 4 (Bead beater)
8369	TMC_DRV7_MOVE_PROTECTION	Move protection for motor with axis ID 7 (Cartridge Lift)
836A	TMC_DRV1_MULTIPLE_CALLS	Multiple cancelable calls: "Command error" + "CALLS warning" on drive with axis ID 1
836B	TMC_DRV2_MULTIPLE_CALLS	Multiple cancelable calls: "Command error" + "CALLS warning" on drive with axis ID 2
836C	TMC_DRV3_MULTIPLE_CALLS	Multiple cancelable calls: "Command error" + "CALLS warning" on drive with axis ID 3
836D	TMC_DRV4_MULTIPLE_CALLS	Multiple cancelable calls: "Command error" + "CALLS warning" on drive with axis ID 4
836E	TMC_DRV7_MULTIPLE_CALLS	Multiple cancelable calls: "Command error" + "CALLS warning" on drive with axis ID 7
836F	TMC_DRV1_UNEXPECTED_RESET	Unexpected reset of drive with axis ID 1
8370	TMC_DRV2_UNEXPECTED_RESET	Unexpected reset of drive with axis ID 2
8371	TMC_DRV3_UNEXPECTED_RESET	Unexpected reset of drive with axis ID 3
8372	TMC_DRV4_UNEXPECTED_RESET	Unexpected reset of drive with axis ID 4
8373	TMC_DRV7_UNEXPECTED_RESET	Unexpected reset of drive with axis ID 7
8374	TMC_DRV1_UNEXPECTED_STATE	Unexpected state (e.g. axis off

8375	TMC_DRV2_UNEXPECTED_STATE	Unexpected state (e.g. axis off)
8376	TMC_DRV3_UNEXPECTED_STATE	Unexpected state (e.g. axis off)
8377	TMC_DRV4_UNEXPECTED_STATE	Unexpected state (e.g. axis off)
8378	TMC_DRV7_UNEXPECTED_STATE	Unexpected state (e.g. axis off)
8379	TMC_DRV1_GUARDING_ERROR	Guarding error of drive with axis ID 1 (timeout of motor position frames)
837A	TMC_DRV2_GUARDING_ERROR	Guarding error of drive with axis ID 2 (timeout of motor position frames)
837B	TMC_DRV3_GUARDING_ERROR	Guarding error of drive with axis ID 3 (timeout of motor position frames)
837C	TMC_DRV4_GUARDING_ERROR	Guarding error of drive with axis ID 4 (timeout of motor position frames)
837D	TMC_DRV7_GUARDING_ERROR	Guarding error of drive with axis ID 7 (timeout of motor position frames)
837E	TMC_DRV1_LIMIT_SWITCH_ERROR	Limit switch error of drive with axis ID 1 (detected in homing)
837F	TMC_DRV2_LIMIT_SWITCH_ERROR	Limit switch error of drive with axis ID 2 (detected in homing)
8380	TMC_DRV3_LIMIT_SWITCH_ERROR	Limit switch error of drive with axis ID 3 (detected in homing)
8381	TMC_DRV4_LIMIT_SWITCH_ERROR	Limit switch error of drive with axis ID 4 (detected in homing)
8382	TMC_DRV7_LIMIT_SWITCH_ERROR	Limit switch error of drive with axis ID 7 (detected in homing)
0x83FF	TMC_ERROR_MAX_ERROR	
8400	MCC_CALC_DIV_BY_ZERO	Division by zero error in calculation with operators DIV or MOD
8401	MCC_INVALID_ARITHMETIC	Invalid arithmetic operation in calculation with operators ADD
8402	MCC_CALC_OVERFLOW	Arithmetic overflow in calculation with operators NEG
8403	MCC_STACK_OVERFLOW	Overflow of calculation stack (too many values on stack)
8404	MCC_STACK_UNDERFLOW	Underflow of calculation stack (not enough values on stack)
8405	MCC_INVALID_VARIABLE_INDEX	Variable index is out of range
8406	MCC_INVALID_VARIABLE_SIZE	Variable size is out of range
8407	MCC_INVALID_ARRAY_INDEX	Array index is out of range
8408	MCC_VARTYPE_CONFLICT	Conflicting data types in variable access (data types do not match)
8409	MCC_STACK_NOTEMPTY	Calculation stack is not empty at begin of a calculation
840A	MCC_AAF_TAG_MISMATCH	Mismatch of xml start-tag and end-tag in AAF
840B	MCC_AAF_UNKNOWN_TAG	Unknown xml start-tag or single-tag in AAF
840C	MCC_AAF_UNKNOWN_OPERATOR	Unknown calculation operator in AAF
8410	MCC_CLAMPING_ERROR	Cartridge/TC clamping has failed

8411	MCC_CLAMPING_ERROR_TC1	Cartridge/TC clamping has failed (TC1)
8412	MCC_CLAMPING_ERROR_TC2	Cartridge/TC clamping has failed (TC2)
8413	MCC_CLAMPING_ERROR_CC	Cartridge/TC clamping has failed (CC)
8414	MCC_CLAMPING_ERROR_BB	Cartridge/TC clamping has failed (BB)
8417	MCC_CLAMPING_ERROR_LL	Cartridge/TC clamping has failed (Lift/Lid)
8418	MCC_CLAMPING_ERROR_TUC	Cartridge/TC clamping has failed (TUC)
0x841F	MCC_ERROR_MAX_ERROR	
8420	TU_ERROR_NO_RESPONSE	No response from Thermal Unit has been received
8421	TU_ERROR_UNDERVOLTAGE	Measured supply voltage is too low
8422	TU_ERROR_OVERVOLTAGE	Measured supply voltage is too high
8423	TU_ERROR_HARDWARE	Invalid hardware version has been detected
8424	TU_ERROR_FIRMWARE	Invalid firmware version has been detected
8425	TU_ERROR_INVALID_RESPONSE	Invalid command response received from Thermal Unit
8426	TU_ERROR_DATA_TIMEOUT	No or insufficient data received in low latency mode (TU should send continuously)
8427	TU_ERROR_UNCALLED_DATA	Data received without request
8428	TU_ERROR_MONITOR_CARTRIDGE	Cartridge monitoring was not enabled properly
8429	TU_ERROR_TEMP_VALUE_INVALID	Measured temperature values are invalid (not plausible)
842A	TU_ERROR_NOT_AVAILABLE	Semaphore is blocked by another task
842B	TU_ERROR_UNKNWON_CODE	TU reports an unknown (unspecified)
842C	TU_ERROR_SEND_FAILED	Failed to send data to TU
842D	TU_ERROR_RESPONSE_TOO_SHORT	TU response is too short
842E	TU_ERROR_DATA_INVALID	TU received data are invalid (not plausible)
842F	TU_ERROR_SERNO_INVALID	TU reported serial number is invalid
8430	TU_ERROR_SVC_01	TU Error of supervisor controller: CAN receive FIFO overflow error
8431	TU_ERROR_SVC_02	TU Error of supervisor controller: CAN message transmission error
8432	TU_ERROR_SVC_03	TU Error of supervisor controller: CAN transmit mailbox error
8433	TU_ERROR_SVC_04	TU Error of supervisor controller: CAN circular command buffer overflow error
8434	TU_ERROR_SVC_05	TU Error of supervisor controller: UART circular command buffer overflow error
8435	TU_ERROR_SVC_06	TU Error of supervisor controller: Command length mismatch (e.g. due to missing data on UART interface)
8436	TU_ERROR_SVC_07	TU Error of supervisor controller: Power fail signal connection to power supply is missing or latched down (checked only once after reset)

8437	TU_ERROR_SVC_08	TU Error of supervisor controller: Temperature sensor could not be accessed
8438	TU_ERROR_SVC_09	TU Error of supervisor controller: I2C EEPROM access error
8439	TU_ERROR_SVC_0A	TU Error of supervisor controller: The last reset was caused by a watchdog counter overflow
843A	TU_ERROR_SVC_0B	TU Error of supervisor controller: Send value is out of range
843B	TU_ERROR_SVC_0C	TU Error of supervisor controller: Invalid Calibration data
843C	TU_ERROR_SVC_0D	TU Error of supervisor controller: Command length mismatch (e.g. due to missing data on CAN interface)
843D	TU_ERROR_SVC_80	TU Error of supervisor controller: No supported hardware revision detected
843E	TU_ERROR_SVC_81	TU Error of supervisor controller: Temperature sensor reports an over-temperature error of the block
843F	TU_ERROR_SVC_82	TU Error of supervisor controller: No communication to peltier frame board of left block established
8440	TU_ERROR_SVC_83	TU Error of supervisor controller: No communication to peltier frame board of right block established
8441	TU_ERROR_SVC_84	TU Error of supervisor controller: No communication to power board of left block established
8442	TU_ERROR_SVC_85	TU Error of supervisor controller: No communication to power board of right block established
8443	TU_ERROR_SVC_86	TU Error of supervisor controller: Heat sink cooling fan of left block is blocked
8444	TU_ERROR_SVC_87	TU Error of supervisor controller: Heat sink cooling fan of right block is blocked
8445	TU_ERROR_SVC_88	TU Error of supervisor controller: Power supply reports power fail
8446	TU_ERROR_SVC_89	TU Error of supervisor controller: Start loop with no End
8447	TU_ERROR_SVC_8A	TU Error of supervisor controller: End loop with no Start
8448	TU_ERROR_SVC_8B	TU Error of supervisor controller: Nested loop
8449	TU_ERROR_SVC_8C	TU Error of supervisor controller: No command found
844A	TU_ERROR_SVC_8D	TU Error of supervisor controller: I2C recovery error

844B	TU_ERROR_SVC_8E	TU Error of supervisor controller: Error in the temperature measurement
844C	TU_ERROR_SVC_8F	TU Error of supervisor controller: Software over-temperature detected
844D	TU_ERROR_SVC_90	TU Error of supervisor controller: UART CRC error detected
844E	TU_ERROR_SVC_93	TU Error of supervisor controller: No valid operating points data available
844F	TU_ERROR_SVC_94	TU Error of supervisor controller: UART parameter mismatch
8450	TU_ERROR_PFB_L_01	TU Error of left peltier frameboard: CAN receive FIFO overflow error
8451	TU_ERROR_PFB_L_02	TU Error of left peltier frameboard: CAN message transmission error
8452	TU_ERROR_PFB_L_03	TU Error of left peltier frameboard: CAN transmit mailbox error
8453	TU_ERROR_PFB_L_04	TU Error of left peltier frameboard: Circular command buffer overflow
8454	TU_ERROR_PFB_L_05	TU Error of left peltier frameboard: Power fail signal connection to power supply is missing or latched down (checked only once after reset)
8455	TU_ERROR_PFB_L_06	TU Error of left peltier frameboard: I2C EEPROM access error
8456	TU_ERROR_PFB_L_07	TU Error of left peltier frameboard: I2C recovery error
8457	TU_ERROR_PFB_L_08	TU Error of left peltier frameboard: The last reset was caused by a watchdog counter overflow
8458	TU_ERROR_PFB_L_80	TU Error of left peltier frameboard: No supported hardware revision detected
8459	TU_ERROR_PFB_L_81	TU Error of left peltier frameboard: Temperature sensor reports an over-temperature error
845A	TU_ERROR_PFB_L_82	TU Error of left peltier frameboard: One or more calibration values are out of range
845B	TU_ERROR_PFB_L_93	TU Error of left peltier frameboard: No valid operating points data available
8460	TU_ERROR_PFB_R_01	TU Error of right peltier frameboard: CAN receive FIFO overflow error
8461	TU_ERROR_PFB_R_02	TU Error of right peltier frameboard: CAN message transmission error
8462	TU_ERROR_PFB_R_03	TU Error of right peltier frameboard: CAN transmit mailbox error
8463	TU_ERROR_PFB_R_04	TU Error of right peltier frameboard: Circular command buffer overflow

8464	TU_ERROR_PFBR_05	TU Error of right peltier frameboard: Power fail signal connection to power supply is missing or latched down (checked only once after reset)
8465	TU_ERROR_PFBR_06	TU Error of right peltier frameboard: I2C EEPROM access error
8466	TU_ERROR_PFBR_07	TU Error of right peltier frameboard: I2C recovery error
8467	TU_ERROR_PFBR_08	TU Error of right peltier frameboard: The last reset was caused by a watchdog counter overflow
8468	TU_ERROR_PFBR_80	TU Error of right peltier frameboard: No supported hardware revision detected
8469	TU_ERROR_PFBR_81	TU Error of right peltier frameboard: Temperature sensor reports an over-temperature error
846A	TU_ERROR_PFBR_82	TU Error of right peltier frameboard: One or more calibration values are out of range
8470	TU_ERROR_PWRL_01	TU Error of left power module: CAN receive FIFO overflow error
8471	TU_ERROR_PWRL_02	TU Error of left power module: Command buffer overflow warning
8472	TU_ERROR_PWRL_03	TU Error of left power module: Low stack memory warning
8473	TU_ERROR_PWRL_10	TU Error of left power module: Power-board temperature warning
8474	TU_ERROR_PWRL_28	TU Error of left power module: Power output not stable warning
8475	TU_ERROR_PWRL_80	TU Error of left power module: Independent watchdog error
8476	TU_ERROR_PWRL_81	TU Error of left power module: I2C recovery error
8477	TU_ERROR_PWRL_90	TU Error of left power module: Unsupported board revision error
8478	TU_ERROR_PWRL_91	TU Error of left power module: Flash memory CRC checksum error
8479	TU_ERROR_PWRL_B0	TU Error of left power module: Power-board over-temperature error
847A	TU_ERROR_PWRL_B1	TU Error of left power module: Frameboard over-temperature error
847B	TU_ERROR_PWRL_B2	TU Error of left power module: Temperature sensor error
847C	TU_ERROR_PWRL_C8	TU Error of left power module: Power stage error
8480	TU_ERROR_PWRR_01	TU Error of right power module: CAN receive FIFO overflow error

8481	TU_ERROR_PWRR_02	TU Error of right power module: Command buffer overflow warning
8482	TU_ERROR_PWRR_03	TU Error of right power module: Low stack memory warning
8483	TU_ERROR_PWRR_10	TU Error of right power module: Power-board temperature warning
8484	TU_ERROR_PWRR_28	TU Error of right power module: Power output not stable warning
8485	TU_ERROR_PWRR_80	TU Error of right power module: Independent watchdog error
8486	TU_ERROR_PWRR_81	TU Error of right power module: I2C recovery error
8487	TU_ERROR_PWRR_90	TU Error of right power module: Unsupported board revision error
8488	TU_ERROR_PWRR_91	TU Error of right power module: Flash memory CRC checksum error
8489	TU_ERROR_PWRR_B0	TU Error of right power module: Power-board over-temperature error
848A	TU_ERROR_PWRR_B1	TU Error of right power module: Frameboard over-temperature error
848B	TU_ERROR_PWRR_B2	TU Error of right power module: Temperature sensor error
848C	TU_ERROR_PWRR_C8	TU Error of right power module: Power stage error
8490	TU_ERROR_GET_SVCGT	No response to command SVCGT : Get 8 actual temperatures of cycling zones
8491	TU_ERROR_GET_SVCGA	No response to command SVCGA : Get ambient temperature
8492	TU_ERROR_GET_SVCGH	No response to command SVCGH : Get heat sink temperature (thermal cycling)
8493	TU_ERROR_GET_SVCRP	No response to command SVCRP : Get ramp rate
8494	TU_ERROR_GET_SVCGC	No response to command SVCGC : Get calibration values
8495	TU_ERROR_GET_SVCGX	No response to command SVCGX : Get 5 heater temperatures
8496	TU_ERROR_GET_SVCBD	No response to command SVCBD : Get PWM duty cycle of beat beater heater
8497	TU_ERROR_GET_SVCGL	No response to command SVCGL : Get liquid sensor raw data
8498	TU_ERROR_GET_SVCSL	No response to command SVCSL : Get liquid sensor currents
8499	TU_ERROR_GET_SVCSO	No response to command SVCSO : Get liquid sensor offset compensation
849A	TU_ERROR_GET_SVCRD	No response to command SVCRD : Get delay time between LED toggle and measurement

849B	TU_ERROR_GET_SVCGK	No response to command SVC GK : Get ADC sampling time used for liquid sensors measurement
849C	TU_ERROR_GET_SVCRS	No response to command SVC RS : Get liquid sensor calibration values
849D	TU_ERROR_GET_SVCGE	No response to command SVC GE : Get error codes
849E	TU_ERROR_GET_SVCGS	No response to command SVC GS : Get device state
849F	TU_ERROR_GET_SVCGB	No response to command SVC GB : Get block information: ID
84A0	TU_ERROR_GET_SVCGF	No response to command SVC GF : Get flash CRC checksums
84A1	TU_ERROR_GET_SVCGN	No response to command SVC GN : Get serial number
84A2	TU_ERROR_GET_SVCGO	No response to command SVC GO : Get output power values
84A3	TU_ERROR_GET_SVCGP	No response to command SVC GP : Get power supply voltage and actual peltier current values
84A4	TU_ERROR_GET_SVCGV	No response to command SVC GV : Get hardware and firmware versions
84A5	TU_ERROR_GET_SVCGW	No response to command SVC GW : Get bootloader versions
84A6	TU_ERROR_GET_SVCP R	No response to command SVC RP : Get progress values of operating points determination process
84B0	TU_ERROR_SET_SVCPE	No response to command SVC PE : Set power enable of thermal cycling
84B1	TU_ERROR_SET_SVCST	No response to command SVC ST : Set nominal temperature of cycling zones
84B2	TU_ERROR_SET_SVCRP	No response to command SVC RP : Set ramp rate
84B3	TU_ERROR_SET_SVCWC	No response to command SVC WC : Write calibration values to EEPROM
84B4	TU_ERROR_SET_SVCHE	No response to command SVC HE : Set heater enable
84B5	TU_ERROR_SET_SVCSX	No response to command SVC SX : Set 3 heater temperatures
84B6	TU_ERROR_SET_SVCBD	No response to command SVC BD : Set PWM duty cycle of beat beater heater
84B7	TU_ERROR_SET_SVCSL	No response to command SVC SL : Set liquid sensor currents
84B8	TU_ERROR_SET_SVCSO	No response to command SVC SO : Set liquid sensor offset compensation

84B9	TU_ERROR_SET_SVCLL	No response to command SVCLL : Start low latency mode
84BA	TU_ERROR_SET_SVCBP	No response to command SVCBP : Set liquid sensing bandpass / lowpass configuration
84BB	TU_ERROR_SET_SVCLD	No response to command SVCLD : Set delay time between LED toggle and measurement
84BC	TU_ERROR_SET_SVCSK	No response to command SVCSK : Set ADC sampling time used for liquid sensors measurement
84BD	TU_ERROR_SET_SVCWS	No response to command SVCWS : Write liquid sensor calibration values
84BE	TU_ERROR_SET_SVCSM	No response to command SVCSM : Set standard mode (end low latency mode)
84BF	TU_ERROR_SET_SVCFE	No response to command SVCFE : Set fan enable
84C0	TU_ERROR_SET_SVCCE	No response to command SVCCE : Clear error codes
84C1	TU_ERROR_SET_SVCSB	No response to command SVCSB : Set block information
84C2	TU_ERROR_SET_SVCSN	No response to command SVCSN : Write serial number to EEPROM
84C3	TU_ERROR_SET_SVCSP	No response to command SVCSP : Set power values
84C4	TU_ERROR_SET_SVCSR	No response to command SVCSR : Software reset
84C5	TU_ERROR_SET_SVCEB	No response to command SVCEB : Enter bootloader mode
84C6	TU_ERROR_SET_SVCEC	No response to command SVCEC : Enable communication with CRC
84C7	TU_ERROR_SET_SVCBO	No response to command SVCBO : Begin operating points determination process
84C8	TU_ERROR_SET_SVCEO	No response to command SVCEO : Abort operating points determination process
84D0	TU_ERROR_UPDATE_COMM	TU is in invalid state: STM bootloader is active
84D1	TU_ERROR_UPDATE_CT_INVALID	TU is in invalid state: STM bootloader is active
84D2	TU_ERROR_UPDATE_SVC	SVC firmware is invalid
84D3	TU_ERROR_UPDATE_PWR	PWR firmware is invalid
84D4	TU_ERROR_UPDATE_PFB	PFB firmware is invalid
#####	TU_ERROR_TEMP_CYCLER_LOW	Measured cycler temperature is too low (< 0 C)
#####	TU_ERROR_TEMP_CYCLER_HIGH	Measured cycler temperature is too high (> 110 C)
#####	TU_ERROR_TEMP_AMBIENT_LOW	Measured ambient temperature is too low (< 0 C)
#####	TU_ERROR_TEMP_AMBIENT_HIGH	Measured ambient temperature is too high (> 110 C)

#####	TU_ERROR_TEMP_HEATSINK_LOW	Measured heatsink temperature is too low (< 20 C)
#####	TU_ERROR_TEMP_HEATSINK_HIGH	Measured heatsink temperature is too high (> 110 C)
#####	TU_ERROR_TEMP_HEATER_LOW	Measured heater temperature is too low (< 20 C)
#####	TU_ERROR_TEMP_HEATER_HIGH	Measured heater temperature is too high (> 120 C)
#####	TU_ERROR_POWER_ON	Thermal unit cannot be switched on
#####	TU_ERROR_SVC_95	TU Error of supervisor controller: Liquid sensing initialization failed (>)
84EA	TU_ERROR_WRONG_CHECKSUM_RECEIVED	Wrong checksum received in response from TU
84EB	TU_ERROR_NAK_RECEIVED	TU has responded with negative acknowledge to a (hopefully
0x84FF	TU_ERROR_MAX_ERROR	
0x8501	DB_ERROR_DEVICE_TABLE	
0x8502	DB_ERROR_SETUP_TABLE	
0x8504	DB_ERROR_ID_TABLE	
0x8508	DB_ERROR_SC_TABLE	
0x8510	DB_ERROR_CALIB_TABLE	
0x8520	DB_ERROR_RESULT_TABLE	
0x8540	DB_ERROR_MECH_CALIB_TABLE	
0x8580	DB_ERROR_EEPROM_HW_VERS	
0x858f	DB_ERROR_MAX_ERROR	
8605	I2C_LPC4337__ERROR_OUT_OF_ORDER	general I2C device error LPC4337 I2C CPU error detected (requires immediately shutdown)
8606	I2C_TCA6424_1__ERROR_OUT_OF_ORDER	general I2C device error 24 Bit IO Expander device have failed in init read or write --> anyway device is no more working
8607	I2C_TCA6424_2__ERROR_OUT_OF_ORDER	general I2C device error 24 Bit IO Expander device have failed in init read or write --> anyway device is no more working
8608	I2C_PCA9534_1__ERROR_OUT_OF_ORDER	general I2C device error 8 Bit IO Expander device have failed in init read or write --> anyway device is no more working
8609	I2C_PCA9534_2__ERROR_OUT_OF_ORDER	general I2C device error 8 Bit IO Expander device have failed in init read or write --> anyway device is no more working
860A	I2C_LM73_1__ERROR_OUT_OF_ORDER	general I2C device error temperature sensor 1 type LM73 have failed in init read or write
860B	I2C_LM73_2__ERROR_OUT_OF_ORDER	general I2C device error temperature sensor 2 type LM73 have failed in init read or write

860C	I2C_ADT7420_1__ERROR_OUT_OF_ORDER	general I2C device error temperature sensor 1 type ADT7420 have failed in init read or write
860D	I2C_ADT7420_2__ERROR_OUT_OF_ORDER	general I2C device error temperature sensor 2 type ADT7420 have failed in init read or write
860E	I2C_ADS1115_1__ERROR_OUT_OF_ORDER	general I2C device error pressure sensor 1 type ADS1115 have failed in init read or write
860F	I2C_ADS1115_2__ERROR_OUT_OF_ORDER	general I2C device error pressure sensor 2 type ADS1115 have failed in init read or write
8610	I2C_LP5520__ERROR_OUT_OF_ORDER	general I2C device error LED backlight type LP5520 have failed in init read or write
8611	I2C_TCA6424_1__SOFTWARE_ERROR	Illegal parameters for access of IO expander 1 TCA6424 or I2C semaphore is blocked
8612	I2C_TCA6424_2__SOFTWARE_ERROR	Illegal parameters for access of IO expander 2 TCA6424 or I2C semaphore is blocked
8613	I2C_PCA9534_1__SOFTWARE_ERROR	Illegal parameters for access of IO expander 1 PCA9534 or I2C semaphore is blocked
8614	I2C_PCA9534_2__SOFTWARE_ERROR	Illegal parameters for access of IO expander 2 PCA9534 or I2C semaphore is blocked
8615	I2C_LM73_1__SOFTWARE_ERROR	Illegal parameters for access of temp. sensor 1 LM73-1 or I2C semaphore is blocked
8616	I2C_LM73_2__SOFTWARE_ERROR	Illegal parameters for access of temp. sensor 2 LM73-1 or I2C semaphore is blocked
8617	I2C_ADT7420_1__SOFTWARE_ERROR	Illegal parameters for access of temp. sensor 1 ADT7420 or I2C semaphore is blocked
8618	I2C_ADT7420_2__SOFTWARE_ERROR	Illegal parameters for access of temp. sensor 2 ADT7420 or I2C semaphore is blocked
8619	I2C_ADS1115_1__SOFTWARE_ERROR	Illegal parameters for access of AD converter 1 ADS1115 or I2C semaphore is blocked
861A	I2C_ADS1115_2__SOFTWARE_ERROR	Illegal parameters for access of AD converter 2 ADS1115 or I2C semaphore is blocked
861B	I2C_LP5520__SOFTWARE_ERROR	Illegal parameters for access of LED backlight LP5520 or I2C semaphore is blocked
86EF	I2C_ERROR_MAX_ERROR	
86F0	ACC_READ_ERROR	
86FF	ACC_ERROR_MAX_ERROR	
8700	TRF_ERROR_NO_RESPONSE	
8701	TRF_ERROR_INCOMPLETE_RESPONSE	
8702	TRF_ERROR_GENERAL	
8703	TRF_ERROR_WRONG_CRC_RECEIVED	
8704	TRF_ERROR_BUFFER_FULL_REPORTED	
8705	TRF_ERROR_NO_MEMORY_REPORTED	
8706	TRF_ERROR_TOO_MANY_DIGITS_REPORTED	
8707	TRF_ERROR_INVALID_HEX_DIGIT_REPORTED	
8708	TRF_ERROR_MISSING_PARAM_REPORTED	

8709	TRF_ERROR_UNDEF_TOKEN_REPORTED	
870A	TRF_ERROR_VALUE_TOO_LARGE_RECEIVED	
870B	TRF_ERROR_WRONG_CRC_REPORTED	
870C	TRF_ERROR_SET_CONFIGURATION_FAILED	
870D	TRF_ERROR_NOT_SPECIFIED_REPORTED	
877F	TRF_ERROR_MAX_ERROR	
8780	QPCR_ERROR_NO_RESPONSE	No response from qPCR module
8781	QPCR_ERROR_NO_RESPONSE_COM	No response from COM task
8782	QPCR_ERROR_INVALID_COMMAND	
8783	QPCR_ERROR_INVALID_FUNCTION	
8784	QPCR_ERROR_INVALID_PARAMETER	
8785	QPCR_ERROR_MOTOR_POSITIONING	
8786	QPCR_ERROR_VERIFY_MEMORY	
8787	QPCR_ERROR_MAX_CYCLES_REACHED	Maximum number of measurement cycles has been reached
8788	QPCR_ERROR_MEASUREMENT_MISMATCH	Number of measurements received from qPCR is not as expected
8789	QPCR_ERROR_CMD_READ_VERSION	
878A	QPCR_ERROR_CMD_READ_MEMORY	
878B	QPCR_ERROR_CMD_WRITE_MEMORY	
878C	QPCR_ERROR_CMD_WRITE_FLASH	
878D	QPCR_ERROR_CMD_CALL_FUNCTION	
878E	QPCR_ERROR_CMD_GET_BASE_ADDRESS	
878F	QPCR_ERROR_CMD_RESET	
8790	QPCR_ERROR_FCT_START_BOOTWARE	
8791	QPCR_ERROR_FCT_START_FIRMWARE	
8792	QPCR_ERROR_FCT_SWITCH_OFF	
8793	QPCR_ERROR_FCT_FLASH_ERASE	
8794	QPCR_ERROR_FCT_GET_FREE_HEAP_SIZE	
8795	QPCR_ERROR_FCT_TRACING	
8796	QPCR_ERROR_FCT_MEASUREMENT_INIT	
8797	QPCR_ERROR_FCT_MEASUREMENT_SINGLE	
8798	QPCR_ERROR_FCT_MEASUREMENT_SINGLE_WP	
8799	QPCR_ERROR_FCT_MEASUREMENT_TEMPERATURE	
879A	QPCR_ERROR_FCT_MEASUREMENT_CANCEL	
879B	QPCR_ERROR_FCT_EEPROM_STATUS	
879C	QPCR_ERROR_FCT_SET_LED_CURRENT	
879D	QPCR_ERROR_INVALID_STATE	
879E	QPCR_ERROR_EEPROM_CRC_WRONG	
879F	QPCR_ERROR_CHANNEL_MISMATCH	Channel number received from qPCR is not as expected
87FF	QPCR_ERROR_MAX_ERROR	
8800	LIGHT_BARRIER_ERROR	
8001	LIGHT_BARRIER_ERROR_TC1	
8802	LIGHT_BARRIER_ERROR_TC2_IN	

8803	LIGHT_BARRIER_ERROR_CC	
8804	LIGHT_BARRIER_ERROR_NOT_USED1	
8805	LIGHT_BARRIER_ERROR_QPCR_IN	
8806	LIGHT_BARRIER_ERROR_QPCR_OUT	
8807	LIGHT_BARRIER_ERROR_SYRINGE_OUT	
8808	LIGHT_BARRIER_ERROR_SYRINGE_IN	
8809	LIGHT_BARRIER_ERROR_LIFT_DOWN	
880A	LIGHT_BARRIER_ERROR_LIFT_UP	
880B	LIGHT_BARRIER_ERROR_TUC_OPEN	
880C	LIGHT_BARRIER_ERROR_NOT_USED2	
880D	LIGHT_BARRIER_ERROR_DUST_FILTER	
880E	LIGHT_BARRIER_ERROR_TC2_OUT	
881F	LIGHT_BARRIER_ERROR_MAX_ERROR	

Table 11.2-16 AM error codes.

12 Spare Parts

12.1 Spare Parts - General

SAP #	Name	Comment	Picture
9243305	Power Cord, India		
9232546	Power Cord, Swiss		
9232547	Power Cord, UK		
9232548	Power Cord, Australia		
9232568	Power Cord, USA		
9022569	Power Cord, European		

Table 12.1-1 Spare Parts - General

12.2 Spare Part List - Product Covers

SAP #	Name	Comment	Picture
9021835	Flex-Air Tube, AM, QSTAT	Flexible duct for the thermal unit	
9026379	Foil and Seal	Foam seal for the AM top cover and foil seals for the front panel	
9026534	Power Switch (Silver), OM, QSTAT	Switch Cover Plate	
9021836	Lid Sealing, QSTAT	Seal for the sliding door	
9026535	Topcover,(Silver) OM, QSTAT		
9026536	Side Cover, Right, OM QSTAT		
9026537	Side Cover Left, OM , QSTAT		

Table 12.2-1 QIAstat-Dx Product Covers Spare Parts List

SAP #	Name	Comment	Picture
9026538	Cover AM, QSTAT	Silver top Cover	
9026539		Side Cover Right, AM, QSTAT	
9026540		Side Cover Left, AM, QSTAT	

Table 12.2-1 QIAstat-Dx Product Covers Spare Parts List

12.3 Spare Parts List - Analytical Module

SAP #	Name	Comment	Picture
9026380	Springs, AM	Three springs on the thermal plates	
9026381	FFC 16pol, AM	Flex cable Auxiliary board	
9026382	Match Pins, AM	Metal key pins between Clamping Module and Thermal Module	

Table 12.3-1 QIAstat-Dx Analytical Module Spare Parts List

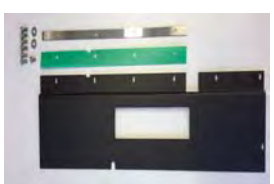
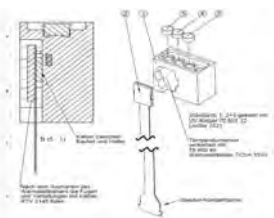
SAP #	Name	Comment	Picture
9026383	Tube-Diafragma-Pump-Manifold, AM, QSTAT		
9026384	Tube Silencer Out	Tube from pump to back panel	
9026385	Silencer Tube In	Tube from the bottom plate to the bac panel	
9026386	QPCR Complete	Not available as a spare part (Not available as a spare part)	
9026387	Syringe Module	Complete Syringe Module	
9026388	Lid Mechanism	Does Not include the door.	
9026474	TechnoSoft Driver, 2A, AM, QSTAT	2 Amp driver board	
9026475	TechnoSoft Driver 4A, AM, QSTAT	4 Amp driver board	
9026533	Internal qPCR Standard, AM, QSTAT		

Table 12.3-1 QIAstat-Dx Analytical Module Spare Parts List

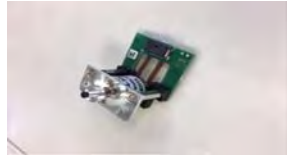
SAP #	Name	Comment	Picture
9026541	Mainboard assembly, AM, QSTAT		
9026542	Auxillary Board, AM, QSTAT		
9026543	Internal Barcode Reader, AM, QSTAT		
9026544	Voice coil, AM, QSTAT		
9026545	TC and Clamping Module, AM, QSTAT		
9026546	Thermal Unit Complete, AM, QSTAT		
9026547	Pressure Pump, AM, QSTAT		

Table 12.3-1 QIAstat-Dx Analytical Module Spare Parts List

SAP #	Name	Comment	Picture
9026548	Pressure Manifold, AM, QSTAT		
9026549	Tilt Sensor, AM, QSTAT		
9026550	Power Supply, AM, QSTAT		
9026551	Chassis Fan, AM, QSTAT		
9026552	Power Inlet, AM, QSTAT		
9026553	Power Switch, AM, QSTAT		

Table 12.3-1 QIAstat-Dx Analytical Module Spare Parts List

SAP #	Name	Comment	Picture
9026554	Light Barrier 1, AM, QSTAT		
9026555	Light Barrier 2 (For QPCR) , AM, QSTAT		
9026556	Motor TC1, AM, QSTAT		
9026557	Motor TC2, AM, QSTAT		
9026558	Motor CC1, AM, QSTAT		
9026559	Motor Lifter/Lid, AM, QSTAT		

Table 12.3-1 QIAstat-Dx Analytical Module Spare Parts List

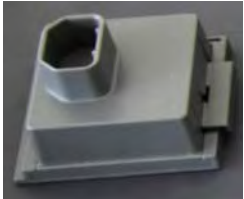
SAP #	Name	Comment	Picture
9026561	Cog Belt, AM, QSTAT		
9026562	Accessory Bag, AM, QSTAT		
9026743	Bridge OM-AM, OM, QSTAT		
9026744	Bridge AM-AM, AM, QSTAT		
9026745	Termination Bridge, AM, QSTAT		

Table 12.3-1 QIAstat-Dx Analytical Module Spare Parts List

12.4 Spare Parts List - Operational Module

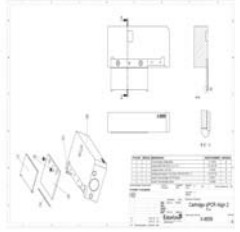
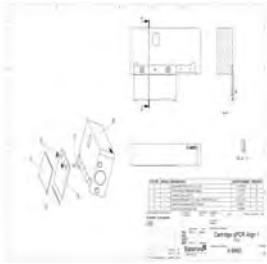
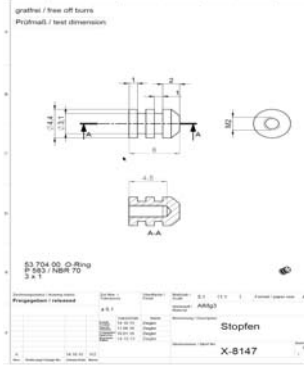
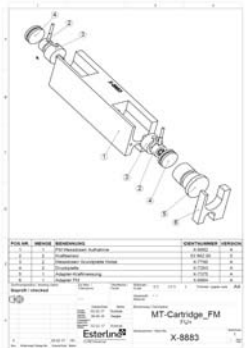
SAP #	Name	Comment	Picture
906389	Touch Screen	Complete Touch Screen Assembly without bar code reader	

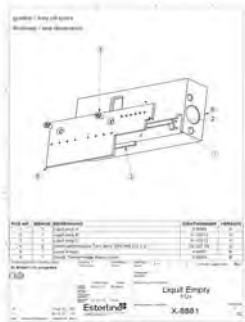
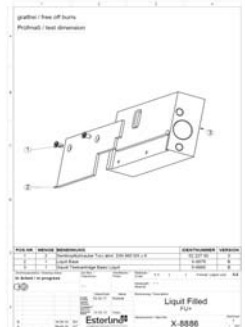
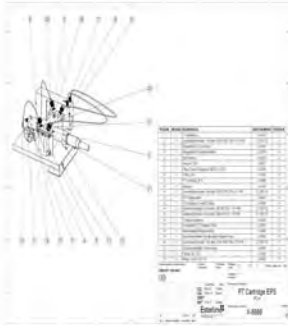
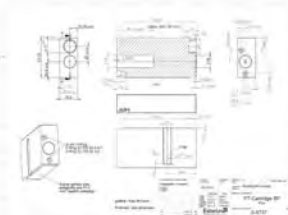
Table 12.4-1 QIAstat-Dx Operational Module Spare Parts List

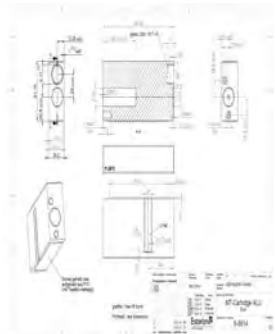
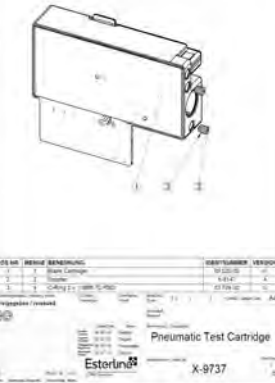
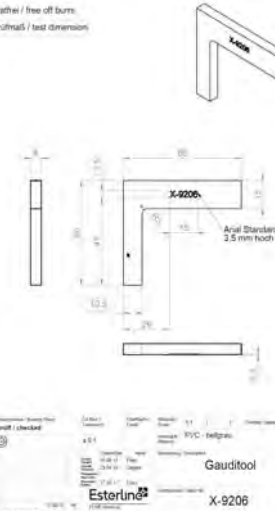
SAP #	Name	Comment	Picture
9026390	LCD-PCB Mamba Lizard Board	PCB for touch screen and LCD display	
9026391	Gaudi Gecko Module	PCB for the USB ports, bar code reader, LAN, and interconnection with the Analytical Module	
9026392	Gaudi Front USB Module	USB port under the touch screen	
9026393	Bar Code Reader	Bar code reader located on the lower edge below the touch screen	

Table 12.4-1 QIAstat-Dx Operational Module Spare Parts List

12.5 Spare Parts- Repair Center Tools

SAP #	Name	Comment	Picture
9026426	Test Mean qPCR_Align2 (Cal)		
9026427	Test Mean qPCR_Align1 (Cal)		
9026428	Plug , AM ,QSTAT		
9026429	Measuring box adapter, AM, QSTAT	Fixture for clamp pressure testing	

SAP #	Name	Comment	Picture
9026430	Test Mean Liquid Empty		
9026431	Test Mean Liquid Filled		
9026432	PT Cartridge EPS FU		
9026433	Cartridge End test/Leaktest		

SAP #	Name	Comment	Picture
9026434	MT-Cartridge		
9026435	Cartridge with contacts	Pneumatic test cartridge	2 Filter im Bereich der Stopfen vorher entfernen  2 Filter im Bereich der Stopfen vorher entfernen ESTERLIN BEWEGUNGSTECHNIK VERFAHRENSTECHNIK 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 Pneumatische Testcartridge ESTERLIN X-9737
9026436	Distance support angle		größer / free off burrs Prüfmaß / test dimension  größer / free off burrs Prüfmaß / test dimension ESTERLIN BEWEGUNGSTECHNIK VERFAHRENSTECHNIK 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 Abstandshalter Winkel ESTERLIN X-9208

SAP #	Name	Comment	Picture
9026437	Position Lifter		
9026438	Pressure plate	Part of the clamp test fixture	
9026439	Adapter force measurement	Part of the clamp test fixture	

SAP #	Name	Comment	Picture
9026444	Fixture QPCR Optic Alignment	Power distributor	
9026445	Control unit TC1-TC2-CC-TU Lever	Power distributor	
9026446	Control device for valves		
9026447	Load Cell, Force Measurement	Force measurement tool, USB connected	

Table 12.5-1 Repair Center Tools

12.6 Field Tools

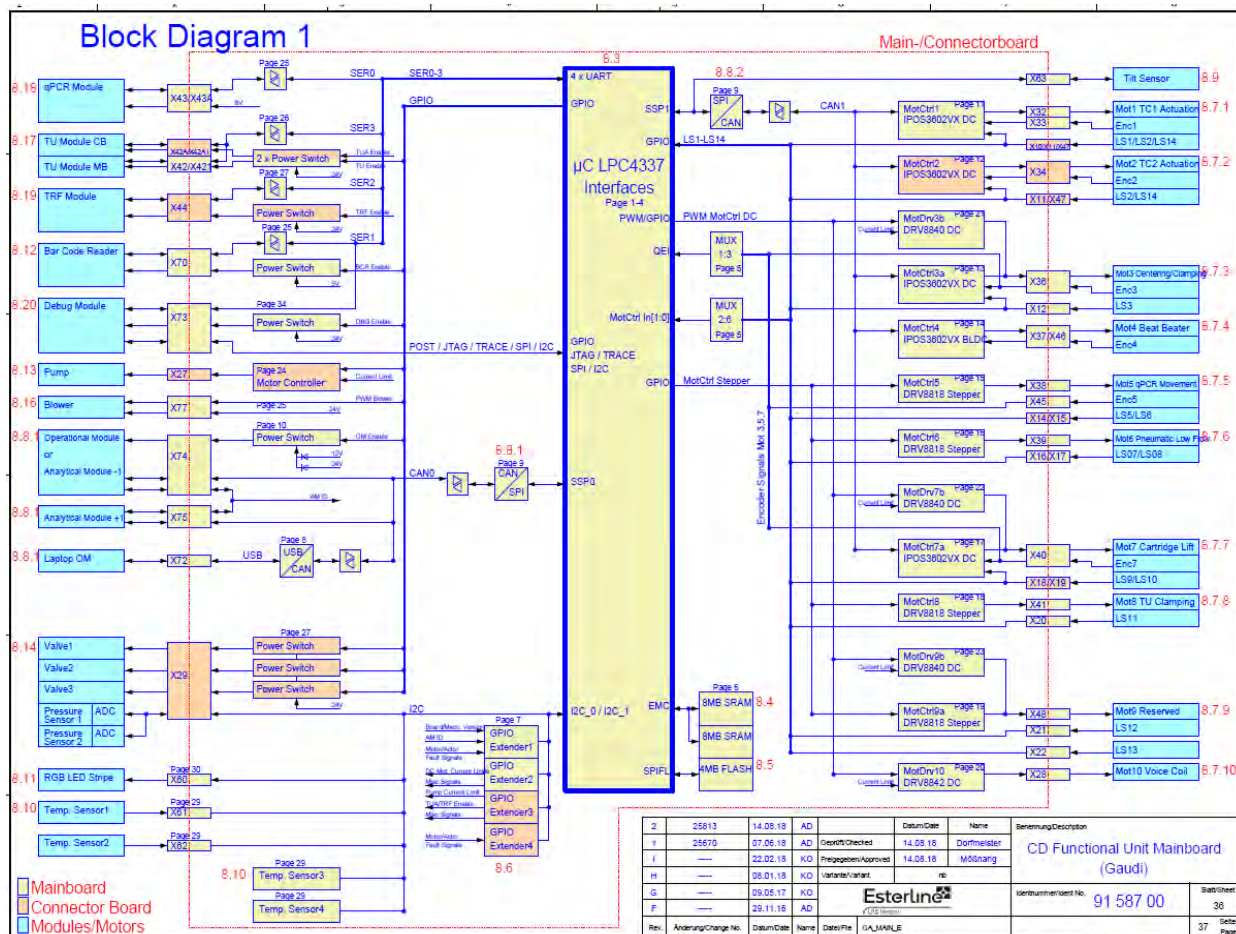
SAP #	Name	Comment	Picture
9026464	Tool-Cartridge Service, AM, QSTAT		
9026465	CAN Connection Cable	Cable CAN SUB-D 15 pin macho SUB-D 9 pin	
9026466	ACC, AM, QSTAT	ACC Cartridge	

SAP #	Name	Comment	Picture
9026467	USB to Can Adapter, AM, QSTAT	IXXAT Can Bus Adapter	

Table 12.6-1 Field Tools

13 Electronic Schematics

13.1 Main Board



Picture 13.1-1 Analyzer main board

14 ISI Notes

14.1 Purpose

The Instrument Service Information (ISI) note is an instrument history file containing important and urgent information about QIAstat-Dx. The ISI note is a way to keep everyone up to date on the changes to QIAstat-Dx between service manual releases.

14.1.1 Procedure

Global Product Support is responsible for creating and distributing ISI notes. All members of the QIAGEN support organization are requested to send comments and suggestions about QIAstat-Dx that may result in the creation of an ISI note.

The following conditions must be fulfilled to justify the creation of an ISI-Note:

- Important information that will reduce operating expenses
- Hardware/Software bug(s) affecting the system reliability

14.1.2 Format

The ISI note will always be delivered as a PDF, and will be available from the document control system.

14.2 Implemented ISI-Notes

This section lists ISI notes that have been integrated into this manual.

ISI No.	Subject	Rev.	Upg. Class	See Page	Relevant
1					

Table 14.2-1 Implemented ISI-Notes

14.3 Actual attached ISI-Notes

This section allows you to keep track of new ISI notes released after the publication of this manual.

ISI No.	Subject	N° of pages	Related to Chapter	Revision of ISI

Table 14.3-1 Actual attached ISI-Notes